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Vowel Reduction and Deletion in Italic: Effects of Stress

A dissertation submitted in partial satisfaction of the

requirements for the degree Doctor of Philosophy

in Indo-European Studies

by

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2008

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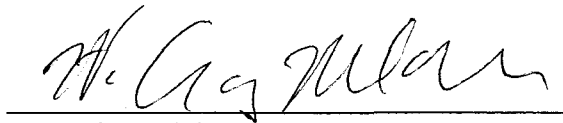
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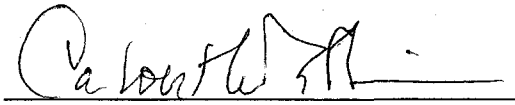
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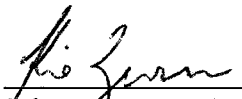
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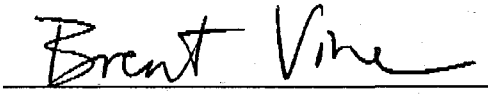
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LIST OF ABBRIVIATIONS

- Languages

Att.	Attic	OIc.	Old Icelandic
Av.	Avestan	OLat.	Old Latin
CLat.	Classical Latin	OSp.	Old Spanish
Dor.	Doric	Osc.	Oscan
Etr.	Etruscan	Pael.	Paelignian
Falsc.	Faliscan	Paleo-Umb.	Paleo-Umbrian
Fr.	French	PIE	Proto-Indo-European
Gk.	Greek	PItal.	Proto-Italic
Goth.	Gothic	Pre-Sam.	Pre-Samnite
Ion.	Ion.	Proto-Sab.	Proto-Sabellic
Ital.	Italian	Skt.	Sanskrit
Lat.	Latin	SPi.	South Picene
Lith.	Lithuanian	Umb.	Umbrian
OCS	Old Church Slavic	Ved.	Vedic
OE	Old English	Volsc.	Volscian
OFr.	Old French		

- Authors and Works

See the *Oxford Latin Dictionary*

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ABSTRACT OF THE DISSERTATION

Vowel Reduction and Deletion in Italic: Effects of Stress

by

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This dissertation is devoted to the clarification of phonetic and phonological mechanisms of vowel reduction and deletion in Italic, both synchronically and diachronically. The majority of the previous studies have focused only on cataloguing the forms before and after vowel reduction and deletion. As a result, we do not really understand the more detailed processes of these sound changes, linking inputs and outputs.

To this end, we need to capture what operating system was necessary to activate vowel reduction and deletion; and the posited system must typologically agree with general principles across languages. The knowledge accumulated by phoneticians and phonologists tells us that vowel reduction (and deletion, as its extreme form) typically appears in stress-timed languages. It is thus undeniable that the Italic languages had dominant stress accent when they underwent vowel reduction and deletion; and to judge

from the fact that these changes occurred in non-initial syllables, the initial-stress system emerges as a solid accentual rule. With this as our starting point, we can work on elucidating how these sound changes functioned or did not function on the basis of our operating system. Further incorporation of theoretical, experimental, and typological information — from the synchronic standpoint — of phonetics and phonology into the diachronic ground enables us to shed light on various aspects, which have never, or insufficiently, been elucidated.

The major suggestions of this dissertation are: 1) Latin vowel reduction occurred in multiple places within a word synchronically by means of different systems; 2) it is necessary to assume a stage of stress under the Penultimate Law in order for Latin vowel reduction to have reached completion; 3) Latin vowel deletion shifted targets throughout history, from two successive light syllables to metrically unparsed syllables; 4) though vowel reduction is alleged to have been rare in Sabellic, it did occur, especially in a labial context, even earlier than in Latin; 5) Sabellic vowel deletion potentially targeted more than one syllable of a word synchronically, with priority on final syllables over medial syllables.

1. Introduction

This work is devoted to elucidating the ways in which vowel reduction (traditionally called "vowel weakening") and vowel deletion ("syncope", in absolute word-final position "apocope") occurred, or did not occur, in the Italic languages, phenomena that have long drawn much scholarly attention: e.g., [[vowel reduction]] Lat. *cōn-ficiō* 'complete' vs. *faciō* 'make'; Lat. *dēligō* 'select' vs. *legō* 'gather'; [[vowel deletion]] OLat. *balineum* 'bath' (< **balaneom* ← Gk. βαλανεῖον) > *balneum*; Pre-Lat. **pri(s)mo-kap-s* > *prīnceps* 'chief' (as either n. or adj.); Osc. *actud* 'plead a case' (fut.impv.3.sg.) vs. Lat. *agitō*;[1] Lat. *aut* vs. Osc. *auti*; etc.[2]

Most of the previous studies have focused on cataloguing forms before and after vowel reduction and deletion. Several linguistic aspects, such as syllable structure and adjacent consonants, have been used as criteria for sorting out examples. The knowledge thereby accumulated is truly useful, but remains superficial. We do not really understand the more detailed processes of these sound changes. In other words, it is necessary to clarify mechanisms linking inputs and outputs; the massive literature on this topic has left many problems unsolved.

Before confronting the essential points, we need to capture what operating system

[1] Umbrian **aitu** (≈ **ai-t**^{o} < **ak-t*^{o} < PItal. **ag-e-t*^{o}) is traditionally connected with these forms. Untermann has a different etymology (2000: 72), appealing to the PIE root *h$_1$ai-* 'to give' and the reconstruction **aḭ-i̯e-tōd*.

[2] Vowels are only speculatively reconstructed in some cases, which are controversial and need caution: e.g., *claudō* 'close' < **klāu̯i-dō*? (Gk. Ion. κληίς, Att. κλείς 'bar; key'); see Ciardi-Dupré (1901: 206), Leumann (1977: 97), and Monteil (1979: 100), but cf. Ernout-Meillet (1985: 126).

activated vowel reduction and deletion. First, we examine the qualitative aspect of Italic accent in light of the information provided by phonetics and phonology (1.1). Second, we discuss the accent distribution rule, on which vowel reduction and deletion were based (1.2). We thereby reaffirm the necessity of the traditional initial-stress theory; arguments against it are refuted (1.3). Finally, I present an overall view of the work (1.4).

1.1. Nature of Accent and Vowel Reduction/Deletion

The qualitative nature of Italic accent, especially of Latin, has been a subject of much discussion, splitting scholars between the "German School" and the "French School" (see Leumann 1977: 244-254; less debate, however, for other Italic languages). The "German School" insists on the intensive, stress nature of the accent in Classical Latin, expanded to Pre-Classical Latin (= Old Latin) and, further, Pre-historical Latin,[3] while the "French School" argues that Classical Latin had pitch — i.e., a musical accent as Greek did —

[3] Skutsch (1913: 188) argues that the positional coincidence of verse ictus and word accent in the Pre-classical theatrical works proves the stress nature of the Latin accent (already in Lindsay 1894a: 412-14; also in Allen 1973: 153), since "ein musikalischer Accent konnte unmöglich irgendwie an den Iktus gebunden sein." Though some doubt has been cast on the existence of ictus itself in Roman poetry by, e.g., Lepschy (1962: 206-215) and Oniga (1990: 217-225), the tendency emerging from statistical works is undeniable (see also footnote 20).

Likewise, some scholars, such as Lindsay (1894a: 412) and Liénard (1977: 611-615), maintain that persistent stress nature of the Latin accent is justified by the regular coincidence of ictus and word accent in the fifth and sixth feet in hexameter, as is suggested by Sturtevant (1919b: 385) with statistical data, and again more explicitly by Sturtevant (1940: 184-189), who considers pitch accent borrowed from Greek. Allen (1973: 154, 1978: 92) also argues that the fifth foot mirrors normal spoken stress, and Sihler (1995: 241-242) similarly considers that "vigorous stress accent" lies in that position, based on the distribution of ictus. The fact that the frequent cadence ∪∪—∪# in Homer is rarely found in Latin, except as blind Hellenic imitation (Liénard 1977: 617-619), may also be regarded as a purely Latin profile emerging in the fifth and sixth feet. Yet, on the other hand, Leumann (1977: 250-253) suggests that this perhaps originated only from some poets' aesthetic taste, especially Virgil's, stating that "die Feststellungen über eventuellen beabsichtigten Zusammenfall von Wortton und „Versiktus" sind nach meiner Meinung nicht schlüssig" (1977: 253). Laurand is also cautious (1938: 137-138).

and that this phonological feature could also be projected back to earlier periods.[4] As Mignot (1975: 422) says, phonetic science in the nineteenth century rigidly distinguished stress accent and pitch accent, and scholars tended to pick one of them for a language, as the German School and French School did. But progress in the field illustrated that these two types of accent can coexist based on different phonetic parameters, amplitude (intensity) and frequency, and cross-linguistically, pitch accent often accompanies some stress in its high peak, as stress accent may induce some rise in pitch (see Allen 1973: 74, 94-95 and Pulgram 1975: 87, 113 et passim).[5] Thus, Mignot's remark (1975: 422): "l'opposition des thèses française et allemande perd beaucoup de son importance."

However, though they do not exclude each other, one is relatively more essential than the other. On the evidence of recent typological studies provided by phoneticians and phonologists, it is most likely that the Italic accent had dominant stress nature in the period when vowel reduction (and deletion as its extreme form) occurred — a conclusion that partly favors the "German School". Lindblom (1963: 1773) states that "[v]owel reduction is said to be a characteristic feature of languages with heavy stress, such as, for instance, English and Swedish." Similar remarks are found in recent works by theoretical phonologists, e.g., Crosswhite (2001: 53, 2004: 223): "prominence-reduction appears to

[4] The main basis on which the "French School" relies is in the descriptions made by old grammarians, such as Cicero and Quintilian, which are, however, generally taken as imitative practice of Greek grammar or imperfect observation (or perhaps arbitrarily interpreted by modern scholars; see Laurand 1938: 136), and so disguise the facts (but cf. Pedersen 1906: 232); for an evaluation on these lines, see, e.g., Lindsay (1894a: 407-408), Brugmann (1897: 975), Palmer (1954: 211), Lepschy (1962: 200-204), Pulgram (1975: 77, 82 et passim), Leumann (1977: 237), Laurand (1938: 137-138), Liénard (1977: 609), Allen (1978: 84), and Sihler (1995: 241).

[5] As already pointed out by Sturtevant (1919a: 244) and Laurand (1938: 136-137). Liénard (1977: 615 n. 50) also addresses that "Il est de toute façon admis qu'un accent n'est jamais purement intensif ou purement musical."

occur only in stress-timed languages (or dialects)"[6] and Flemming (2005: 18): "If stress is marked only by pitch movements..., no vowel reduction is expected." Recall also Meillet's description (1900: 169) that any kind of vowel in Latin was subject to vowel reduction.[7] Despite the need of some elaboration and restatement, his remark is somewhat true. See also his other statement (1918: 107) that "[l]es langues où les voyelles inaccentuées s'altèrent sans considération du degré d'ouverture sont celles où l'accent est, de manière dominante, un accent d'intensité: l'allemand, l'anglais, le grand russe".[8] Moreover, as Hayes (1995: 49) suggests, languages with pitch accent may have contrasts of accentual pattern within the syllable: for a heavy syllable, two kinds of prominence {rising/falling}, as seen in Ancient Greek, Lithuanian, and so forth. There seems to be no such linguistic trait in Latin, or in Italic in general.[9]

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[6] Prominence-reduction is "based on the desire to avoid particularly long or otherwise salient vowel qualities in unstressed positions" (Crosswhite 2001: 34, 2004: 204). Given such a state of affairs, the occurrence of unstressed [a], a highly sonorous vowel, is remarkably disfavored. This type of reduction is distinguished from "contrast-enhancing reduction," whose object is "elimination of noncorner vowels, especially mid vowels" (Crosswhite 2004: 192); unstressed [a] is then favored. Parker (1988: 234-236) claims different patterns of vowel reduction, like *i, u* > *e* and *a* > *e, e* > *i*. Typologically, his hodgepodge of various kinds of vowel reduction is hard to justify.

[7] Note that there is some difference in frequency among vowels targeted by vowel reduction. Though the "apophonie" *a* ~ *i*, *a* ~ *e*, and *e* ~ *i* is frequently observed, reduction of back vowels like *o* and *u* is relatively limited. See also Oniga (1990: 210-217). Rix (1966: 162) also suggests that *o*- or *u*-vocalsim is often restored by analogy. Juret (1919: 94) says that "les composés dont la voyelle radicale est *ŏ, ŭ*, ont presque toujours conservé ces timbres, sans doute parce que la soudure des éléments de ces composés a été tardive et s'est faite alors que la transformation de ces timbres n'était plus exigée : *invocō, colloquor, ēducāre*, etc." Note, however, that the history of attestation and the date of creation are two different things; Juret's view is thus circular. The infrequency in vowel reduction of *o* and *u* will be discussed in Chapter 2.

[8] See also Laurand (1938: 135-139). In contrast to syncope, Skutsch (1913: 194) ascribes vowel reduction to musical accent, which is without foundation.

[9] As to the use of the terminology such as acute, grave, circumflex, by Roman grammarians, critical comments are provided by Lindsay (1891: 373), Pulgram (1975: 92), and Allen (1978: 83-84). But cf. Pisani (1930: 169).

Schmid (1954: 33 n. 1) argues that anaptyxis is a feature of languages with stress accent (Osc. **aragetud** 'money' = Lat. *argentō*, Osc. **teremníss** 'boundary stones' = Lat. *terminibus*). But cf. Hall's work (2003)

This does not imply total exclusion of "the presence of associated pitch-phenomena" (Allen 1969: 193; already in Brugmann 1897: 975). Pitch accent emerges in the historical scenario claimed by Niedermann (1953: 11) from the Literary period (from ca. 250 B.C.E.) onward[10] based on ancient grammarians' descriptions, though Allen (1973: 151-154) accuses them of slavish misapplication of Greek principles, thus assuming the maintenance of stress feature throughout the history of Latin. Leumann (1977: 248-254) also proposes that pitch accent became dominant, accompanied by stress accent, in Literary Latin (cf. the reverse in Lindsay 1894a: 408).[11] From a sociolinguistic standpoint, Kent (1932: 66), Pulgram (1975: 118-122), and Deroy (1981: 227-228) argue that stress accent was subsumed by the "high-style" Greek pitch accent during the Literary period, though it survived in the form of vulgarism.[12] Vowel deletion sporadically continued in the Literary period, which may imply a predominance of stress accent over pitch accent; but it is hard to know how to evaluate this.

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on vowel intrusion, especially Chapter 5 concerning Hocank (Winnebago), a Siouan language with *pitch* accent.

[10] I adopt a dichotomy between "Pre-Literary (Latin)" and "Literary (Latin)" (cf. Clackson 2004: 790). Note, however, that this distinction is in fact misleading, as Liénard (1977: 602 n. 9) correctly points out, since there should have been some forms of literary works, like Saturnian verses before, ca. 250 B.C.E. "Literary Latin" will be utilized for the sake of convenience, in order to name the period from Livius Andronicus onwards, including more familiar appellations of stages like "Old (Pre-Classical) Latin", "Classical Latin", and "Post-Classical Latin".

[11] Both Niedermann (1953: 11) and Leumann (1977: 246-248), on the other hand, suppose stress accent on initial syllables in Pre-Literary period. Maniet (1957: 28) notes that the nature of accent is a matter of degree, and is varied depending on factors like linguistic substrata and social ranks.

[12] On the later resurgence of stress nature over pitch in the development to the Romance languages, see also Biville (1990: 19).

1.2. Location of Accent: The Initial-stress Rule in the Italic Languages

From the discussion above, it follows that vowel reduction and deletion in Italic were undoubtedly triggered by stress accent, and took place productively in unstressed syllables.[13] A question, then, arises as to which syllables were stressed or unstressed in words. It demands a diachronic perspective, since the accent distribution was determined by different rules in separate periods.

Leumann (1977: 246) relates that since Verner's Law was claimed in 1877, people have tried to trace several sound changes in Latin (maybe in Sabellic as well) back to an accentual system reconstructed for Proto-Indo-European (PIE). In the parent language, the position of accent is morphologically connected with the ablaut among root, suffix, and ending, and the accent may fall on any syllable of the word according to PIE morphophonemic rules. Though Leumann reports that such attempts mostly failed, it is now believed by some scholars that Proto-Italic may have kept at least some remnants of the PIE accentual system.[14] For instance, Vine (2004: 623, 2006a) demonstrates that

[13] Oniga (1990: 215 et passim) objects that the vocalic alternation in Latin discussed here was not motivated by accent but a morphological factor, claiming that only in the syllable after a morpheme boundary (e.g., after a prefix or a reduplicated syllable) is the vowel eligible to qualitative change, as in the compound *con+ficiō* (cf. *cale#faciō*, in which two elements are not so univerbated as to be qualified for the vowel change) (1990: 215-216). However, his view is not satisfactorily grounded. If such a development had nothing to do with the phonetic feature of accent, we could have utterly different types of sound changes, for example, *i* > *e* or *u* > *o*. Based on the behavior of unstressed vowels, there is no doubt that stress (or more precisely, the absence of stress) played a role in our cases.

[14] Though Brugmann (1897: 971) considers his evidence insufficient, Hirt (1895: 42-43) already suggests the same idea, adducing the following examples: **-mi*, **-si*, **-ti* > *-m*, *-s*, *-t*; *tot* 'so many', *quot* 'how many; as many as' (≈ Ved. *táti*, *kátí*); *per* 'through' (≈ Ved. *pári*, Gk. πέρι); *et* 'and' (≈ Gk. ἔτι) vs. *rūre* 'in the country'; *pede* (≈ Gk. ποδί).

*-ou̯- > -au̯- ("Thurneysen-Havet's Law") in Latin was sensitive to PIE accentuation.[15] Some scholars, such as Meiser (1998: 74) and Meier-Brügger (2002: 158), ascribe the retention of the final syllable of Latin *pede* 'foot' (abl.sg.) to the PIE accent (< **ped-í*, and also *nocte* ≈ Gk. νυκτί 'night', *ante* ≈ Gk. αντί 'opposite'; cf. apocope in unaccented verbal endings **-mi*, **-si*, **-ti*, **-nti* > *-m*, *-s*, *-t*, *-nt*). In addition, Dybo's Law possibly belongs here: e.g., **u̯īró-* > *vĭr* 'man' (see Meiser 1998: 75). Yet, what we know about this period is rather scant; vowel deletion seems marginal and vowel reduction is hardly identified.

The accentual system of Literary Latin, the so-called Penultimate Law[16] (or

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[15] The very early application of this change in Italic, prior to *-eu̯- > -ou̯-*, is also pointed to by Tadani (2005: 91-93), who however does not mention the name of the law.

[16] The terminology is in fact misleading, as Allen (1983: 1) points out; for if the penult is syllabically light, the accent is shifted to the antepenult. Regarding the possible participation of the ultima in accentuation, see the same work of Allen and Kent (1932: 66); according to Allen, the overweight of the ultima may have rendered words oxytone, as in *audĭt*, *fūmăt*, *nostrās*, *tantŏn*, *illĭc*, and *illínc*, which come from *audĭvit*, *fūmăvit*, *nostrātis*, *tantŏ-ne*, *illĭce*, and *illínce*, respectively (1983: 2; see also Maniet 1957: 158 and Allen 1978: 87). But as another factor, paradigm leveling to keep the accent on the same syllable may have restricted words from becoming oxytone (Allen 1983: 8-10). Leumann (1977: 238-240, 601), on the other hand, suspects that the major part of the description by ancient grammarians concerning the existence of oxtyones is but their arbitrary invention, or inaccurate observation; see also a similar comment by Lindsay (1894a: 410). Leumann also takes another type of divergence from the Penultimate Law as dubious, as with *trĭgintā* (Consentius 5.392.4 K). But some forms from Romance languages prove antepenultimate accent in the higher decads: e.g., **quadrá(g)inta* '40' > CIL XIII 7645 **QVARRANTA** > Ital. *quaranta* and OFr. *quarante* (see Weiss 2005: 396, and also Lindsay 1891: 407-408). Menéndez Pidal (1973: 243-244) notes that while Spanish and Portuguese usually continue Classical Latin accent (OSp. *veínte*, *treínta*), other Romance dialects reflect one of Vulgar Latin, assuming a historical development something like *-ā(g)intā* > *-āíntā* ("Classical" type) > *-ăintă* ("Vulgar" type) > *-anta*. As Menéndez Pidal adds, the "Vulgar" type encroached in Iberia as well, appearing in Catalan *seixanta*, *vuytant*, etc., Leonese *cinquanta*, *novanta*, etc., Aragonese *quaranta*, *xixanta*, etc.; for the vulgarity in 'twenty' and 'thirty', see Portuguese *vinte* and Portuguese Leonese *trinta* (< **vī́-*, **trī́-*). For further implications of the evidence, see footnote 85 in Chapter 2.

Prosdocimi (1986: 603) argues that an accentual system with no morphological motivation should be fixed in location "per sua natura," and that languages like Latin where the syllabic weight of penults determines whether they will take accent or forward it to antepenults are not normal. Typologically, however, the accentual pattern found in Latin is also seen in a number of languages (see Hayes 1995: 92). His rigid attribution of accent to penults at some point in the history of Italic is not well-grounded, and undermines his whole argument.

three-syllable rule), is the best understood part of the accentual history of Italic: stress falls on the penult if it is syllabically heavy, otherwise on the antepenult in words of more than two syllables; monosyllabic words bear stress on their only syllable. This rule obviously does not continue the PIE accentual rule, as is pointed out by many scholars, such as Dietrich (1852: 545), Sommer (1914: 84), Pisani (1930: 151-152), Leumann (1977: 246), Prosdocimi (1986: 613), Sihler (1995: 59), and Meier-Brügger (2002: 157-158).[17] It is securely established by evidence internal to Literary Latin, that is, testimony provided by ancient grammarians and later manifestations in the Romance languages. Under this system, vowel deletion was relatively more productive than when the PIE accentual rule was still operative. But many of the forms that underwent vowel deletion cannot be explained by this rule; as a matter of fact, most of them appear already syncopated when the Literary period started. Vowel reduction also seems to have been mostly completed (see more details in 1.2.1).

It follows that there was an intermediate stage between PIE and Literary Latin, where vowel reduction and deletion were productively occasioned. Indeed, scholars have long attempted to elucidate what the accentual system looked like in Pre-Literary Latin. The Sabellic dialects have no direct offshoot themselves, as Latin has the Romance languages; hence the evaluation of their limited number of linguistic remains is the only way to consider their accentuation system. This difficulty may be partly reduced by application of findings in Latin to the Sabellic context.

[17] Nyman (1983: 34) argues for a certain similarity of accentual rule between PIE and Latin, so as to suggest that the accent of Pre-Literary Latin is relatively free in terms of its location (see also Oniga 1990: 203-4); but the evidence he provides is mostly dubious.

The most remarkable feature of vowel reduction and deletion in Italic is that the overwhelming majority of examples have those changes in non-initial syllables, as is observed by many scholars, such as Thurneysen (1883-1885: 313), Meillet (1933: 56, 133), Sturtevant (1940: 28, 177), Palmer (1954: 211), Benediktsson (1960: 281), and Jiménez Zamudio (1980: 149).[18] This fact strongly suggests that initial syllables bore stress at some point in the history of Italic, thus triggering vocalic changes in non-initial unstressed syllables. Indeed, the most common idea among scholars is the initial-stress theory,[19] initially supported by the "German School"; according to this, the word accent of Pre-Literary Latin and Sabellic was placed on the initial syllable of each word.[20] In

[18] Ballester (1990: 41) claims that the vocalic alternations in question are marginally seen in initial syllables as well, presenting the following forms and their reconstructions: "¿ —*bĭtūmen* < **betu-*; *cĭcātrix*, cf. **cecare*?; *cĭcĭndēla* < **cecand-*?; *cĭcōnia*; *cīsōrium* < **caesōrium*; *ĭmāgō* cf. *aemulus*?; *lĭdūna* cf. *ledō*; *Nĕptūnus* < **năp-*?; *sĭmītū* < **semeitus*?; *lŭcūna* = *lacuna*—?"; but none of the examples are well-grounded enough to support his argument. Some are based only on vague etymologies or late attestations, and others can be explained by assimilation to the vocalism in following syllables. The form *imāgō* currenly explained as the zero-grade of the root (**h₂eim-*), equated with Hitt. *ḫimma-* « substitut »; see Martzloff (2007: 125).

[19] This theory is said to have originated from Dietrich (1852: esp. 546), though his definition is not purely phonological, but rather morphologically motivated (see also Möller 1922: 169 n. 1, 3). The concept of "initial stress" seems to have been "mistakenly" deduced from Dietrich's theory by someone, maybe Thurneysen (1883-85: 313); see Pederson (1895-96: 67; 1905: 338 "die Thurneysen'sche hypothese"), and also Oniga (1990: 203). Though Corssen is honored as the author of the theory by Lindsay (1891: 406; 1894a: 408), Corssen (1870: 892-906) makes in fact no contribution to its establishment, rather claiming a different (less binding and unnatural) system. Since Dietrich (or Thurneysen), it has acquired so many followers that it has repeatedly appeared in most of the standard (hand)books of Latin: e.g., Lindsay (1894: 157-159), Sommer (1914: 84-86), Palmer (1954: 212), Leumann (1977: 246-48), Allen (1973: 93-94, 133, 151; 1978: 83), Sihler (1995: 239), Meiser (1998: 53, 66, et passim), Baldi (2002: 269); non-Latin Italic: e.g., von Planta (1892: 589), Conway (1897: 495), Buck (1928: 101), Bottiglioni (1954: 23), Poultney (1959: 30), Meiser (1986: 32-33), Wallace (2006: 16); and of Indo-European: e.g., Hirt (1895: 41), Brugmann (1897: 213), Vineis (1994: 298-299), Villar (1997: 476), and Meier-Brügger (2002: 157-158); as well as in specialized studies of Latin and Sabellic prosody: e.g., Thurneysen (1908: 240-42), Skutsch (1913: 187), Götze (1923: 80), Kent (1931: 177, 188-89), Sturtevant (1940: 177-178), Kuryłowicz (1958: 381), Benediktsson (1960: 281-283), Mignot (1975: 425-26), Liénard (1977: 609-611), Allen (1983: 2), Deroy (1981: 223-224, 228, 232-234), Biville (1990: 19-20), and Parsons (1999: 133-134).

[20] Besides vowel reduction and deletion, some scholars appeal to metrical evidence. The podic ictus in iambo-trochaic verse has been regarded as evidence for initial stress (but cf. Pedersen 1905: 337-338). Poets of Pre-Classical Latin, such as Plautus and Terence, are said to have tried to make word accent and ictus coincide with each other. This traditional view is found in many scholarly works, as in Lindsay

this hypothesis, Latin is assumed to have changed accentual placement before (ca. 4th c. B.C.E., by Leumann 1977: 246; cf. "im frühen dritten Jahrhundert" in 1977: 253) or around the beginning of the Literary period (ca. 250 B.C.E., by Baldi 2002: 269). A possible shift in accent placement in Sabellic will be briefly examined in Chapter 5. Initial stress might possibly be interpreted as a Proto-Italic feature, as is assumed by Brugmann (1897: 971), Kent (1932: 65), Sturtevant (1940: 178), and Benediktsson (1960: 281). In Brugmann's view, initial stress developed as a secondary accent under the PIE accentuation, the latter of which was then abandoned. Skutsch (1913: 199) does not accept the Proto-Italic origin, because it cannot explain the difference of accentual effects between Latin (mostly vowel reduction) and Sabellic (mostly vowel deletion).

Some scholars have suggested that linguistic contact with a neighboring language (or languages) with strong initial stress secondarily induced Latin and Sabellic to gain the same or a similar stress system. Among others, Etruscan is often adduced as the source of

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(1894a: 158, 1894b: 408-409), Sommer (1914: 86), Sturtevant (1919a; 1940: 181-186), Allen (1978: 83), and Palmer (1988: 213). For the history of research on this issue, see Corssen (1870: 892-893) and Fortson (1996: 9-12). As noted in footnote 3, theories based on ictus are not entirely impeccable (see Ernout 1929: 112 and Gratwick 1993: 59-60). Yet, among the arguments for such theories, despite Corssen's objection (1870: 893, 906), the fact that forms of ∪∪∪⏑#, as is *facilius* (*facilis* 'easy'), *sequiminī* (*sequor* 'follow'), *ceciderō* (*cadō* 'fall'), *mulierem* (*mulier* 'woman'), *voluerat* (*volō* 'want'), *capitibus* (*caput* 'head'), *miseria* 'distress', *meminerō* (*meminī* 'remember'), *parietem* (*pariēs* 'wall'), *hariolum* (*hariolus* 'soothsayer'), are often so arranged in iambo-trochaic verse for ictus to be posed on the initial syllable, is noteworthy. The reason seems to be that the word accent is situated there, i.e., *fácilius*, etc.; see Lindsay (1892a, b). Sturtevant (1919a), following Lindsay's formulation (1894: 158) on the *fácilius*-type accent, conducts a statistical study on harmony between accent and ictus. The accentuation has been interpreted as a relic of initial-stress period (e.g., Brugmann 1897: 973-974), eliminated toward the Classical period; this, however, needs some restatement (see 3.3.2).

Alliteration has been invoked as evidence for initial stress by many scholars; more recently, e.g., Meiser (1998: 53) and Meier-Brügger (2002: 158-59). Even Havet (1889: 13), who disagrees with the initial-stress theory, employs alliteration as evidence for his theory of intensive pronunciation of initial syllables (see 1.3.1). But Meillet and Vendryes (1948: 115) are not convinced by such a correlation, saying that "l'allitération n'est pas en latin un procédé essentiel comme elle l'est dans la plus ancienne versification des Irlandais, des Anglo-Saxons et des Scandinaves"; nor are Pulgram (1975: 95 n. 43) and Oniga (1990: 226). See also Sturtevant's citation (1940: 180) of Enn. *Ann.* 104Sk: *O Tite tute tibi tanta tyranne tulisti*; then, "It is perhaps open to question whether this feature of early Latin was a survival from the time of the prehistoric initial stress or whether it was still favored by the accent of Ennius' own time."

such a system; in Etruscan, vowels in unstressed position are believed to have all merged into [ə], represented graphically in a haphazard way from the first half of the fifth century, and later syncopated: e.g. Gk. Ἀχιλλεύς → *Axale* ~ *Axile* ~ *Axele* ~ *Axule* vs. *Axle*.[21] However, the discussion does not seem to surpass the level of speculation. Rather, in the case of Sabellic, influence from Etruscan is almost inconceivable chronologically, as will be discussed at some length in 4.1.3.

1.2.1. Vowel Reduction in Latin

Latin exhibits a wide range of vowel reduction in non-initial unstressed syllables, which requires a fairly detailed description. Chronologically, most scholars agree that it began later than the first layer of Latin attestation and predated the Literary period (since ca. 250 B.C.E.). Though its absolute chronology cannot be easily specified, there are a few inscriptions that are of some interest for this purpose. CIL I^2 4 ("the *Duenos* inscription"; Rome, ca. 600-575 B.C.E.) provides the forms **IOVESAT** '*iurat*, swears' and **FECED** 'made', instead of ****IOVISAT** and ****FECID**, respectively. This is also the case of **IOṾE[S]A** '*iura*, oaths' in the Corcolle Altar fragments (ca. 500 B.C.E.), based on Vine's interpretation (1993: 76-78), followed tentatively by Hartmann (2005: 167). The form **FHE : FHAKED**

[21] For general descriptions of the Etruscan accent, see De Simone (1970: 91-92), Leumann (1977: 247), Cristofani (1991: 42), and Rix (2004a: 949). Concerning other hypotheses, with reference to different sources, such as Germanic and Celtic (e.g., Thurneysen 1883-1885: 313), see a brief survey by Leumann (1977: 247-248). Similarity of Latin with Germanic, as well as, Etruscan in terms of accentuation is already mentioned by Dietrich (1852: 545). Skutsch (1913: 188) reports that connection to Celtic and/or Germanic was generally abandoned. He (1913: 190, 198-199) then strongly argues for Etruscan influence over the Italic languages (cf. Pisani 1930: 150, who assumes the opposite direction), though the idea is refuted by De Simone (1970: 92) on the basis of the different patterns of subsequent sound changes.

'*fecit*, made' in the disputed Praenestine Fibula can perhaps be added here (i.e., neither **FHE : FHIKID nor **FHE : FHIKED). Further, MAMARCOM (Gnade and Colonna 2003; Satricum, 575-525 B.C.E.)[22] and MAMARTEI in the CIL I^2 2832a ("Lapis Satricanus"; Satricum, ca. 500 B.C.E.), contrasting with Classical *Māmertīnī*, etc., may indicate that reduction in closed (and perhaps diphthongal) syllables was a historical event.[23]

More important is to acknowledge that vowel reduction has its origin in Pre-Literary Latin, which means that the Penultimate Law is in principle irrelevant to the instantiation of this vocalic change (yet, I will propose that this accentual rule played a certain role in finalizing the whole process of vowel reduction; see 2.4). True, as Nyman (1983: 33) points out, arguing against the initial-stress theory, the majority of the cases of vowel reduction can also be explained by accent on initial syllable assigned regularly even within the Penultimate Law, because trisyllabic words with a light penult are very common: e.g., **pér-fakit* > *pérficit* 'finish', **kón-tenet* > *cóntinet* 'hold', **ánamos* > *ánimus* 'mind' (cf. Gk. ἄνεμος 'wind'). But we have a group of forms in which heavy medial syllables underwent vowel reduction; they are difficult to explain under the Penultimate Law (as already pointed out by Corssen 1870: 896): e.g., *aestimō* (< **ai-*) 'estimate' vs. *exīstimō* 'id.' (< **eks-aistimō*); *Tarentum* [toponym] (< **Tárantum* ← Gk. Τάρας, Τάραντος). Initial stress may better account for such vowel reduction in the following heavy syllables.

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[22] I am grateful to G. Rocca for informing me of this form.

[23] The retention of original vowels in these cases could perhaps only be graphic, concealing the real phonetic values. Phonemically non-neutralizing vowel reduction (cf. 2.2) is difficult to identify from the spelling alone. However, as there is no other linguistic symptom of such a reduction, it is better not to entertain speculative thoughts.

It is also less likely that the change took place under the PIE accentuation; though some sound changes, such as apocope, may have been realized by that system, **pedí*, for instance, never became ***pidí* (> ***píde*). Thus, vowel reduction is regarded as a linguistic event in a period not much earlier than Literary Latin.

With initial stress as a conditioning factor, vowel reduction and deletion followed somewhat regular patterns which have been repeatedly described in many handbooks and relevant articles. Different vowels in the root of simplex verbs and prefixed verbs (the latter's derivatives as well) are frequently cited, as has already been done at the beginning of this work. In the case of *faciō* 'make' and its prefixed verb *cōn-ficiō* 'complete', we notice the vocalic alternation *-a-* ~ *-i-*, one type of the so-called "apophonie latine," in French-School terminology. Within the initial-stress theory, this phenomenon could be historically explained as follows: *cōn-ficiō* begins as something like **kón-fakiō*, with a stress accent on the first syllable; the vowel *a* in the following syllable is weakened to *i*; later, the accent is transposed to the antepenult in accordance with the newly established Penultimate Law, and *-ficiō* remains as a relic of the earlier stage. There are many other examples of simplex/prefixed verbs that can be explained by assuming the same type of diachronic process: e.g., *agō* 'lead' vs. *ex-igō* 'drive out' (< **éx-agō*); *cadō* 'fall' vs. *ac-cidō* 'fall upon' (< **ád-cadō*); *capiō* 'take' vs. *in-cipiō* 'I take in hand' (< **ín-capiō*); *legō* 'collect' vs. *dē-ligō* 'single out' (< **dḗ-legō*); *rapiō* 'snatch' vs. *sur-ripiō* 'snatch away' (< **súb-ripiō*); *arguō* 'declare' vs. *rederguō* (~ *redarguō*) 'disprove'; *scandō* 'climb' vs. *inscendō* 'climb in' (< **en-skandō*), etc. A vocalic alternation seen in Pre-Classical Latin subjunctive *tagam* vs. *at-tigās* (< **th₂ĝ/g-*, cf. *tangō* 'touch') most

likely belongs here; see LIV (2001: 616-617). We certainly have quite a few examples that exhibit no vowel shift: e.g., *edō* 'eat' vs. *comedō* 'eat up'; *gemō* 'groan' vs. *ingemō* 'groan over'; *medeor* 'heal' vs. *remedium* 'remedy'; *petō* 'seek' vs. *appetō* 'strive for'; *veniō* 'come' vs. *subveniō* 'come to help', etc., besides doublets with or without a reduced vowel: e.g., *arguō* 'declare' vs. *rederguō* 'refute' (cf. *redarguō*); *canō* 'sing' vs. *occinō* '(of birds) to break in with a song' (cf. *occanō* '[of trumpets] to interpose a call'), *occentō* 'sing at' (cf. *occantō*); etc. It might seem that there is analogical retention of the root vowel of simplex verbs.[24] But restrictions on such an operation will be provided in 2.4.

Reduplicated perfect forms can also be taken to show the same vocalic alternation. This could also be explained by supposing a stress accent on the initial syllable, whence vowel reduction: *canō* 'sing' vs. *cecinī* < **ké-kan-aḭ*; *cadō* 'fall' vs. *cecidī* < **ké-kad-aḭ*; *meminī* 'remember' (< **mémon-*); *pangō* 'fasten' vs. *pepigī* (< **pépag-* < **-ph$_2$ĝ-*); etc. The forms such as *dedī* (*dō* 'give') and *stetī* (*stō* / *sistō* 'stand [intr./tr.]') are also formed by reduplication, but have only two syllables, one syllable less than the other forms above. However, when they are prefixed, the whole number of syllables is increased. Given that the stress accent falls on the prefix, the vowel in its following syllable that is originally reduplicant is normally reduced: *perdō* 'destroy' vs. *perdidī*; *perstō* 'stand firmly' vs.

[24] This view is repeated in the literature, such as Juret (1919: 5), Niedermann (1953: 3), and Biville (1990: 9-10). But Oniga (1990: 212) claims that "si suole affermare che i composti non apofonici sono tutti formazioni recenti o ricomposizioni analogiche, ma queste affermazioni sono in molti casi sostenute solo dall'ipotizzata legge fonetica, per cui il ragionamento è circolare," adducing the word *accentus*, which he argues was formed in a relatively late period.

As Lindsay (1891: 407) indicates, accent seems to have shifted to the stem in Vulgar Latin: e.g., *recípit* 'hold back' (Ital. *riceve*, Fr. *reçoit*), *dēmóra* (Ital. *dimora*, Fr. *demeure*). Thus, Lindsay argues that "[w]ith this we may connect the tendency in the spelling of post-classical Inscriptions and of our earliest MSS. to restore the Vowels in Compound Verbs to their undecayed form, e.g. *consacro*, *compremo*." In Chapter 2, I will claim that there is no need to wait until Vulgar Latin to explain the vocalic restorations.

perstitī; *resistō* 'resist' vs. *restitī*; *trādō* 'deliver' vs. *trādidī*; etc. Note that the formation of the perfect by means of reduplication is no longer productive in Latin, so the number of this type of example is quite limited.

In addition to verbal categories, the paradigm of consonant-stem declension can also be regarded as showing vocalic alternation due to initial stress. In the paradigm, the number of syllables in the nominative singular is different from that of other case forms, which motivates vowel reduction. That is, when a nominative singular consists of two syllables, other case forms have more than two. Therefore, provided that a stress accent falls on initial syllables, vowels in the following syllables in non-nominative cases become reduced: nom.sg. *caput* 'head', gen.sg. *capitis*; nom.sg. *manceps* 'legal purchaser', gen.sg. *mancipis*; nom.sg. *pecten* 'comb', gen.sg. *pectinis*; etc. This does not take place in all consonant-stem nouns; nom.sg. *anas*, gen.sg. *anatis* 'duck'; nom.sg. *seges*, gen.sg. *segetis* 'cornfield'; etc. It is likely that such cases involve analogy with the nominative singular form or an assimilative effect caused by the vowel of the initial syllable, as in *alacer* 'cheerful', *alapa* 'box on the ear', *calamitās* 'disaster', *elementum* 'element(s)', and *vegetus* 'lively'.[25]

These are categories that productively show vowel reduction. There are some other sporadic cases, such as prefixed nouns or adjectives and univerbated prepositional phrases, whose reduction process is parallel to that of the prefixed verbs: e.g., *medius*

[25] This assimilation process is presented by many works, such as Juret (1919: 94, 104), Kent (1932: 99-101), Leumann (1977: 81, 100), and Biville (1990: 10). In the manuscripts of Plautus and Cicero, we have *anites* (*Capt.* 1003) and *anitum* (*N. D.* 2.124), respectively, as Biville notes. The form *alacer* must also have had variants like **alicer* and **alecrem*, which were taken over by the Romance languages, as in Sardinian *allirgu* (< **-i-*), Old French *aliegre*, Italian *allegro*, Spanish *alegre* (< **-e-*). See more details in Juret (1919: 105) and Biville (1990: 11).

'middle' vs. *dīmidius* 'half'; *datus* (*dō* 'give') vs. *-ditus*; *peregrī* 'abroad' < **per* + **agro-*; *prōdigium* 'portent' < maybe **-ag-* (Brugmann 1897: 221, Ernout and Meillet 1985: 538); *inimīcus* 'unfriendly' vs. *amīcus* 'friendly'; *facilis* 'easy' vs. *difficilis* 'difficult'; *ēminus* 'at a distance', *comminus* 'hand to hand' (cf. *manus* 'hand'); *prōtinus* 'straightforwards' (~ *prōtenus*; cf. *tenus* 'reaching to'); OLat. *propitius* 'favorable' < **pró-petios* ← *petō* 'seek' (or *prope* 'near'); *assiduus* 'persistent' < **ád-seduos* (cf. *admodum* 'up to the measure', *affatim* 'sufficiently', *advena* 'newcomer'); *locus* 'place' vs. *īlicō* 'on the spot' (< **en stlokōd*); *sēdulō* 'diligently'; *dēnuō* 'again'; *profectō* 'actually'. Some nominal derivatives and compounds are also added here: e.g., *genitor* 'father' (γενέτωρ, Skt. *janitar-*); *capitālem* 'mortal' < **kaput-* (cf. the archaized **CAPVTALEM** in CIL I^2 581 "Senatus Consultum de Bacchanalibus" [= "SCdB"]; see Wachter 1987: 295-296); *hospitem* 'host' < **hosti-potem*; *Iuppiter* [theonym] vs. *pater* 'father'. Further, some cliticized forms: e.g., *quīlibet* 'no matter who/what' < **-lubet* (see Havet 1889: 16 n. 2); *ita* 'thus' vs. *itidem* 'in the same way';[26] *undique* 'everywhere' vs. *unde* 'whence'. More importantly, reduction of the thematic vowel is remarkably common in the third conjugation: e.g., *agite*, *agitō* 'drive' vs. Gk. ἄγετε, ἀγέτω.

Further, borrowings from Greek form a massive group of examples of vowel reduction:[27] ἀγχόνη → Lat. **ángona* > *angina* 'angina'; Ἀκράγας → Lat.

[26] Cf. Kent (1932: 104) for an explanation invoking analogy with words in *-idem*, such as *tantidem* and *indidem*. The retention of *-a-* in *itaque* is probably due to analogy with *ita*. Leumann (1977: 240) suggests the accent differentiation between *ítaque* 'daher' vs. *itáque* 'und so' is an invention of the grammarians. Cf. the treatment of this problem by Probert (2002: esp. 201-205). The suggested presence of *itáque* (with stress dragged rightward by *-que*) may have contributed to the retention of *-a-*; but see difficulties with this view, as poited out by Blumenfeld (2007).

[27] The following examples are selected from Sommer (1914: 85), Juret (1919: 94), Ernout and Meillet

*Ágrag(-entum) > *Agrigentum* [toponym]; Ἀλαλία → Lat. *Aleria* [toponym];[28] καμάρα → Lat. *cámara*[29] > *camera* 'vault'; καστάνεια → Lat. *castanea* ~ *castinea* 'chestnut (tree)' (cf. κάστανον); Κατάνᾱ (Dor.) → Lat. **Cátana* > *Catina* [toponym]; Μασσαλίᾱ → Lat. **Mássalia* > *Massilia* [toponym]; μᾱχανᾱ́ (Dor.) → Lat **mă̄chana* > *máchina* 'machine'; Σικελίᾱ → Lat. **Sícelia* > *Sicilia* [island name]; Τάρας, Τάραντος (gen.) → Lat. **Tárantom* > *Tarentum* [toponym]; τρυτάνη → Lat. **trútana* > *trutina* 'balance'; φάλαρα → Lat. **phálarae* > *phalerae* 'metal plates'.

Whether borrowings underwent vowel reduction or not partly depends on the time of their entry into the Latin vocabulary. Leumann (1977: 81) opposes ancient ones with vowel reduction (*camera*, *phalera*) to recent ones without it (*hilaris* ~ *hilarus* 'cheerful' ← ἱλαρός, *barbarus* ← βάρβαρος). But as Biville (1990: 10-16) claims, the situation is far more complicated than it may seem to be. Results of vowel reduction became a normative vowel alternation somehow, which restricts the appearance of low and mid vowels in non-initial syllables. Hence a certain number of words, borrowed into Latin even from the Literary period onward, copied such vocalic alternation synchronically. While Classical Latin represented Greek καστάνεια and κέρασος as *castanea* 'chestnut (tree)' and *cerasus* 'cherry (tree)', respectively, their variants like *castinea* and *ceresia* appeared in a later stage of Latin, and were then inherited by the Romance languages (see

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(1985), Leumann (1977: 81, 373, 457), Biville (1990: 9, 16), Oniga (1990: 198), and Poccetti (1999: 80). Allen (1973: 132) does not think of Lat. *Philumina* (← Φιλουμένη) as a case of vowel reduction. Rather, the substitution of *-i-* for short *-é-* is understood as reflecting an open quality of Latin *-i-*; thus, this type of example should be carefully excluded.

[28] For the consonant dissimilation in Latin, see Leumann (1977: 231).

[29] As for comments on this form by ancient grammarians, see Biville (1990: 20-21).

Biville 1990: 11). Unless *castinea* and *ceresia* already existed in the Pre-Literary period, and then submerged during Classical Latin under the influence of educated forms like *castanea* and *cerasus* (which, though less likely, is theoretically possible), the *i*-vocalism in the non-Classical forms may have arisen as a secondary synchronic phenomenon.[30] The examples provided by Biville (1990: 16), such as *monisterium* (vs. *monasterium*), *episcupus* (vs. *episcopus*), and *monichus* (vs. *monachus*), are no doubt recent creations in the Christian period, even if they exhibit the analogical "apophonie"-type variation.

1.2.2. Vowel Deletion in Latin

Vowel Deletion, particularly syncope, is not as widely operative in Latin as in Sabellic, but the number of its occurrences is not at all negligible. The precise conditions for syncope are known to be fairly complicated.[31] Part of the reason is that its date cannot be precisely determined, as Juret (1919: 96-97, 99) points out; it occurred a number of times throughout the history of Latin, not only in the Pre-Literary period but also during the Literary period, in contrast to vowel reduction, which is a relatively short-term event (see Ballester 1990: 42). Therefore, it is most likely that different conditions were involved in this sound change in different periods, as Lindsay (1894a: 170-171) and Exon (1906a:

[30] See also other possible cases provided by Biville (1990: 13): *cithara* 'lyre' [Var.+] vs. *citera*, *Tartarus* 'underworld' [Lucr.+] vs. *Tarterus*, *adamās* 'adamant' [Verg.+] vs. **adimans* (Fr. *aiment*).

[31] On this point, see the pessimistic remarks by Sommer (1914: 133-134), Buck (1928: 57), Moorhouse (1940: 308), and Leumann (1977: 249). In this situation, Rix (1966) does not seek a systematic description of the conditions for syncope, but instead, he tries to clarify its phonetic (phonological) mechanism which will be discussed in detail in Chapters 3 (for Latin) and 4 (for Sabellic).

119-121) have suggested.[32]

As for the Pre-Literary stage, people have traditionally considered syncope to be an effect of initial stress, which would have triggered the total loss of a vowel in the immediately following syllable. As Dietrich (1852: 544), Lindsay (1894a: 158), and von Planta (1892: 589-590) indicate, syncopated vowels often occupied the third syllable from the end of the word; if the Penultimate Law had operated then, such vowels could have been accented in accordance with its principles, and thus might be expected to have been preserved. Therefore, it seems desirable to assign these changes to a linguistic stage when accent location was determined by a rule different from the Penultimate Law, such as the one with initial stress, as is commonly thought: e.g., *ūndecim* '11' [Pl.+] < **ói̯no-dekem*; *quīndecim* '15' [Pl.+] < **k^{w}ínk^{w}e-dekem*; *dexter* 'right' [Pl.+] < **déksiteros* (Gk. δεξιτερός);[33] *prīnceps* 'chief' [Pl.+] < **prí(s)mo-kap-s*; *auceps* 'bird-catcher, fowler' [Pl.+] < **avi-cap-s*. In a later period, on the other hand, the Penultimate Law may also account for syncope: e.g., **kalēfákiō* 'make warm' > *calĕfáciō* [Pl.+] > *calfíciō* [Cic.];[34] *validē* [Pl.+] > *valdē* 'very' [Pl.+] vs. *validus* 'strong'.[35]

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[32] This seems to be true, despite Anderson's statement (1965: 71) in opposition to Lindsay. Though Exon eventually asserts that all instances can be explained under a single law (1906: 143), the identification of two different types of syncope (for one of them, see Mester 1994) in terms of both chronology and conditioning will be presented in Chapter 3.

[33] Cf. Ernout and Meillet (1985: 171), who insist on a reconstruction **dexteros* without *-i-*; but the idea is properly criticized by Ciardi-Dupré (1901: 209). If reconstructed as **dêkstero-*, the consonant cluster would have ended up as *-st-*, as in *Sestius* [personal name] < **sextios* (cf. *sextus*), *illūstris* 'bright' < **-l(e)uks-tri-* (see also in Götze 1923: 123, Benediktsson 1960: 213 n. 61, and Leumann 1977: 203).

[34] On the details of this process, which involves iambic shortening and syncope that was perhaps triggered by the secondary stress on the initial syllable, see Leumann (1977: 109, 248). As regards the syncope of a secondarily shortened vowel, see Rix (1966: 156).

[35] If the initial stress is taken to have triggered vowel deletion systematically, *válidus* 'strong' should

1.2.3. Vowel Reduction in Sabellic

Vowel reduction is said to be poorly attested in Sabellic (see, e.g., Solmsen 1894: 153). But as Buck (1928: 55-57) describes, the position before or, in some cases, after a labial serves as a locus of this change, which results graphically in either *-u-* (or *-i-* in some cases). It is somewhat likely that the vowel in the preceding or following syllable also played a certain role in this vocalic alternation: Osc. **últiumam** '*ultimam*, furthest' < **-tomo-* < **-tm̥mo-*;[36] Osc. *pertumum* '*perimere*, prevent' < **pert-emom*; Umb. **prehubia** '*praehibeat*, provide' (perhaps influenced by forms such as **prehubust*; cf. **prehabia**);[37] Osc. *praefucus* '*praefectus*, governor' (cf. *facus* '*factus*, be made'); etc. The occurrence of this phonological change in non-initial syllables strongly supports the initial-stress theory. We will discuss vowel reduction differing in conditioning context or type (4.4 and 4.5), which also occurs in all non-initial syllables and agrees with the initial-stress system.

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also have ended up as ***valdus*. The proclitic use of *validē* (> *valdē*) may have been responsible for the special deletion.

[36] The reconstruction involves a prevocalic **-m̥-*; thus, **-tm̥(m)o-* would be a more accurate representation with a homorganic glide in parentheses. The outcome **-om-* is based on Vine (1993: 247-249). This issue, resumed in 2.2.2, will help in understanding detailed phonetic mechanisms of vowel reduction.

[37] Meiser (1986: 268-271), adducing this and some other Umbrian forms, seems to consider a treatment peculiar to the language, but the Oscan forms above, and well as others from different Sabellic dialects, lead us to consider the phonological development from a wider perspective, maybe a Pan-Sabellic standpoint, as in 4.1.

1.2.4. Vowel Deletion in Sabellic

While in Sabellic we do not know of vowel reduction on as large a scale as in Latin, vowel deletion, especially syncope, is remarkably common in non-initial syllables. Traditionally, this frequency has also been explained by positing a linguistic phase with initial stress (see Thurneysen 1908: 240-242 and Pisani 1964: 6-7). Buck (1928: 57) remarks that "for Oscan-Umbrian, with the limited amount of material before us, it is almost useless to speculate upon the original conditions of the syncope." Yet, Benediktsson was the first to quite successfully conduct an extensive study of syncope in non-initial syllables in Oscan and Umbrian and confirm the conventional idea, saying that "a stress accent on the initial syllable is the only probable explanation of the syncope" (1960: 283). Examples may include **deḱsitero-* (Gk. δεξιτερός) > Osc. **destr-**, Umb. **testru**, etc. 'right'; **medes-to-* or **medos-to-* (cf. Lat. *modestus*) > Umb. *mersto*, etc. 'propitious(?)'; PIE **pt-n-h₂-s-eh₁-nt* > PItal. **pᵊt-na-s-ē-nt* > Osc. **patensíns** '*aperirent*, open' (subj.impf.3.pl.); etc.[38]

[38] Note that some of his phonological and morphological treatments are obsolete. For instance, see Benediktsson's scenario (1960: 208) for Osc. **patensíns**: PItal. **pátenesē-* > **pátnesē-* > **patn̥sē-* > **patensíns**; based on the reconstruction above, the two medial *e*'s cannot be explained.

1.3. Objections to the Initial-Stress Theory

1.3.1. "Intensité initiale"

Throughout the history of research, many objections or other options have been raised against the initial-stress theory, particularly by "French School" scholars, presupposing that the accent placement was basically unchanged from an archaic stage to Literary Latin. But most of their points seem to be groundless. Havet (1889: 11) names the initial-stress theory as "chimérique", though claiming a special intensity of initial syllables that may affect the following syllables (1889: 11-17).[39] Vendryes (1902: 41), supporting Havet's view, elaborates the concept of "intensité initiale", which coexisted with the pitch accent ("ton", "hauteur") inherited from PIE in his "période préhistorique" before his "période classique", set in the second century B.C.E. (1902: 99, 316).[40]

[39] Juret (1919: 99 n. 1), however, reports that Havet later changed his view to be close to that of Juret; see footnote 42. Meillet (1900) also already uses the term "intensité", which seems to imply *stress* accent, as the title of his article ("De l'accent d'intensité") indicates. But as is clear from other terms used in Meillet (1933: 56, 129-130, 241) without reference to stress, such as "le rôle particulier de l'initiale", "l'importance particulière" (of the initial), and "la valeur de l'initiale," as well as his description of the quality of the Latin accent (1933: 128-129), he seems to side with more traditional French-School views. The same idea is recapitulated in Meillet and Vendryes (1948: 115, 125) as "caractère spécial" and "valeur spécial", though Mignot (1975: 421) refers to these terms as more ambiguous than "intensité initiale". Kent (1931: 180 n. 4) suggests that the term "intensité initiale" inevitably implies the existence of stress accent. He further addresses that vowels in initial syllables are not always exempt from sound changes if they are unaccented (1931: 184), listing and discussing examples from Romance, Germanic, Slavic languages, and Modern Greek. However, that there is a certain stability of the position is still somewhat true, as based on Flemming's claim (2005: 4, 29-30), making the whole argument more complicated.

[40] Regarding the location of pitch accent, Vendryes provides no decisive remark, only raising the question: "était-il [= the pitch accent] déjà restreint à la pénultième et à l'antépénultième ou jouissait-il de la liberté qu'il a en sanskrit védique?" (1902: 99-100). According to him, the origin of the initial intensity is to be sought in neighboring non-Indo-European languages, such as Etruscan (1902: 52, 100). Juret (1919: 103-107), on the other hand, rejects Vendryes' "intensité initiale", insisting that Latin purely continued the PIE musical accent.

According to him, this coexistence ended in favor of the pitch accent by the start of his "période classique".[41] These two different types of accentual elements are phonetically distinguishable (cf. Maniet 1957: 27), but his theory seems to involve too much speculation, without decent description of how the two were *phonologically* operative in different positions of the word when they did not coincide. Leumann (1977: 247, 254) denies the possibility that high-pitched position and stress-accented position can coexist in different places within a word. This view is already addressed by the French scholar, Mignot (1975: 422-424), who takes an anti-French-School stance, and is also followed by Oniga (1990: 204). As Sturtevant (1940: 178 n. 4) suggests, "intensité initiale" can hardly be methodologically differentiated from "initial stress" as conceived by the German School. It can therefore be said that Vendryes was, or may have been, somewhat influenced by the initial-stress theory.[42] In fact, Niedermann (1953: 13) was more deeply influenced by the German School idea; while endorsing the French-School stance for Literary Latin, he fully accepted the initial-stress stage for Pre-Literary Latin.

[41] Vendryes (1902: 48-51) reports that von Planta (1892: 589) gives an analogous picture in Sabellic, and underscores that von Planta's discussion is an unreliable compromise between the theory of initial accent and the one of penultimate accent. But since von Planta makes no reference to the co-existence of two different types of accent, and rather suggests that there would be a diachronic gap between them, Vendryes' criticism is off the mark. This issue will be briefly discussed in 5.2.2.

[42] Juret, rejecting Vendryes' "intensité initiale" (1919: 99, 103), argues that initial syllables hold special stability or distinctness ("netteté"), both phonetically and psychologically (1919: 103, 107). The same terminology is used by Maniet (1957: 143). Mignot (1975: 422 n. 9) asserts that Juret's idea became "l'objet d'un certain ostracisme" among scholars (the same tone is found in Lepschy 1962: 228-229). However, recent typological studies, as presented by Flemming (2005: 4, 29-30) and Barnes (2006: 94), show that in some languages initial syllables are durationally stabilized, thus allowing Juret's view a certain amnesty.

1.3.2. "Dynamique de mot"

Monteil (1979: 91) also denies the initial-stress theory and proclaims "la dynamique de mot" ("dynamique phonatoire" in 1979: 99) to be a cause of vowel reduction and deletion. His notion is summarized as a set of phonological principles: 1) length provides vowels with stability; 2) openness of vowels is proportionate to length; 3) the word-length effect (cf. 1.1); 4) the nearer a vowel is to word-initial position, the longer it is; and 5) closed syllables are more stable than open syllables (1979: 91-92). While most of these seem to be reasonable, item 4 is questionable, since the position of accent is not considered at all. Monteil (1979: 91) maintains that accent fell on either initial or medial syllables (perhaps bearing in mind the Penultimate Law). If medial syllables had been accented, vowels there would have been expected to be strengthened and prolonged phonetically, thus not being targeted by vowel reduction or deletion. Item 5 is not without its own problems as well, since the duration of vowels in closed syllables may be shorter than in open syllables. His discussion does not include that issue, undermining his entire argument.[43]

1.3.3. Accentual Rule for Composite Formations

A morphological factor has also been incorporated by some other scholars to explain vowel reduction and deletion. Möller (1922: 170-178) deals with northern Germanic languages, in which simplex verbs were provided with grave accent, while such an accent

[43] Nor is Liénard (1977: 610) convinced, though his disagreement is only intuitively expressed.

was altered to acute accent in prefixed verbs; vowels with such an acute accent experienced a sound change. Möller (1922: 179-185) applies the same process to the Latin "apophonie," ascribing vocalic changes in medial syllables to acute accent distributed there. Möller's idea is followed by Bonfante (1930: 55-56).

The apparent similarity with Germanic is, in fact, already mentioned by Dietrich (1852: 545). His definition is not as purely phonological as occasionally purported, but refers to morphological information. According to him, an accent fell on syllables of the stem or, in the case of composite (i.e., morphologically complex) words, syllables of the first element (i.e., prefixes, etc.). Pedersen (1895-96: 67), rejecting the initial-stress theory (as in 1905: 338-339, 1906: 232-235), suggests that the PIE accentuation was responsible for vowel reduction, which he implicitly assigns to the prefix (e.g., *inimīcus* < **én-amīkos*). Similarly, though not in the same way, Exon (1906a: 129), alongside the Penultimate Law (with some additional rules of his own) in his analysis of syncope in Republican Latin, acknowledges another type of accentuation that operated in Prehistoric Latin: "the accent of all compounds lay on the first syllable." His examples include the following nominal compounds (1906: 138-39): *anculi* 'servants' (P. F. 18.18L) < **ámbicolo-* (cf. Gk. ἀμπίφολος), *nūncupō* 'declare' < **nṓmocapō* (see footnote 106 in Chapter 2), *naufragus* < **nā́vifragos* (but cf. footnote 6 in Chapter 3), and *officium* 'duty, obligations' <**ópifacium*. Such accent distribution is entirely incompatible with the normal Penultimate Law.

There is, however, a crucial problem in these scholars' discussions. Based on their theory, as Skutsch (1913: 188 n. 4), Juret (1919: 98-99), and Leumann (1977: 247)

properly point out, we must acknowledge a typological distinction of vocalic changes between complex words and non-complex words, such as third declension nouns (*nōmen*, gen.sg. *nōminis* < *-*men*-) and Greek loan words (*angina* 'angina' < Lat. **ángonă* ← Gk. ἀγχόνη). However, there is no difference between these two categories in patterns of vocalic changes, which makes such a dichotomy unnecessary. Even Pedersen (1895-96) admits that for perfect forms like *pepercī* (pres. *parcō* 'spare') and *fefellī* (pres. *fallō* 'deceive'), an accent supposed to be on the initial syllables to explain vowel reduction in second syllables cannot be Indo-European. He insists that such evidence is not sufficient to claim initial stress in the pre-history of Latin, but when combined with other counterexamples, these examples give the final blow to his theory.

More recently, Nyman (1983) likewise argues for the same type of accentual rule for nominal compounds in Latin from a wider Indo-European standpoint,[44] though misleadingly (even mistakenly) scrambling all types of compounds, except coordinating ones (*Dvandva* in Sanskrit terms), into a single semantic category as "determinative compounds" (encompassing the *Tatpuruṣa*, *Karmadhāraya*, *Dvigu*, and *Bahuvrīhi* types), and automatically distributing the accent on the first element (1983: 35-36). While the *Bahuvrīhi* compound generally has the accent on its first member (though not always on the *initial* syllable) as in *sū́ryatejas*- 'possessing the brightness of the sun' and *yajñá-kāma*- 'having desire of sacrifice', the *Tatpuruṣa* and *Karmadhāraya* compounds are basically opposite to it, e.g., *nṛ́-páti*- 'lord of man' and *agni-taptá*- 'heated by fire' (Whitney 1889: 501-511, Jamison 2004: 693-694). Note also his abuse of the

[44] Already suggested by Pisani (1930: 148), appealing to Celtic and Germanic.

terminology "determinative compound", which should strictly refer only to the *Tatpuruṣa* type. He even attempts, too simplistically, to include verbal forms, such as reduplicated perfect forms, collapsing composition and reduplication under the one term to explain vowel reduction there: e.g. *pepercī* < **pé#park-ai*, *fefellī* < **fé#fall-ai* (1983: 42). Compare Vedic forms that carry no accent in reduplicated syllables: *ca-kắr-a*, *ca-kr̥-máhe*, *ca-kr̥-úr*, *ca-kr-vắṃs*, *ca-kr-āṇá* (← *kr̥-* 'make'; for exceptions in some modal forms, see Whitney 1889: 293).

1.3.4. Emphatic Initial Stress

Pulgram (1975: 100-109), appealing to prosodic features in French sentences, such as *c'est un événement ȉncroyȁble* and *j'ai dit prȏdulre, et non pas cȏndulre* (⁺ is emphatic accent and ' is grammatical accent in his notation), hypothesizes that an accentuation like **cȏnfȁcere* with emphatic accent on the initial syllable was also possible in Latin. Then, he elaborates that "its original task was merely to distinguish... the various compounds of *facere* whenever such emphasis was necessary or helpful" (1975: 104). Such an accent driven by semantics (or, perhaps more precisely, pragmatics) was, according to him, strong enough to cause vowel reduction in unstressed syllables. He also adduces **dȇ⁺xȉteros* and **quȉ⁺nquèdecem*, invoking a semantic contrast with *sȉ⁺nȉster* and "a general tendency to accent the first syllable of the numeral in counting," respectively. The theory has exactly the same problem as the preceding one that claims an alleged rule for composite formations. The lack of sufficient mention of non-complex words with vowel

reduction, such as, again, *nōminis* and *angina*, undermines his entire argument. It is scarcely possible to claim such a pragmatic contrast for all nouns as a factor of phonological changes. Misgivings are also raised by Poultney (1978: 396), Liénard (1977: 832), and Nyman (1983: 34).

1.3.5. Leftward Scansion with Exceptional Rules about Syllable Weight

For the Penultimate Law in Classical Latin, metrical scanning to distribute accent starts from the right edge of words, and final syllables are extra-metrical. In some of the previous studies, this direction from right to left has been tacitly agreed on as a fundamental factor in accentual distribution for Pre-Literary Latin as well. As already mentioned in 1.2.1, Nyman (1983: 33) argues that many cases of vowel reduction took place in trisyllabic words with a light penult: e.g., **pér-fakit* > *pérficit*, **kón-tenet* > *cóntinet*, **ánamos* > *ánimus*. For such a claim, however, vowel reduction in heavy medial syllables is the crux: e.g., *aestimō* vs. *exīstimō* (< **éks-aistimō*), *Tarentum* (< **Tárantum* ← Gk. Τάρας, Τάραντος). Likewise, syncopated vowels often occupied the third syllable from the end of the word, which might be expected to have been accented and preserved if the Penultimate Law had operated then: e.g., *ūndecim* '11' < **ói̯no-decim*, *bálineum* 'bath' (← Gk. βαλανεῖον) > *balneum* (see Dietrich 1852: 544, Lindsay 1894a: 158, and von Planta 1892: 589-590).

For a possible solution, as already suggested by Dietrich (1852: 543), the distinction between short vowels in syllables long by position (.CVC.) and other short vowels (.CV.)

might not have always been made. A similar presumption can be found in Prosdocimi (1986: 617), who claims that there was a linguistic period when syllables with long vowels (.CV:.) and closed or diphthongal syllables (.CVC. and .CVY.) were treated distinctively in terms of syllable weight. Ballester (1990), who considers the initial-stress theory to be without foundation, presents such an argument more explicitly. He describes the above cases, like *exīstimō* and *Tarentum*, as deviant from the Penultimate Law and tries to formulate exceptional rules for them. He hypothesizes that in Pre-Literary Latin, though accent placement was determined by scanning syllables from the word end as in the standard Penultimate Law, it was not syllable weight but only vowel length that played a role in its determination (1990: 36-37). In other words, even when the penult was a heavy syllable (except with a long vowel), accent could go further to the antepenult, which was likely to be the initial syllable of the word. Later on, according to Ballester, this special treatment vanished as the standard Penultimate Law became established. With this claim of an "earlier version" of the Penultimate Law, the vowel reduction in *inscendō*-type forms could be explained in the same way for "$\breve{\sigma}$ ŏ ‹σ›"-type words such as *subigō* 'move below' (< **súb-agō*).

Ballester's claim could be considered beneficial to a certain extent, in explaining the vocalic reduction in closed or diphthongal medial syllables. Typologically, distinctive treatment in terms of syllable weight between long-vowel syllables (heavy) and closed/diphthongal syllables (light) is not surprising at all; see Gordon's studies (2004). Yet, it is not certain that there has been such distinction throughout the history of Latin. There seems to be no conclusive trace of it in what we have of the language. In the verse

of Latin comedy, for instance, both long-vowel syllables and closed syllables may be scanned as short: PY. *quid ĭllúc quod díco?* AR. *ĕhĕm, scĭŏ iam quíd uis díceré* (Pl. *Mil.* 36); the *-ĭ-* in *ĭllúc* and the second *-ĕ-* in *ĕhĕm* before the initial consonant of the following *scĭŏ* are scanned as short in anapestic (∪∪—) and proceleusmatic (∪∪∪∪) feet, resepectively; the *-ŏ-* in *scĭŏ* loses its original length in a proceleusmatic foot; both cases result from iambic shortening or its subtype. The so-called "*Iuppiter*-rule" would be another example, whereby *Iuppiter* with gemination was generated from *Iūpiter* by change of syllable *structure*. Despite a certain recentness of this alternation (see Leumann 1977: 183) and some restriction on vowels qualified for this process (see Weiss 2005: 168), such occasional interchangeability between a long vowel and a short vowel + a geminate consonant would reveal an inherent nature of syllable weight in Latin, that is, equal treatment of .CV:. and .CVC./.CVY.

1.4. Prospective of the Work: Application of General Linguistics

I have not exhaustively cited the immense literature on the topic. But in viewing commonly used literature, I have provided an overview of the theory that Pre-Literary Latin and Sabellic located *stress* accent on the *initial* syllable, and have examined supporting evidence to conclude that this theory is reasonable and indispensable. Given the fact that vowel reduction and deletion are well explained by means of this hypothesis, we cannot do without "initial stress" in dealing with the data cited above.

On the other hand, the initial-stress rule is only the background setting the stage for

vowel reduction and deletion in the Italic languages. As mentioned at the beginning of this chapter, our goal is to understand how these sound changes functioned, or did not function, on the basis of such an operating system as the initial-stress rule. For this purpose, it is highly useful to incorporate theoretical, experimental, and typological information of phonetics and phonology into the historical perspective. We have already seen a measure of the applicability of such knowledge to our work in 1.1, in regard to the qualitative nature of accent in languages that show vowel reduction and deletion. It will be widely employed for more crucial points, as I briefly explain below. Even in languages that are only historically documented, there are several clues that reveal phonetic and phonological mechanisms and help us uncover some hidden facts

Cross-linguistically, more than one unstressed syllable in a word may host vowel reduction, as shown and discussed by Crosswhite (2001, 2004), Flemming (2005), and Barnes (2006). In Latin, medial syllables immediately following initial stressed ones have traditionally drawn far more attention from scholars than other syllables. But vowel reduction productively occurred in other syllables, including final ones: e.g., *artifex* 'skillful' < **-fak-* (cf. *faciō*).[45] It can then be said that the lack of stress triggers vowel reduction. Recognizing synchronically multiple occurrences of the change allows us to make a comprehensive analysis of the multiple places potentially targeted by vowel reduction (2.1). The same principle will be applied to vowel deletion in Sabellic (4.3.2). On the other hand, as Barnes (2006: 37) indicates, a single language may have more than

[45] Compare also vowel reduction in final syllables in Sabellic: e.g., acc.sg. *-om* (e.g., Osc. **húrtúm** 'protected place') ~ *-um* (e.g., Osc. *dolum* 'guilty intention', cf. *dolom*). Further, note vowel deletion in final syllables in both Latin and Sabellic: e.g., Lat. *calcar* 'spur' < **kalkari*, Paleo-Umb. **setums** (< **septomos* < **septm̥(m)os* 'seventh'; personal name derived from the ordinal number). Etruscan also shows vowel deletion in final syllables: e.g., *Leθamsul* > *Leθnsl* [theonym], *macstr-* (← Lat. *magister*).

one pattern of vowel reduction. This should be kept in mind so as to avoid overly simplistic generalizations.

Rix (1966: 161) suggests that the reduction process in Latin first changed vowels into schwa-like sounds, prior to the frequent end-result /i/. The initial step, the detailed pattern of which is theoretically predicted by Crosswhite (2001), may account for the oscillatory treatment of [ə] in terms of orthography. The importance of the surrounding phonetic context is highlighted by Lindblom (1963: 1773) and Flemming (2005: 6). Further, F1 reduction, that is, the neutralization of [high][low] contrasts significantly alters the low tongue and jaw position for the low vowel, as indicated by Flemming (2005: 6). This fact will be employed to explain frequent reduction of *a*-vocalism in Latin (2.4.3), and tentatively Sabellic (4.5).

In terms of underlying causes, vowel deletion is more complex than reduction. As Zuraw (2002: 387) notes (and also Rix), vowel deletion can be regarded as an extreme form of prominence reduction (2.2.1, 3.2). Which vowels are subject to the process may rely somewhat on relative prominence based on vowel sonority hierarchy. The model schematized by Crosswhite (2001: 39) is handy for considering this. The enhancement in sonority by the rounding gesture, mentioned by Zuraw (2002: 390), will also be provisionally considered for syncope in Sabellic (4.2). On the other hand, there is at least another conditioning factor of vowel deletion. Operating with the premise that Latin is metrically based on bimoraic trochees (⏑⏑ = LL or — = H) in terms of foot structure, Mester (1994: 37-43) explains post-tonic syncope in Literary Latin (Pre-Classical and Classical) as deletion of vowels in unparsed syllables (3.3.1). For considerations of

metrical parsing in earlier periods of Latin, Hayes (1995: 180), Kager (1999: 169), and Parsons (1999: 122-123) are of great interest; the direction of parsing and which foot was assigned primary stress will be discussed. Afterwards, Zawaydeh's "*Light Light constraint" (1997: 202), which disfavors two consecutive light syllables and triggers syncope in Arabic dialects, will be invoked for Latin; the same metrical configuration seems to have triggered vowel deletion, in combination with prominence reduction (3.2). The metrical template (*Light Light) may also apply to a sequence comprising -*s*T(R)- (with syllable boundary before *s*). To understand the inconsistent behavior of -*s*- in terms of syllabification, Solé's brief description (1995: 24-25) is of a certain interest (3.3.4). Though the metrical factor cannot clearly be defined in Sabellic, the peculiarity of -*s*- is also somewhat functional there (4.1.2, 4.3.2). We will also discuss syncope in the sequence *C*r*V (3.4), the phonetic mechanism of which will be better understood by Hall's observation of vowel intrusion (2004: 3)

The above considerations have been employed only sporadically before, and almost never applied systematically to the analysis of vowel reduction and deletion in the Italic languages. We will start with vowel reduction in Latin (Chapter 2), followed by the discussion of vowel deletion in Latin (Chapter 3) and both in Sabellic (Chapter 4). Chapter 5 comprises a summary of previous chapters, and a brief discussion on the Italic accent of no direct bearing on vowel reduction and deletion.

2. Patterns of Vowel Reduction in Latin: Phonetics and Phonology

This chapter focuses on phonetic/phonological details of vowel reduction in Latin. Faliscan materials are also discussed, though only marginally, because the presence of the phenomenon is not orthographically explicit and has long been disputed (cf. 3.3.5). It is traditionally assumed that vowel reduction is a process anterior to deletion. This is partially true. But as Rix (1966) has already pointed out, more microscopic mechanisms should be taken into account to lay out a more accurate scenario, not only diachronically but also synchronically. Another conventional view to reconsider is that vowel reduction is a relatively short-term event (i.e., a direct merger of some vowels with /i/), in comparison with vowel deletion. This is also partially true. But "short-term" does not mean "very short". What lies underneath the surface will turn out to be more complicated than previously thought.

2.1. Chronology between Word-internal and Word-final Vowel Reduction

As is pointed out in Chapter 1, most of the previous studies have focused on word-internal vowel reduction. But I have already emphasized that word-final vowel reduction should not be dismissed. Both must be incorporated in a more synthetic analysis. Yet scholars have occasionally suggested a distinction between medial and final position in terms of the dating of Latin vowel reduction (and also Sabellic syncope). What follows is an attempt to show that such a claim is tenuous, thus providing a certain

chronological uniformity to those non-initial vocalic changes.

Vowel reduction in Latin is thought to have begun about the fifth or fourth century B.C.E. and to have been completed by about the third century B.C.E. Some inscriptions seem to show that vowel reduction gradually diffused from medial to final syllables. In CIL I^2 2391 (late 3rd – early 2nd c.), **N͡MISIO** (= **NVMISIO**[1] < **Numas-*; CLat. *Numeriō*) shows vowel reduction in the medial syllable, while in **DONO** (≈ *-om*; CLat. *dōnum* 'gift') final vowel reduction seems not to have occurred yet. As for **TREBIBOS** (< **-ubos*) in CIL I^2 398 (uncertain origin, 3rd c.; cf. **FECIT** < **-ed* and **LAINIO** ≈ *-os* in the same text), Wachter (1987: 459) takes **-I-** in the medial syllable as a result of vowel reduction, assuming something like the so-called "sonus medius", /ü/, as its phonetic value, while **-O-** in the final syllable is retained. In CIL I^2 37 (Praeneste, before 150 B.C.E.), the divine name **APOLONES**[2] (gen.sg.) preserves **-E-** (at least graphically) in contrast to CLat. *Apollinis* (gen.sg.), and **VICESMA** (cf. CLat. *vīcē(n)sima* 'twentieth') in the same text seems to be a product of vowel reduction as well (if so, the weakened vowel *-u-* or *-i-* is even omitted).

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[1] For the ligature of **N-** and **-M-**, see Vine (1993: 289, 328).

[2] Some inscriptions from the third and second century B.C.E. widely provide a lexeme for the divine name "Apollo". Wachter (1987: 397) provides **APLONE** (Pocc 218; with **-e-* hidden between **-P-** and **-L-** by means of so-called syllabic notation), **APOLON[E]** (CIL I^2 59), and **APOLONES** (CIL I^2 37) as examples, asserting that **-O-** between **-L-** and **-N-** should be long in these instances, otherwise it would have been weakened as in CLat. *Apollĭn-*. As a matter of fact, to judge from a nearly contemporary text, CIL I^2 581 ("SCdB"; Bruttium, 186 B.C.E.), which attests **NOMINVS** and **HOMINES** with a weakened medial vowel, and the Pre-Classical works by Plautus (around 200 B.C.E.) consistently with *Apollĭn-*, Wachter's statement about the vowel length is convincing, as seen by Vine (1993: 241). The long-vowel version of this lexeme is actually found in Oscan: **appelluneís** (Ve 18), α]ππελλουνηις (Ve 197a), απελλουνηι (Ve.196). As for the *e*-vocalism in the second syllable, see Wachter (1987: 397 n. 909); this vocalism is taken to have arisen from a Doric form. On the other hand, the origin of *-ĭ-* in *Apollĭn-* is perhaps **-ŏ-*. The traditional explanations concerning the creation of the variant **Apóllŏn-* are briefly outlined by Vine (1993: 242), who in turn adds his own view based on the morphological parallelism between "Apollo" and "homo".

Bader (1960: 238) is more extreme, positing a "loi de « limitation de l'apophonie »" particularly for compounds, whereby the "apophonie" happens only in one place in the word: "un même mot ne peut avoir qu'une seule voyelle apophonique." She entirely ignores vowel reduction in final syllables, which is not vowel reduction in her definition.[3] More restrictively, Bader (1960: 238-239, esp. 239 n. 12) believes that the "apophonie" does not pass over second syllables, by adducing contrastive forms, such as *agri-cola* 'farmer' vs. *in-quilīnus* 'inhabitant' / *an-cilla* 'female slave'; *arci-tenēns* 'holding a bow' vs. *con-tinuus* 'continuous' / *con-tinēns* 'holding together'; *hosti-capas* ('hostium captor' in P. F. 91.15L) vs. *man-ceps* 'purchaser'. A crucial factor for Bader as to where vowel reduction occurs is the number of syllables of first members in compounds. Vowel reduction in non-second syllables in forms such as *tībīcen* 'flute-player', *tubicen* 'trumpeter' is explained by analogy with forms such as *oscen* 'foreboding bird' (cf. *canō* 'sing, play'), where the second syllable underwent vowel reduction (Bader 1960: 240). The same is true, according to her, of *particeps* 'partaking', which needs an analogical support from *auceps* 'bird-catcher' in her theory (though *auceps* was originally trisyllabic, i.e., **aṷi-cap-s*).

The inscriptional data and Bader's argument might seem to support the relative chronology that vowel reduction in medial syllables preceded the one in final syllables, implying that only syllables right after initial stressed syllables in early Italic were targeted by vowel reduction. However, it is not necessarily correct to assume that unaccented vowels immediately after stress are the only ones licensed for sound changes.

[3] Weiss (2005: 165) also excludes some of the sound changes in final syllables, i.e., *-os* > *-us* and *-om* > *-um*, from those triggered by stress.

Typological studies, provided by scholars such as Crosswhite (2001, 2004), Flemming (2005), and Barnes (2006), tell us that there are a number of manners of vowel reduction, and demonstrate that in many languages, more than one unstressed syllable may host the change, no matter where they are located in the word, albeit some positional restrictions or variations of degree of reduction apply in individual languages.[4] That is, vowel reduction may take place synchronically in more than one place in a single word[5] and/or in (either immediately or remotely) pre-tonic position: e.g., Russian /tsexovój/ as [t͡sixavoj] 'shop' (adj.) (cf. [t͡séx] 'shop'), /sadovód/ as [sədavót] 'gardener' (cf. [sát] 'garden'), and /gorodók/ as [gəradók] 'little city' (cf. [górət] 'city') (see Crosswhite 2001: 60-61 and Flemming 2005: 27); Etruscan *mulvanice* ~ *mulvenece* ~ *mulvunuke* 'gave as a present' (see Rix 2004a: 948) and *parθanapaes* ~ *partinipe* ← Gk. Παρθενοπαῖος (see Deecke 1878: 176, Giacomelli 1962: 359, De Simone 1970: 10). Latin should be re-examined in light of such typological evidence. A simplistic division, like medial syllables vs. final syllables, might be useful for the sake of descriptive clarity, but such a dichotomy should not be introduced so naïvely to disconnect the relationship between these two positions. Proper comparison between them will strengthen our inquiry.

Moreover, inscriptional data provide evidence which draws our attention to reconsider the alleged chronology of vowel reduction between medial and final syllables. CIL I^2 5 (ca. 350-300) includes L[---]NIBVS, which is most likely a dat./abl.pl. form of an

[4] Cf. many dialects of South Italy, where [a] to the left of the stressed vowel is not usually reduced: e.g., Agnone (Molise) [marga'rojtə] < *[marga'rita] 'daisy'. See Maiden (1997: 10).

[5] The same is true of vowel deletion, e.g., in Etruscan *clutmsta*, *cluθumusθa* ← Κλυταιμνήστρα and *Vestrcna* < *Vestiricina*.

i- or consonant stem (< **-ibos*; e.g., **L[ECIO]NIBVS** 'armies' as accepted by Wachter 1987: 370). Given the fact that this text was discovered in Marsian territory, it may well involve some dialectal features. But the ending of **-BVS** unequivocally reflects Latin morphology (cf. Osc. **luisarifs** [month name], **anafríss** [theonym], Umb. *fratrus* 'brothers', SPi. **manus** 'hands', with syncope [**-b^{h}os* > **-fos* > *-fs* > *-s(s)*] and the fricative reflex of **-b^h*-), thus there seems to be no problem in treating this ending in a Latin phonological context. It follows that the *u*-vocalism in the final syllable strongly suggests that vowel reduction was already an on-going process in as early as the fourth century B.C.E. Therefore, Sturtevant's assigning (1940: 178) of *-o-* > *-u-* in final syllables to late in the third century should be abandoned. A fragmentary tail of a perfect form in CIL I^2 360 verso (4th c.?; see Wachter 1987: 460-461), **-VIT** (< **-u̯et*), followed by a punct, can perhaps be explained in the same way. One might claim that **-I-** continues not **-e-* but **-ei̯-*, with recourse to **DEDIT** 'gave' in CIL I^2 561 ("Cista Ficoroni"; Praeneste, ca. 300 B.C.E.), reconstructed with **-ei̯-* by Meiser (1998: 217) and de Melo (2007). However, in view of the persistent use of ‹ei› in CIL I^2 581 ("SCdB"; 186 B.C.E.) — though the text is somewhat archaizing —, **PLOIRVME** (= *-ē̯* < **-ei̯* < **-oi̯*; not *plūrimī* yet) in CIL I^2 9 (ca. 200 B.C.E.), and **VTEI**, **QVEI** (> *utī*, *quī*), etc. in CIL I^2 614 (189 B.C.E.; cf. **INPEIRATOR** '*imperator*' in the same text), it is less likely that **-ei̯-* > *-ī-* was completed in such early periods. If this argument is correct (though it requires further inquiry), and if we allow for the reconstruction with **-e-* and its subsequent reduction (see Wachter 1987: 126), then the dating by Brugmann (1897: 225) of *e* > *i*/__*s*, *d*, *t*# to the third century B.C.E. would likewise turn out to be too late. Note that these texts also show unaltered *o*-vocalism, in

CANTOVIO (≈ *-os*) and **-BOS** (dat./abl.pl.?), respectively. I suggest that such orthography was already archaizing and such conservatism lingered throughout the history of Latin.[6] And there is another caveat about *-om#* > *-um#*. As is well-known, this vocalic shift is not seen in Pre-Literary Latin (Leumann 1977: 424, Weiss 1998: 713; cf. Wachter 1987: 285 for **ITERVM** in CIL I^2 610 [ca. 200 B.C.E.]). The reason that this sequence did not participate in the early vowel reduction is reflected in orthographic variation like *-o* instead of *-om*: e.g., CIL I^2 9 (Rome, 3rd c.) **OINO, OPTVMO, VIRO** (≈ *-om*). In such cases, the final vowel was most likely nasalized, thus phonetically (not phonemically) prolonged enough to be resistant to vowel reduction for a while.[7] This is the case with

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[6] The spelling **-IO(S)** in gentilicia is somewhat traditional: e.g., CIL I^2 561 **NOVIOS, PLAVTIOS** vs. **FECID, DEDIT** (see Sommer-Pfister 1977: 115); CIL I^2 9 (Rome, 3rd c.) **FILIOS** vs. **CEPIT**; CIL I^2 412a/c (Tarquinii, ca. 250) **GABINIO** (cf. 412b **GABINIVS**) vs. **RETVS, CALEBVS**. Wallace (2005: 177) dates CIL I^2 412b (maybe a/c as well) to ca. 300, which seems too early on account of the consistent use of ‹g›. But as Mommsen (1883: 885) assumes "propter litteraturam vetustissimam" (such as ||| for ‹e›, |' for ‹f›, oblique-stroked ‹a›, angled ‹l›), these inscriptions may be dated to a period before the Second Punic War.

Wachter (1987: 495) focuses on the variation of the gen.sg. ending of consonant stems, *-es* (> *-is*) vs. *-os* (> *-us*). He says that there are five attestations from Latium of *-es* (> *-is*) before 150 B.C.E., starting from the third century B.C.E.: **DIOVIS** (CIL I^2 361), **CERERES** (973), **IVNONENES** (444), **SALVTES** (450), **VENERES** (451). It may be possible to add here **APOLONES** in CIL I^2 37 (of uncertain origin but probably from Latium as well, as suggested by Vine 1993: 241) and CIL I^2 612 **HONORIS** (probably from 193 B.C.E.; see Sommer-Pfister 1977: 115). While most of these cases show **-ES** as the ending, **DIOVIS** from the third century reveals a more modern spelling. This, in turn, allows for a hypothesis that the sound shift **-e-* > *-i-* before final *-s* was completed by the period of **DIOVIS**, albeit with the traditional orthography **-ES** continued.

[7] Cross-linguistically, nasalized vowels are phonetically longer than their oral counterparts, all things being equal; see Barnes (2006: 38). Further, cf. the French *brune* [bʁyn] 'brunette' vs. *brun* [bʁœ̃] 'brown', where the nazalized vowel of the latter is lower than the non-nasal counterpart of the former; see Ohala (1993: 242-243).

Note that we also have another pair of orthographic variants, namely *-u* and *-um*. But as Porzio Gernia (1974: 145-151) indicates, the following edge of the line may be the reason for the omission of final *-m*, e.g., CIL I^2 10 **INGREMIV|**, 61 **DONV|**, 626 **SIGNV|**, 1831 **PLOSTRV|**, etc., though there are naturally some exceptions, e.g., CIL I^2 1531 **DONV DANVNT|**. (B. Fortson kindly brought Porzio Gernia's work to my attention.)

As a ramification of this consideration, the question arises whether or not *-om* > *-um* (regarded as a feature of "il latino urbano" by Porzio Gernia 1974: 162-163) was part of the Latin vowel reduction (pointed out to me by M. Weiss), due to the fact that it occurred at a chronologically later stage. As I suggest in 2.2 and 2.4, the completion of vowel reduction in medial syllables was achieved under the Penultimate Law in Literary Latin. This may indicate that the change *-om* > *-um* in final position can also be defined as vowel reduction, despite its relative recency. The parallel development in Sabellic (see

CIL I^2 9 **CONSENTIONT** and *nequīnont* [Andr.] (> *-unt*), with *o*-vocalism preserved (cf. CIL I^2 378 **DEDROT**, 379 **DEDRO** [near Pesaro, ca. 175 B.C.E.] ≈ CLat. *dedērunt*; see Vine 1993: 339 for the so-called "syllabic notation"). Taken altogether, we must not miss a group of old forms with reduced vowels behind such archaizing orthography. There is little justification for historically differentiating vowel reduction according to its location in the word, and I propose that this fact should be construed to mean that the process targeted both medial and final syllables synchronically.

In light of this chronological uniformity of word-internal and word-final vowel reduction, explanations argued by Bader for other cases, relevant to her "loi de « limitation de l'apophonie »", also come to be less grounded. For instance, for Bader (1960: 238), one of the reasons that *circum-(i)iciō* (< disyllabic *circum-* + *iaciō*) shows vowel reduction in the third syllable is provided by appealing to analogy with other compounds of the same verb with monosyllabic prefixes. Then, she explains: "elle [= la voyelle apophonique] sera transférée... sur la syllabe initiale du second élément." In my theory, however, vowel reduction is not limited to a single syllable of a word, that is, each non-initial unstressed syllable may be targeted by the phonological change. Thus, we do not need to assume that "apophonie" is transferred ("transférée") from one syllable to another. (Some other arguments of Bader will be tested in Chapter 3.)

One may raise the issue that patterns of vowel reduction are not entirely equal between medial and final syllables. As far as similarity is concerned, **-a-* in closed and diphthongal syllables developed into *-e-*, whether it is located word-internally or

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footnote 12 in Chapter 4), which is no doubt a pattern of vowel reduction, would also support this view.

word-finally (see more details in 2.4.4): [[word-internally]] *caedō* (< **cai-*) 'cut' vs. *incīdō* 'cut into' (< **-kei-* < **en-kaidō*); *ars* 'skill' vs. *inert-* 'unskillful' (< **in-art-*); *captus* 'taken' vs. *conceptus* 'taken completely' (< **kom-kaptos*); etc. [[word-finally]]: *artifex* 'skillful' < **-fak-* (cf. *faciō*); *rēmex* 'rower' < **-ag-* (cf. *agō*); etc.[8] On the other hand, positionally different patterns of vowel reduction are seen concerning *e*-vocalism in closed syllables. It is reduced to *i*-vocalism word-finally as in the genitive ending in the third declension (*nōminis* 'name' < **-es*) and the 2nd and 3rd pers.sg. ending in the third conjugation (*agis, agit* 'drive' < **-es*, **-et*), while it is preserved word-internally as in *dēfendō* 'ward off' and *elementum* 'element(s)'. This fact might lead us to doubt positional independence of vowel reduction.

But it is not surprising that a single language has more than one pattern of vowel reduction, as clarified by recent theoretical phonologists based on typological works. According to Barnes (2006: 37), in Brazilian Portuguese the vowel inventory /i, e, ɛ, ɔ, o, u/ in stressed position is reduced in unstressed position. In pre-tonic syllables, the contrast between high mid and low mid (or tense mid and lax mid) is neutralized, thus yielding [e, o]. In post-tonic syllables, on the other hand, the mid vowels are utterly eliminated and /a/ is raised to schwa, resulting in the inventory [i, u, ə]. Such a dual treatment in vowel reduction is paralleled by most dialects of Russian. Crosswhite (2001: 52-55) describes that /a/ and /o/ are neutralized as [a] in immediate pre-tonic position, while they are realized as [ə] in other unstressed syllables (though the actual formulation

[8] Cf. the change **-o-* > *-u-* in the same contexts, although it is in need of a restatement; see 2.3 (word-finally) and 2.4.4 (word-internally).

is more complex; see also Barnes 2006: 37).[9]

Barnes (2006: 37) also provides a case study of Belarusian. In the immediately pretonic syllable, whose vowels are characteristically longer than other unstressed syllables (like Russian), /a/ is realized as [a]. In other unstressed syllables, /a/ is realized somewhat differently. Barnes follows a study by W. Czekman and E. Smułkowa, who symbolize the vowel in question as Greek 'α' and describe it as weaker and higher than stressed [a], contrary to the claims of many other descriptions. This case is also understood to demonstrate multiple manifestations of vowel reduction in a single language.

Turning to Latin, different treatments of *-e-* depending on its position thus does not prove diachronic diversity of vowel reduction at all. Locational distinction in patterns of vowel reduction is certainly explained on a synchronic level, as typological studies instantiate. Taken altogether, what I have deduced from inscriptional materials will not contradict any linguistic principles, but rather comprise a phonological reality. That is, vowel reduction is understood to have synchronically targeted both word-internally and word-finally.

Remember, however, that this is a simplistic version of a conclusion about the problem of chronology between medial and final syllables of vowel reduction. By "simplistic", I do not mean "inaccurate" at all. Rather, the simultaneous initiation of vowel reduction in more than one place in a word seems to be very likely, as just laid out.

[9] Note also Barnes's argument (2006: 47) against Crosswhite: "The analysis of Russian as possessing two distinct degrees of vowel reduction is in fact the standard treatment in the Slavic grammatical tradition. Generally, however, the difference is treated as one of degree, rather than one of kind." Barnes insists that vowels in pre-tonic position are in fact reduced in height, though the outcomes are [a]-like vowels. At any rate, what Russian possesses is not a uniform pattern of vowel reduction at all.

But as I have already repeated several times, vowel reduction is not indispensably accompanied by a *phonemic* neutralization in the most strict sense, whereby reduced vowels merge with existent phonemes in the vowel inventory. Vowel reduction comprises a wide range of degree *per se*, not being a unitary phenomenon. For instance, word-final vowel reduction in Latin is a case where phonemic neutralization was attained by an uncomplicated process (see more details in 2.3). On the other hand, the mechanism underlying medial vowel reduction includes a certain complexity, which I will clarify in the next section.

2.2. Vowel Reduction in Medial Syllables

We have just confirmed that there are different mechanisms of vowel reduction between medial and final position. What comes next is to lay out at length how different those phonological processes are. More detailed procedures will be disclosed in what follows. In this connection, it is worth citing Coleman's statement (1962: 103): "[i]t is sometimes asserted that, as orthographic systems tend to be phonemic, we are condemned to ignorance of the sub-phonemic phenomena of dead languages. However, even where we have only written texts to guide us, provided that these are authentic, as they are in the case of inscriptions, we are still able to make some inferences about allophonic composition of phonemes from spelling confusions and hypercorrections and from the sound changes reflected in changes of spelling." Reality of phonology in dead languages is often disguised by orthography, which is frequently incomplete particularly in

inscriptions. (Even in modern languages, such as English and Russian, the orthography rarely reflects such phonetic alternation caused by vowel reduction.) With Coleman's warning and expectation, let us start with vowel reduction in medial syllables.

2.2.1. Outline of the Mechanism: Phonetics of Reduced Vowels

The oft-cited final outcome of vowel reduction is convergence of targeted vowels into *-i-* in medial open syllables: e.g., *conficiō* 'complete' (< **-fak-*), *cecidī* 'fell over' (< **-kad-*), *pectinis* 'comb' (< **-ten-*), *dīmidius* 'half' (< **-med-*), *angina* 'angina' (< **-gon-*). This simple formulaic pattern has been recapitulated in many standard handbooks. On the other hand, the reconstruction of an earlier process of vowel reduction, prior to the "/i/-merger," has been presented by several scholars.

The idea is already found in Skutsch (1913: 192-193). He first tries to explain a historical development of *conscendō* 'go on board', envisaging the following scenario with syncope: **conscandō* > **conscn̥dō* > *conscendō*, on the basis of Etruscan *atlnta* ~ *atlenta* (← Gk. Ἀταλάντη), *elχsntre* (← Gk. Ἀλέξανδρος), and *caśntra* (← Gk. Κασσάνδρα). But in contrast to these vowel-less stages (at least graphically) in Etruscan, there is less likelihood of a stage in Latin like **conscn̥dō*,[10] because the same type of vocalic change as in *conscendō* is found in forms without sonorants, such as *tractō* 'handle' vs. *dētrectō* 'decline' (~ *dē-tractō*), *aptus* (p.p. of *apiō* 'fasten', *apīscor* 'grasp') vs. *ineptus* 'foolish', *captus* 'taken' vs. *conceptus* 'taken completely', and *factus* 'made'

[10] Leumann (1977: 85) also disagrees with Skutsch's scenario.

vs. *confectus* 'completed'; it is hardly conceivable that *-fectus*, for instance, originated from **-*fktos*. Skutsch also provides an alternative to the syncope scenario, namely, "Färbung eines übrig gebliebenen Minimalvokales unter Einfluß der umgebenden Laute" (1913: 194), attractively suggesting a possible relic of the vocalic element before *-n-*, which he also calls "Restvokal" (1913: 195) or "polymorphe Schwa" (1913: 196). In another case of a different phonological context, he clearly represents such a minimal vowel as *-ə-* in **artəmis* (← Gk. Ἄρτεμις) for Etruscan *Artumes* (1913: 195), as opposed to "ein direkter Übergang eines Vokals in einen andern." This view firmly presupposes the existence of an intermediate stage in the process of vowel reduction.

In more detail, Rix (1966: 161-162) lays out phonetic implications of vowel reduction as well as syncope. According to his theory, syncope is not a process subsequent to "vowel weakening" — a traditional term defining *a/e/o* > *i/u*, *u* > *i* in medial open syllables and *a* > *e/u* in medial closed/diphthongal syllables (note that Rix only focuses on the former) —, which *a priori* implies that terminal products are realized as vowels existing in the phonological inventory.[11] Rather, vowel reduction and syncope are recognized as two distinct processes, though sharing the same inceptive step. They start by undergoing a common development into [ə] (or [ɪ])[12] or "open transition" (like, e.g.,

[11] See also Leumann's skepticism (1977: 84) of the linear chrolonogical process, "vowel weakening" → syncope.

[12] This is followed by Meiser (1998: 66-67), though his view of syncope is different from Rix's in that [ə] is not conceived of as a stage prior to syncope. Pulgram (1975: 101 n. 59, 105) also posits a phonetically intermediate stage [ə]. However, his argument comes from recognizing two different outcomes of vowel reduction, *-i-* and *-e-*, as "freely varying weak grades" regardless of syllable structure (open or closed): *-i-* in *-ficiō* (vs. *facere* 'make'), *-tingō* (vs. *tangō* 'touch') and *-e-* in *-scendō* (vs. *scandō* 'ascend'), *peperī* (vs. *pariō* 'give birth to'). He entirely overlooks the effects of surrounding consonants (such as *-r-* and *-ng-* [ŋ]) on preceding vowels.

French *fenêtre* /f + nɛtr/),[13] which is defined as "eine[r] Stufe der Neutralisation" by Rix (1966: 161). Their unstable ("labil" in Rix 1966: 161) phonetic status is replaced by either total deletion ("syncope") or more sonorous vowels, available in accented syllables, like /i/, /e/, and /u/. The latter process is the one traditionally recognized as "vowel weakening", which necessitates phonemic neutralization in unstressed position.[14]

Rix's theoretical approach (preceded by Skutsch's) might seem to be speculative, but his claim about the phonetically intermediate stage ([ə], [ɪ], "open transition") is supported by studies of phonetics and phonology, such as the oft-cited experimental work by Lindblom (1963) on target undershoot, which is reinforced by Moon and Lindblom (1994). The author(s) underscores three points as primary factors to vowel reduction (1963: 1778-80; 1994: 40-41, 53-54): 1) vowel duration, 2) "locus-target" distance (that is, articulatory or, more precisely, formant transition between a consonant and a vowel), and 3) "style" or "articulatory effort" (to promote perceptual intelligibility). 1') Duration insufficient for ideal articulation, 2') sizable "locus-target" distance, and 3') less effort to reorganize acoustic patterns result in the so-called "undershoot" effect, that is, "systematic shifts away from hypothetical target values" (Moon and Lindblom 1994: 40). "Undershoot" may be used as an equivalent of vowel reduction in a broad sense, but the extreme case that leads to phonemic neutralization, namely *phonological* vowel reduction

[13] Cf. the case in Central Alaskan Yupik (Gordon 2001: 210-211), where consonant clusters engendered by syncope show different behavior from other clusters. First, they resist regressive devoicing (except in a specific dialect; see Gordon 2001: 211 n. 8): e.g., /kuvəciquq/ → [kúvciqúq] 'it will spill', cf. /ajaʁtuq/ → [ajáχtuq] 'he leaves'. Secondly, in the Hooper Bay-Chevak dialect, codas that result from syncope are released. That is, the first stop consonant of the cluster in question explodes, instead of being implosive. See also schematic representations in Hall (2004: 19-21), describing close/open transitions on the basis of gestural phasing of articulation.

[14] Godel (1961: 60) argues that the change **-o-* > *-i-* had **-e-* as an intermediate stage, although he provides no statemant about the phonological status of **-e-*.

(as described in Flemming 2005: 17-18; see also Pettersson and Wood 1987: 261), is defined as a default value for the term "vowel reduction" by Crosswhite (2001: 6): "[t]he neutralizing vs. non-neutralizing qualities of vowel reduction and vowel undershoot (respectively)". Further, as Zuraw (2002: 387) maintains, the phenomenon may reach a radical form of prominence reduction, that is, vowel deletion (see 3.2). This standpoint mostly matches Rix's hypothesis of a phonetically gradient process.

The motivation for the stage [ə] ([ɪ], or "open transition"; hereafter I mostly use [ə] as a cover term), that is, for a non-high vowel, as opposed to /i/, is the occasional attestation in inscriptional materials of spellings with ‹e› or ‹o› for medial reduced vowels; see the examples in (2.2.2, 2.2.3, 2.2.4). As already suggested by Rix (1966: 160-162), this fact is taken to support the idea that in an early stage of vowel reduction, reduced vowels were still not raised so as to phonemically merge with /i/ (see 2.4.1). Vowel reduction as a process of several vowels into [ə] may raise an essential question as to the phonemic status of [ə]. It seems less likely that Latin obtained a new phoneme /ə/ as a result of vowel reduction, since the evidence from the Romance languages does not necessitate the presence of /ə/. In this connection, vowel reduction in Bulgarian is contrastive (Crosswhite 2001: 33-43, 2004: 204-213). In this language, /e/ and /o/ coalesce with /i/ and /u/ respectively, while /a/ is reduced to /ə/. As understood from the representation, Bulgarian has a phoneme /ə/ (cf. Pettersson and Wood 1987: 261-265). Thus, the vowel reduction here is a case of phonemic neutralization. This is most likely not the case with Latin. Barnes's remark (2006: 19) is worth citing here: "I maintain the distinction between the phonological and the phonetic here, arguing in fact that a number of UVR [=

unstressed vowel reduction] patterns commonly identified as phonological in the literature are in fact phonetic in nature".[15]

The phonological ambiguity of [ə] or the like in Latin may explain the reduced vowels' greater susceptibility to surrounding phonetic contexts, reflected by the arbitrary notation, better than a phonemic /i/ would, although the /i/ has often been conceived of as a direct result of vowel reduction (or traditional "vowel weakening'). Then, describing what phonetic influnce reduced vowels underwent is important, as Lindblom insists that research on speech generation should aim at elucidating "the various modifications that speech sounds suffer in context" (1963: 1773). In fact, Rix's definition (1966: 159) of "vowel weakening" ("Schwächung") reflects this attitude: "In offener Binnensilbe vor Konsonant kann ein vorlateinischer Kurzvokal... erscheinen... in der Qualität durch die Lautumgebung bestimmt..." (a similar remark recurs in 1966: 160). That is, the phonemic/graphic identification of an intermediate stage like [ə] with more sonorous vowels that proceed in association with surrounding phonetic contexts is implied. Flemming (2005: 6) also notes this concisely: "as vowel duration decreased, the vowel formants were displaced towards values characteristic of the neighboring consonants".[16] A good example is provided by Flemming (2005: 28) about an effect of palatalization on the pattern of vowel reduction. Standard Russian has five vowels in stressed syllables, [i,

[15] Barnes also says that "we can be truly certain a given process involves changes in phonological feature specifications only if output feature specifications can be shown to be phonologically relevant to some other process. Where there is no evidence that the features in question are phonologically relevant (for the linguist and the learner alike), there is no compelling reason to assume that any change in the category membership of the phonological entity in question has taken place" (2006: 14).

[16] A similar stance is found in Godel (1961: 57, and n. 10), inspired by F. de Saussure's lecture "Grammaire historique du grec et du latin," though Godel solely focuses on vowel reduction in medial syllables.

e, a, o, u], and three, [i, ə, u], in most unstressed syllables. But only two, [i, u], of the latter appear after palatalized consonants. This is a result of the strong raising effect of such consonants, which makes the F1 (i.e., height) contrast between the central vowel [ə] and [i, u] no longer tenable. The nearby consonants may in turn be exposed to and influenced by adjacent vowels (e.g., velar *-l-*), thus the indirect effect of remote elements also comes into view.[17]

This line of reasoning will be broadened and deepened for the Italic languages, presently Latin, and then Sabellic in Chapter 4. Despite their material limitation as dead languages, attention to dynamic correlation between phonetics and phonology will lead to a better understanding and reconstruction of language history. Traits of mechanism(s) are occasionally visible, but have yet to be explained and settled in a larger picture. Vowel reduction especially leaves more evidence than deletion, which totally removes vowels.

2.2.2. Orthographic Fluctuation before Labials

In a pre-labial context, how much phonetic coloring a central vowel underwent is exhibited in a remarkable way by its oscillating orthography. The vowel may well have obtained more or less roundedness from labial consonants, and was no doubt phonetically represented by either [ɵ] or [ʉ]. Examples that recur in many handbooks and articles are (here mainly from Brugmann 1897: 224, Parker 1988: 234-235, and Meiser 1998: 68):

[17] Thus, Hall's statement (2003: 6): "Gestures do not simply fall in a linear order like segments or features; they overlap one another greatly." A gesture is defined as "an instruction for an articulator to achieve a particular constriction in the vocal tract: for example, an opening of the velum, a closing of the glottis, or a raising of the tongue body" (2003: 4).

CIL I^2 756.14 **SVRVPVERIT** ~ *surripuerit* (cf. *rapuerit*); CIL I^2 1488 **PONTVFEX,** *pontufices* [Var. *L.L.* 5.83] 'high priest' ~ *pontifex* (< **ponti-*); *obstupēscō* 'become senseless' ~ *obstipēscō* (*stupeō* 'be stunned' ← PIE **(s)teu̯p-*); *contubernālis* 'comrade' ~ CIL III 10506 **CONTIBERNALIS** (cf. *taberna* 'hut'); *monumentum* ~ *monimentum* 'monument' (< **mone-mentum*);[18] and many others like *vituperō* 'declare invalid', *aurufex* 'goldsmith', *quadrupēs* 'four-footed', *magnuficus* 'magnificent', *maritumus* 'of the sea', *novissumus* 'latest'. Etruscan *Artumes* (← Gk. Ἄρτεμις) also belongs here, as Skutsch (1913: 195) suggests. Some complexity would be added by the occasional involvement of vowels in the preceding syllable in coloring the central vowel. Front vowels as well as *-a-* (recall *-a-* > *-e-* [+front] in *arguō* 'declare' vs. *-erguō* ~ *-arguō*) may or may not favor a front vowel, and back vowels a back vowel: e.g., *regimentum* [cf. Fest. 348.16L], *alimentum* 'food', *benificus* 'beneficent' (~ *beneficus*) vs. *monumentum* 'monument', *documentum* 'example'; *legimus* 'gather', *agimus* 'drive' vs. *possumus* 'be able', *volumus* 'want'; *condemnāre* 'condemn' (normal) ~ *condumnāre* (see Brugmann 1897: 224, Maniet 1957: 144, and Godel 1961: 63). Likewise, forms like *animus* 'mind', *anima* 'breath', and *victima* 'animal offered in sacrifice' are generally with *-i-* (Godel 1961: 63); but cf. *recuperō* 'recover', *decuma* 'tithe', *tegumentum* 'protection', *intumus*

[18] As presented in Schrijver (2003: 71), presents in **-i̯e-* and causative-iteratives in **-ei̯e-* came to possess a stem in **-e-* serving as a non-present base. A morphological make-up of this sort is thought to have spread to statives as well, after the inflectional merger of statives (e.g., *tacēre* 'to be silent'; *-ē-* < **-eh$_1$-*) and causative-iteratives (e.g., *monēre* 'to advice'; *-ē-* < **-ei̯e-*) through phonological changes: Umb. **taçez**, *tases*, Lat. *tacitus* 'silent' < **tak-e-to-s*. Godel (1961: 59) already takes *mone-* or *mere-* (*mere-trīx*, **MERETO(D)** > *meritō*) as the basis of derivation.

Alternatively, *mone-* may be a product of analogical re-segmentation. To explain *monitus* (p.p.), Jasanoff (2003: 103 n. 30) postulates a process **mon-éi̯e/o-* →**moné-i̯e/o-*, based on **kap-i̯e/o-* (thus, **mone-tos*, *kap-tos* > *monitus* 'warned', *captus* 'taken'). The derivation of **mone-mentum* may thus be based on this *mone-*. The form *documentum* also belongs here.

'inmost', personal name *Hecuba* ~ *Hecoba* (Quint. 1.4.16; Gk. Ἑκάβη), CIL VI 2059.40 **VICTVMA** (~ *victima*; see Coleman 1962: 86 n. 2), etc.[19] On the other hand, the following syllable, particularly if it has *-i-*, seems more important in terms of vowel harmonic effects; the same vocalism is favorably selected to represent the result of vowel reduction (see Monteil 1979: 97): e.g., *occipiō*, *incipiō* 'begin' (3rd conj.), *eximius* 'excepted, exempt' (< **-em-*, cf. *emō* 'buy' < *'take'), *mūnicipium* 'self-governing community', *abripiō* 'take hold of', etc; cf. *occupō* 'seize' (1st conj.), *aucupō* 'go bird-catching'; *manubiae* 'prize-money' > *-i-* (attested relatively late; see Coleman 1962: 90, and n. 3).

Influence from surrounding vowels on the pre-labial vowel is paralleled in Sabellic to a certain extent. For superlative forms, Buck (1904: 55-56) suggests that whether the vowel before *-m-* is a front or a back one is determined by the kind of vowel in the preceding syllable: e.g., Osc. **últiumam** '*ultimam*, furthest' (< **h₂ol-tm̥mo-*), Umb. *hondomu* '*infimo*, lowest' (< **ĝʰom-tm̥mo-*), and Osc. **nessimas**/Umb. *nesimei* 'nearest' (< **ned-tm̥mo-*),[20] though the Paleo-Umbrian personal name **setums**[21] makes his explanation less decisive.[22]

It is to be noted (as traditionally so) that vowel reduction seems not to have been a

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[19] Godel (1961: 63) adds a possible condition that [velar]__*m* particularly feeds *u*-vocalism, based on *drac(h)uma* (later *drachma*) 'drachma', personal names *Alcumēna* and *Tecumēssa* (later *Tecmēssa*), *arcubus* (dat./abl.pl. of *arcus* 'bow'), *quercubus* (dat./abl.pl. of *quercus* 'oak'). If this is the case, forms with *-u-*, such as *recuperō*, *decuma*, *tegumentum*, *Hecuba*, are perhaps explicable, albeit with a front vowel in the preceding syllable.

[20] About the choice of this suffix, instead of **-ism̥mo-* in the traditional view, see Nishimura (2005: 171-173).

[21] Notice the recurrent typo of this form in Nishimura (2004: 244): ˣ**setuns** → **setums**. The discussion there is not affected by this error.

[22] See also the change **-e-* > *-u-* in Osc. *pertumum* '*dirimere*, prevent' (< **pert-emom*), instead of **pertimum*, as in Osc. **nessimas** and Umb. *nesimei*, with a front vowel in the preceding syllable.

momentary process. The gradual transition in Latin of *-u-* > *-i-*, known from the oscillating orthography in the above cases, reflects this fact. Further, it is of great value to explore what the origin of *-u-* is. Let us take an example from the superlatives and ordinal numbers, which are very illustrative for this purpose. Some of the superlative suffixes (**-(t)m̥mo-*, **-(i)s-m̥mo-*) and ordinal numbers (e.g, **septm̥mo-* 'seventh', **dekm̥mo-* 'tenth', **u̯ī-k̂m̥t-tm̥mo-* 'twentieth') are reconstructed with the prevocalic **-m̥-* (thus, **-m̥(m)o-* would be a more accurate representation with a homorganic glide in parentheses). In previous studies, the prevocalic **-m̥-* is often said to have changed into **-em-* or **-am-* and undergone vowel reduction to reach the attested sound *-um-* (> *-im-*). However, there are several problems in the diachronic scenario of both **em-* and **am-*hypotheses, as pointed out by Nishimura (2004: 232-241). Based on Lat. *homō* 'human being' and *humī* (see Vine 1993: 247-249, 2001: 120, and Livingston 2004: 37) as well as perhaps Osc. (ϰ)homoι '*humi*, on the ground' (see Del Tutto Palma 1989a/b, 1991, Lejeune 1990: 21, 55, and Untermann 2000: 395-96) and Umb. **sumel** (TIg. IIa 27) '*simul*, at the same time' (Meiser 1986: 59, and n. 2, and Livingston 2004: 33-34; this and Osc. (ϰ)homoι were presented as evidence by Nishimura 2004: 245-246), it is most likely that **-m̥-* changed into **-om-* in the first place,[23] then, as Vine (1993: 248) suggests (followed by Livingston 2004: 33 and Weiss 2005: 142), was reduced to *-um-*, which is exemplified by **mag-is-m̥mo-*[24] > **maksomo-*[25] > *maxumo-* > *maximo-*. Now based on

[23] This development is already, though very briefly, mentioned by Leumann (1977: 89). The notation of **-°m-* (> **-um-*) in Godel (1961: 62) may suggest the same. Coleman (1962: 87) uses a similar notation in **opt$_o$mos*, from which he has already deduced the intermediate stage **optomos*. See some details on Faliscan **maxomo** in 3.3.5.

[24] The root part **mag-* is traced back to **m$_e$ĝh$_2$-* (see Weiss 1998: 705, and nn. 12, 13). Its reflex **mag-*

my theory appealing to the presence of a central vowel in the reduction process, it can be said that a certain phonological complexity lies behind the result of the change, that is, a sound notated as "*-u-*".

To clarify this, we have rather informative forms in an inscription that dates to approximately 300 B.C.E., namely, **DECIMIO** and **DECEMIA** in CIL I^2 2848.[26] The former is erased on the surface and thus not perfectly legible, but there is some general agreement on its paleographic interpretation (see Solin 1972: 184-185, 1974: 167). Both exemplars are personal names derived from **dek̂m̥mo-* 'tenth'. One might suspect that the prevocalic **-m̥-* in the reconstruction developed into *-em-*, which is thus continued in **DECEMIA**. Then, with a vowel reduction **-E-** > **-I-**, this would yield **DECIMIO**. However, the development from the prevocalic **-m̥-* into *-em-* is made implausible by Lat. *homō*, *humī*, Osc. (ϰ)homoι, and Umb. **sumel**. Moreover, **DECIMIO** was presumably engraved earlier than **DECEMIA** on the evidence of its erasure. If this is the case, we have to do with an anachronism in terms of phonological change. Finally, both forms, with **-I-** or **-E-**, show an orthographic peculiarity as early as about 300 B.C.E., since a normal Republican form *Decumius*, later than those forms, is usually understood as a conservatism and as a precursor of *Decimius*. Then, the question arises how to interpret a linguistic stage with

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in the superlative can be explained either directly from the proto-form (via loss of prevocalic laryngeal) or by means of transplantation of the secondary **mag-*, which was realized by syncope from **maga-* in preconsonantal position (e.g., **meĝh₂*-no- > **maga-no-* > *magnus* 'big').

[25] For the outcome **om* of prevocalic **m̥* in Italic, see 2.2.2. Following the stage **maksomo-*, original **-g-* may have been secondarily restored, and then triggered the lengthening of the preceding vowel (i.e., according to Lachmann's Law). See more details in Jasanoff (2004). In the present work, I will follow the conventional notation of the word.

[26] A discussion of these forms has already been given in Nishimura (2004: 242-243), but what follows includes some modifications to it.

DECIMIO and **DECEMIA** that precedes a later stage with Republican *Decumius*.

I propose that these inscriptional forms cannot be taken at face value for the purpose of delineating phonological change, but instead, that their orthographic fluctuation reflects a sound, more precisely, the allophonic realization of a phoneme, which Latin speakers felt puzzled in transcribing, but eventually succeeded in codifying as ‹u› rather uniformly. It was most likely a central vowel, [ɵ] or [ʉ], rounded by the following labial. Some native speakers did not interpret this roundedness as the full rounding normally associated with ‹u› and were uncertain as to the [front]/[back] status of the vowel. As a result, they represented this central vowel by copying the same vowel in either the preceding or following syllable, which they "perceived" to be approximate to the reduced vowel. Coleman (1962: 96) presumes a shift, [u] > [ɨ] > [i], in (**optm̥mo-* > **optomo-* >) *optumo-* > *optimo-* (i.e., /u/ > /i/, for him), noting that [u] > [ɨ] is concealed under the standardized spelling ‹u›. But in my theory, to set up [u] as a starting point is phonetically dubious; we should instead start with [ɵ] or [ʉ] (though with no disagreement on [ɨ] > [i]; see below), thus its phonemic status as /u/ would also become less clear.[27] Sabellic examples I have adduced above, such as Osc. **últiumam**, Umb. *hondomu*, Osc. **nessimas**/Umb. *nesimei* 'nearest', and Palaeo-Umb. **setums**, also exhibit orthographic fluctuation of the vowel emerging from prevocalic **-m̥-*. It too would be identified as a phonetic entity, which was hard to transcribe for native Sabellic speakers.[28]

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[27] The nature of this phonological process must be understood as discrete from the so-called "*umerus*" rule in Latin, which is formulated as **-o- > -u-/#(C)__m(C)e/i-* (e.g., *umerus* 'shoulder', *umbilīcus* 'navel', *humī* 'on the ground', *cumera/cumerum* 'bin, basket', and perhaps *numerus* 'measure'; see Vine 1993: 249-250). This rule completely merged a part of the phonetic realization of original */o/ to the sphere of /u/ without regard to whether it was stressed or unstressed.

2.2.3. Corner High Vowels -i- *and* -u-*: Before Palatal* -l- *vs. Velar* -l-

As seen in the samples above, vowels targeted by reduction are *-e-*, *-a-*, and *-o-* in most cases. The upper corner vowels in the vowel inventory, *-i-* and *-u-*, show very few cases of orthographic fluctuation before labials: e.g, *pontufices*, **PONTVFEX**; *obstipēscō* (< *opstupēscō*; cf. *stupeō*); and perhaps *lacrima* < *lacruma* (cf. Andr. *dacrima*). The intermediate stage [ə] stemming from *-e-*, *-a-*, and *-o-*, eventually merged with /i/ (see more details in 2.3). It is thus not hard to imagine that the same categorical /i/-merging process may have included /u/ in some cases (*maxumo-* > *maximo-* could be added here, though the phonemic status of *-u-* as /u/ is not so clear, as mentioned above). Following this observation, the question arises whether the corner vowels /i/ and /u/ were also reduced to [ə] like /e, a, o/, or firmly stable except in a few cases.

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[28] Vowel reduction in adjacency to another labial sound, [w], has occasionally been mistaken as syncope due to the ambiguous notation for a reduced vowel and [w]; e.g., *dēnuō* 'anew' < **-nou̯ō*; *ab-*, *al-*, *col-*, etc. + *-luō* (cf. *illuviēs* 'dirtiness', *malluvium* [Fest. 152.25L, P. F. 153.13L], *malluviae* [Fest. 152.27, P. F. 153.13L, 225.12L] < **-lau̯ō* or **-lou̯ō*; *dēpuviō* 'beat thoroughly', *terripuvium* [etymologizing form of *tripudium* 'ritual dance in triple time'] vs. *paviō* 'thump'; *trīduum* 'a period of three days' ← **tridiu̯om*. Skutsch (1913: 197) erroneously posits syncope in *dēnuō* and *-luō* with subsequent vocalization of the original **-u̯-*. See Godel's correct interpretation (1961: 57), with vowel reduction.

The verb *concutiō* 'shake, strike' (*quatiō* 'shake') was also understood as a result of syncope by Skutsch (1913: 197), Kent (1932: 98), Maniet (1957: 150), Rix (1966: 158 n. 8), and Meiser (1998: 67). However, interpretation by means of vowel reduction is offered by Godel (1961: 57-58), Leumann (1977: 90), and Monteil (1979: 97, 101), with plausible influence from the preceding [^{w}]: **kom-k^{w}atiō* > **kon-k^{w}utiō* > *concutiō* (for delabialization before *-u-*, see Leumann 1977: 138-139), parallel to **-i̯aciō* > *-i̯iciō* > *-iciō* (see Schrijver 2003: 63-64). The gerundival adjective *secundus* 'following' (< **sek^{w}ondos*) also supports this view.

When adjacent to [w], a reduced vowel rarely shows orthographic fluctuation, unlike before other labials. This is understood as the early (and complete) merger of reduced vowels with /u/, particularly in this position; see a similar remark in Maniet (1957: 144-145). Such a radical assimilation is also manifest in the rounding of even /i/, which is generally resistent to vowel reduction: cf. *trīduum* < **-diu̯om* (Skt. *-divam* 'day'). Furthermore, in initial syllables, it may also be responsible for the vocalic change in PIE **penk^{w}e* > Italic **k^{w}enk^{w}e* > Sabellic **k^{w}o-* > *po-* (e.g., Osc. **pumperias** [calendar term], Umb. **pumpeřias** '?').

A solution comes from consideration of the phonemic realization of [ə] before *-l-*. Latin is well-known to have two phonetically different *l*'s, that is, *l exilis* (palatal *-l-*) and *l pinguis* (velar [so-called "dark"] *-l-*). The former appeared before *i*-vocalism or another *-l-*, motivating the preceding high front vowel /i/ to surface as a phonemic outcome of vowel reduction or anaptyxis. Before a velar *-l-*, on the other hand, the high back vowel /u/ emerged instead. Most of the following examples are well-known cases:[29] *epistula* ← Gk. ἐπιστολή 'letter'; *sēdulō* 'diligently' vs. *dolus* 'wile'; *rotula* 'little wheel' vs. *rota* 'wheel'; *famulus* 'servant' (< **famelos*; cf. Paelignian *famel*) vs. *familia* 'family' (< **fameliiā*); *pepulī* 'pushed, struck', *(te)tulī* 'carried', *sustulī* 'picked up' < **pepalai̯* or **pepolai̯* (< **-pl̥h₂-* or *-polh₂-*; see Meiser 1998: 210), etc.;[30] *occulō* 'conceal' < **ob-kel-ō*;[31] *Siculus* 'Sicilian' (← Σικελός) vs *Sicilia* 'Sicily'; *adulēscō* 'grow up', *adulēscēns* 'youthful' (but cf. *o*-vocalism in *abolēscō* 'wilt', *adolēscō* 'grow up', *indolēs* 'innate character', *subolēs* 'offspring')[32] vs. *alō* 'feed', *alēscō* 'be nourished' (*coalēscō*

[29] Examples are selected from Niedermann (1953: 4, 9), Monteil (1979: 97, 111), Sihler (1995: 174), and Meiser (1998: 68).

[30] Some clarification may be needed for these forms. In the 1.sg.pf., the point of departure would have been **pépalai̯* (or **pépolai̯*), and then, **pep*[ə]*lei̯*, reflecting vowel reduction in both medial and final syllables. Note that *-l-* was not palatalized by *-e-* (see Weiss 2005: 101 n. 8), thus the reduced vowel before *-l-* did not fail to emerge with [+back]. (For a similar gradient as a conditioning factor among front vowels, see Tuttle 1997: 30 concerning palatalization in Italian, where [i] stands before [e] as a triggering factor.) In the 3.sg.pf., a process like **pépole* + *-t* > **pep*[ə]*lit* > ***pepilit* would be expected, the result of which might have exerted an analogical effect on the 1.sg.pf. Recall also the variation of the ending of the 3.sg.pf., as in Pl. *Capt.* 9 *vendidīt*, 34 *ēmīt*, CIL I^2 1297 (= Diehl 615) **FVVEIT**, CIL I^2 584 ("Sententia Minuciorum") **POSEDEIT** (see Leumann 1977: 607). This would lead us to assume the variant **pep*[ə]*leit* (> **pepulīt*), where [ə] had a chance to bear [+back].

[31] On the other hand, prefixed verbs based on *colō* 'cultivate' (< **k*^{w}*el-*), such as *incolō* 'inhabit', *accolō* 'dwell by', and *recolō* 'cultivate again', do not exhibit any sound change in root vocalism. This may be reckoned as a result of the avoidance of "homonymie fâcheuse" with forms in *-culō*. But cf. *anculus* 'man-servant' [P. F.] (< **k*^{w}*el-*), with the original sense of the root « circuler » frozen, thus becaming associated less with *-colō*; see Ernout-Meillet (1985: 32).

'grow together', cf. *cōlēscō* [Var., Lucr.]; *prōlēs* 'offspring');[33] *ex(s)ulō* 'spring up' (< **-saltō*) vs. *exsiliō* 'spring forth' (< **-saliō*); *simulāre* 'pretend' vs. *similis* 'similar'; *catulus* 'young dog' vs. *Catilīna* (cognomen); *facultās* 'ability' (later *facilitās*) vs. *facilis* 'easy', *difficilis* 'difficult'; *umbilīcus* 'navel' (and OIr. *imbliu* < $*h_3n̥b^h$-*el*-; cf. Gk ὀμφαλός < $*h_3n̥b^h$-*l̥*-, OHG *nabulo* < $*h_3nob^h$-*l̥*-; see Meiser 1998: 107); *sepeliō* 'bury' vs. *sepultus* (p.p.); *nebula* 'mist' (← Gk. νεφέλη).

Interestingly, Juret (1919: 94 n. 2) offers examples with *-i-* before a velar *-l-*: *mutilus* 'mutilated', *nūbilus* 'cloudy', *petilus* 'thin', *rutilus* 'ruddy', *pestilēns* 'unhealthy', *postilēna* 'crupper(?)', *cantilēna* 'refrain'. Some of these most likely have *i*-vocalism to begin with in the derivation (e.g., *pesti-*, *posti-*), because their base words belong to the third declension (with some morphological overlap acknowledged between *i*-stem and consonant-stem nouns, as in *nūb-i-lu-s* [*nūbēs* 'cloud'; cf. Gk. νέφος, Ved. *nábhas*]). André (1959: 83) maintains that *-i-* remained intact before a velar *-l-*, following Niedermann (1953: 22) and Maniet (1957: 142-43, 145). Then, Juret's question (1919: 100): "Dans *nubilus* [*sic*], *mutilus*, etc., un *ĭ* se maintient devant une *l* vélarisée. Comment concilier ce fait avec l'affirmation que le timbre des voyelles aurait été à la merci des consonnes suivantes?" Returning to our question above, when combined with Juret's, a larger picture emerges. That is, descriptively, one of the corner vowels, high front /i/, may have been phonemically preserved in vowel reduction. In other words, /i/ in

[32] The variation seems to imply the phonetics of an older stage, that is, a relatively free-varying height of the vowel, [ɵ] ~ [ʉ], before a velar *-l-*. The mid-height is more compatible with ‹o›, while the high position is so with ‹u›: e.g., **POCOLO(M)** 'cup' in CIL I^2 439-452, **CONSOLVERVNT** 'took counsel' (3.pl.pf.) vs. CLat. *cōnsuluērunt* (< **konsel-*), etc.

[33] See some details in André (1959: 85-86).

unstressed position was presumably not reduced as [ə] unlike /e, a, o/. In this view, it seems plausible that the other corner vowel, high back /u/, also remained phonemically unchanged (cf. most conservative forms, e.g., *coniugem* 'consort', *pecudis* 'livestock'), and it was only a further development, namely, a rather wholesale merger with /i/ that blurred such a retained stage in some cases (*capitis* 'head' < **caput*-, *manicae* 'handcuffs' < **manu*-, *corniger* 'having horns' < **cornu*-). But the persistence of the spelling ‹u› in Literary Latin before labials (e.g., *obstupēscō*, *lacruma*; cf. *obstipēscō*, *lacrima*) seems to favor the reconstruction of a linguistic phase when /i/ and /u/ were retained side by side. Thus, we can now visualize a rather clear picture of the opening process of vowel reduction, which I refer to as "Process I" in this work:[34]

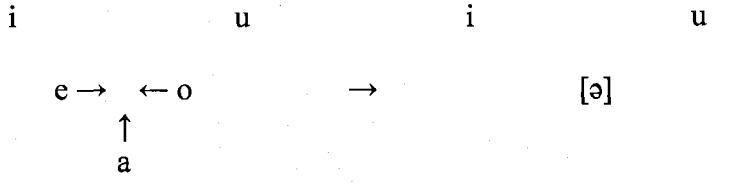


Fig. 2.1: Vowel reduction in medial syllables: Process I.

Crosswhite's Appendix B (2001) presents the development as theoretically possible, though to her knowledge, it is unattested; note relatively similar patterns, attested in Balear Calatan, Rossellenes Calatan, and Russian.

The position before the other liquid, *-r-*, is also known to have incurred great

[34] Janson (1979: 52) restricts medial vowel reduction in open syllables to non-back vowels *-e-* and *-a-*. But his exclusion of *-o-* seems too extreme, disregarding a certain number of examples.

influence on an adjacent vowel. Phonemes /e, a, o/ may have been centralized, but no longer evolved in terms of height (i.e., not toward /i/), being realized as either /e/ or /o/. Examples include *pariō* 'bring birth' vs. *-periō*, pf. *peperī* (cf. Falsc. **pepara‹i›**);[35] *imperō* 'demand the payment of" < **-parō*; *legere* 'to gather' < **legese*; *dēvorō* 'devour'; *exorior* 'arise'; *memorem* 'mindful' (Godel 1961: 55, 57); *s*-stem nouns, in which the suffixal element *-or-* is preserved (i.e., not changed into *-er-*).[36] This vocalic shift or retention can be construed thus: while /a/ lost its phonemic independence in pre-rhotic position, merging with /e/, the other two phonemes /e/ and /o/ kept their original phonemic status due to the effect of the following *-r-*. From this conjecture, it follows more naturally that a vowel targeted by reduction did not merge phonemically with /i/ at all, but rather evolved into a central vowel like [ə].[37]

It is true that high vowels in unstressed position might appear to have been reduced to central mid vowels as well, and then merged with front or back mid vowels: *capere* 'to

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[35] For the morphological shift from the third to the fourth conjugation in *re-periō*, an analysis from a standpoint of metrical phonology is presented by Mester (1994: 27-28).

[36] As Bader (1960: 236-237) remarks, a group of *s*-stem neut. nouns with *-er-* (< **-es-*) in oblique cases had **-es-* (not **-os-*) from the beginning, e.g., *vulnus, -eris*; *tempus, -erī* (*-or-* was analogically obtained from *s*-stem animate nouns such as nom.sg. *honōs* < **-os-s*, gen.sg. **honŏses* [→ *-ōris*, from the strong stem]).

[37] This reveals a methodological problem with Chart I in Parker (1988: 230). His scenario **genesem* > **genisem* > **genirem* > *generem* might theoretically be possible, but he really needs *-i-* as a product of vowel reduction for his argument, which is implausible based on what I have claimed above. If he is right, a scenario like **maxomos* > **maximus*(?) > *maxumus* > *maximus* would also be possible; the principle of economy Parker seeks (1988: 225) seems violated by him. Parker (1988: 234 n. 39) also admits that "*u* is a conditioned reflex in labial environments," taking phonetics into consideration; rhotic environments should be treated analogously from the phonetic standpoint.

Parker (1988: 239-240) appeals to CIL I^2 32 **NVMISIO**, 33 [**NV**]**MISIO** [theonym] to corroborate his relative chronology, *a* > *e* > *i* → rhotacism → R-Lowering (*i* > *e*). However, the appearance of *-i-* in the theonym must be regarded as a result of non-application of rhotacism (and subsequent R-Lowering in Parker's theory). Conservatism commonly seen in proper nouns is most likely responsible for the default of *-s-* > *-r-*. This case is exactly paralleled by the co-existing family names, *Numerius* and *Numisius*.

take', *facere* 'to make' < **-i-r-* < **-i̯e-s-* (or **-i-s-*?; see Schrijver 2003: 67); *lēgerim* < **lēg-is-ī-m* (pf.suj.1.sg. of *legō* 'gather'); *cinis* 'ashes' (Gk. κόνις) vs. *cineris* (gen.sg.); *socer* 'father-in-law' < **su̯ekuros* (Gk. ἑκυρός); *vetus*, *veteris* 'old'; *veternus* 'morbid state of torpor' < **u̯eterinos* < **u̯etus-ino-s* (cf. *Falernus* [toponym] < **Falerinos* < **Falis-ino-s*); *peiierāre* 'forswear, swear falsely' < **ped-i̯us-ā-se* (for *-ur-* > *-er-*, see Parker 1988: 223-224). Recall, however, that the high corner vowel was salient in its resistance to vowel reduction, as demonstrated above (2.2.2). As a matter of fact, even *stressed* high vowels occasionally became mid vowels: *serō* 'plant' < **sisō*; *fóre* (fut.inf. of *sum* 'be') < **fu-se* (cf. *fu-tūrus*); *nora* 'daughter-in-law' < **snusā* (Skt. *snuṣā*, OCS. *snŭxa*, Gk. νυός); *fórās* 'out of doors' ≈ Gk. θύρᾱς (see Maniet 1957: 145, 147, Monteil 1979: 96, and Parker 1988: 225).[38] Given this observation, the mid-vocalizing effect of *-r-* is understood to have operated regardless of whether the vowel in question was stressed or unstressed.[39] Thus, the forms given, such as *lēgerim* < **lēg-is-ī-m* and *cineris*

[38] This did not occur in closed syllables, as in *circum* 'round'; and even the opposite effect, namely, raising, occurred in some forms (*fīrmus* 'strong' vs. *fermē* 'almost'; see Leumann 1977: 45). Non-application of *i* > *e* in *dirimō* 'pull apart' (< **dis-emō*) and *diribeō* 'dispense' (< **-(h)abeō*) can be analogical to *dis-* + consonants. According to von Planta (1892: 98), there is no counterpart of the *serō*-type change in Sabellic. An old etymology of **vereiiai**, **vereias**, **verehasiúí** that von Planta mentions, in possible connection with Lat. *vir* 'man', was replaced by Rix (1999).

[39] The mid-vocalizing effect of *-r-* might have been operative remotely from the targeted vowel. As Sen (2006: 156-159) remarks, consonant clusters like *-Tl-* and *-Tr-* are in general heterosyllabically divided in Pre-Literary Latin, unless a morpheme boundary (if to some extent explicit to speakers) comes before those clusters. The suffix *-trīx* is counted as a case of the latter. A sequence of CV[mid](s)C*e.trīx* (not °C*i.*) is frequently found: e.g., *genetrīx* 'mother', *meretrīx* 'courtesan', *obstetrīx* 'midwife' (cf. *stator* 'official servant of provincial governors'). According to Sen, *e*-vocalism is accounted for by means of the mid vowel in the preceding syllable and *-r-* in the following syllable, each of whose phonetic background (a vowel harmonic effect and the coarticulatory effect of *-r-*, respectively) seems to be plausible. While his argument is attractive in many respects, the longer life as an independent suffix assumed for **-tro-* (thus, **-.tro-*, to explain vowel reduction in *tālitrum* 'flick with the joint of a finger'; cf. *tālus* 'knuckle, heel') than **-dʰro-*/**-dʰrā-* (> **-b.ro-*/**-b.rā-*, via development of the consonant; e.g., *palpebra* 'eyelid', *scatebra* 'bubbling spring') seems to be arbitrary ("a limited survival" in 2006: 157 for **-tro-*?); the discussion does not entirely avoid being circular.

(< **kinis-es* [via vocalic assimilation] < **kenis-*), cannot be defined as the outcome of vowel reduction. A similar development can also be found in Umbrian: e.g., *furfant* '?', *tursitu* 'drive away', etc. < **-or-* (see Meiser 1986: 116).

The lowering of *-ur-* > *-er-* needs some more clarification, since the vowel shifted from [+back] to [+front] as well as from a high to a mid vowel. The noun *socer* (< **sụekuros*) can be solved by analogy with other kinship terms, such as *pater* 'father', *mater* 'mother', *gener* 'son-in-law', etc., as in Götze (1923: 108), Kent (1945: 102), and Sommer-Pfister (1977: 83).[40] In fact, *satur* 'well-fed' (< **saturos*) retains its original medial vowel (see Ciardi-Durpé 1901: 220).[41]

2.2.4. Vowel Harmonic Effect

The posited linguistic stage with [ə] facilitates understanding oft-cited vowel harmonic phenomena, that is, a *phonemic* retention of a string of historically identical vowels, which we have seen in 1.2.1: e.g., **MERETO[D]** (CIL I^2 9, ca. 200 B.C.E.), **MERETO** (CIL I^2 1531, ca. 150 B.C.E.) = CLat. *meritō* 'deservedly' (cf. CIL I^2 62 **MERTO**, 3rd c. B.C.E.); *alacer* 'cheerful'; *alapa* 'box on the ear'; *anatis* 'duck' (~ *anitis* [Pl., Cic.]; gen. of *anas*);[42] *Calaber* 'Calabrian'; *calamitās* 'disaster'; *calamus* 'reed pen' (Pl.+; Gk. ←

[40] Cf. Prisc. II 27.17K: "... antiqui 'auger' et 'augeratus' pro 'augur' et 'aurguratus dicebant."

[41] The *e*-vocalism in verbs such as *dēierat* 'swear'/*ēierāre* 'to reject upon oath on the grounds of partiality' (← *iūs iūris* 'law'), *peiierāre* 'swear falsely' (*peius* 'worse' < **ped-ịus*; see Leumann 1977: 546), is presumably explained by an assimilatory process caused by the same vocalism in initial syllables.

[42] Ernout (1909: 108-109) suggests *anas anatis* was a rural form, and this would be the reason why it preserved the original vocalism. This claim is, however, problematic; one would have to extend the claim

κάλαμος); *celeber* 'busy, festive' (~ f.[/m.] *celebris*, n. *celebre*); *elementum* 'element(s)'; *elephantus* 'elephant'; *genetīvus* 'of birth' (~ *genitīvus*); *hebetis* 'blunt, dull' (gen. of *hebes*); *neglegō* 'disregard' (vs. *colligō* 'collect'); *percolopō* 'cuff' [Petr.] (← Gk. κόλαφος); *perpetuus* 'continuous'; *segetis* 'cornfield' (gen. of *seges*); *Seneca* [personal name]; *sepeliō* 'bury'; *teretis* 'smooth and rounded' (gen. of *teres*); *vegetus* 'lively'; *vehemēns* 'violent'; etc.[43]

The orthographic selection of the same vowel for vowels in both first and second syllables might appear to imply that they also have *phonetically* the same value. But it is more likely that vowels in non-initial unstressed syllables are actually [ə], caused by vowel reduction, and their original *phonemic* status, on the other hand, is generally preserved (orthographically, as well) due to a vowel harmonic effect from preceding vowels of the same phonemic value. This operation is not so consistent as to entirely prevent the later wholesale merger of reduced vowels with /i/. Some of the examples, where a set of doublets stand side by side (*anat-* ~ *anit-*, *neglegō* vs. *colligō*, **MERETO** ~ *meritō*), imply that the /i/-merging process also took place in such cases, which is more easily deduced from a stage of [ə] than directly from /e, a, o/.

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to all the forms above, and even having done so, fail to capture the fact that all these forms contain the identical vocalism in two successive syllables.

[43] We could add *ēnecō* 'kill' (~ *ēnicō*) here. See Barsby (1999: 190) on Ter. *Eun.* 554 *enicet*: "Don. [= Donatus] vouches for the spelling *enicare*, which he regards as more euphonious than *enecare*." See also Sardinian *mármaru* 'marble' (< Lat. *mármor*) and *cábadu* 'head' (< Lat. *cáput*), as pointed out by Y. Kanazawa (p.c.), and Etr. *calanice* (← Gk. Καλλίνικος) and *parθanapae* (← Gk. Παρθενοπαῖος), cited by Skutsch (1913: 195).

Havet (1889: 26-27) suggests that a phonological context adjacent to *-l-* favors a harmonic effect, but not all of the examples can be accounted for in this way.

2.3. Vowel Reduction in Final Syllables

In the preceding section, we have reviewed vowel reduction in medial syllables, profiling its phonetics and phonology by way of something like [ə]. As has been suggested above, central vowels engendered by the initial step of vowel reduction would eventually merge phonemically with /i/ in most cases. A more detailed analysis of this process will be postponed until the following section (2.4), since the object of this section, vowel reduction in final syllables, is helpful for understanding features of medial as well as final vowel reduction.

Vowel reduction in final position has traditionally drawn less attention than the process in medial syllables. But given the certain chronological uniformity of vowel reduction between these two positions demonstrated in 2.1, it has become more pivotal to clarify the phonological mechanism of vowel reduction in that location for a more general understanding of this sound change. Here, positional differentiation of the process emerges as an especially interesting problem.

One specific feature of final vowel reduction is that vowels are generally in closed or diphthongal syllables. Thus, a pattern of reduction such as *-a-* > *-e-* is comparable to its counterpart in the medial position, which has been briefly surveyed in 2.1:[44] *artifex* 'skillful' < **-fac-*; *rēmex* 'rower' < **-ag-*; *inlex* 'decoy' < *-lac-*, cf. *inliciō* 'allure'; *auceps* 'fowler', *princeps* 'chief' < **-cap-*; 1st decl. dat.-abl.pl. *-īs* < **-ei̯s* < **-ai̯s* (cf. dat./loc.sg.

[44] The following are selected from Thurneysen (1883-85: 313), Brugmann (1897: 227), and Monteil (1979: 111).

*-āi̯- > -ai [> -ae]); etc.[45] Another reduction pattern from -o- to -u- is seen in forms such as *maius* 'bigger') < **mag-i̯os*; nom.sg.m. *Novius*, *Plautius* [personal names] < *-i̯os (cf. **NOVIOS PLAVTIOS** in CIL I^2 561); **CERERVS** [theonym] (CIL I^2 677; Capua, 106 B.C.E.), **NATIONV** (CIL I^2 60; Praeneste, before 200 B.C.E.) < *-os (gen.sg., instead of the Standard Latin *-es [> -is]); acc.sg.m/nom.-acc.sg.n. -um < -om (e.g., *dominum* 'owner', *dōnum* 'gift");[46] 3.pl.pres. -unt < -ont; *is-tud* 'that which you have' < *-tod; *iecur* 'liver' < *-or < *i̯ek^{w}r̥t (Skt. *yakṛt*); **hospus* (→ *hospes* 'host'; cf. Paelignian *hospus*) < **hosti-pot-s*; 3.sg. pass. -tur < *-tor (e.g., *agitur* 'is driven'); nom./acc.sg. of some *s*-stem neut. nouns -ur < *-os (e.g., *fulgur* 'lightening' [cf. gen.sg. *fulguris*;[47] m. *fulgor*, *-ōris*], *ebur* 'ivory', *guttur* 'gullet' [cf. m. *gutturem* in Pl.], *sulphur* 'sulphur' [cf. *aequor* 'expanse', *ador* [a kind of grain], *marmor* 'marble' (cf. Quint. 1.6.22 *marmur*)]; see Leumann 1977: 94, 379).[48] The lack of this pattern in Faliscan, on the other hand, is confirmed by forms such as **lartos** [personal name] (Ve 245b) and **loifirtato** '*libertatis*, freedom (as either the abstract noun or the divinity)' (Ve 253), which might be construed as a part of the evidence for the general absence of vowel reduction in the language (but

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[45] The development -i- < -a- is conceived of by Meiser (1998: 71) in *prōdit* 'project' < **prō-dat*, which is however not entirely secure. A secondary thematization, something like **didh$_3$-ti* > *didati* →**dideti*, outlined by Meiser himself (1998: 185, 188; see also 2003: 52-55), also seems to underlie the base verb *-dit* (thus, from **-det*).

[46] For some chronological details of this change, see 2.1 and foonote 7.

[47] For assimilatory seriation of *-u-*, see Kent (1932: 99, 101).

[48] The vocalic change in the nom.pl. and dat./abl.pl. of the second declension, *-oi̯ > -ei (e.g, **VIREI** in CIL I^2 581, text with archaizing spelling) > -ē̯ (e.g, **PLOIRVME** in CIL I^2 9) > -ī (not *-oi̯ > *-ui̯) and *-ōi̯s > -ois (e.g, **RIVOIS** in CIL I^2 4, **QVROIS** in CIL I^2 2833) > -eis (e.g, **CASTREIS** in CIL I^2 614, **FACTEIS** in CIL I^2 10) > -īs (not *-oi̯s > *-ui̯s), respectively, is an exceptional case (cf. dat.sg. *-ōi̯ > -ō). The development -o- > -e- is perhaps the result of fronting triggered by the following -i̯-. Umbrian shows a similar development in the dat./abl.pl. (**-es**, etc.), which however took place in a later period (cf. Osc. **-úís**, etc.), and is thus not Proto-Italic.

cf. 2.4.1). And thirdly, as mentioned in 2.1, *-e-* is reduced to *-i-*, unlike the usual retention in medial closed syllables: *agis*, *agit* 'drive' < **-es*, *-et*; *ped-is* 'foot', *homin-is* 'man' < **-es*.[49]

What immediately draws our attention is that the [front]/[back] contrast in this position was retained, when we look particularly at *-o-* > *-u-* and *-e-* > *-i-*. In medial position, we have seen such a retention of contrast particularly before liquids (e.g., *familia* 'family' vs. *famulus* 'servant') in 2.2.2. Here, as it seems, we need not consider such phonetic factors caused by surrounding consonants. True, patterns of *-o-* > *-u-* and *-e-* > *-i-* are not totally parallel: the former is hardly found before *-t*; before *-r* and *-m*, the latter was arrested (e.g., *ager* 'land, *patrem* 'father'). But the reduced outcomes of *-o-* and *-e-* remain distinct in phonemic terms. Vowel reduction in final syllables can therefore be understood as a process that did not neutralize the [front]/[back] contrast, unlike the process in medial syllables. Along this line of reasoning, a similar picture would also emerge from another pattern of change *-a-* > *-e-*. We can adhere to the view of retention of [front]/[back] by hypothesizing that at the very beginning of the reduction process, not

[49] Cf. an oft-cited vocalic shift *-i-* → *-e-*, as in **-dik-s* → *index* 'informer', *iūdex* 'judge', and **com-i-t-s* → *comes* 'companion' (cf. *itāre* 'be in the habit of going'); see Monteil (1979: 102). The shift is descriptively not vowel reduction but lowering from a high vowel to a mid vowel. But to phonologically interpret the coincidence of the outcomes between *-a-* > *-e-* and *-i-* > *-e-* is unfruitful; the nouns in *-e*C(C)# (< **-a-*) form a rather uniform semantic category, names of professions or the like (e.g., *artifex* 'craftsman' < **-fak-s*; *antistes* 'overseer' < **-sta-t-s* [< **-sh₂e-t-*; see Nussbaum 2004: 8]; *tībīcen* 'piper', *tubicen* 'trumpeter' < **-can-s*; see also *mūni-fex* 'soldier who does not have exemption from fatigues or sim.' [Plin.; P. F.] ← *mūni-fícus* 'dutiful'). The forms above that continue **-i*CC# also belong to this semantic group, and may thus have phonologically aligned to *-e*C(C)# (see Leumann 1977: 94). Already Brugmann (1897: 226) regards *iūdex* and *comes* as results of later modification. This process seems to be the case with *hospes* (< **g^{h}osti-pot-*s), in spite of Parker's alternative (1988: 232-234). Other nouns of profession, such as *mīles* 'soldier', *vēles* 'skirmisher", *satelles* 'companion', *eques* 'horseman' (← **equo-t-*?; see Brugmann 1897: 226 and Parker 1988: 233), may have participated in this semantic association with *e*-vocalism. I am indebted to A. Nussbaum (p.c.) for reminding me of contrastive *i*-retention in different semantic fields, such as *salix* 'willow'.

[e] directly but rather [ə] resulted from /a/. This process consistently preserved [−front]/[−back]. In this position, unlike the medial one, the /i/-merging reduction (cf. 2.4) was not applied. Hence [ə] was no longer raised, but was dissociated from /a/, then merged with /e/, with which it is more similar in terms of tongue height. (See more phonetic details on unstressed /a/ in 2.4.3.). Taken altogether, it can be said that the whole process was more moderate in terms of the [front]/[back] contrast than the one in medial position that neutralized /e, a, o/ into [ə] wholesale in most cases (cf. 2.2.3):

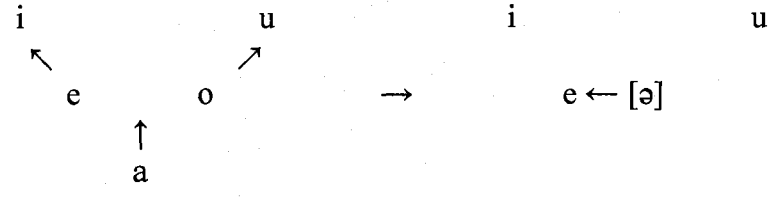


Fig. 2.2: Vowel reduction in final syllables

The coexistence of two different systems of vowel reductions and the distribution of more and less restrained vowel reduction in final and medial position, respectively, are both typologically common. By way of a type of Russian ("Old Muscovite"), Crosswhite (2001: 149-152) describes a partial block of the reduction process. In principle, /a/ and /o/ are reduced into [i] after palatalized consonants. But in morphological endings, the process only proceeds to [ə]: e.g., [kápl^{j}ə] 'drop' (nom.sg.f.), [brát^{j}jəf] 'brothers' (gen.pl.m.), etc. If accented, these endings are realized with [a], [o]: e.g., [zim^{j}á] 'earth' (nom.sg.f.), [katóf] 'cats' (gen.pl.m.), etc. This partial reduction can be ascribed to the

constraint that a single morphological ending must emerge with the same specification in terms of [front]. Hence, the vowel in question is not reduced into [i], which is [+front] in contrast to [a] and [o]. This example involves morphological information (case ending) as well as a phonetic factor (consonant palatality), and is thus more complex than our situation in Latin. Still, the Russian case conclusively proves that vowel reduction of a lesser degree coexists with the more extreme one.

In the example above, morphological endings, that is, final syllables of words, have drawn our attention. We are then led to expand on this positional characteristic, since this location is cross-linguistically known to reveal phonological peculiarities. Barnes's work (2006) is of great value in this matter. He provides a compact but sufficiently detailed description of several languages in which final syllables show partial resistance to vowel reduction due to final lengthening. For instance, in Nawuri, which belongs to the Kwa family, phrase-final vowels is not affected by centralization (long vowels and absolute-initial vowels also remained unaffected), all of which are associated with increased duration (Barnes 2006: 94). Yet, such resistance is only partial. In Swedish, despite undeniable lengthening of final-syllable unstressed vowels up to the approximate duration found in stressed vowels in initial syllables, vowels in final position are still somewhat reduced (Barnes 2006: 95-96). We must thus recognize the gradient reduction pattern carried out in final syllables, where "[i]n principle..., a language could come to generalize not the more reduced, but rather the less reduced variant of the final syllable vowel" (Barnes 2006: 107).[50]

[50] We can add a similar case, which would actually justify my conception more succinctly. According

2.4. Further Vowel Reduction in Medial Syllables

2.4.1. Phonological Merger with /i/ in Medial Open Syllables

Now we turn to vowel reduction in medial position, as promised in 2.3. The linguistic stage of vowel reduction we have reached via Process I was a central vowel like [ə], which must have had phonetic variation, depending on adjacent phonological contexts (e.g., [ɵ] or [ʉ] before labials). Many of the reduced vowels eventually merged with /i/ in open syllables (cf. /u/ before velar *-l-*). As I have already implied, this convergence may be regarded as a second step of vowel reduction, which partly corresponds to the second half of *V > [ə], [ɪ], or "open transition" > /i, u/ (or Ø, via syncope), stratified by Rix (1966); above, I have specified *V as */e, a, o/ and added [ɵ] or [ʉ] as an intermediate stage before labials.[51] This process is mirrored in the phonological transition, for

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to Maiden (1997: 10-11), an Italian dialect spoken in Servigliano (Marche) shows different degrees of a vowel harmonic process, concomitant with stress-driven reduction. These are classified in accordance with relative location to stress: 1) complete harmony of post-tonic vowels: e.g., [do'mennaka, do'menneke] < *[do'menika, -e] 'Sunday, Sundays', ['prɛdoko, 'prediki, 'prɛdaka] < *['prɛdiko, -i, -a] 'I/you/he preach(es)', ['alama, 'aleme] < *['anima, -e] 'soul, souls'; 2) partial ("vertical" = keeping [front][back] contrasts) harmony in pre-tonic vowels: e.g., [pu'timo] < *[po'timo] 'we can', [fjuril'litti] < *[fjorel'litti] 'little flowers' (or no harmony in pre-tonic position: e.g., [ɲenok'kjo] 'kneeling', [mari'ta] 'marry'). This typological evidence can corroborate my description of positionally differentiated vowel reduction in Latin.

[51] Janson (1979: 53) insists that the same picture, *e, a, o* > [ə] > *i*, is found in Leumann (1977: 81), which is, however, not true. Rather, Leumann posits a process *e, a, o* > *e* > *i* (similarly, *a* > *e* > *i* in Götze 1923: 130, Kent 1932: 103, and André 1959: 86; see also *o* > *e* in Parker 1988: 232). Janson refutes any two-step procedure, but only with cursory comments. He argues that a stage of [ə], if it really existed, should have been attested as ‹e›. In fact, we have already seen **DECEMIA** above, to which the following can be added: **APOLENE** (CIL I^2 384; Mosciano, 289-264 B.C.E.; see Wachter 1987: 438), **APOLENEI** (CIL I^2 368; Pisaurum, ca. 200 B.C.E.), **ESQELINO** (CIL I^2 416; ca. 220 B.C.E.), **HEC** (CIL I^2 9; late 3rd c. B.C.E.) < *-*o*- (unaccented when cliticized), *acced-* (Enn. *scen.* 247, etc., instead of *accid-*), and perhaps **PROIECITAD** (CIL I^2 401 "Lex Luceria", early 3rd c. B.C.E.) < *-*i̯ak*-; also in later periods, **CONIECIANT**, **OPPEDEIS** in

instance, from *maxumus* (with [ʊ] or [ʉ] before the labial *-m-*) to *maximus* 'greatest', being relatively slow due to the existence of *-m-*. Where there was no impeding factor from surrounding sounds, this phonological merger would have been completed more quickly.

In phonological terms, the second step in medial syllables can be considered total vowel reduction in both F1 and F2, that is, a complete neutralization in the contrast of height, and [front]/[back] and [round] features, respectively. In the case of Latin, only a vowel with [+high] and [+front]/[−round] is allowed to emerge; thus, /i/.[52]

We need to clarify what the synchronic and diachronic status of such a sequential process was, namely, reduction of a vowel to a centralized one, and then to /i/ in most cases. I propose that these two steps of reduction were two synchronic facets of a diachronic change that seems to have taken a while to be completed as a whole, following Rix's prediction (1966: 160-162). As briefly mentioned in 2.2.1, he suggests that in the beginning of the literary period, reduced vowels were still not changed so as to phonemically merge with /i/, invoking graphic fluctuation such as *optumus* ~ *optimus* 'best' and ‹i› ~ ‹e› (*hic* ~ CIL I^2 9 **HEC** 'this'; unaccented when cliticized);[53] note that I have construed ‹e› as a default spelling for [ə] in footnote 51. This can be interpreted as

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CIL I^2 583 [as late as 123/2 B.C.E.] (see some details in Wallace 1988: 219-220) and perhaps *dēiĕcit* [Quadrigarius (1st c. B.C.E.) apud Gell. 7.9.2-6], if it is historical present (see Courtney 1999: 149). The letter ‹e› can be regarded as a default spelling for [ə], a mid unrounded vowel, closest to /e/ in terms of phonetic features. Though Crosswhite (2001: 144) asserts that [ə] is unspecified for color features like [front]/[back] and [round], it differs from [ʊ] in [round] feature.

[52] Cf. a brief summery of the traditional terminology (F1 and F2) and its potential problem in Wood and Pettersson (1988: 242).

[53] Cf. the case of Bulgarians who occasionally fail to spell unstressed vowels, unless they have sufficient etymological knowledge; see Pettersson and Wood (1987: 267).

telling us that the phonemic merger indicated by the arrows in the following diagram ("Process II") came about through a complicated reanalysis of the relationship between phonemes and allophones (parentheses means partial retention, as in *sēdulus* 'diligently' [vs. *dolus* 'wile'] and *anatis* 'duck'):

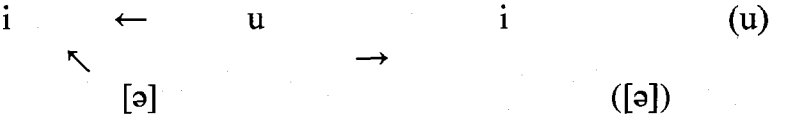


Fig. 2.3: Vowel reduction in medial syllables: Process II

During the phonological neutralization, phonetic ambiguity between [i] and [ə] and between [i] and [u] in unstressed syllables seems to have perplexed native speakers. Such confusion in perception (and perhaps articulation) can be found in inscriptions and fluctuating forms in literary texts:[54] e.g., *ponti-* ~ **PONTVFEX** (CIL I^2 1488) 'high priest';[55] *vitip-* (in rare occasions) ~ *vituperāre* 'declare invalid' (cf *vitium* 'defect'); *testi-* ~ **TESTVMONIVM** (CIL I^2 582) 'validating testimony'; **LVSIMACV(S)** (CIL I^2 2393) ← Gk. Λυσίμαχος; *-ibus* ~ **TEMPESTATEBVS** (CIL I^2 12, ca. 260) 'weather' (personified as a deity); pf. *-ripuī* ~ *surrupuī* [Pl.], **SVRVPVERIT** (CIL I^2 756.14) 'steal'; *mancipis* (class.) ~ *mancupis* (arch.) 'purchaser' (gen.sg. of *manceps*); *lacrima* ~ *lacruma* 'tear' (cf. Andr. *dacrima*); as well as *hic* ~ **HEC**, *optumus* ~ *optimus*, and *maxumus* ~ *maximus*.[56]

[54] The following examples are mainly selected from Niedermann (1953: 23-24) and Meiser (1998: 68).

[55] See also **PONTFEX** (CIL I^2 2835; Rome, early 3rd c. B.C.E.), where the omission of the medial vowel implies an inarticulate pronunciation like [ə]; see Wachter (1987: 344), followed by Vine (1993: 339).

As seen from these examples, such fluctuation is often in pre-labial position.[57] But it dwindled and eventually ceased even in this phonological context: e.g., *sapiō* 'have a flavour/a sense of taste, to be sensible' vs. *-sipiō*; *rapiō* 'snatch' vs. *-ripiō*; *capiō* 'take' vs. *-cipiō* (but cf. *occupō* 'seize'); *emō* 'take' vs. *-imō*; *premō* 'press' vs. *-primō*; *adeps -ipis* 'soft fat (of animals)'; *prīnceps -ipis* 'chief'; etc.,[58] as *capitis* < **caput-*, *manica* < **manu-*, *corniger* < **cornu-* (cf. *coniugem*, *pecudis*) in other phonological contexts.

It has traditionally been alleged that Faliscan demonstrates no vowel reduction (nor deletion). Sommer (1914: 86) maintains no such development in the language on the basis of **cuncaptum** 'received' (Lat. *conceptum*), found in a Latinized Faliscan inscription CIL I^2 365. Wachter (1987: 453) also acknowledges to some degree the local Faliscan feature of a lack of vowel weakening, admitting that there is no secure evidence to prove this. Giacomelli (1978: 516-517) argues the same claim, appealing to **pepara[i]** (≈ Lat. *peperī*) as well as **cuncaptum** and interpreting **-i-** in **aciptum** 'welcome' (≈ Lat.

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[56] Cf. **PARISVMA** (CIL I^2 7; similarly in Pl. *Curc.* 506), **SAI[PI]SVME** (CIL I^2 364b, written in Latin but with Faliscan features), and some literary forms like *ipsissumus* 'the very same' [Pl.], *occīsissumus* '(very) dead', etc. (see Meillet and Vendryes 1948: 386).

[57] Safarewicz (1960: 86-87), representing the phonetic value of reduced vowels in pre-labial position as *ü* (see also Kent 1932: 100), provides it with no phonemic status. Though the written language was so influenced by Greek as to render *ü* as a "pedantic" phoneme (i.e., /ʉ/) written as ‹i›, ‹u›, ‹y›, or ├ (see Leumann 1877: 12, 89) the spoken language seems to have never deployed this sound as a natural phoneme. See a similar remark on German borrowings in English by Sturtevant (1940: 23). As Safarewicz (1960: 88) succinctly explains, the Romance languages show no reflex of such a *phoneme*, instead having attributed the *allophonic* value of [ʉ] to either /i/ or /u/.[57] As Barnes (2006: 12) correctly argues, it is natural that realizations of two independent phonemes phonetically overlap without phonological merger: the selective affiliation of [ʉ] to either /i/ or /u/ is due to such phonetic overlap in realization of these two phonemes. In passing, one should not equate *ü* with [y]; see Coleman's reasonable objection (1962: 97-98), with his reconstruction of [ɨ] for *sonus medius*. The exact phonetic value of the vowel in question is impossible to determine. We should be satisfied with its central feature as the most plausible.

[58] Note also the following group of forms (see Ernout and Meillet 1985: 668, and LIV 2001: 540): *supat*, *-āre* 'throw' [P. F. 407.9L, 252.10L] vs. *-sipat -āre* [P. F. 407.9L, 93.17L with *dis-*, *ob-*, *in-*; with *ob-* in Pl. *Cist.* 579] (but cf. arch. *dissupō* > *dissipō* 'disperse', which is the only compound attested in CLat.), *insipiō -pere* [Cato+; P. F. 93.17L] 'throw in' (< **sup-ɟé-*? ← **seup-*).

acceptum) as a hypercorrection modeled after Latin vowel alternation (in this case, *-cipiō* would be the better candidate than *-ceptum*). But for **cuncaptum**, Vine (1993: 109 n. 69) advocates a process of recomposition, restoring the original vocalism, which is a common feature of documents in a legal context; it is thus not advisable to use this form in order to eliminate a possible existence of vowel reduction from the language. More essentially, however, the apparent lack of vocalic change in *notation* does not necessarily mean the entire absence of vowel reduction in Faliscan. It is likely that this language underwent Process I (or at least target undershoot), but not Process II productively like Latin. If this is the case, it is understandable that we do not have much explicit evidence for vowel reduction. For instance, the second syllable in **pepara[i]** may have been pronounced as a schwa-like sound, which was presumably represented by phonemic notation due to forms such as the present 1.sg. ***paro**. One might recall that Faliscan lacks a reduction pattern from *-o-* to *-u-* in final syllables (see 2.3), suggesting the exceptionless absence of vowel reduction in the language. But positional difference of reduction systems or that between presence and absence of the change should be taken into account in future inquiries. This issue is resumed in 3.3.5, regarding the possible presence of vowel deletion in Faliscan.

2.4.2. Possible Effect of Adjacent i*-vocalism*

I have laid out a twofold vowel reduction in medial position through a central vowel to /i/. The cause of the entire process of vowel reduction was essentially the lack of sufficient duration for articulation triggered by absence of stress. The eventual phonemic merger

into /i/ may have already been destined at the outset of vowel reduction. However, there are some factors that may have delayed or halted this completion, such as surrounding sounds and/or recomposition. On the other hand, there seems to have been a phonological factor aiding the neutralization process of the [front]/[back] contrast. In previous studies, such as Meillet and Vendryes (1948: 113) and Weiss (2005: 143), it has been said that *i*-vocalism in adjacent syllables seems to have promoted the merger with /i/, which has been briefly mentioned in 2.2.2: e.g., *-cipiō* (< **kap-*, cf. *occupō* 'seize', *aucupor* 'go bird-catching'), *eximius* 'excepted' (< **-em-*, cf. *emō*), *mūnicipium* 'self-governing community', *abripiō* 'snatch away', etc. Such examples abound in a pre-labial context, as already noticed in the tradition. This context otherwise persistently resists Process II vowel reduction, though eventually yielding to it in many cases. In view of *i*-vocalism as an enhancing factor of Process II in pre-labial context, one may assume that the same factor played a certain role in facilitating the /i/-merger in other phonological contexts. In fact, in regard to *inclitus* 'celebrated' (vs. *inclutus*, cf. *clutus* [P. F.]), Meillet (1933: 134) already suggests a possible influence from the initial *in-* (< **en-* ≈ EN in CIL I^2 4). Such an assumption may be verified to a certain extent by what follows.

Bader (1960: 236) remarks that words prefixed by privative *in-* (< **en-* < **n̥-*) and *dis-* frequently show vowel reduction, such as *inimīcus* 'unfriendly' (vs. *amīcus* 'friendly') and *difficilis* 'difficult' (vs. *facilis* 'easy'). The list of examples is lengthened by *irritus* 'invalid' (vs. *ratus* 'reckoned'); *inficētus* 'boorish' (vs. *infacētus*; cf. *perfacētus* 'very clever'); *īnsipiēns* 'unwise' [Pl.+], whose finite forms (**insipiō*) are missing in Plautine Latin (cf. *insapiēns* in Catul. 43.8; *persapiēntis* 'very wise' [Cic.] and *nesapius*

'ignorant' [Petr.+]); *indigus* 'needing' (cf. *egeō* 'suffer want'); *inermis* 'unarmed' (vs. *arma* 'weapons'); *inīquus* 'unfair' (vs. *aequus* 'equal, even'); *imberbis* 'beardless' (vs *barba* 'beard'); *iners* 'inactive' (vs. *ars* 'art, skill'); *īnsulsus* 'unattractive' (vs. *salsus* 'witty'); *integer* 'not touched' (cf. *tangō* 'touch'); *dīmidius* 'half' (< **dis-*; vs. *medius* 'central', cf. rare forms like *intermedius*, *permedius*, *submedius*; see Ernout and Meillet 1985: 393); *displiceō* 'displease' (vs. *placeō* 'please', cf. *complaceō* 'take the fancy', *perplaceō* 'be thoroughly pleasing');[59] etc. (but cf. *illepidus* 'lacking grace', *inanimus* 'lifeless'). These forms commonly possess a negative sense due to prefixes with *i*-vocalism; the latter may thus be phonetically strengthened by a certain emphasis, as Pulgram (1975: 108-109) points out, whereby the following elements presumably underwent vowel reduction, and were influenced by the *i*-vocalism.

In this connection, *īlicō* 'on the spot; at once' (< **en-slocō*, cf. *locus* 'place') is worth mentioning, though its prefix is not privative but locational/temporal. This adverb continues a univerbated and fossilized prepositional phrase. As Lindsay (1891: 403-404) remarks, in prepositional phrases in Latin, prepositions are generally unstressed, unless governed elements are pronouns with semantic emphasis (e.g., *ád mē*, *ábs tē*, *ín sē*, etc. in Pl./Ter., just as the regular accentuation in Gk. πρός με, πρός σε). This observation is reinforced by the absence of vowel reduction in forms such as *affatim* 'sufficiently' [Andr.+]; *praemodum* 'exceedingly' (Andr. ad Gel. 6(7).7.12: *Livius in Odyssia 'praemodum' dicit quasi admodum...*); *admodum* [Pl.+] 'to a great extent'; *commodus*

[59] The evidence from Romance languages presupposes the existence of **displacēre*. See Meyer-Lübke (1935: 243 no. 2681), Sommer-Pfister (1977: 76-77), and Ernout and Meillet (1985: 511).

'convenient' [Pl.+] (← **com modō* 'mit Maß'; see Leumann 1977: 402);[60] *propemodum* 'virtually' [Pl.+]. These forms perhaps reflect the pre-univerbated accentuation, like **ad fátim* (cf. separate spelling *ad fatim* in Pl. *Men.* 91), **prae módum* (or *módō*), etc. (In the case of *affatim*, vowel harmonic effect can be considered.) On the other hand, vowel reduction in *īlicō* must be ascribed to stress on the prefix *in-*, which was provided only in a post-univerbated stage. This form differs from *affatim*, etc. in possessing /i/ in initial syllable. It can also be argued that while the morphologically main component *locus* only contributed to setting up a locational or temporal notion, the prefix *in-* determined the more important deictic content about 'where *or* when' (→ 'on the spot; at once') and thus bore the accent due to its semantic saliency. In this way, the emphatically stressed *i*-vocalism in prefixes, privative *in-*, *dis-*, and the other *in-* in *īlicō*, seems to have promoted centralized vowels in medial position to merge with /i/.[61]

[60] Cf. Kent (1932: 99), for recomposition on the model of *modus*, thus not ***commidus*.

[61] This might further lead us to the detection of similar cases. As we have reviewed in 2.3, the contrast of [front]/[back] was distinctively retained in vowel reduction in final syllables, so that /i/ and /u/ (and /e/ secondarily for [ə] originating from /a/) gradually became dominant. As demonstrated in 2.1, vowel reduction was synchronically operative in both medial and final positions. But section 2.2 laid out a complicated process particularly for medial vowel reduction, which we may regard as spanning a long range of time. This would allow us to hypothesize a linguistic stage when [ə] in medial position and /i/ in final position were standing side by side. It is thus possible that the schwa may have been affected by the following /i/, and then phonemically merged with /i/. This scenario could perhaps explain some important categories in instances of vowel reduction, i.e., prefixed third conjugation verbs (e.g., **sub-ages* > **sub*[ə]*gis* > *subigis* 'drive under'), consonant-stem nouns (e.g., **homones* > *hom*[ə]*nis* > *hominis* 'human being'), and reduplicated perfects (e.g., **ke-kadet* > **kek*[ə]*dit* > *cecidit* 'fell'). Note, however, that some problems are left unsolved. True, the phonemic merger with /i/ abounds in the third conjugation, but it is also found in other verbal classes; and consonant-stem nouns do not always possess *i*-vocalism in case endings (e.g., acc.sg. *-em*, nom.pl. *-ēs*). Accordingly, this issue is open to question and needs further study.

2.4.3. Completion and Incompletion of Vowel Reduction

Out of the two reduction processes targeting medial vowels—the one in height and the other in [front]/[back] features, the first one was no doubt initiated under the initial-stress rule, together with the final vowel reduction, though on different phonological patterns (2.1). The question in this section concerns the second one. This process may have come into effect soon after the first one, still under the same initial-stress rule. However, I will tentatively suggest that the second half of medial vowel reduction was entirely completed not within the era of initial stress but under the Penultimate Law. The prefixed verbs I will employ in order to prove this are unfortunately not so straightforward that a clear-cut conclusion can be deduced. This reveals that Process II, which finalized vowel reduction, emerged gradually around the end of the initial-stress rule and/or the outset of the Penultimate Law. Again, recall Rix's prediction (1966: 160-162) that in the beginning of the literary period, reduced vowels had not yet merged phonemically with /i/. This prediction will be verified to a certain extent by what follows.

We immediately notice that not all of the prefixed verbs underwent Process II. It is possible that examples such as *subigō* 'drive under' (< **sub-agō*) with vowel reduction are old creations, formed under the still active operation of this phonological change, and others such as *concrepō* 'crack' (< **kom-krepō*) without vowel reduction are of a more recent origin and postdate the end of the reduction process. Despite its effectiveness in some cases, this line of reasoning must be the last resort, as Janson (1979: 55) also points out. I have already mentioned possible analogical operation of simplex verbs towards

their prefixed verbs as a factor restricting the occurrence of vocalic alternation (1.2.1). Though this standpoint is refuted by Janson, I consider Janson's inclination on lexical diffusion (1979: 55-56) neither attractive nor compelling in our case. If there is a correlation between distribution of medial vowel reduction and some phonological and/or morphological conditions, we ought not to dismiss it. Indeed, such a correlation does exist in prefixed verbs. The notion of analogy needs also to be expanded.

In most of the previous studies, medial vowel reduction in prefixed verbs has been discussed in the present tense, except in the case of reduplicated perfects (e.g., *tetigī* 'touched', *fefellī* 'deceive'). But most recently, Garrett's work (2005), followed by Weiss (2005: 146-147), offers a wider perspective on this issue by taking the formation of perfect stems (not only reduplicated ones) into account. He presents some correlation between distribution of medial vowel reduction and morphology of perfect stems.

Garrett (2005: 2) distinguishes two different types of verbs according to their perfect stem formation: strong verbs vs. weak verbs. The former form the perfect by internal stem change, e.g. *fēcī* (by lengthening; pres. *faciō* 'make'), *cecidī* (by reduplication, though not strictly an internal process; pres. *cadō* 'fall'). The latter, on the other hand, form the perfect stem by suffixation, e.g. *scrīpsī* (by suffix *-*s*-; pres. *scrībō* 'write'),[62] *amāvī* (by suffix *-*u̯*-; pres. *amō* 'love'). According to his framework, strong verbs tend

[62] The form *ē-maneō* 'stay away' [pf. *mānsī*] is the only example given by Garrett (2005: 2) for an *s*-perfect verb without vowel weakening. But actually, this form is of no value in this matter, as it intersects with the problem of the three homonymous roots with the shape **men*- (see LIV 2001: 435-437), all of them inflected in the second conjugation by Latin: 1. **men*- 'einen Gedanken fassen' (Lat. *moneō* 'advise'), 2. **men*- 'bleiben, warten' (Lat. *maneō* 'stay'), and 3. **men*- 'emporragen' (Lat. *ē-mineō* 'project'). The third root has nominal reflexes in Latin, e.g., *mōns* 'mountain' (< **món-ti*-) and *mentum* 'chin' (with the substantivized meaning from the adjectival **mn̥-to*- 'projecting'); see Vine (2006b: 154-155). The retention of *a*-vocalism in *ē-maneō* can be explained by the avoidance of a potential "homonymie fâcheuse"; see Ernout and Meillet (1985: 383).

to retain vowel reduction in composition (*con-ficiō* 'complete', *in-cidō* 'fall into', etc.), while weak verbs tend to level it out or resist it (*de-amō* 'love passionately', *in-gemō* 'groan (at)' [pf. *gemuī*], etc.).

Garrett's insightful framework, however, does not go far beyond a classification of verbs, which actually confronts several counterexamples (e.g., *-veniō*, *-vēnī*, strong verb without vowel reduction; *habeō* 'have', *habuī* vs. *-hibeō*, *-hibuī*, weak verb with vowel reduction). Nor does it provide any consideration of phonological factors. It may thus be worth considering such matters in prefixed present and perfect forms. In the first place, a difference in vocalism must be taken into account, and its importance will be clarified as the discussion proceeds. It should suffice here to invoke the descriptive facts that *-a-* is more subject to vowel reduction than *-e-* and *-o-*. We will postpone the discussion of *a*-vocalism until the other vowels have been treated. In the second place, we need to work on undiscovered factors of accent location, since both strong and weak verbs must have been straightforwardly susceptible to vowel reduction if the entire process had operated under the Pre-Literary initial-stress rule. As mentioned above, I will thus examine some of the prefixed forms, both present and perfect, as to how they behave in terms of accent distribution under the Penultimate Law.

I will start with weak verbs with either *e-* or *o*-vocalism in the root, continuing either PIE *e*-grade or *o*-grade.[63] Those in the present tense, such as *concrepō* 'crack' [Pl.+] (pf. *concrepuī*) and *ēdoceō* 'instruct thoroughly' [Pl.+] (pf. *ēdocuī*), could be targeted for

[63] Some of the instances of *o*-vocalism are secondary developments in Latin (or Italic). The verb *colō* 'inhabit' and *sonō* 'sound' are further thought to have had the PIE *e*-grade (*$*k^{w}élh_{1}$-e-* and *$*s\underline{u}énh_{2}ti$*, respectively); see LIV (2001: 386-388, 611). In *vomō* 'vomit', the original PIE *e*-grade (Gk. (Ϝ)εμέω, Lith. *vemiù*) was replaced by sound change *$*e > o$*; see Nussbaum (2003; #(C)$\underline{u}$__NV[−front][+back]). The direct inheritance of *o*-grade, proposed by Leumann (1977: 101), is criticized by Vine (1991a: 223 n. 12).

vowel reduction by "initial" accent on the prefix, which was realized under either the initial-stress rule or the Penultimate Law. We might ask why they could not merge vowels with /i/. Consideration of the perfect forms solves this problem. Despite being realized with initial stress in Pre-Literary era, they do not continue the same accentuation under the Penultimate Law:[64]

sg.	1.	*concrépuī*	*ēdócuī*
	2.	*concrepuístī*	*ēdocuístī*
	3.	*concrépuit*	*ēdócuit*
pl.	1.	*concrépuimus*	*ēdócuimus*
	2.	*concrepuístis*	*ēdocuístis*
	3.	*concrepuḗrunt*	*ēdocuḗrunt*

Tab. 2.1: Perfect paradigms of *concrepō* and *ēdoceō*

It is remarkable that vowels targeted by reduction under the initial-stress rule, in turn, bear accent or are not located in post-tonic position, where otherwise Process II applied. It may thus follow that the [ə] caused by the first step of the whole reduction process did not lose its status as an allophone of /e/ or /o/, and that under stress (or located in pre-tonic position) later under the Penultimate Law, /e/ and /o/ regained their ideal phonetic representations [e] and [o]. Location in pre-tonic position must also have attenuated a degree of reduction in the vowel in question (cf. the immediate pre-tonic

[64] Accent distribution like *ingémuimus* and *ēdócuimus* in the following paradigms is based on the OLat. *fácilius*-type accentuation (i.e., ◡◡◡◡#); see footnote 20 in Chapters 1 and 3.3.2 for further explanations and references. In the 2.sg. and 2.pl., there may have been variants like - *´...istī* and - *´...istis*, respectively (cf. *amāstī* 'you loved'). The consonant cluster type -*s*C- is not straightforward with regard to syllable boundaries, which will be examined in 3.3.4.

reduction in Russian, which does not attain so centralized a sound as [ə]). Given the fact that medial syllables were no longer susceptible to vowel reduction in the perfect forms under the Penultimate Law, the present forms could maintain the phonemic status of the original vowel on the model of the perfects, thus not activating Process II, that is, the merger with /i/ (not ***concripō*, ***ēdiceō* but *concrepō, ēdoceō* ← pf. *concrépuī, ēdócuī*). Other examples include *metuī* (*metuō* 'fear'), *genuī* (*gignō* 'bear' with zero-grade root), *fremuī* (*fremō* 'roar'), *strepuī* (*strepō* 'make a loud noise'), *petīvī* (*-e-* in pre-tonic position; later occasionally *petiī*; *petō* 'seek to obtain'),[65] *vetuī* (*vetō* 'forbid' < *votō* [Pl.]; see Leumann 1977: 47), *sonuī* (*sonō* 'sound'), *tonuī* (*tonō* 'thunder'), *vomuī* (*vomō* 'vomit'), *oluī* (*oleō* 'smell').[66]

Bear in mind that such an "analogical" process must have been enforced by simplex verbs; *gemō* 'groan' (pf. *gemuī*) and *doceō* 'instruct' (pf. *docuī*) are all marshaled to strengthen (or at least maintain) the original root vocalism. In this connection, prefixed verbs *reminīscor* 'recollect' and *comminīscor* 'invent' [Pl.+] reveal a different aspect. The accent distribution of these forms under the Penultimate Law will be something like *reminíscor*. If this continues **ré-menīscor* (or **ré-meneįscor*?; see Schrijver 2003: 74-75), the *-e-* in the second syllable would be reduced to [ə], but its phonemic status /e/ would be retained in pre-tonic position in the Penultimate Law. But in fact, the merger with /i/

[65] Cf. *competum* ~ *compitum* (with vowel reduction) 'cross-roads', as being outside the analogical networking of verbs.

[66] There are a few marginal exceptions: *necō* 'kill' (pf. *-āvī*) vs. *-nicō* (*ē-* [relular in Pl. and Ter., Var.]; but cf. *ē-*, *inter-*, *per-necō*), *secō* 'cut' (pf. *-uī*) vs. *-sicō* (*re-* [Cato, Var.], *prae-* [Var.], *sub-* [Var.]; but cf. *cirum-*, *con-*, *dē-*, *dis-*, *ex-*, *in-*, *inter-*, *per-*, *prae-*, *prō-*, *re-*, *sub-secō*). The cases in Old Latin (i.e., except the cases in Var., which probably represent archaizing forms) might be real archaisms, which demonstrate the relatively recent nature of the analogy with perfect forms.

was achieved. A related form *meminī* 'remember' (< **memonaị*) may have been partly responsible for *i*-vocalism in *reminīscor*. More crucially, however, this form has no exact counterpart in simplex like **menīscor*. This absence made the medial unstressed **-e-* more subject to the reduction process. The same is the case with prefixed verbs such as *applicō* 'apply (to)' [Pl.+] and *complicō* 'fold together' [Pl.+] (pf. *-plicāvī* ~ *-plicuī*). The original *e*-vocalism of the root **plek̂-* (cf. *plectō* 'plait') is not preserved in the simplex *plicō* [Lucr.+] 'fold, bend', either, which is thus understood as backformed from its prefixed verbs, as Ernout and Meillet (1985: 514) state. This may well lead us to suppose that when vowel reduction took place, the present simplex like ***plecō* did not contribute to retention of the original vocalism in prefixed verbs.[67]

Weak verbs with *s*-perfects are taken to level out or resist vowel alternation, but it may be better to state this differently. As Garrett (2005: 3) acknowledges, there are some counterexamples to this claim: e.g. *speciō* 'look' vs. *-spiciō* (pf. *spexī*), *premō* 'press' vs. *-primō* (given by Garrett; pf. *pressī* < **pret-sī* or **pres-sī*; see Ernout and Meillet 1985: 534). It is true that not all weak verbs with *s*-perfect of this class exhibit such vowel alternation: e.g., *trahō* 'draw' (pf. *trāxī*), *vehō* 'convey' (pf. *vēxī*), *coquō* 'cook' (pf. *coxī*), but these cannot be regarded as the majority.[68] Suffixation of *-s-* in the perfect creates a

[67] Philological information provided by Ernout and Meillet (1985: 514) makes the end of the story more interesting. The perfect form (-)*plicuī* is not attested before Cicero; given the fact that the *āvī*-type perfect verb rarely shows Process II, the original (-)*plicāvī* entails an exceptional make-up. Presumably, the more natural form (-)*plicuī* came into being secondarily, because the *u*-perfect could coexist with /i/ (cf. *habuī* 'had' vs. *-hibuī*, etc.; see below). Incidentally, as Ernout and Meillet suggest, this verb seems to have no derivational connection with the nominal *-plex*, *-plicis* from the semantic standpoint. Thus the transportation of a reduced vowel *-i-* from the nominal forms to verbs need not be considered here.

[68] For the retention of the original vowels in *-trahō* and *-vehō*, see below in this section. The verb *coquō* is not an indisputable example. According to Bader (1960: 237), a labiovelar impedes vowel reduction (i.e., Process II): *-loquus* (*loquor* 'speak'); *praecoquis* ~ *praecox* 'early ripening', *ollicoquus*

different syllable structure from the one in the present, namely, closed syllables. This factor may be responsible for such haphazard application of analogy between the present and the perfect; cf. *concrepō* (pf. *concrepuī*) and *ēdoceō* (pf. *ēdocuī*), consistently showing open syllable structure.

The strong verb with *e/o*-vocalism, on the other hand, shows more complicated patterns of accent location under the Penultimate Law and therefore different results of vowel reduction. The ones with lengthening (occasionally involving *-s-* of weak verbs) in the perfect stem may demonstrate a pattern like **kón-legō* > *cól-ligō* 'complete' (3.sg.pres.) vs. *con-lḗgit* (pf.), which has varied accent placement, as in weak verbs. Note that the perfect contains a long vowel (*-ē-*), which is exempt from phonological vowel reduction in Latin. The difference in vowel quantity may or may not have been sufficient to restore original phonemic conditions and halt Process II in the present, which consistently kept the reduced vowel in post-tonic position: [[/i/-merging]] *legō* 'gather' vs. *-ligō* (pf. *lēgī*);[69] *emō* 'buy' vs. *-imō* (pf. *ēmī*); *sedeō* 'be seated' vs. *-sidet* (pf. *sēdī*);[70] *regō* 'direct' vs. *-rigō* (pf. *rēxī*, with *-s-* of weak verb; but cf. Andr. ad P. F. *surrēgit*, briefly discussed by Meiser 2003: 153 n. 7); [[not /i/-merging]] *edō* 'eat' (pf. *ēdī*);[71] *veniō*

'cooked in a pot'; *īnsequō* ~ *īnsecō* 'tell' (Gk. ἔννεπε), *-sequor*, *-sequus*, *sequiae*, *-sequium* (< **sek^{w}-*) (cf. *īnsicium* 'rissole' < **sekH-*). The failure of the change seems to be limited to *pre*-labiovelar position: cf. *-cutiō* < **-k^{w}atiō* (see footnote 28).

[69] According to Ernout and Meillet (1985: 349), the selection between *-lēgī* vs. *-lēxī* of a perfect formation in prefixed verbs hinges on the semantic relationship (close or detached, respectively) with the simplex *legō*; for the latter, see *dīligō*, *-lēxī* 'choose apart from others, love'. But cf. the different treatment in LIV (2001: 276, 397): **leĝ-* 'sammeln, auflessen' vs. **h$_2$leĝ/g-* 'sich kümmern um'

[70] Cf. nominal compounds from the same root **sed-*, e.g., *dēses* 'sitting down', *praeses* 'sitting before as a guard', *reses* 'that remains seated', which regularly show vowel reduction in the oblique cases, e.g., *-sidis* (gen.sg.) and *-sidī* (dat.sg.).

'come' (pf. *vēnī*); *tegō* 'cover' (pf. *tēxī*); *fodiō* 'pierce' (pf. *fōdī*). The latter are also counterexamples to Garrett's classification. (There are no good examples of reduplicated perfect for verbs with *e/o*-vocalism in the simplex; see below for some details about OLat. *tetinī* for *teneō* 'hold'.)

Combining Garrett's idea with the Penultimate Law thus explains both completion and elimination of phonological vowel reduction somewhat effectively in prefixed verbs with *e/o*-vocalism. There are, however, a certain number of cases yet to be solved, that is, those with *a*-vocalism. As a particularly problematic group for Garrett's classification, we have a dozen *u*-perfect weak verbs that underwent Process II: e.g. *habeō* 'have' vs. *-hibeō* (pf. *habuī* vs. *-hibuī*);[72] *taceō* 'be silent' vs. *-ticeō* (pf. *tacuī* vs. *-ticuī*); *rapiō* 'snatch' vs. *-ripiō* (pf. *rapuī* vs. *-ripuī* ~ *-rupu-*);[73] *sapiō* 'have a flavor/a sense of taste, to be sensible' vs. *-sipiō*, *resipiscō* 'recover one's senses' (pf. *-sipuī*); *saliō* 'leap' vs. *-siliō* (pf. *saluī*, *-siluī* ~ *-suluī* ~ *-silīvī* ~ *-saluī*); *placeō* 'please' vs. *-pliceō* (pf. *placuī* vs. *-plicuī*); *ē-mineō* 'project' (pf. *ē-minuī*);[74] *statuō* 'set up' vs. *-stituō* (pf. *statuī* vs. *-stituī*);

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[71] Leveling in prefixed verbs with *-edō* is also promoted by the irregular 2./3. sg. forms *-ēs/-ēst*, as Garrett (2005: 3) demonstrates.

[72] Janson (1979: 51) suggests that prefixed verbs of *habeō* with vowel reduction, such as *inhibeō* and *perhibeō*, went far from the original meaning of the simplex *habeō*, whereas *ante-habeō* [Tac.+] and *post-habeō* [Ter.+] kept the original meaning and vocalism together. It seems that the latter forms are not phonologically univerbated, that is, they somehow retain individually assigned accent (i.e., *póst-hábeō*), as reflected in CIL II 6278.23 **POST HAVITA** (written as two words); perhaps the same goes for *calefácis* (cf. *-fíciō*, *-fícis*; see Kent 1932: 66). This assumption will be expanded to the failure of vowel reduction in nominal compounds, discussed in 3.3.

[73] For inscriptional material of *-rupu-*, see Coleman (1962: 90 n. 1). He argues that *u*-vocalism in the root was triggered by the vocalism of the following syllable. This must partly speak to that; however, in view of other verbs with *Cuī*-type perfects that retain their original root vocalism, e.g., *metuī* (pres. *metuō* 'fear') and *sonuī* (pres. *sonō* 'sound'), the whole of responsibility should not be granted to it.

[74] We cannot immediately decide the vocalism of the simplex for *ē-mineō*. The problem of the three homonymous **men-* roots (*maneō* 'stay', *moneō* 'advise', *ē-mineō* 'project') should be recalled, as

latēscō 'hide oneself' vs. *dē-litēscō* (~ *-litīscō*; pf. *dē-lituī*); etc. These instances form a class of °C*uī*-type perfect verbs, with a different type of syllabic configuration from °V̄*vī*-type perfect verbs (*parō* 'provide', *parāvī*; *petō* 'seek to obtain', *petīvī* [~ *petiī*]). In view of *ingemō* (pres.) / *ingemuī* (pf.), also classified under the C*uī*-type perfect verbs, the compounds of *habeō*/*habuī*, etc. might be expected to restore or retain their original vocalism in the simplex by analogy with the perfect (**-hábuī* etc. like *-gémuī*). But in fact both the present and the perfect exhibit vocalic alternation, as in pres. *habeō* vs. *-hibeō* and pf. *habuī* vs. *-hibuī*, etc.

To solve this conundrum, I propose that the root *a*-vocalism in these forms (*habeō*/*-hibeō*, etc.) may be significant. A phonetic factor that proves a strong tendency of this vocalism to be reduced in prefixed verbs, unlike *e*/*o*-vocalism, would be required, in addition to Garrett's framework. In fact, there seems to be a phonetic background to account for such a propensity of *a*-vocalism. (What follows is also applicable to **-a-* > [ə] > *-e-* in final syllables, discussed in 2.3.) Flemming (2005: 6) states that "much more F1 undershoot is observed in low vowels than in high vowels... because producing the low tongue and jaw position for a low vowel requires greater articulator movement from the constricted positions of adjacent consonants than the production of a high vowel." Thus, "[i]n particular, the lowest unstressed vowel is significantly higher than its stressed counterpart" (Flemming 2005: 18). See also the following chart in Flemming (2005: 7), based on Lindblom (1963), which shows the formant frequencies of five Swedish vowels

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mentioned in footnote 62. But *a*-vocalism is morphologically expected in the unattested simplex of *ē-mineō*, whose root was originally in zero grade (**mn-*), and is thus expected to have triggered either the so-called "schwa secundum" (**m₍ₔ₎n-* > **man-*) or the prevocalic *n̥* (**mn-ē-* > **mn̥(n)-ē-* [via Lindeman's Law] > **man-ē-*); I prefer the former option (2004: 236-240).

in the context [g_g] for three vowel durations: 200 ms, 125 ms, and 100 ms. (the values shown below are not the measured formant frequencies, but are derived from Lindblom's model):

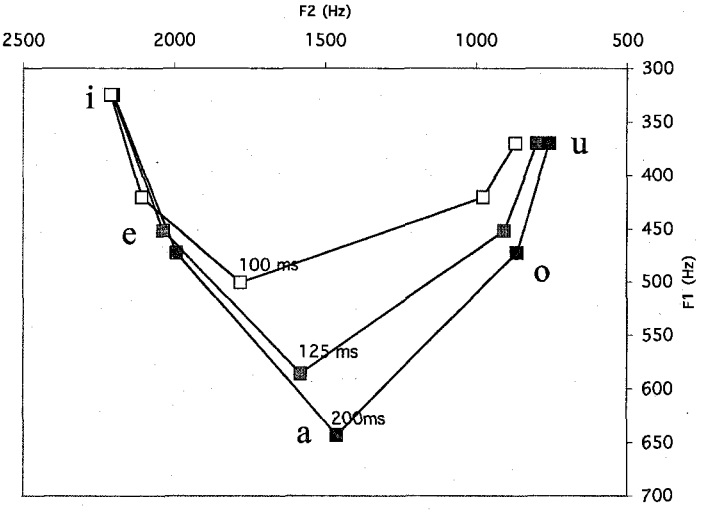


Fig. 2.4: Formant frequencies of five Swedish vowels in the context [g_g]

(Flemming 2005)

Likewise, Wood and Pettersson's description (1988: 240) of Bulgarian vowel reduction is noteworthy: "The actual occurrence of reduction in everyday speech is subject to normative, social, dialect and morphological constraints. Unstressed /a/ is reduced in all dialects".[75]

The significant reduction-induced raising of /a/ in unstressed position seems to have

[75] This passage is also cited by Barnes (2006: 33). The same idea is already claimed by Pettersson and Wood (1987: 261, 266, 268). Increased jaw position depression in Bulgarian in non-high vowels (vs. high vowels), as reported by Wood and Pettersson (1988: 252-253), is also noteworthy.

been so extreme in Latin as to facilitate the completion of Process II. Phonetic realization of /a/ as [ə] may already have threatened its phonemic status due to drastic height change. When the [ə] started to proceed further toward /i/, phonological association of [ə] (~ maybe [ɪ]) with /a/ was jeopardized, due to the greater phonetic divergence from [a] in stressed position. Thus, not only presents but also perfects underwent such a radical reduction (e.g., pres. *′*-habēt* > *′*-h*[ə]*bĕt* > *′-hibet*; pf. *′*-habuet* > *′*-h*[ə]*buit* > *-híbuit*) that the analogical restoration of vocalism from the latter to the former was less motivated than instances of *e/o*-vocalism.[76]

Other examples of root *a*-vocalism include strong verbs, which mostly retain phonological vowel reduction, regardless of the make-up of their perfect forms: *agō* 'drive' vs. *-igō* (pf. *ēgī*); *cadō* 'fall' vs. *-cidō* (cf. *concadō* 'fall at the same time' [Sen.]; pf. *cecidī*, *-cidī*); *canō* 'sing' vs. *-cinō* (cf. *occanō* ~ *occinō*;[77] pf. *cecinī*, *-cecīī* ~ *-cinuī*); *capiō* 'take' vs. *-cipiō* (cf. *antecapiō* ~ *anticipō* [< *ante-*] 'anticipate', *occupō* 'seize'; pf. *cēpī*); *faciō* 'make' vs. *-ficiō* (pf. *fēcī*); *gradior* 'proceed' vs. *-gredior*; *iaciō* 'throw' vs. *-iciō* (cf. *inter-*, *per-*, *prae-*, *subter-*, *super-*, *suprā-*, *trāns-*; pf. *iēcī*);[78] *laciō* 'draw gently'

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[76] Another verb of the same morphological pattern, *caveō* 'beware' (pf. *cāvī*), produces just a couple of prefixed verbs, among which only *praecaveō* 'beware' [Pl.+] is common (Ernout and Meillet 1985: 107). Notice that the root *a*-vocalism is preserved, just as in its participle *praecautus* (not **praecūtus*, as in *conclūsus* [pres. *conlūdō* 'shut up']). With the prefix *prae-* as well as *per-* as an intensifier (see Bader 1962: 350), the stem generally does not show Process II: e.g., *per-magnus* 'very large' [Ter.+], *per-facilis* 'very easy' [Pl.+], *per-cautus* 'very circumspect' [Cic.] (no **per-caveō*). It seems that the stem serves as the main semantic domain and thus resists phonological reduction. See also 2.6.1.

[77] The pair *occinō* (pf. *occinuī*, *occecinī*) [Liv.+] ~ *occanō* (pf. *occanuī*) [Pl.+] shows a rather complicated morphology in terms of perfect formation. Semantically, the former is used for birds and the latter for trumpets. The retention of *-a-* in the latter would be a result of semantic differentiation from the former.

[78] As Monteil (1979: 101) stresses, *-iciō* (vs. *iaciō*) is not a result of syncope but attained through *-*i̯ikiō* with a later loss of *-i̯-* after a consonant of a preceding preverb. See also footnote 28.

(cf. P. F. 103.25L, 104.16L, 25.14L) vs. *-liciō* (pf. *illexī* < **illaxī*); *pariō* 'give birth to' vs. *-periō* (pf. *peperī*, *-pperī*); *patior* 'be subject to' vs. *-petior*; *quatiō* 'shake' vs. *-cutiō* (pf. *-cussī*); *apīscor* 'grasp' vs. *-ipīscor*; *fatīgō* 'tire out' vs. *dēfetīgō* 'exhaust in mind *or* body' (~ *-fat-*), *fatīscor* (*-ō*) 'become tired' vs. *dēfetīscor* 'become tired out' (cf. Christian Latin *dēfatīscō*); *fateor* 'confess' vs. *-fiteor*.[79]

It is noteworthy that prefixed verbs with *a*-vocalism do not always show vowel reduction. Yet, we can specify some additional factors that protect *a*-vocalism from Process II in open syllables. First, the conjugation type makes a difference. The prefixed verbs with reduced *a*-vocalism belong to the second, third, or fourth conjugation,

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[79] The participial *impatiēns* 'impatient' and its related derivatives, which lack finite forms, are most likely late creations [Verg.+]. Among prefixed verbs of *valēscō*, the only *con-valēscō* possesses a relatively old attestation [Cato+], and may still be a sporadic creation in that period. Prefixed verbs of *valeō* also have only late attestations (*per-* [Lucr.+], *prae-*[Liv.+]). The verb *coalēscō* is a result of recomposition in the Classical period, restored from *cōlēscō* [Var., Lucr.]. Given the fact that there is no attestation in Pre-Classical Latin, it is most likely a recent creation. The source of this recomposition, especially recovery of *a*-vocalism, is the perfect *coáluī* (so accented under the Penultimate Law to begin with, if the creation of this prefixed verb was not terribly old). In contrast, others, such as *ab-*, *ad-*, *ex-*, *in-olēscō* (with a type of vowel reduction before a velar *-l-*), have perfects in *-olēvī* as default, which were less useful for the recovery of the original *a*-vocalism.

Some forms exhibit *e*-vocalism in place of expected *-i-*, a matter long in dispute (see Leumann 1977: 83-84 for various ideas and their problems). For *gradior* vs. *-gredior*, the involvement of *-r-* (or liquids, in general) is perhaps possible, as Skutsch (1913: 197) assumes, but other cases like *-rigō*, *-primō*, *-liciō* would not accord with this. Maniet (1957: 146) explains *e*-vocalism by means of vocalic assimilation in *ē-gredior* 'come out' and then analogical spread of *e*-vocalism to others like *aggredior* 'advance'. This explanation also does not seem entirely compelling, because many other verbs may also be prefixed by *ē-* or *per-*, and do not necessarily have *e*-vocalism in the root. However, a vowel harmonic effect of this sort (see 2.2.4) could underlie at least *fatīgō* vs. *dēfetīgō* and *fatīscor* vs. *dēfetīscor*, since only *dē-* is prefixed to these verb stems (cf. *īnfatīgābilis* 'tireless' [V. Max.+]). For *gradior* vs. *-gredior*, and also *patior* vs. *-petior*, we should turn our attention to their inflectional forms. The prefixed verbs in the 1/2/3.sg. and 1.pl. succeeded in attracting stress to the stem vowel, which had reduced to [ə] via F1 reduction (Process I): e.g., *′*-gr*[ə]*dior*, *′*-gr*[ə]*deris*, *′*-gr*[ə]*ditor*, *′*-gr*[ə]*dimor* > *-grédior*, *-gréderis*, *-gréditur*, *-grédimur*. Compare this to *fateor* vs. *-fiteor* (2nd conj.): *′*-f*[ə]*tĕor*, *′*-f*[ə]*tēris*, *′*-f*[ə]*tētor*, *′*-f*[ə]*tēmor* > *-fiteor*, *-fitḗris*, *-fitḗtur*, *-fitḗmur*, where the stem vowel frequently continued to be unstressed. Due to this accentual shift, the stem vowel in **-gradior* and **-patior* may not have undergone Process II, and so not have been raised to /i/. On the other hand, we should also recall the phonetic features of /a/ in unstressed position. The sound underwent a significant reduction-induced raising while the initial-stress rule was operative. It is thus probable that the phonetic realization of /a/ as [ə] rendered its phonemic status ambiguous, due to the drastic height difference. The phonemic status of [ə] as /a/ was eventually unsustatined, even after it became stressed under the Penultimate Law, so that the schwa collapsed with the mid vowel /e/. The same will be claimed in 2.4.4 of prefixed verbs with *a*-vocalism in closed/diphthongal stem syllables.

featuring either -*ĕ̄*- or -*ĭ̄*- as a stem-forming vowel. Verbs in the first conjugation are formed with -*ă̄*- (a suffix originally, in most cases), which is widespread and salient in the entire paradigm of each verb: e.g., *ap-parō* 'prepare', *ap-parās*, *ap-parat*, *ap-parāmus*, *ap-parātis*, *ap-parant*. The *a*-vocalism in this verbal category may have undergone reduction to a certain extent, reaching [ə]-like sounds, but the further process, that is, phonological merger with /i/ was blocked by a kind of vowel harmonic effect caused by extremely widespread -*ă̄*- in the paradigm. Hence we see the retention of the root *a*-vocalism in prefixed verbs, derived from, e.g., *amō* 'love', *arō* 'plough', *calō* 'proclaim', *lavō* 'wash', *parō* (*com-* 'match, compare', *dis-* 'divide'), *vador* 'accept sureties from (the other party) for his appearance or reappearance in court at an appointed date',[80] unless some phonetic factor is involved in the reduction process, as in *proculātō* 'provocato, summon' (P. F. 293.10L, from *calō*, by velar -*l*-). Of particular interest among these is *lavō* 'wash', since when employed as a third conjugation verb ("marquant l'action et s'employant transitivement"; see Ernout and Meillet 1985: 344), it shows a pattern of vowel reduction seen before -*u̯*- in prefixed verbs, thus, -*luō*.[81]

In the second place, verbs derived from adjectival roots (perhaps nominals, in general), have a very strong (almost absolute) tendency of keeping the original

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[80] Cf. CIL I^2 585 **PRAEVIDES** (CLat. *praedēs*; sg. *praes* 'one who acts as surety' ← *vas vadis* 'one who guarantees the appearance in court of a defendant'); vowel reduction could occur in this exemple, since its inflected forms, as nouns, do not contain *a*-vocalism.

[81] Verbs such as *iūr(i)gāre* 'quarrel' [Pl.+] and *lītigāre* 'litigate' [Pl.+] (composite verbs with nouns as first members; root *ag-*; 1st conj.) might seem to be exceptions. But as Leumann (1977: 96, 550) analyzes them, nominal (agentive) compounds most likely underlie these verbs (and are thus denominative: e.g., **i̯ou̯z-agos* 'bringing the oath' < **i̯eu̯(o)s-h$_2$ĝo-s*, following Dunkel 2000: 94; cf. *rēmigāre* 'row' and *rēmex* 'rower' < **-ag-s* < **-h$_2$ĝ-s*?), in view of their shift in inflection from the third to first conjugation. Hence, vowel reduction presumably took place in nominal compounds, and was then retained in the denominatives.

a-vocalism. Prefixed verbs of *-parō* (*com-* 'match, compare', *dis-* 'divide'), mentioned just above, also belong here: e.g., *pār* 'equal', *compār* 'similar', *dispār* 'unequal', etc. Analogical recomposition in support of these adjectives (i.e., outside the verbal category) is most likely responsible for such vocalic preservation. Other examples include prefixed verbs derived from *maceō* 'be thin', *macēscō* 'become thin', *madēscō* 'become wet' (pf. *maduī*); they have clear connections with *macer* 'thin' (< **mak-ro-*), *madidus* 'drenched'. Prefixed verbs in *-stituō* (< **-statuō*) might appear to be exceptions, if the base verb is a denominative from the *u*-stem noun *status* 'being in a standing position' (see Leumann 1977: 539). This could be explained by a rather old origin of this verb. First, many of its prefixed verbs are in use from the Pre-classical period: e.g., *dēstituō* 'set up, fix' [Pl., Naev.+], *īnstituō* 'set in operation' [Pl.+], *praestituō* 'fix (a point of time *or* conditions)' [Pl.+], *prōstituō* 'prostitute' [Pl.+], 'dishonor' [Cato+], *restituō* 'set up again, restore' [Pl., Enn.+]. In addition, if the equation of *restituō* with an Umbrian 3.sg.fut.impv. **restatu** 'restore, offer an additional sacrifice' (< **re-statue-*, via haplology; see Nishimura 2006) is correct, this allows us to posit a composition in **-statuō* at a very early period (perhaps in Proto-Italic). Further, the semantics of the prefixed verbs are no longer clearly connected to the cardinal meaning of *status* 'being in a standing position', which was bleached in extended and/or metaphorical use throughout the long tradition of these verbs. It is such semantic opacity that permits vowel reduction to apply.

There are some other sporadic cases that keep original vocalisms including *-a-* and also *-e-* vocalism. A tentative solution would be commonly applicable to them. The retention of the original vocalism in *-trahō* and *-vehō* is apparently mysterious. Dietrich

(1852: 552) remarks that the segment *-h-* caused retention of the original vowel, but this should be restated differently in order to capture a generalization. As is known, a feature of Vulgar Latin was that *-h-* was pronounced less distinctly, which caused a type of hiatus, i.e., *-tra(h)ō* and *-ve(h)ō*. A typological study by Crosswhite (2001: 141-147) indicates that the hiatus often serves as a factor blocking vowel reduction. In fact, prefixed verbs of *eō* 'go' retain the original vocalism very well, e.g., *adeō* 'approach', *intereō* 'die', etc. (cf. *ambiō*, semantically restricted in political contexts and thus detached from *eō*; see Ernout and Meillet 1985: 26). Forms, such as *aureus* 'golden', *alveolus* 'basin', and *fīliolus* 'littlte son', can be added here as well (see Kent 1932: 99, 101, 103 and Maniet 1957: 145). Hence the vocalic retention in *-trahō* and *-vehō* can be explained likewise.

Lastly in this section, some clarification is required for *teneō* 'hold' vs. *-tineō* (pf. *tenuī* vs. *-tinuī*), which apparently runs counter to the expected retention of *e*-vocalism (**-*teneō* ← **-*ténuī*). But the *e*-vocalism in *teneō* calls for an additional note. This *-e-* does not actually continue a PIE *e*-grade. The PIE stative suffix *-*eh*$_1$- (followed by *-*i̯e/o*-), securely reconstructed in this verb, is normally attached to zero-grade roots (but cf. *sedeō* 'be seated'), as clearly seen in *videō* 'see' (< **u̯id-eh*$_1$-*i̯e/o*-, cf. Ved. *véda*, Gk. (Ϝ)οἶδα < **u̯oi̯d*-), apart from "denominative" stative verbs such as *albeō* 'be white' (← *albus* 'white'),[82] which preserve the root structure of their base. Thus, the expected

[82] The verb *indigeō* (pf. *indiguī*) 'stand in need of' (← *egeō* 'be needy') is also a denominative stative based on *indigus* 'needing' (< **indegos*). The *e*-grade in *egeō* is secondarily motivated by the *s*-stem **eges*- (n.) in *egestās* 'indigence' (f.). See LIV (2001: 231).

The make-up of a deponent verb, *fateor* 'admit' (vs. *-fiteor*), which seemingly belongs to this morphological class, has now been clarified by previous studies. As Leumann (1977: 555) says, it is likely that this verb derived from **fātus* (see Chantraine 1968: 1196; cf. Ernout and Meillet 1985: 219, considering the verb as derived from an agent noun **fat*-, albeit with no such attestation), the original past passive participle of *fārī* 'speak'. The truncation **fāt*- served as a pseudo-stem, from which the past passive

reconstruction would be **tn-eh$_1$-i̯e/o-*, which in Latin would have yielded **tnēō* (1.sg.), **tnēs* (2.sg.), **tnēt* (3.sg.) in the present tense, after laryngeal lengthening (**-eh$_1$-* > **-ē-*), the loss of intervocalic *-i̯-* (**-ē-i̯e/o-* > **-ē-e/o-*), and the contraction of like vowels (**-ē-e-* > **-ē-* in 2./3.sg). These forms had a consonant cluster **tn-* in initial position, which came to be disallowed in Latin phonology. The monosyllabic forms **tnēs*/**tnēt* either underwent a (secondary) schwa secundum insertion or Lindeman's Law or its subtype applied, which would have resulted in **t$_ə$nēs*/**t$_ə$nēt* and *tn̥(n)ēs*/**tn̥(n)ēt*, respectively (see footnote 74). The sequence **-$_ə$n-* or **-n̥(n)-* can be expected to have spread throughout the paradigm by leveling. The former option would regularly yield the stem **tanē-*. Based on the latter, it is to be noted that the **-n̥-* is located in prevocalic position. Though its reflex in that context has been much disputed as with prevocalic **-m̥-* (see 2.2.2), one would also expect the development of **tn̥(n)ē-* into **tanē-*. But eventually this **tanē-* was replaced by *tenē-*, due most likely to *tentus* (p.p. < **tn̥-tó-*) or *tendō* 'progress; stretch out' [Pl.+] (with disputed reconstruction; see LIV 2001: 626-628).[83]

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participle, *fassus* (< **făt-to-*) was formed (see Vine 1998: 52, and n. 121). The root of this lexeme is traced back to **b^{h}eh$_2$-*; **fătus* is thus reconstructable as **b^{h}h$_2$-tó-*. For the make-up of the verb *fateor*, Vine (1998: 52-55) considers **(n̥-)b^{h}h$_2$-etó-* '(not) to be spoken' as underlying the verb, as well as *īnfitiās* (+ *īre* 'deny'), like *cognitus* < **-genoto-* < **(n̥-)ĝn̥h$_3$-etó-* '(not) to be known, unknowable' (1998: 37-38).

[83] There is more to say about the verb *teneō*/*-tineō*. In Pre-Classical Latin we have *tetinī*, *abstinī* [Pac.] alongside *tenuī*, *abstinuī* [Pl.], as cited in Leumann (1977: 587). The former is formed by reduplication, making it a strong verb in Garrett's terminology. In this sense, there is no problem with the fact that the compound *abstinī* retains a reduced vowel. Nor is it problematic to have a set like *tenuī* (← **tanuī*), *abstinuī*, the latter of which also presents vowel reduction, since the original stem had *a*-vocalism. More interestingly, a certain analogical effect derived from *teneō* with *e*-vocalism outhg be acknowledged, namely, *teneō* : x = *gemō* : *-gemuī*, x = *-tenuī*, which actually underlies CIL I^2 15 **OPTENVI** (Rome, ca. 135 B.C.E.).

2.4.4. Outcome of Vowel Reduction in Closed/Diphthongal Syllables

In the preceding subsection, I have laid out a process of /i/-merging Process II, which was completed in the historical period, with or following the shift to the Penultimate Law. Vowels in medial syllables that were reduced to /i/ remained unaccented under the Penultimate Law as well. Vowels accented under this rule (or located to left of the accent) may have exercised an analogical effect on related forms to block Process II reduction, whose vowels had been unaccented since the initial-stress period. This theory further leads us to re-consider vowel reduction in medial, closed or diphthongal syllables,[84] since these syllables are consistently accented under the new rule due to their syllable weight. This fact may well explain the failure of the /i/-merging process in such syllables. Thus, the phoneme /e/, unaccented in the initial-stress stage (thus, reduced to [ə]) but later accented under the Penultimate Law, restored its ideal phonetic value [e] in most cases: e.g., *scelestus* 'calamitous'; *necesse* 'essential (adv.)'; *cōnsentiō* 'to be in harmony'; etc.[85] Nor did the phoneme /o/ participate in /i/-merger, but instead, was

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[84] I use the traditional terminology "diphthong(al)" just for the sake of convenience. The consonantal feature of the second element of the sounds in question is concisely explained by Cser (1999: 182-191). Vowel sequences conventionally called "diphthong(s)" can thus be compared to syllables closed by consonant(s).

[85] In footnote 16 in Chapter 1, we have reviewed a problem in *trǐgintā* [Consentius 5.392.4K], about whose accentuation Leumann (1977: 238) has doubts. As cited above, there is much evidence from the Romance languages that higher decads were allowed to have antepenultimate accent. Hence Leumann's suspicion should be attenuated to some extent. Interestingly, Corssen (1870: 901-902) seeks the origin of such an antepenultimate accent in vulgar pronunciation of Old Latin, assuming that such an accentuation may have appeared off and on. Presumably, Latin decads consistently had accentual variation in different dialects, social ranks, and more so in the speech of various contexts in daily life (see also 2.6.1); cf. Pl. *vǐgintǐ*, *trǐgintǎ* (′ = ictus), mostly limited to the position before *minae* [Greek unit of weight], *diēs* 'day' (see Lindsay 1891: 408). Even after the shift to the Penultimate Law had begun, these words may have been allowed to be pronounced with antepenultimate accent, which accordingly made the heavy penult

preserved (e.g., *opportūnus* 'serviceable' [cf. *portus* 'port']; *cōnsortium* 'association' [cf. *sors* 'lot']; *at-*, *dē-tondeō* [cf. *tondeō* 'cut (hair, beard)']; *praecordia* 'vitals' ← *prae corde*) or later secondarily changing into /u/ (see more details below).[86]

With regard to /a/, we can recall its phonetic feature in unstressed position, as seen in the preceding subsection. This sound tends to undergo significant reduction-induced raising in this position. Phonetic realization of /a/ as [ə] may have made its phonemic status ambiguous due to the drastic height change. Then, phonetic divergence from [a] may have made it difficult for [ə] to maintain the phonemic status as /a/ even if it became stressed later under the Penultimate Law. Thus, the schwa collapsed with /e/ by default, phonetically closer than /o/ due to the lack of rounding.[87] On the other hand, surrounding consonants, such as velar *-l-* and *-u̯-*, may have provided the [+round] feature for reduced central vowels to be realized as [ɵ], which merged with /o/. These changes may square with a nearly absolute chronology of vowel reduction in open syllables, which may be

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unstressed. Whereas Leumann (1977: 100) and Weiss (2005: 395) explain **-e-* > *-i-* as a result of assimilation caused by *i*-vocalism in the preceding (as in *trīgintā*) or following (or both, as in *vīgintī*) syllable, I tentatively suggest that it is a further case of exceptional vowel reduction in medial syllables, formulated as **´-en-* > *-in-*. The *i*-vocalism in *cicindēla* 'firefly; candle' (cf. *candeō* 'shine') might be explained likewise. This form is only attested later than Pliny, who considers it a rustic term (see Ernout and Meillet 1985: 119); other Italic languages that kept the initial-stress system may be responsible for the vocalism.

[86] Exceptionally, /o/ was not preserved in pre-yod position: e.g., *relīquī* 'left' < **re-(le)loik^{w}-ai* (Gk. λέλοιπα), but cf. Leumann (1977: 560): **leloik^{w}e* > *lelīqui-t* > *relīquit* (by dissimilation); CIL I^2 1603 **INTERIEISTI** 'you died' < **-iį-oį-staį* < **h$_1$i-h$_1$oį-* (Meiser 1998: 223); but *pōmērium* 'bare strip of land round the walls' (not ***pōmīrium*) < **post-moiriom*. An intermediate stage like [ę̄] (Meiser 1998: 71) would explain the multiple results. Before reaching that stage, the *-o-* must have been fronted by the following *-į-*. This process seems not to have affected most initial syllables: e.g., CIL I^2 581 **OINVORSEI** 'all together', **FOIDERATEI** 'allied' (after **u̯-*, however, *vīcus* 'village' < *u̯ę̄-* (CIL I^2 756 **VEICVS**, 391 **VECVS**, 388 **VECOS**) < **u̯eį-* < **u̯oį-*; cf. Gk. Ϝοῖκος). Given the fact that final syllables were subject to this change (e.g., nom.pl. *-ī* < **-oį*, dat./abl.pl. *-īs* < **-oįs*; see footnote 48), it is likely that the lack of stress under the initial-stress rule occasioned this phonological development.

[87] See Maniet (1957: 147). For more detailed conditions governing *-a-* > *-e-* based on following consonants, see Garrett (2000: 7-9; 2002: 3-4).

deduced from inscriptional sources (see 1.2.1). On the basis of instances such as **MAMARCOM** (Gnade and Colonna 2003) and **MAMARTEI** in the CIL I^2 2832a, which contrast with Classical *Māmertīnī*, etc.,[88] changes in closed/diphthongal syllables can be considered to have been as a historical event, roughly simultaneous with vowel reduction in open syllables. For this reason, there is no need to adduce separate causes for both the former and the latter processes. Note also that analogical recomposition was also more prevalent in closed and diphthongal syllables than open syllables, because the non-application of the /i/-merging process in the former made schwas remain [–high].

Typical examples of *-a-* > *-e-* or *-o-* (> *-u-*), accompanied by occasional recomposition, include pairs of simplex and prefixed verbs:[89] *causor* 'plead a cause' vs. *accūsō* 'accuse' (< **-au̯-*); *arceō* 'enclose' vs. *coerceō* (cf. *coarctō*) 'shut up together'; *arguō* 'declare' vs. *rederguō* 'disprove' (~ *redarguō*); *carpō* 'pluck' vs. *discerpō* 'tear into pieces'; *damnō* 'condemn' vs. *condemnō* 'id.' (cf. CIL I^2 582 **CONDVMNO**, Ital. *condannare*); *scandō* 'climb' vs. *inscendō* 'climb in'; *tractō* 'handle' vs. *dētrectō* 'decline' (~ *dē-tractō*); *aestimō* 'estimate' vs. *exīstimō* 'id.' (< **-ai̯-*; cf. **EXAESTIMAVERIT** in CIL I^2 583 ["Lex Acilia Repetundarum"; 123 B.C.E.], *inaestimābilis* 'impossible to estimate'); *caedō* 'cut' vs. *incīdō* 'cut into'; *laedō* 'injure' vs. *allīdō*

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[88] The form usually restored as **mamar[kos** in the "Tita Vendia" inscription (ca. 600 B.C.E.) might be considered here. But the letter interpreted as **r** may be a **d** (whose shape is something like D); the other **r** in the text (**urna**), on the other hand, has a very long vertical line (like P). Recall the Faliscan name **mama** (Ve 241), whose abl.sg. might be ***mamad**. This problem is reserved for future study.

Moorhouse (1940: 318) states that there is no counterpart in Oscan and Umbrian, but Osc. μαμερτει may perhaps be a candidate; see some details in 4.4.3.

[89] The following examples are selected from von Planta (1892: 590), Juret (1919: 94-95), Meillet (1933: 137), Niedermann (1953: 27), Maniet (1957: 147-148), Leumann (1977: 81), Nyman (1983: 33), Ernout and Meillet (1985), and Weiss (2005: 212).

'dash'; *quaerō* 'seek' vs. *inquīrō* 'search out'; *fraudō* 'defraud' vs. *dēfrūdō* 'id.' (cf. *dēfraudō*; pf. *dēfrūdāvī* ~ *dēfraudāvī*); *claudō* 'close' vs. *conclūdō* 'conclude';[90] *tangō* (pf. *tetigī*) 'touch' vs. *attingō* (< **ad-tangō*); *frangō* 'break in pieces' vs. *perfringō* 'break'.[91] Nominals are also an important part of our material: *altus* (p.p. of *alō* 'nourish') vs. *adultus* 'adult'; *aequus* 'even' vs. *inīquus* 'uneven' (cf. *inaequus*, *inaequālis* 'id.'); *ager* 'land' vs. *peregrē* 'abroad'; *annus* 'year' vs. *biennium* 'a period of two years' (< **bi-annio-*; cf. Umb. *peracni-*, *seuacni-*); *aptus* (p.p. of *apiō* 'fasten', *apīscor* 'grasp') vs. *ineptus* 'foolish' (cf. *exaptus* 'that is fastened'); *ars* 'skill' vs. *inert-* 'unskillful'; *barba* 'beard' vs. *imberbis* 'beardless' (< **in-barbis*); *captus* 'taken' vs.

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[90] As a first element of diphthongs, **-a-* changed into **-e-*, and later followed general rules of monophthongization in Latin, as seen in *exīstimō*, *incīdō*, *illīdō*, and *inquīrō*; see Meiser (1998: 70-71). However, the intermediate stage between **kom-klaṷdō* and *conclūdō* has been disputed (as with **akkaṷsō* > *accūsō* and **dēfraudō* 'defraud' > *dēfrūdō*). Several scholars have argued a phase with **-eṷ-* (e.g., Monteil 1979: 110). Some Italic forms, such as a Faliscan gentilicium **evios** (Ve 241), perhaps corresponding to Osc. **úviis** (Vetter 1953: 281-282) or Gk. Εὔιος (epithet to *Liber* = Διόνυσος; see Scherer 1956: 118, followed by Peruzzi 1964: 157-159), and dialectal Latin **NEVEN** 'novem' (CIL I^2 455; Ardea, 3rd c.) and **NEVNA** 'Nonae' (CIL I^2 2845-6; Lavinium, early 3rd c.), seem to retain a diphthong *-eṷ-*, as Vetter (1953: 281-282) suggests, whereas the change **-eṷ-* > *-ou-* is understood by many scholars to be one of the oldest sound changes in Italic, as per Moorhouse (1940: 307, 310), Monteil (1979: 110), and Vine (1993: 63 n. 56, 2006: 212). If Vetter's view about its preservation in the historical period is correct, there would have been an intermediate stage, such as **konkleṷdō*, in the process **konklaṷdō* > *conclūdō*, *-eṷ-* being phonotactically still acceptable in that period. In Peruzzi (1964: 173 n. 75), as already in Scherer (1956: 118), prevocalic position is recognized as a locus of retention of **-eṷ-*, e.g., **NEVEN**, **NEVNA** (< **neṷen-*?), and perhaps Faliscan **nevo** (Ve 242B) 'novus, new' (so segmented by Peruzzi 1964: 172), in contrast to Ve 276a **[l]ọifirtạ** '*liberta*, freedwoman' and Ve 253 **loifirtato** '*libertatis*, freedom' (more recently Ve 322a **loferta**). (The same sort of condition is also necessary to explain Vetter's equation of Falsc. **evios** with Osc. **úviis**, according to Scherer 1956: 118). If this is the case, then pre-consonantal *-eṷ-* in **konkleṷdō* would be much less likely. On the other hand, Giacomelli (1978: 514), refuting Peruzzi's interpretation **nevo** (see also 1978: 527), insists that *-eṷ-* otherwise appears in foreign personal names and divinities and the retention of *-eṷ-* in the historical period would hardly be possible. Meiser (1998: 59) and Rix (2004: 437), convinced by R. Lipp, consider **NEVEN**, **NEVNA** continuing **-ēṷ-*, as *neu* < **nēṷ(e)*, and Rix regards **LEVCESIA** in "Carmen Arvale" as 'grecismo' (Gk. Λεύκιος, cf. **leuk-* > Lat. **loukiios* > *Lūcius*), thus understanding **-eṷ-* > *-oṷ-* as Proto-Italic, or at least pre-Latin (see also Solmsen 1908: 2059), whereby the stage **concleudō* becomes less feasible. Godel (1961: 59) is also suspicious of **-eṷ-* in **konklaṷdō* > *conclūdō*. Allen (1973: 134) also excludes such a stage. In my theory, this prefixed verb evolves as follows: **´-klaṷdō* > **´-kl*[ə]*ṷdō* > *-clóudō* (stress shift + phonological slide of [ə] from /a/ to /o/, being assimilated to *-ṷ-*) > *-clūdō*. No appeal to **-eṷ-* is required.

[91] In *attingō* and *perfringō*, the **-e-* further developed into *-i-*, due to the phonetic effect of the following *-ng-* ([ŋ]). See Meillet (1933: 140), Palmer (1954: 215, 220), and Maniet (1957: 147).

conceptus 'taken completely'; *factus* 'made' vs. *confectus* 'completed'; *lautus* 'washed' vs. *illūtus* 'unwashed' (~ *illautus*, *illōtus*); *olīva* 'olive (tree)' (< **óleiu̯a*[92] < Lat. **élaiu̯a* ← Gk. ἐλαίϝᾱ); *Pompeiānus* 'of the town of Pompeii' (< **Pompa* + **-iio-*, cf. Osc. **púmpaiians**; see Rix 2000: 239 n. 12);[93] *salsus* 'salted' vs. *īnsulsus* 'unsalted'; *taesum est* 'be tired' vs. *pertīsus* (p.p. of *pertaedeō* 'fill with disgust'; cf. *pertaesus*); *tangō* 'touch' vs. *integro-* 'not touched' (cf. Umb. **antakres**); *Tarentum* [city name] (< Lat. **Tárantum* ← Gk. Τάρας, Τάραντος).[94]

Another pattern of change *-o-* > *-u-* has often been equated with *-a-* > *-e-* (or *-i-*, *-u-*, depending on phonological contexts). But as per Juret (1919: 94) maintains that "en syllable intérieur fermée, seule la voyelle brève la plus ouverte, *-ă-*, est altérée devant toute consonne" (see also Juret 1919: 101, 105 for the same claim), the change *-o-* > *-u-* has occasionally been challenged in contrast to the more common one *-a-* > *-e-*. Oniga (1990: 209) also suggests that examples cited for *o* > *u* are "rarissimi e spesso discutibili".

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[92] The original initial vowel **e-* was changed to *o-* by operation of the following dark *-l-*; see Meillet (1933: 138). It is allowed to occur before either the original **-ai-* or the reduced result **-ei-*.

[93] Though Giacomelli (1962: 366-367) refers this phonological treatment **-aii-* > *-eii-* to Umbrian (and Etruscan), the hypothesis seems to be weak. When followed by another consonantal *-i-*, an *i*-diphthong is said to have been preserved even in Umbrian, where monophthongization was remarkably productive (see Buck 1928: 42-43, Meiser 1986: 122-124: **pernaies** 'appearing ahead' < **pernaii-io-* < **pernaii-iio-*); but cf. **teteies** (TIg. Ib 45, IIa 44; gent., nom.sg.) < **tettaiio-* (Lat. *Tetteius*, *Tettaius*, Etr. *tetaia* ET Cr 2.46), which is formed with an individual name *Tetta* (Etr. *Teta*, *Teθa*) and "Gentilnamensuffix" **-iio-* (see Rix 2000: 239 n. 12; but cf. 4.3.2). Note, however, that the development **-aii-* > *-eii-* in Umbrian was probably not stress-driven vowel reduction, since it happened in initial syllables as well (e.g., **fakitōd* > **faχ'tǫd* > **fajtu* > *feitu* 'facito'; see Meiser 1986: 124). Therefore, it is not advisable to appeal to Umbrian in order to explain the vocalic change in unstressed syllables in Latin.

[94] One might add *anhēlus* 'out of breath' [Lucr.+] here, which is resconstructed as **aneslo-* < **anaslo-* < **h₂enh₁-s-lo-* (← **h₂enh₁-* 'blow'; cf. Lat. *anhēlāre* 'pant', *hālāre* 'exhale') by Schrijver (1991: 44-45). However, in his forthcoming paper, Vine points out that there is no strong evidence for an *s*-stem noun **h₂énh₁-(e)s-*, and suggests a thematic derivation **h₂n̥h₁-ēl-o-* 'characterized by breathing' from a hysterokinetic *l*-stem abstract (or collective) noun **h₂n̥h₁-ḗl-*/**h₂n̥h₁-(e)l-* 'breathing'. We can *anhēlus*, directly from **h₂n̥h₁-ēl-o-*, without recourse to vowel reduction.

We can add here Biville's remark (1990: 9), too, that "nous ne devrions donc trouver, à l'intérieur des mots latins, que des voyelles brèves fermées, essentiellement *i* et *u*, *e* et *o* selon les contextes phonétiques, mais en aucun cas on ne devrait trouver de *a*," with minor exceptions recognized by herself. True, there are apparently several instances that might support *-o-* > *-u-* in closed syllables: e.g., *feruntur* 'carry' (3.sg.pass.; cf. Gr. φέρονται); *euntis* (gen.sg. of pres.ppl. of *eō* 'go'; ≈ Gk. ἰόντος); *volunt-* (old pres.ppl. stem of *volō* 'want', seen in *voluntās* 'will, wish'; see Leumann 1977: 347, 526); *secundus* (< **sek*^{w}*ondo-*, gerundive of *sequor* 'follow'); **onos* (> *onus* 'load') vs. *onustus* 'loaded' (cf. *honestus* 'honest' and *modestus* 'modest'); **venos* (**VENOS**, used as a divine name, in CIL I^2 558 [Orbetello, ca. 300 B.C.E.] and 550 [Praeneste, ca. 250 B.C.E.]; ≈ CLat. *Venus*) vs. *venustus* 'attractive'; *arbor* 'tree' vs. *arbustus* 'tree-covered'; *alumnus* '(foster-)son' < **alo-*; *cothurnus* 'high boot' (← Gk. κόθορνος); *amurca* 'olive-juice' (← Gk. ἀμόργη);[95] *amphora* 'jar' vs. *ampulla* 'flask'; *sepultus* < **-pol-* (p.p. of *sepeliō* 'bury'); *industrius* 'diligent' < *indostruus* [P. F. 94.15L]; *monti-* 'mountain' vs. *prōmunturium* 'promontory' (but cf. *prōmonturium*, Ital. *promontorio*); *dupondius* 'the sum of two *asses*' vs. *dupund-* [Lucil.+].

However, these cases can be considered results of secondary innovations. Forms, such as *feruntur* 'carried (3.sg.pass.) and *eunt-* 'go (pres.ppl.) may have been reformed by analogy with *ferunt* and *eunt* (*-o-* > *-u-* in the *final* closed syllable), respectively. The medial syllable of *secundus* continues **-k*^{w}*on-*; the labial element may thus have helped the following *-o-* to turn into *-u-* (cf. *volgus* 'general public' > *vulgus*, *avonculus*

[95] This is perhaps via Etruscan, as the devoicing of -γ- implies; see Sen (2006: 145).

'maternal uncle' > *avunculus*). Adjectives derived from *s*-stem nouns, like *onustus*, *venustus*, *rōbustus* 'made of oak', and *eburnus* 'made of ivory' follow an analogous morphological reshaping, motivated by *onus*, **venus*, *rōbur* 'oak-tree', and *ebur* 'ivory', as laid out by Monteil (1979: 98); *arbustus* may have been analogically modeled after these. The example *alumnus* did originally not contain a closed syllable, reconstructed as **h*$_2$*el-o-mh*$_1$*no-* > **alomano-*, continuing PIE mediopassive present participle like *fēmina* 'woman' < * *fēmanā-* < **dʰeh*$_1$*-mh*$_1$*neh*$_2$- (Meiser 1998: 183). Unless there is evidence that proves that the earlier application of syncope yielded an intermediate stage like **alomno-*, the vocalic change **-o-* > *-u-* is parallel to that in **maksomo-* > *maxumo-*, a pattern of vowel reduction in open syllables in pre-labial context. The Greek borrowings *cothurnus* and *amurca* might allow for positing *-o-* > *-u-*, standing independent of any other words. However, this type of vocalic change is commonly observed before *-r-* (as well as *-l-*, *-m-*, *-n-*) even in *initial* syllables (see Leumann 1977: 48): *furnus* vs. *fornus* [Var.] 'oven', *furnāx* vs. *fornāx* 'furnace', *curvus* 'bent' vs. *corvus* 'raven', *furmīca* vs. *formīca* [App. Probi] 'ant', *ursus* 'bear' (< **or-* < **r̥-*). This development may thus be irrelevant to vowel reduction. It is also possible that *ampulla* underwent the same pre-rhotic change (**ampor-la*; see Leumann 1977: 74). For pre-lateral postion, *sepultus* (p.p., vs. *sepeliō* 'bury') is compared with *multa* 'penalty', *culpa* 'blame' (< *-ol-*). Finally, *industrius* (< *indostruus*) requires clarification. Benveniste (1948: 119), analyzing its nominal form *industria* « application, zèle » [Pl., Enn.+] (cf. *ex/dē industriā*, *ob industriam* « à dessein », interprets its underlying meaning as something like « machiner à l'intérieur » and thus construes *industrius/indostruus* as loan translation from Homeric

βυσσοδομεύων « machinant secrètement ». If this is the case, the creation of *indostruus* must be later than Livius Andronicus's adaption of Homeric works into Latin (soon followed by the nominal derivation *industria*) and thus is not old enough to undergo vowel reduction.[96]

I am not tempted to retain the change *-o-* > *-u-* as a pattern of vowel reduction solely for the limited number of forms such as *prōmunturium* and *dupund-*. Besides the conditional application of *-o-* > *-u-* described just above, the absolute chronology conceived of from inscriptional data seems to betray that this change was irrelevant to the reduction process. Meiser (1998: 70) assigns this development to as late as the second century B.C.E., a period of time which was already in the literary period.[97] The Latin stress system in that era must have been that of the Penultimate Law, in which case it is hardly likely that *-o-* > *-u-* was triggered by initial stress like other patterns of vowel reduction. CIL I^2 581 ("SCdB"; 186 B.C.E.), a text well-known for very proper archaizing orthography, preserves **OQVOLTOD** 'concealment' (> *occultō*) and **CONSOLVERVNT** 'consulted' (> *consuluērunt*).[98] But in terms of vowel reduction, many other forms are writen with reduced vowels, such as personal name **MARCIVS** (< **-os*), **BACANALIBVS** 'Bacchic cult' (< **-bos*), and again **CONSOLVERVNT** (< **-ont*). Among others, **INCEIDERETIS** 'engrave' (> *incīderetis*) is of particular interest, since the medial

[96] The development *-uus* > *-ius* is either analogical (to *glōria* 'praise, honor'/*inglōrius* 'lacking renown', *iniūria* 'unlawful conduct'/*iniūrius* 'unjust', etc.; see Ernout and Meillet 1985: 136) or phonological (**-strou̯-iu̯ā-* > **-strūi̯ā-* > *-stria*; see Weiss 2005: 167).

[97] In regard to *-ol-* > *-ul-*, Götze (1923: 131) also sees the change as "ein junger Vorgang," based on inscriptional data.

[98] The **QV**-spelling in **OQVOLTOD** (< **-kl̥to-*) possibly resulted from an incorrect archaizing process, which was triggered by the following *o*-vocalism (cf. *quom* for *cum*). See Leumann (1977: 49).

diphthongal syllable of this form is only partially archaized with the spelling **-EI-** (the outcome of vowel reduction, not the ultimate origin, i.e, **-ai-*; cf. *caedō*). Therefore, it would be reasonable not to assume that the **-O-** in the medial closed syllable of **OQVOLTOD** and **CONSOLVERVNT** resulted from intentional archaism to show pre-reduction stages, but that the change *-o-* > *-u-* did not affect **OQVOLTOD** and **CONSOLVERVNT** until vowel reduction had been (entirely or almost) completed. Other forms in different inscriptions, though these are personal names, such as CIL I^2 555 **LVQORCOS** (> *Lycurgus*; Praeneste, maybe ca. 300-250 B.C.E.; cf. *Apollō*, not ***Apullō*) and 561 **MACOLNIA** (Praeneste, ca. 300 B.C.E.) with the old **-O-**, are chronologically compatible with this idea.

2.5. Vowel Reduction and Vowel Contraction

In this section, I focus on a problem that is related to the chronological relationship between vowel reduction and vowel contraction. The crux of this issue is how some compositions, such as *cōgō* 'drive together' (< **ko-agō*) and *dēgō* 'spend one's time, live' (< **dē-agō*), came about from a stage with uncontracted vowels. In the traditional framework of simple formulaic patterns of vowel reduction (*a/e/o* > *i/u* in medial open syllables, etc.), if vowel reduction preceded contraction, the proto-form **ko-agō* would have developed into ***coigō*, following other prefixed verbs derived from *agō* 'drive', such as *adigō*, *exigō*, and *subigō*, and later ***cūgō* (via monophthongization type *cūra* 'worry, care' < **koisā*) or ***coegō* (with the phonologically permissible diphthong like

poena 'punishment'). One might thus be forced chronologically to put vowel contraction before reduction. Sommer's comment (1914: 117) reflects such a state of affairs: "Den historischen Akzentsitz für die Differenz von *coḗgī* — *cōgō* verantwortlich zu machen..., ist nicht ratsam, weil des letzteren Vorform **cóagō* mit ungeschwächtem a beweist, daß die Kontraktion im Praes. noch vor die Schwächungen unter dem prähistorischen Akzentgesetz fällt."[99] In his view, vowel contraction would have occurred under the fixed accent distribution on the first syllable of the word, as in **kóagō* and **kóēgī*.

This hypothesis is theoretically possible. Given the fact that a string of *like* vowels, whether created in an older or more recent period, contracted to a simple long vowel of the same quality throughout the history of Latin (e.g., *cōpia* 'supply, stock' < **ko-opiā* [simple contraction; cf. *in-opia* 'lack of wealth']; *trēs* '3' < **treies* [via the Proto-Italic *i*-loss in intervocalic position]; *bīmus* 'two years old' < **dui-ĝʰim-o-* [via *h*-loss]; *nīl* < *nihil* 'nothing' [via *later h*-loss]; *lātrīna* < *lavātrīna* 'washing-place', *sīs* < *sī vīs* [via *u*-loss]),[100] constant contraction of *unlike* vowels throughout the history (perhaps further back to Proto-Italic) might also be conceivable.[101] Yet later treatment of such a sequence is not always straightforward. It may have been preserved as such: e.g., *diēī* 'day'; *fidēī* 'trust' (cf. *fidĕī*, *rĕī* 'wealth'); *fīō*, *fiam* 'take place' (cf. *fīs*, *fīmus*); OLat. *fūimus* ~ CLat.

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[99] See also Leumann (1977: 121): "Ob die Vokallänge oder der historische Akzent für die Verschiedenheit (**có-agō* > *cōgō*, aber *coāctus*) verantwortlich war, ist unklar."

[100] Cf. occasional dissimilation of like vowels in hiatus: e.g., *hietō* 'open the mouth wide' and *societās* 'partnership' (< **-i-i-*). See Ernout-Meillet (1985: 295) and Weiss (2005: 148).

[101] The first conjugation verb may be the oldest case: *laudās* 'praise (2.sg.)' < **laudāes* < **laudāiesi*. The same pattern is found in Sabellic (e.g., Osc. **faamat** 'order' < **fāmāet* < **fāmāieti*). Sabellic is said to have usually preserved the hiatus (Umbr. **puemune** [dat.sg.; divine name], cf. *pōma* 'fruit' < **pó-ema*); the development of **faamat**, together with *laudās*, may thus belong to a very early stage.

fŭimus 'we were'; personal names *Menelāus*, *Alphēus* (← Gk. Μενέλᾱος, Ἀλφειός). Alternatively, a hiatus with change in length may have come into being: *proavus* 'great-grandfather' (< **prō-avus*); *dĕamat* 'be in love with' (< **dē-amat*); *flĕō* < **flēō* 'weep' (cf. *flēre*, *flēbilis* 'lamentable'), *seorsum* 'separately' (< **sē-(v)órsum*; cf. *sorsum* below); this is the so-called "*vocalis ante vocalem corripitur*", discussed in many handbooks like Kent (1932: 144-145), Meillet and Vendryes (1948: 120), Leumann (1977: 118-119), Sihler (1995: 531), and Weiss (2005: 153). Thirdly, they may have also contracted, the first member of unlike vowels determining the quality of the outcome: e.g., again, *cōgō* (< **ko-agō*), *dēgō* (< **dē-agō*); *pōma* 'fruit' (< **po-ema*); perhaps *tantus* 'such' (< **tānt-* < **tāu̯ont-*, cf. *quantus* 'as' < **kʷāu̯ont-*; see Szemerényi 1956: 100-102); *cōmō* 'arrange' (< **ko-emō*); *cōpula* 'union' (< **có-apula*, *cf. apiō* 'fasten'); *combūrō* 'burn up' (< **co-amb-ūrō*); *dēbeō* 'have from, owe' (< **dē-habēi̯ō*, cf. *dĕhíbeō*); *prōmō*/*prōmptus* 'bring forth' (< **prō-emō*/**prō-emptus*);[102] *cōlēscō* 'grow together' (Var., Lucr.; later restored as *coalēscō*; see footnote 79), *prōlēs* 'offspring' (< **co-*, **prō-al-*, cf. *alō* 'feed').

Among other cases of vowel contraction, there are some that show or at least suggest a relatively late application of vowel contraction to sequences of unlike vowels. The verb *mālō* 'would rather' resulted from *māvolō* (< **magu̯olō* < **mag(i)s-u̯elō*), whose common attestation in Pre-Classical Latin would indicate later participation of this form in a contraction process. The variation between the standard *quŏad* 'to that extent?; until' and *quōd* in Varr. *L.L.* 5.23, **QVOD** in CIL III 6593 (Alexandrea) also provides indications

[102] For a secondary *-ēmptus*, found in some inscriptions with apex, see Leumann (1977: 113).

of a later contraction process (see Leumann 1977: 121). Further, with regard to the contraction process in *sorsum* 'separately' (vs. *sĕorsum* < **sē-vorsum*), because the second vowel (*-o-*) determined the quality of the result, *seórsum* accented according not to the initial stress rule but to the Penultimate Law is a likely immediate pre-stage of *sorsum*.

In view of these examples, the relative chronology Sommer (1914: 97) insists on, namely, where vowel contraction (particularly of unlike vowels) precedes vowel reduction, has become less robust. Both changes were effective over a certain length of time, not coinciding with each other. Vowel contraction of unlike vowels seems to have continued through the literary period after vowel reduction had been completed. If this is the case, the reverse order, reduction → contraction seems more plausible. In fact, the theory I have laid out in this chapter enables this chronology to be a possible candidate. By means of an intermediate stage like [ə] between Process I and Process II, vowel contraction in *cōgō* is analyzed as follows: **kó-agō* > **kó*[ə]*gō* > *cōgō* (not **kó-agō* > ***kóigō* > ***cūgō* or ***coegō*). Recall that the high corner vowels such as *-i-* and *-u-* were not reduced to [ə], and thereby *coetus* (< **ko-itos*) instead of ***cōtus* is not an obstacle to this idea at all.[103]

This appeal to an intermediate stage in the contraction process is already suggested by Schrijver (2003: 61). He posits a historical development that Pre-Latin **e/o* in non-initial open syllables changed through a short closed front vowel (*ï* in his representation) into *i* ultimately, then explains a change commonly observed in Latin **-iie/o*(-) > *-ī*(-) by

[103] Allen (1973: 145) adduces *praetor* 'leader' as an example of vowel contraction (< **praĭ-itor*), though he provides no clear process. In my opinion, the proto-form reached *praetor* via syncope of *-i-*.

contraction through the stage *-*iį̈ï*(-).[104] My theory may restate *ï* as [ə] (or [i] as in final closed syllables), keeping Schrijver's view of contraction and most of the examples with the process of *-*iį̈e/o*(-) > *-*iį̈*[ə/i](-) > -*ī*(-): *fīlie* (Andr.) > *fīlī* 'son' (cf. *fīliolus* 'little son'); **tībio-kan-* > *tībīcen* 'flutist'; gentilicia **Lūciolios* > *Lūcīlius* (cf. *Lūciolus*), **Māniolios* > *Mānīlius*; tribal name **Samni-etēs* > *Samnītēs*.[105] We may also add *nītor* 'lean' here (< **ni-į̈et-e/o-*; see LIV 2001: 314). The intervocalic homorganic glide -*į̈*- would not impede this scenario; whatever its phonetic/phonological status, *illīus* 'that' and *istīus* 'this' underwent "*vocalis ante vocalem corripitur*" (thus, *illĭus* and *istĭus*, respectively), and the process was not blocked by an expected existence of -*į̈*- (i.e., [-ījju-]).

2.6. Non-application of Vowel Reduction in Nominal Compounds

Vowel reduction may happen in any unstressed vowel of a word: e.g., **súbagomos* > *subigimus* (1.pl. of *subigō* 'subdue'). However, there is a set of forms that do not meet this expectation, namely, nominal compounds, which have generally been neglected or have not been properly treated by scholars regarding this matter. In many instances of this morphological category, the second member of the compound resists vowel reduction in its first syllable. One might claim an analogical influence from its independent use. But

[104] The direct contraction process has been presented in many handbooks, such as Sommer (1914: 117), Kent (1932: 151-152), Leumann (1977: 122), and Sihler (1995: 63).

[105] For *fatīgāre* 'assail; make tired' [Pl.+] derived from the nominal compound, it is likely that the element which follows the first member **fati-* is **-h_2ĝ-o-*, a thematized zero-grade of **h_2eĝ-*, instead of **-ag-*; if so, the -*ī*- is not an outcome of vowel contraction, but **-i-* lengthened by the following laryngeal. See Dunkel (2000: 93-94).

this seems not to be the case. Rather, in the same way as phonetic/phonological factors determined the occurrence of vowel reduction in prefixed verbs, it turns out that a certain system underlies the non-application of reduction in nominal compounds.

2.6.1. Accentual Independence of Compounded Elements

Like verbs, nominal stems (including adjectival ones) are productively prefixed (see Fruyt 2005: 34-37), though the combination is more limited than in prefixed verbs: e.g., *dē-mēns* 'out of one's mind, mad' [Pl., Enn.+], *in-ūtilis* 'useless' [Pl.+], *per-magnus* 'very large' [Ter.+], *con-servus* 'fellow slave' [Pl.+], and *prae-ceps* 'headlong' [Pl., Cato, Enn.+]. In addition, they are preceded by other nominal elements far more frequently than verbal stems: e.g., *ūndecim* '11' [Pl.+], *agricola* 'farmer' [Pl., Cato+], *indi-gena* 'native' [Verg.+], *mīlifolium* 'marsh plant' [Plin.], and also *auceps* 'bird-catcher, fowler' [Pl.+], if we are allowed to interprete the second member as a root noun derived from a verbal root (cf. *capiō* 'take').

Among previous studies of nominal compounds (prefixed nouns included), some scholars, such as Bader (1960), Garrett (2000, 2002), and Fruyt (2005), deal with as many examples as possible in order to disclose their features. Yet, the corpus is inevitably a mixed one, consisting of formations from various periods throughout the history of Latin. First attestation is of more importance for nominal compounds than prefixed verbs. Components of the latter, prefix and base verb, are so variously combined as to form a frequent category of composition. Hence, even if PREFIX_1 + VERB_1 has only a late

attestation, $PREFIX_{2, 3...}$ + $VERB_1$ of old pedigree could guarantee the long history of the involvement of $VERB_1$ in composition. Nominal compounds reflect a different state of affairs, since the selection of components is rather restricted, and so the participation of some of them in the compounding is hardly datable. Thus, instances with only attestation from Classical Latin (onward) need to be handled with much caution.[106] Needless to say,

[106] In addition, artificial creation of compounds, particularly by poets, was bolstered by Greek influence, as Fruyt (2005: 42) points out. Fruyt also emphasizes the strong tendency to produce choriambic words (—∪∪—; the first ∪ is *-i-* in many cases), or ones with a configuration of ∪∪—. This is proved by compounds like *vulni-ficus* 'causing wounds' [Verg.], (oblique cases of) *odōri-fer* 'fragrant' [Verg.+], *sopōri-fer* 'bringing sleep' [Verg.+], and *labōri-fer* 'enduring toil' [Ov.+]. Others, whose first attestation is found in prose, align with this inclination somewhat: *foedi-fragus* 'treacherous' [Cic.+] and *honōri-ficus* 'honorific' [Cic.+].

Incidentally, the formation of the first members of compounds, as in *foedi-fragus* and *vulni-ficus*, as well as *homi-cīda* 'homicide' [Cic.+], has long been the focus of much discussion, since part of the suffix (of [*no*]*s*- and *n*-stem) is truncated (i.e., not ***foederi-*, ***vulneri-*, ***homini-*; cf. *foedus* 'treaty', *vulnus* 'wound', *homō* 'human being' [gen.sg. *hominis*]). Cf. Gell. 19.7.5: "hostes qui foedera frangerent 'foedifragos' non 'foederifragos' dixit Laevius."

Fruyt (2005) invokes several factors in order to solve the problem, but none of them are compelling. For instance, haplology in **homini-* > *homi-* is not well motivated, due to the difference in consonants of adjacent syllables (on the other hand, her interpretation of *venē-ficus* 'of or concerned with sorcery; (as n.) sorcerer, poisoner' [Pl.+] < **venēni-ficus* by means of haplology is possible). She also adduces *vet-ulus* 'elderly' [Pl.+] and *corp-ulentus* 'corpulent' [Pl.+] as cases reflecting suffix truncation (cf. *vetus -eris* 'old' and *corpus -oris* 'body'), claiming that "la raison de cette truncation serait... d'ordre phonologique ou rythmique" (2005: 48); but the conditions of the process are only vaguely defined, solely on the basis of word length or the like. For *vetulus*, see Leumann (1977: 308), with contamination of *vetus* + *catulus* 'young dog' (Var. *R.* 2.9.3 *catuli* et *vetuli* [sc. *canes* 'dogs']). The adjective *corpulentus* may be construed as a result of haplology from **corpor-ulentus*. A verb *nūn-cupō* 'declare' [XII Tab.+], presumably derived from an unattested nominal compound **nōmi-cep-s* (see Ernout and Meillet 1985: 451), involves the same problem (cf. *nōmen -minis* 'name') as *foedi-fragus*, *vulni-ficus*, and *homi-cīda*. Fruyt (2005: 48-49) compares this with elision in *lītigāre* 'litigate' [Pl.+] (← *līt*[*em*] *agere*(?); but cf. Leumann 1977: 96, 550, and also footnote 81) and *vēnīre* 'be sold' (< *vēn*[*um*] *īre*, cf. *vēndō* 'sell'; see Leumann 1977: 224), but her hypothesis is hardly justified due to the lack of, e.g., ***nōmigō* or ***nōmeō*.

A more attractive solution would be imitation of a Greek morphological process, a pattern concisely laid out by Nussbaum (2007: 2-3). Some Greek **-o/es*-stem nouns behave like masculine **o*-stem nouns for various derivational purposes. One of the aims is to yield the first member of compounds: εἰρο-κόμος 'working in wool' (← εἶρος, -εος 'wool'), θυο-σκόος 'haruspex' (← θύος, -εος 'sacrifice'), modeled after ξεινο-δόκος 'one who receives strangers' (← ξεῖνος 'stranger'), etc. vs. old type σακεσ-πάλος 'shield-brandishing' (← σάκος 'shield'), τελεσ-φόρος 'bringing fulfillment' (← τέλος 'end, accomplishment'). The inherited counterpart of the process is only marginally found in Latin (see Heidermanns 2002: 193 n. 13): e.g., *opi-fex* 'craftsman' [Pl.+] (**opo-* ← *opus* 'work'); Heidermanns's other example *mūni-ceps* [Lucil.+; but not in Pl./Ter.] 'native of the same *municipium*' is ambiguous as to whether its first member is an avatar of *mūnus* 'task', or an *i*-stem variant seen in *com-mūnis* 'common'). Given the fact that *foedi-fragus* and *vulni-ficus* (as well as *homi-cīda*) are only attested beginning in the Classical period, it is less likely that the formation of the first members are of an old pedigree, despite

the lack of attestion in Old Latin could be solely accidental. And it cannot be ruled out that newer compounds were formed on the model of ones of older origin; we could thus learn something about the older formations by looking at these new instances. But for the sake of as much certainty as possible, we focus on examples that are attested in Old Latin. It might be better to exclude hapax legemona, which are perhaps ephemeral creations by Latin authors. But their phonological and morphological make-up would reflect traits of the Old Latin compounding system.

In such a reduced corpus, we still find a lot of forms that reveal a common feature: the second member of the compound retains its original vocalism in the first syllable. We have already seen some of them, such as *per-magnus* 'very large' (cf. footnote 76), *ūndecim* '11', and *agricola* 'farmer'. As mentioned above, these might be analogical to *magnus* 'large', *decem* '10', and *colō* 'cultivate', repectively. But several other instances draw our attention, revealing a common peculiarity; the second member, derived from a verbal root, retains its original vocalism, even if the counterpart in prefixed verbs shows vowel reduction: *prō-seda* 'prostitute' [Pl.; P. F.] (*sedeō*) vs. *-sideō*, *bū-caeda* 'ox-slaughterer' [Pl.] (*caedō*) vs. *-cīdō*; *sacri-legus* 'temple-robber' [Pl.+] (*legō*) vs. *-ligō*; *busti-rapus* 'grave-robber' [Pl.] (*rapiō*) vs. *-ripiō*.[107] One might argue that these forms

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Lindner (2002:15-18). At the same time, the marginal presence (**opo-* and **mūno-*[?]) could not have been a productive morphological model. It is therefore likely that Roman authors learned this type of morphological process through their learning of Greek texts and then echoed it in their works, substituting the expected *-o-* with the common linking vowel *-i-* in compounds (cf. *arti-fex*, etc.). Other stem classes, especially *n*-stems, likely followed the same morphological pattern (*homi-*, **nōmi-* < **homo-*, **nōmo-*). Metrical convenience must also have been a spur to imitation.

[107] Whether *sacri-* (< **sakro-*), *busti-* (< **busto-*), and many other cases are the result of vowel reduction has been a perennial problem. Without unamious agreement, such forms have been seen as exemplifying vowel reduction by some scholars, as with Exon (1906) and Nyman (1983). Though Stolz (1887: 317) strongly supports vowel reduction, his recourse to Greek parallels such as ὀμβρο-φόρος

mirror the phonological shape of simplex verbs, disregarding the one of prefixed verbs, which I consider to be less likely. It seems significant that the verbal root *sed-* shows regular vowel reduction in athematic nominal compounds: e.g., *obses -sidis* 'hostage' [Naev.+], *praeses -sidis* 'guardian' [Pl.+]. From this observation, the examples above are most likely governed by a certain phonological system that allows them to preserve the vowel in question.

My proposal is that compounded elements were only loosely univerbated for a while after composition, whereby each element had its own stress (there may have been a difference of primary and secondary stress between compounded members): **prṓ-séda*, **bṹ-káiḍa*, **sákro-légos*, and **bústo-rápos*. This can perhaps be construed as reflecting the application of the initial-stress rule to each compound member. I suggest that such accentuation blocked vowel reduction in the position in question.[108] The same system of separate accent may also underlie some other compounds, such as *agricola* 'farmer' [Pl., Cato+] (*-colō*), *alti-sonus* 'that sounds high up, sublime' [Enn.+] (*-sonō*), and *mōri-gerus*

'rain-bringing' and μηλο-φόρος 'bearing apples' provides only circumstantial evidence. In fact, Bader (1960: 239-240, 1962: 16-17) maintains that *-i-* came to function as "voyelle de liason" in nominal compounds. This view is also found in Godel (1961: 56). The key factor in the argument is the presence of a number of nominal compounds which have *i*-stem nouns or consonant-stem nouns (with *-i-* under the close-knit influence of *i*-stem ones) as first members: e.g. *ponti-fex* 'high priest' [Cato+], *parti-ceps* 'participant' [Andr.+], *frūgi-fer* 'productive' [Enn.+], *lūci-fer* 'light-bringing' [Acc.+]. The vowel *-i-* is original in these cases, and its spread to other compounds not based on *i*-stem nouns is fairly conceivable. In fact, if the proto-form of *sacri-*, that is, **sakro-* had undergone the most natural sound change, it would have come out as *sacer-* as in *sacerdōs* 'priest' [Pl.+], *sacellum* 'shrine' [Ter.+] < **sakro-d$^{(h)}$ōs*, **sakre-lo-* (cf. 3.3.6 and 3.4.2). Witness also *agricola* (< **agro-*, cf. *ager*). This replacement of *-o-* by *-i-* seems to have continued in Literary Latin. Biville (1990: 18) cites a pattern of the type *hōrologium* ~ *(h)ōrilogium* 'instrument for showing the time'; the former is a standard form attested since Varro, a transcription of ὡρολόγιον, which later coexists with the latter.

[108] A similar idea is already suggested by Götze (1923: 126).

'compliant' [Pl., Naev.+] (*-gerō*), as well (i.e., **ágro-k^{w}ólă*,[109] **álto-sónos*, and **mŏ́ri-géros*). Even among coinages in Classical Latin, it is plausible that lengthy words such as *per-terri-crepus* 'making a terrifying sound' [*Inc.trag.* 142 apud Cic. *Orat.* 164, Lucr.] and *versūti-loquas* 'speaking crafty words' [*Inc.trag.* 114 apud Cic. *de.Orat.* 3.154] were not tightly univerbated, separating hosts of stress. Compounds with nouns or adjectives as second members, such as *magn-animus* 'generous' [Pl.+], *sicc-oculus* 'dry-eyed' [Pl.], *multi-modīs* 'in many ways (adv.)' [Pl.+], *angi-portus/m* 'alley' [Pl.+], *sēmi-senex* 'semi-old man' [Pl.], *albi-capillus* 'grey-haired' [Pl.],[110] and so forth, can also be incorporated into this framework.

Numerals such as *ūn-decim* '11' [Pl.+] (< **oi̯no-dekem*) and *quīn-decim* '15' [Pl.+] (< **k^{w}ink^{w}e-dekem*) deserve some further clarification. The phonological appearance of *-decim*, which has long been mysterious to many scholars, also enables us to make tenuous a possible analogical operation from *decem* '10' due to the partial disagreement in vocalism. An analogy with ordinal numbers, such as *ūndecimus* '11th', as Kent (1932: 100-102) suggests, to gain *ūndecim* instead of **ūndicem* or **ūndicim*, seems weak to me; one must justify the direction of analogy from ordinal to cardinal numbers. Leumann (1977: 94) argues that *-decim* resulted by a sort of remote metathesis from **-dicem*. But it

[109] As mentioned in 2.1, Bader (1960: 238-239, esp. 239 n. 12) believes that the "apophonie" does not pass over second syllables, which she considers responsible for vowel retention in the third syllable of *agricola* (1960: 241). She further suspects that this retention impeded vowel reduction in *incola* 'inhabitant' [Pl.+] and *incolō* 'dwell' [Pl.+], which, however, I think is unlikely. Bader (1960: 245-256), on the other hand, implies that *agricola* had secondary accent in the third syllable (see 2.6.2), which contradicts her above view, although it is much closer to the truth.

[110] The prefix *per-* intensifies the meaning of the following adjectives, as Fruyt (2005: 37) points out. Thus, in contrast to adjectives with privatives like *in-* and *dis-* (see 2.4.2), the *per*-adjectives have strengthened meanings of base adjectives, which may be a factor in retaining the vocalism as well.

is entirely impossible to verify his idea, until the old form *-*dicem* is actually found somewhere. My theory would be the least problematic. Accentual independency in these numerals is supported by the less-bound morphological order of each element; Untermann (2000: 169) presents Latin *decem duo* and Greek δέκα δύο '12' in the reverse order from *duodecim* and δώδεκα to defend Umbrian *desenduf* (TIg. VIIb 2). Vowel reduction in the final syllable of *-decim* (< *-*dekem*) needs some clarification, since *-e-* before *-m* is usually retained in Latin: *septem* '7', *decem* '10' (cf. Ital. *dieci*), acc.sg. *-em* of the third declension, etc. But compare *enim* < **enem*, already mentioned in Dietrich (1852: 550-551) and Ernout-Meillet (1985: 196) in association with *-decim*. Effortless pronuncation of lengthy numerals may have reduced vowel sonority in quick counting.

2.6.2. Solutions to Counterexamples

There are some instances that appear to contradict my hypothesis, but we can solve them by setting up some phonological and morphological conditions. As a first group, third-declension compounds in *-ceps -cipis* (< *-*kap-*; e.g., *praeceps* 'headlong; impetous'), *-fex -ficis* (< *-*fak-*; e.g,. *artifex* 'skillful'), *-plex -plicis* (< **plek-*; e.g., *duplex* 'double'), *-cen -cinis* (e.g., *tubicen* 'trumpeter' < *-*can-s*), and *-spex -spicis* (< *-*spek-*; e.g., *auspex* 'augur') regularly prove vowel reduction in the first syllable of the second member. Bader (1960: 244) already notes that such forms in contrast with other compounds without vowel reduction in the same position (e.g, *sacrilegus*, *ūndecim*, *agricola*). Her solution is word length (i.e., the number of syllables) and accent

distribution, which I believe is useful for our purpose. Though what she puts as a premise for Latin compounds — they are enliticized forms, whose first member is accentually salient (1960: 244-245) — is not entirely tenable,[111] the ultimate outcome of her analysis, something like *ágri-còla* ("un accent secondaire portant sur un second membre polysyllabique", "le double accent" in 1960: 245) is somewhat compatible with my theory of accentual independence in nominal compounds. Bader does not provide phonological reasons why monosyllabic elements bear no accent as second members, but it is most likely because they occupy word-final syllables that are prosodically extrametrical, and hence unstressed. True, oblique cases of, for instance, *artifex*, have a multisyllabic element as second member: gen.sg. *artificis*, acc.sg. *artificem*, etc., whereby separate accentuation like **árti-fákes* and **árti-fákem* would be conceivable. But synchronically, at least in Literary Latin, such "apophonie" of *-e-* in the nominative singular and *-a-* in other cases is strictly banned; vowel height should be phonemically either equal or differentiated, as the nominative singular possesses a lower vowel (cf. *pecten, pectinis*; *homō, hominis*, etc.), not vice versa. Hence, it is plausible that oblique cases secondarily followed this phonotactic pattern, probably when the vowel in question became unstressed under the Penultimate Law (i.e., **árti-fákes* → **artífakis* → *artificis*, etc.). Much the same is applicable to *iners* 'clumsy', *sollers* 'skilled', and *expers* 'having no share' (gen.sg. *-ertis*). Further, the second member in *puer-pera* 'woman who has been, or is in process of being, delivered of a child' can be explained likewise. We have

[111] Bader (1960: 246) invokes the forms, *cŏnficiō* 'complete', *sĕdulus* 'attentive', and *dĕnuō* 'anew' (also *áffatim* 'sufficiently' and *ádmodum* 'to a great extent' from Gell. 6.7.2) among others; for prefixed verbs, she appeals to Vedic evidence in order to see a vestige of PIE accent. However, we can account for vowel reduction in these forms by means of the initial-stress rule. For accent distribution in the prepositional phrases including the ones reported by Gellius, see 2.4.2.

its masculine counterpart of the second member in *pau-per*, *-eris* 'poor', which continues **pau-paros* « qui produit peu ». Through the shift in inflection type from *o*-stem to consonant stem (see Ernout and Meillet 1985: 490 and Bader 1962: 111) and subsequent relocation of **-par* as final syllable, its *a*-vocalism became qualified to reduce to *-e-*, following the general reduction pattern, and this *-e-* most likely spread to other case and gender forms.

The second group of examples in opposition to my theory is a minor one, but is of morphological interest. Some of the compounds produce further derivatives by means of suffixes. Outcomes of such a process may or may not retain their original vocalism in the second member. Forms, such as *du-pondius* 'the sum of two *asses*', *bi-pālium* 'instrument for double-digging', *prō-geniēs* 'progeny', *sēm(i)-animis* 'half-alive', *inopia* 'lack of wealth' (*in-ops* 'poor'), *reciprocē* 'to and fro',[112] *remedium* 'remedy', and *acu-pedius* [P. F.], preserved original vowels, which must potentially also be retained in pre-suffixed compounds (**du-pondos*, etc.). Meanwhile, some other forms incurred vowel reduction: e.g., *prae-cipuus* (*-ceps*, **PISCICAPVS** 'catcher of fish' [CIL IV 826]; cf. *contiguus* 'adjacent', *propitius* 'propitious'), *mūscipulum* 'mousetrap' (<**mús-kapelom*, cf. *capulus* 'sword-handle'), *adsidēlae* (tables for priests to sit on) [P. F. 18.8L] (cf. *prōseda*), *sōlstitium* 'solstice' (cf. *status* 'fixed'), *im-berbis* 'beardless' (*barba* 'beard'), *per-ennis* 'perennial' / *quinqu-ennis* 'five years old' (*annus* 'year'), *per-emnis* 'auspices

[112] Dunkel (1979: 189 n. 16) induces Cowgill's take (per litt.) on the medial vowels of this form (the *o*-vocalism in the third syllable is more important to our discussion). Cowgill asks him "why not **recupricus*?," and suggests "that at the time of Latin vowel weakening, perhaps there was still a strong stress on the second element, etwa ['rek^{w}e^prok^{w}os], such that the **o* of the second element developed as in initial syllable, and the **e* at the end of the first developed as in medial open syllalbe." Sen (2006: 151-152) also suggests a trace of secondary stress in *reciprocē*.

taken before crossing a river' [Cic.; Fest.] (*amnis* 'river'; probably an *i*-stem derivative, like *per-ennis* / *quinqu-ennis*, in view of the semantic specification). Examples are increased by *dupund-* [Lucil.+] (variant of *du-pondius*; see right above) and *tripudium* 'a ritual dance in triple time' [Cic., Catul.+; Fest.] (*tripudiō* 'perform a *tripudium*' [Acc.+]) / *repudium* 'breaking of an engagement' (cf. *acu-pedius*). Bader's observation (1960: 242, 247) that the "apophonie" is more frequent in derivatives than compounds is noteworthy. Compare *prōseda*, **DOMISEDA** 'she who stays at home' (CIL VI 11602, 34045) with *-sidium* (*ob-*, *sub-*, *prae-*, *dē-*) and *agricola* with *domicilium*, *ancilla*, *inquilīnus*. From such a propensity toward vowel reduction, it follows that forms derived from compounds may have lost the separate-stress system, then evolved as unitary words in terms of prosody. Thus, in contrast to accentual independence posited in, e.g., *[prō][sedă̆] and *[domo][sedă̆], derivatives of the same verbal root noun in the second member could be segmented like *[[[ob][sed]i̯i̯om]]; stress was then assigned in the larger prosodic unit, accordingly *[[óbsedi̯i̯om]].[113] Basically, in order to obtain this post-compositional accentuation, de-agentive suffixes would be required. Note that the element *-o-* in forms like *-capus* (< **-kapo-s* in **PISCICAPVS**; cf. *-ceps* < **-kap-s*) makes the entire compound remain agentive. Presumably, the same is applicable to *-a-* in *prōseda*, *domiseda*, *bū-caeda*, etc. In some cases, these elements are only gender-specifiers (*prōseda*, *domiseda*), thus most likely domain-centered; in other words, not [[[bū][caed]a]] but

[113] One might claim that words of a certain length have secondary accent as well as initial stress, like **dómik*^{w}*òliom*, and vowels with such an accent should be retained. But it suffices to cite Barnes's report (2006: 31): "... in Italian, only vowels under primary stress are immune to vowel reduction, while secondarily-stressed vowels undergo the neutralization of unstressed high and low mid vowels to [e] and [o] just as would an unstressed vowel (Crosswhite 2001: 206). Experimental evidence... provides an explanation for this fact, in that vowels with secondary stress are seen not to differ significantly from unstressed vowels in duration, while the vowels of stressed syllables are regularly longer than both."

[bū][caeda].[114]

[114] It is now possible to comment on the compound *parri-cīda* 'murderer of a near relation' [Pl.+], with which *paricidas* (Fest. 247.19 L) is associated. In contrast to *bū-caeda* 'ox-slaughterer' [Pl.], this compound exhibits vowel reduction and subsequent monophthongization (**-kaidă̆* > **-eĭ-* > *-ī-*). According to Ernout and Meillet (1985: 483), the first element *parri-* is difficult to etymologize (see some traditional views there), though they judge this entire compound as being of very old origin. In fact, the variant *paricidas* in Festus is in a citation from "Lex Numae", as recollected by Marchese (1979: 109). Though the notation ‹i› for *-ī-* may perhaps be a later interpolation, the presence of this compound in a very archaic period seems plausible. The old pedigree can also be verified to a certain extent by acknowledging its old derivative *parricīdium* 'murder of a near relation' [Pl.+], as opposed to the lack of **bū-caedium* (or **bū-cīdium*). The vowel reduction (and subsequent monophthongization) in *parri-cīdium* can be squared with the *domicilium*-type. It is thus plausible that *-ī-* in *parri-cīda* was transmitted from *parri-cīdium*, as well as *-cīdō* in prefixed compounds, during an older period. The vocalism in *bū-caeda*, on the other hand, reveals its recent creation and shows more clearly the default pattern of separate accentuation. Another compound, *homicīda* 'homicide' [Cic.+], was most likely formed in a relatively late stage, copying *-ī-* from relevant words that were already monophthongized. The derivation of *homicīdium* [V.Max.+] may be too late to affect the vocalism of *homicīda*. Incidentally, vowel reduction in *-ficus* (*male-ficus* 'wicked', *bene-ficus* 'beneficent', *mūni-ficus* 'dutiful', *magni-ficus* 'magnificent', *damni-ficus* 'that causes loss') can likewise be explained by the presence of old derivatives in *-ficium* (e.g., *male-ficium* 'misdeed' [Pl.+], *bene-ficium* 'kindness' [Pl., Cato+]).

3. Restatement of Vowel Deletion in Latin: Prominence and Metrical Constraints

In the foregoing chapter, we considered various problems of vowel reduction in Latin. This chapter aims to elucidate some mechanisms of vowel deletion in Latin. Faliscan materials are marginally utilized as supporting evidence. As in the analysis on vowel reduction, phonetic and phonological factors that have been ignored or insufficiently discussed are widely employed here. It is usually said that vowel deletion, especially syncope, is not as extensive in Latin as in Sabellic, but actually, the number of occurrences is inexhaustible.

Vowel deletion is a drastic change, in that the number of syllables of the word may change, resulting consonant clusters may require some adjustments, and morphological components may become less transparent.[1] As mentioned in 1.2.2, it is a difficult task to define clear-cut conditions that trigger or do not trigger the change. The following discussions will shed light on some specific aspects of vowel deletion with representative materials. Eventually, we will come to reconfirm that vowel deletion is loaded with complex mechanisms. But this complexity will be untangled to some extent from both the synchronic and the diachronic standpoint.

[1] It would be relatively easy to know when and where vowel deletion took place by means of the existence or non-existence of vowel letters. But old spellings tend to linger after the phonetic change. Note also Hayes's remark (1995: 289-290) on Asheninca /i/ drop: "Native speakers apparently are unconscious of the deletion, since they spell the deletion site with letter *i*..."

3.1. Brief Stratification and Chronology of Vowel Deletion in Latin

Determining the conditions of vowel deletion is a recalcitrant matter. This sound change took place a number of times throughout the history of Latin, not only in the Pre-Literary but also the Literary period. This state of affairs makes one logically assume that different factors were involved in vowel deletion at different periods. In the Pre-Literary stage, initial stress most likely occasioned this sound change. The evidence marshalled for this conception is that in originally quadrisyllabic words, deleted vowels often occupied the antepenultimate syllable, whose vowel must have been accented and thus preserved if the Penultimate Law already operated then. Some of the other words whose accent location coincides with the one in the Penultimate Law may be interpreted as proving the operation of initial stress, if they appear in Pre-Classical Latin with vowels already deleted: e.g., [[quadrisyllabic]] *ūndecim* '11' < **ói̯no-dekem*; *quīndecim* '15' [Pl.+] < **k^{w}íṇk^{w}e-dekem*; *ōpiliō* 'shepherd' [Pl., Cato+] ← **áu̯-pol-o-* (via syncope) < **áu̯i-pol-o-* (via accentual shift) < **au̯i-pol-ó-* (via Thurneysen-Havet's Law) < **ou̯i-pol-ó-* 'sheep-driving' (< **h$_2$ou̯i-polh$_2$-ó-*);[2] **ópi-fakiom* > *officium* 'duty' [Pl.+] (cf. *opifex* 'craftsman' [Pl.+], *opificīna* 'performance of constructive work' [Pl.]; *opificium* [Var.+] is presumably a later recomposition); *sinciput* 'half-head; fool' [Pl.+] < **sēmi-caput*;[3] *nūper* 'recently' [Pl.+] < **nou̯o-paros*; *dexter* 'right' [Pl.+] < **déksiteros*

[2] For this historical scenario, see Vine (2006a: 234-235). The form *ōpiliō* is considered to be the "rustic" variant (see also Ernout 1909: 209, Götze 1923: 124), as opposed to the "urban" *ūpiliō*. The latter continues a form with restored first member **ou̯i-* (> **ou̯-* > *ū-*).

[3] Alternatively, **su̯īno-caput* 'pig-headed', favored by Ciardi-Dupré (1901: 200-201), Leumann (1977: 97), and Lindner (2002: 29, 221).

(Gk. δεξιτερός); *maxumus* 'biggest' [Pl., Enn.+] < **magisomo-* < **mag-isṃmo-*; *ūsurpāre* 'carry out' [Pl., Enn.+] < **oịsu-rup-ā-* (← PIE **reup-*, Lat. *rumpō* 'cause to break'; see Watkins 1970: 323-328; cf. *ūsū-capiō* 'acquire ownership of (a thing) by virtue of uninterrupted possession' [Cic.+]), *repperī* [Pl.+] < **répeparī* (← PIE **pṛh$_3$-*; pf. of *reperiō* 'discover'); *reppulī* [Pl.+] < **répepalī* (← PIE **-pl̥h$_2$-*; pf. of pres. *repellō* 'push back'); *rettulī* [Pl.+] < **rétetalī* (← PIE **-tl̥h$_2$-*; suppletive pf. of pres. *referō* 'carry back'; cf. *extetulisses* in Ennius; see Pulgram 1975: 111); *reccidī* < **rékekadī* (← PIE **-ḱad-*; pres. *recidō* 'fall back');[4] [[others]] *auceps* 'bird-catcher, fowler' [Pl.+] < **áụi-cap-s*, *prīnceps* 'chief' [Pl.+] < **prí(s)mo-kap-s*, *hospes* 'host' ← **hospus* < **hosti-pot-s* (cf. 2.3),[5] *forceps* 'tongs' [Lucil., Cato+] < **formkap-* < **formo-kap-* (cf.

[4] See Leumann (1977: 587) on this kind of perfect form. For the zero-grade reflexes of roots with laryngeals (*-CR̥H- > -CaR- before -V-) and the appearance of the *CaC-root as it is, see Meiser (2003: 160-161). Exon (1906: 137) suggests that syncope began in forms with an accented heavy syllable and four or more syllables such as *repperistī* (< **repeperístī*), etc., in the Republican period. This idea is not compelling; see footnote 22.

Besides these forms, Leumann (1977: 587) posits the same type of reconstruction and development (but involving the reduction of consonant cluster as well) for *attulī* (pres. *afferō* 'bring near'), *appulī* (pres. *appellō* 'bring to'), and *comperī* (pres. *comperiō* 'find out'), like **at-tetulī* > *attulī*, and so forth. He assumes that due to such forms, reduplication became obsolete (this propensity is significant in Latin: e.g., *fēcī* vs. Osc. **fifikus** < **fe-fēk-* 'feceris, made'; *dīxī* vs. Umb. *dersicust* < **de-dik-* 'dixerit, stated') and that its omission spread to other prefixed verbs: *occidī* 'fell in the way', *attigī* 'touched', *contudī* 'crushed', *occīdī* 'killed', *accurrī* 'ran (*esp.* to help)', *attendī* 'paid attention', *impendī* 'disbursed', *refellī* 'refuted', *abstinī* 'kept away' [Pl. *Amph.* 926] / *continī* 'held together' [Pl. *Asin.* 582; or *continuī*?] (cf. *tetinī* [Pac.]). Note, however, that it is not easy to determine which ones are really analogical results; syncope could account for some of them (cf. probable application of the so-called "*mamilla*-rule" in *refellī* < **reffellī*). Be that as it may, it is true that reduplication in prefixed verbs eventually became obsolete. See also Faliscan **porded** 'obtulit, gave' (Ve 241) < **por-deded*, so analyzed (not by means of syncope) by Peruzzi (1964: 155) and Giacomelli (1978: 519).

Leumann (1977: 587) mentions *reddō*, which continues a reduplicated present **re-didō*. Its perfect form would have arguably been reconstructed as **re-dedī* (or **re-dedaị*). If this had undergone the same type of syncope as the present, the outcome would have been falsely ***reddī*. The real form *reddidī* is peculiar in two aspects: 1) vowel reduction (**-e-* > *-i-*) instead of syncope and 2) the pendant *red-* (the pre-vocalic variant of *re-*prefix; thus, not ***redidī*). The former is perhaps due to a preference for having more than one syllable for the perfect stem like other forms (i.e., not ***-dī* but *-didī*; cf. the monosyllabic present *dō*, thus *red-dō*). The latter would be interpreted as a result of analogy with *repperī*, *reppulī*, and *rettulī*, containing double consonants, which probably spread to the present (*reddō*) as well.

quadrisyllabic oblique cases like gen.sg. **áu̯i-cap-es* > *aucipis*, **prí(s)mo-kap-es* > *prīncipis*); *cette* 'give here' (impv.pl. of *cedō*) [Pl., Enn.+] < PItal. **kedate* < PIE **k̂e-dh₃-te* (pl.); *palma* 'hand' [Pl., Enn., Lucil.+] < PItal. **palamā-* < PIE *pl̥h₂-meh₂-* (Gk. παλάμη, OIr. *lām*); *imber* 'rain' [Enn. *Ann.* 284, 522] < **embris* < **m̥bʰris*; *suprā* 'on top of' [Pl., Cato+] < **SVPERA** (CIL I^2 581; cf. *superā* [Cic., Lucr.]), *suprēmus* 'topmost' [Pl., Enn.+] < **superēmos* (cf. *superus* 'upper'); **posinō* > **posnō* > *pōnō* 'place, set' [Pl., Cato, Naev.+]; *audēre* 'intend, dare' [Pl., Naev.+] < **au̯id-ē-* (cf. *avidus* 'greedy'); Gk. theonym Πολυδεύκης → Lat. **Polud-* > **Pold-* > **PODLOVQVEI** (CIL I^2 2833; Lavinium, ca. 550-525 B.C.E.), perhaps with /ld/ misrepresented (see Leumann 1964: 84), **POLOCES** (CIL I^2 549; Praeneste, ca. 350-300 B.C.E.) > *Pollūx* [Pl.+]; etc.[6]

The following examples should perhaps be assigned to a slightly later period, given that their pre-syncopated forms are chiefly attested in Pre-Classical Latin (see more details in 3.3.2 on accent distribution of this period): e.g., *balineum* 'bath' [Pl.+] (< **balaneom* ← Gk. βαλανεῖον) > *balneum* [Caecil.+] (cf. *balneātor* 'bath-attendant' [Pl.+]);[7] *fericulus* 'tray for food', *praefericulum* [P. F. 293.11L] vs. *ferculum* [Cic.+] (cf.

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[5] For the reconstruction of the consonant stem **pot-* (not the *i*-stem **poti-*), see Heidermanns (2002: 190).

[6] If *naufragus* originated from **nāvifragos* via syncope of *-i-* in the antepenult, initial stress must be required to explain this process. If this was the case, the creation of this compound could be regarded as old. But the recent attestation of this form [Cic., Catul.+] is remarkable, so that one can consider *nau-* as deduced from *nauta* 'sailor' or other forms. It is less likely that *nāvi-fragus* [Verg.+] is an archaic form, since its choriambic shape discloses the purpose for the use in dactylic hexameter. See below for more details on this matter.

[7] Pedersen (1895-96: 67) insists that vowel reduction in this form (*-i-* < **-a-*) can be explained by the original Greek accentuation of the word, implying pre-tonic vowel reduction, which itself is not entirely untenable. But it is unclear how to interpret the development from *balineum* to *balneum* in his theory. It is without foundation that the change presupposes **balinĕum* or the like (cf. Lindsay 1892a: 373 and Pulgram 1975: 96 n. 46).

Gk. φαρέτρα 'quiver', Skt. *bharī-/bhari-* < **b*^{h}*erH-*; see Vine 1989: 128).

Further, the standard principles of the Penultimate Law may also account for syncope in a later period, namely, Classical Latin, as Leumann (1977: 95) points out: e.g., **kalēfákiō* 'make warm' > *calĕfáciō* [Pl.+] (but cf. *cálefacis, -it* or *calefácis, -it*?) > *calficiō* [Cic.]; *válidē* 'very' [Pl.+] > *valdē* [Pl. +] (in enclisis) vs. *válidus* 'strong'; *(ob)iūrigō* 'quarrel' [Pl.] > *(ob)iūrgō* [Pl.+] (syncope after rhotacism and vowel reduction at the stage **i̯ou̯s-agō*); *calidus* 'hot' [Pl., Cato, Enn.+] > *caldus* [Lucil.+] (preferred by Augustus, according to one reading of Quint. 1.6.19; syncope after *-ld-* > *-ll-*, cf. Gk. Πολυδεύκης → Lat. *Pollūx*); *āridus* 'warm' [Pl.+] > *ardus* [Pl., Lucil.] (see Exon 1906: 131); *discipulīna* 'instruction' [Pl.+] > *disciplīna* [Pl.+]; *aevitās* 'age' [Lex XII] > *aetās* [Pl.+], *aeviternus* 'permanent' [Pac.+] > *aeternus* [Pl.+].[8]

This tentative chronological division, Pre-Literary, Pre-Classical, and Classical Latin, will become meaningful (3.3) in the description of the way vowel deletion was realized, in different modes from peried to period.

The examples above all reflect syncope. Latin also exhibits a certain amount of cases of "apocope" throughout its extensive history. As mentioned in 1.2., it is currently believed by some scholars that Proto-Italic may have preserved at least some relics of the PIE accentual system. Apocope, particularly of unaccented **-i* in verbal endings (1./2./3.sg. **-mi*, **-si*, **-ti*, 3.pl. **-nti* > *-m*, *-s*, *-t*, *-nt*), seems to have occurred in such a

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The retention of *-i-* up to Plautine Latin is partly due to a phonotactic reason that the resulting *ln*-cluster was rare in Latin, as Ernout and Meillet (1985: 65) indicate. In the Romance languages, *-ln-* ended up not becoming geminate, but losing *-l-*. The non-application of vowel reduction in *-eum* (i.e., not ***-ium*) is ascribed to hiatus as a blocking factor (see 2.4.3).

[8] In Late Latin, forms such as *oclus* (< *oculus* 'eye') and *anglus* (< *angulus* 'angle') are reproached in the Appendix Probi.

stage, judging from the fact that *-*i* retained as -*e* in absolute word-final position in Italic was accented under the PIE rules (e.g., *pede* 'foot' ≈ Gk. ποδί, *nocte* 'night' ≈ Gk. νυκτί, *ante* 'in front of' ≈ Gk. ἀντί). The only exception is a verb *tremonti* in the *Carmen Saliare*, the linguistic reality of which is itself disputed due to textual difficulties (see Sihler 1995: 467 n. 1; cf. *i*-less forms **IOVESAT**, **MITAT** in the *Duenos* inscription).[9] Deletion of final vowels in adverbs, capable of being prepositions and prefixes, could have taken place when they were associated with the following nouns or verbs, that is, practically in word-internal position. They are thus not pure cases of apocope. Perhaps the change occurred at the Proto-Italic stage, though parallel innovation in the separate languages at a very early stage cannot be excluded (cf. CLat. *post* 'behind; after(-wards)' < OLat. *poste* < **posti*): e.g., **amb*^{h}*i*- (cf. Gk. ἀμφί 'around') > *am-putō* 'cut off', *an-culus* 'man-servant' [P. F.], Osc. **amvíanud** '?', Umb. *anferener* 'circumferendi, carry round'; **apo*- (cf. *ab* 'from') >, *ab-dicō* 'resign', Osc. **aflukad** '?' (Gk. ἀπό, Ved. *ápa* 'away'); **s-upo*- (cf. *sub* 'under', Gk. ὑπό 'id.', Ved. *úpa* 'near') > *subtrahō* 'draw from under', Umb. **sutentu** '?'. "Apocope" at the phrasal or clitic level is also conceivable in conjunctions and demonstrative pronouns, but is more recent given the frequent presence of doublets of old and new forms and/or findings of archaic forms in inscriptions: *ut* 'just as' < **uta* (cf. *aliuta* 'in another way' [P. F.]); *et* 'and' < **eti* (Gk. ἔτι); *nec* < *neque* 'and... not'; *ac* < *atque* 'and... too'; *seu* 'or if' < **sēve* (~ *sīve*) < **seiue* (= **SEIVE** in CIL I^2 401); *aut* 'or' < **auti* (cf. Osc. *auti*, Umb. **ute**); *hāc* 'this' < **hākke* (= **HACE** in CIL I^2

[9] The group of Salian priests may trace its origin back to the reign of King Numa Pompilius (715-673 B.C.E., according to Livy); if this is somewhat reliable, it is possible that the *Carmen Saliare* was composed at that time and that *tremonti* itself was already archaizing.

582) < **hād-ke*, etc.

Still, there are some genuine cases of apocope left, which seem relatively old, albeit hard to date exactly: e.g., **moksu* (Ved. *makṣū́*) > *mox* 'in the (near) future' (see Meiser 1998: 74); **Bakkānāli* > *Bacchānal* 'site where the rites of Bacchus were celebrated' [Pl.+] (cf. **BACANAL** in CIL I^2 581), **u̯ektīgāli* > *vectīgal* 'revenue derived from public property' [Cato+]. We will briefly discuss these cases in 3.5.

3.2. Sonority of Vowels and Metrical Constraints

Vowel deletion may target any kind of vowel, as far as they are short (see Ciardi-Dupré 1901: 189, Rix 1966: 157), remarkable infrequency of *u*-syncope notwithstanding: cf. *Pollūx* (< Lat. **Polud-* ← Gk. Πολυδεύκης), *mox* (< **moksu*).[10] In 2.2.1, we reviewed Rix's theory (1966: 161-162) on vowel reduction and syncope. According to him, syncope is not a process subsequent to "vowel weakening". Rather, these two phonological changes are recognized as two distinct processes, though having the same primary step in common. They start by evolving into [ə] (or [ɪ] / "open transition"). Its phonetic status either causes phonological re-configuration or is totally terminated by deletion. As Zuraw (2002: 387) suggests in her work about vowel reduction in Palauan (Western Malayo-Polynesian), it "could be analyzed as an extreme form of prominence reduction". In fact, some of the Italian dialects give evidence for this view (see Maiden

[10] The numeric adverb *quater* 'four times' [Pl.+] perhaps developed regularly from **quatrus* via *u*-syncope (cf. *quadru-*; Av. *čaθruš*). But the possible influence from *ter* 'three times' cannot be excluded; see Coleman (1992: 417) and Meiser (1998: 176).

1997: 9). In southern Italy and some other neighboring areas, unstressed vowels, except pre-tonic [a], merge as [ə], which is itself frequently deleted in word-final position: e.g, Canosa di Puglia [aˈvaɻnə] 'oats' < *[aˈvena]; [kaˈpiddə] 'hair' < *[kaˈpellu]; [mətəˈtaur] 'reaper' < *[metiˈtore]. Emilia-Romagna also shows deletion in a rather extreme way: e.g., Bolognese [ˈdmaŋga] 'Sunday' < *[doˈmenika], [zbdɛl] 'hospital' < *[ospiˈtale]. Of the ancient languages, Etruscan clearly reflects a diachronic picture of this type of vowel deletion: e.g., Gk. Ἀχιλλεύς → Etr. *Αχale ~ Αχile ~ Αχele ~ Αχule* vs. *Αχle*.[11] Speech rate may be one of the factors. It is fairly changeable in our daily speech acts according to contexts. Fast rate may play a special role in contributing to vowel reduction and further deletion. Solé (1995: 37) remarks that "the faster the speech rate and the more casual the style, the more likely such processes [assimilation, deletion, weakening] are to occur".[12]

Combined with my theory diagrammed in 2.2.3 (Fig. 2.1), this line of reasoning allows for /e, a, o/ to pass through [ə], and on to deletion under certain conditions. On the other hand, I have maintained that /i/ and /u/ were so nearly intact during Process I as not to be neutralized as [ə]. However, these phonemes (esp., /i/) very frequently underwent vowel deletion. A possible solution to this problem is that toward Process II the distinction of [i], [u], and [ə] in unstressed position became ambiguous and then so

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[11] It is generally agreed that Etruscan had stress accent in initial syllables. See, e.g., Leumann (1977: 247), Cristofani (1991: 42), and Rix (2004a: 949).

[12] See also Tuttle (1997: 30) in regard to Italian plural marker -[i] that changes into yod in rapid speech rate.

From a different angle, Monteil (1979: 99) suggests that vowel *reduction* in speakers' articulation facilitates eventual vowel deletion. Vowels vaguely pronounced are acoustically misperceived by hearers, then hearers, taking a turn to speak, do not produce unheard sounds. Such a phonologization process through listeners' reinterpretation of unintended phonetic implementation has been extensively studied (see, e.g., Ohala 1993, Solé 1995: 24). This factor may also be applicable to vowel reduction to a certain extent.

neutralized as to qualify for vowel deletion like [ə]. But this cannot explain deletion of high vowels in Pre-Literary Latin, when Process II was not entirely completed. Alternatively, deletion may have applied to certain vowels based on the vowel sonority (prominence) hierarchy, which is schematized by Crosswhite (2001: 39) as in the first line of the following figure;[13] in the case of Latin, the distinction between low mid [ɜ, ɔ] and high mid [e, o] does not apply ("$_{prom}>$" = "is more prominent than"):

$$\begin{array}{c} a_{prom} > ɜ,ɔ_{prom} > e,o_{prom} > i,u_{prom} > ə \\ \text{no syncope} \leftarrow \mid \rightarrow \text{syncope} \end{array}$$

Fig. 3.1: Vowel prominence and syncope

The vertical line I tentatively added below demarcates the range of application of vowel deletion in Latin. If one may highlight the infrequency of syncope of /u/ and distinguish this from /i/, the enhancement in sonority caused by the rounding gesture can serve as a factor, as Zuraw (2002: 390) suggests. (I will come back to this point in 4.2 for the discussion of Sabellic materials.) This issue requires further inquiry based on theoretical and/or typological studies.

We should extend our attention to another factor of vowel deletion. In order to account for vowel deletion in different Arabic dialects, Zawaydeh (1997: 202) appeals to the "*Light Light constraint" in Optimality Theory (OT), which disfavors two successive

[13] A similar hiararchy of vocalic (and concomitantly, consonantal) sonority is found in Selkirk (1984: 112). See also Plaster (2007: 5) for the applicability of this notion to vowel reduction in Tocharian.

light syllables.[14] It is likely that a similar type of constraint based on the metrical system is also operative in Latin. In fact, many scholars have already conducted research along this line, considering syllable weight in positions adjacent to syllables targeted by vowel deletion.

Accordingly, in contrast to vowel reduction, it is very plausible that these two factors, namely, radical reduction in sonority and metrical constraints, are both involved in vowel deletion. This state of affairs would reflect the complexity involved in describing vowel deletion. In the following sections, I will attempt to amplify more crucial parts, especially by elaborating on metrical conditions of vowel deleition in medial syllables, whereby an unprecedented synchronic and diachronic picture of vowel deletion will emerge.

3.3. Vowel Deletion in Medial Syllables and Metrical Factors

Before presenting new ideas (3.3.3-3.3.6), I certainly need to review some previous work, including a good sample of explaining syncope based on metrical structure (3.3.1) and a historical transition of accent distribution rules, which has to do with metrics (3.3.2).

[14] Davis and Zawaydeh (1997: 56) refer to practically the same constraint as "*OpenOpen," which "disallows adjacent open syllables." Gordon (2001: 206, 216) demonstrates how to explain syncope in a derivational rule-based framework, before proceeding to his own analysis based on the OT model, by using an Arabic example, /ʔin(kásа)rat/ → ʔin(k´sa)rat → [ʔin(ksá)rat] 'she got broken', which is described as "[s]yncope..., stranding the stress which subsequently docks on the remaining syllable of the original foot" (2001: 206). A similar explanation is already in Hayes (1995: 41-43). Though a linguistic stage such as ʔin(k´sa)rat, with accent on an almost (or even totally) zero-sonority lacuna, seems theoretically possible in synchronic analysis, it has little reality as a historical entity in diachronic terms. One pronunciation of English *thank you* as ['ɳkju] (see Jones 1964: 245, cited by Allen 1973: 77-78) or /ˊkju/ (ˊ = low rise tone: see Solé 1995: 22) might be referred to in this connection, but it does not seem that such pronunciation is entirely responsible for vowel deletion in accented syllables. Gordon's own explanation in the OT model includes no such intermediate stage as ʔin(k´sa)rat, though his constraint-based model seems rather complicated, as he (2001: 215) admits.

Though I will add almost nothing new to these, they will be incorporated into the larger picture of vowel deletion I have promised to paint.

3.3.1. Syncope under the Standard Penultimate Law: Remedy for Degenerate Syllables

It has traditionally been alleged that the weight of the syllables preceding and/or following a deleted vowel must have been responsible for the occurrence of syncope. This conventional view, if properly formulated,[15] receives support from a theoretical analysis in Mester (1994: 37-43).[16] With the premise that Latin is metrically based on bimoraic trochees (∪∪ = LL or — = H), he explains post-tonic syncope in Literary Latin (Pre-Classical and Classical) as deletion of "trapping." For instance, *lāridī* 'bacon' (gen.sg.) is parsed as (lā́)ri‹dī›; the heavy syllable *lā-* can stand as an independent foot because it is bimoraic, and then stressed. Final syllables in Latin basically do not participate in accent distribution, and are thus extrametrical (indicated by ‹ ›). In this state of affairs, *-ri-* in the medial syllable is stranded ("trapped"), and the vowel is thus eliminated by syncope in order to reach a full-parsed (lar)‹dī›. Note, however, that if the final syllable is light, it may be associated with the preceding light syllable to secondarily

[15] The metrical weight of following syllable has already been invoked by von Planta (1892: 214). Based on the Umbrian imperative **-etōd* (with syncope: e.g., *deitu* 'say') vs. ppl. **-eto-* (without syncope: e.g., *muieto* 'make a noise'), he claims that whether following syllable was heavy or light should be factored into conditions of syncope, since **-eto-* could have **-e-* in the syllable closed by syncope of the following one (i.e., **-eto-s* > *-ets*), which was generally retained by paradigm leveling. Hence, also in Latin, *caldō* (< *calidō*) and *calidus* (with the vowel retained, in case *-dus* can be scanned short) 'hot', for instance, might have affected each other to yield hybrids *calidō* and *caldus*, etc. The applicability of the Sabellic facts to Latin, however, does not seem to be useful, because Latin demonstrates no exact parallel with the Sabellic final syncope.

[16] The principles of Classical Latin prosody are concisely summerized by Parsons (1999: 120).

form a trochee, that is, (∪∪)#. For this purpose, iambic and cretic shortenings serve to change pu‹tat› 'believe (pres.3.sg.)' and (dī)ci‹tō› 'say (fut.impv.2./3.sg.)' into (putăt) and (dī)(citŏ), respectively, to realize full-word parsing. It follows that syncope is triggered when the final syllable is consistently heavy and so not united with the preceding light syllable.

Mester's other examples include **perregō* > *pergō* 'continue', *porrigō* 'stretch out' > *porgō*, **subregō* > *surgō* 'rise' (cf. *subrigō* 'raise' [Verg.+] 'rise'), *pūrigō* 'apologize; clean' > *purgō*, *iūrigō* 'quarrel' > *iurgō*, *aevitās* 'age' > *aetās*, *ūvidōs* 'wet (acc.pl.)' > *ūdōs*. The list of examples can be increased by forms such as *vernō* 'in spring' < **u̯ērinōd* (cf. *vēr* 'spring'), and *frīgdārium* 'cold store' [Lucil.] < **frīgi-* (cf. *frīgidārius* [Vitr.+]).[17] These examples might have contained both syncopated and non-syncopated forms in the paradigm, due to syllable weight of final syllables: e.g, 1.sg. *(per)regō vs. 3.sg. *(per)(regĕt) → *pergō* vs. **perregit*. Paradigm leveling most likely excluded such variation. Analogical processes could also operate among different paradigms of the same morphological category, as Maniet (1957: 151) lays out: e.g, (cáli)dō 'hot (dat./abl.sg.)' without syncope vs. (ā́)ridō 'dry (dat./abl.sg.)' > (ar)dō with syncope, then affecting each other to yield hybrids *caldō* and *āridō*, respectively.

Note that consonants adjacent to targeted vowels, particularly foregoing sonorants and semivowels, may facilitate the occurrence of syncope (see also Ciardi-Dupré 1901:

[17] To judge from *frīgdārium*, syncope is permitted in pre-tonic as well as post-tonic position, as Ballester (1990: 41) also points out. Such a bilateral operation of the change may be paralleled by Vulgar Latin: *bon(i)tătem* 'kindness', *cap(i)tālis* 'death penalty', etc. vs. *'om'nes* 'men' (> Provençal *omne*; cf. *(h)ómināes* > Ital. *uomini*), *dóm(i)nus* 'master', etc. Grandgent (1962: 99-102), who adduces these examples, marks secondary accent on pre-tonic examples. It is thus possible to consider the cooperative function of primary and secondary accents. See also Hittite iteratives in **-i̯e/o-sḱé/ó-* > *-iške/a-* (see Melchert: 1994: 85, 173).

190, Juret 1919: 95, Maniet 1957: 150, Anderson 1965):[18] e.g., *lāridī* > *lardī*, **perregō* > *pergō*, *aevitās* > *aetās*, *ūvidōs* > *ūdōs*. Part of the reason for this fact is that if the process involves these segments, resulting consonant clusters would be phonotactically legal in most cases or adjustable by further phonological changes to be legitimately represented (but cf. *frīgdārium*). Restrictions on resulting consonant clusters may well be designed to avoid cluster reduction that results in morphological opacity: e.g., *algidus* 'cold' > not ***aldus* (see Ciardi-Dupré 1901: 205).[19]

3.3.2. The Early Version of the Penultimate Law

There are, however, quite a number of cases of syncope that are not conditioned by the metrical weight of the preceding syllable in Pre-Classical Latin and earlier: e.g., **deksiteros* > *dexter*, *balineum* > *balneum*, and **posinō* > **posnō* > *pōnō*. Based on Mester's metrical framework, none of these is well motivated: *(dek)(síte)‹ros› > (dex)‹ter›, deleting a stressed vowel in the full-parsed word; ba(líne)‹um› > (bal)ne‹um›, deleting a stressed vowel in the parsed foot and leaving a degenerate syllable; and *(pósi)‹nō› > *(pos)‹nō› > *(pō)‹nō›, with entirely unnecessary syncope in a fully parsed word.

Note that Mester's insightful theory is grounded on the principles of the *standard*

[18] Cf. Deecke (1878: 176-178) for locations of syncope in Etruscan: between two *sonorants*; between a *sonorant* and a stop or *-s-*; and between *-s-* and a stop (but cf. *hectamei* < **-cat-* ← Gk. Ἑκάτομνος).

[19] Cf. restrictions on syncope in Tonkawa due to resulting consonant clusters (see Kisseberth 1970: 122-128). This language does not tolarate triconsonantal clusters. In addition, glottalized consonants cannot form clusters with other consonants, except for sonorants. Geminates are also banned. See also a brief comment on other languages in Myers (1991: 199).

Penultimate Law. It should be remembered that in Pre-Classical Latin, we have evidence for accentuation patterns that deviate from the norm. As mentioned in footnote 20 in Chapter 1, among arguments based on the theory that the podic ictus in iambo-trochaic verse frequently coincides with word accent, the one for ⏑⏑⏑⏓# regarding forms like *facilius*, *sequiminī*, *ceciderō*, and *mulierem*, seems to be relatively reliable (i.e., *fácilius*, etc.). Vowel reduction in the initial syllable of *igitur* 'then' < **agetor* also indirectly proves this accentuation, because it is usually explained as resulting from its enclitic use, as in *quid igitur?* 'what then?' (e.g., Ter. *Eun.* 854), treated as a single prosodic word (i.e., **k*^{w}*ídăgĕtor*), as opposed to the secondary use as an independent sentence-initial conjunction already in Pre-Classical Latin.[20] Of the instances of vowel deletion cited above, *balineum* actually exhibits such a syllable configuration. It is thus unfair to place these exceptional cases under Mester's framework. We should first explore what metrical system, if any, underlies such forms. With this established, we might be able to include them in the same or a similar metrical framework. It is also beneficial to consider this matter in the discussion in 3.3.6, where we will deal with a number of originally quadrisyllabic words.

Brugmann (1897: 973-974) interprets the accent distribution ⏑⏑⏑⏓# as a relic of the initial-stress period. Leumann (1977: 251), however, is skeptical about this view, observing that the same accent assignment is possible in Classical Sanskrit without any

[20] Though Jacobs (2000: 336, 339-340, 345, 350) asserts that a quaternary stress system never exists in Latin, his argument (and his ranking of constraints in 2000: 334-335 based on an OT model) would be undermined by this evidence. I note in passing that in his analysis on vowel deletion (2000: 345-350), Jacobs entirely ignores examples, such as *dīce* > *dic* 'say!' and *face* > *fac* 'do!', which also stand to undermine his argumentation for 'uneven trochee' (HL). In fact, Blumenfeld (2007: 7) also disagrees with such a podic unit in explaining the (dis)preference in foot structure of encliticized words.

preceding linguistic phase with initial accent. Another cautious remark is found in Allen (1978: 83). Further, Oniga (1990: 224-225), who however does not trust the *fácilius*-type accent to begin with, questions how such an accent, if a relic of the initial-stress system, survived into the Plautine period, when the quantitative accentual rule was already in order. Presumably, the *fácilius*-type accent would rather be defined as a part of an early version of the Penultimate Law in Pre-Classical Latin, if we have reliable principles to guide stress to the initial syllable of the ⏑⏑⏑⏑̲# -patterned word.

One of the normal functions of the Penultimate Law is that if the penult is syllabically light, accent is moved toward the antepenult. For the purpose of incorporating the *fácilius*-type accent, Exon (1906: 142) posits a different rule in the Republican period, whereby accent can fall on the fourth mora (not *syllable*) from the end of the word if its antepenult is also syllabically light.[21] According to him, words such as *amīcitia* 'friendship', with a long vowel in pre-antepenult, covering both fifth and fourth morae, are not qualified for this rule, and thus bear accent on the third mora (i.e., neither ***amĭ́citia* nor ***amiícitia*, but *amīcítia*). True, Exon's formulation might seem well designed for explaining syncope in the following examples: *bắlĭnĕum* > *balneum*, *cŏ́lŭmĭne* > *colmine* (cf. *columen* 'summit'), *tĕ́gĭmĭne* > *tegmine* (cf. *tegimen* 'cover'), and a disputed case *puĕ́rĭtĭa* > *puertia* 'boyhood' (see footnote 25).[22] However, this view

[21] Nyman (1983: 34, 42) also recognizes this type of accent placement as "four-syllable rule" (e.g., *benevolus* 'benevolent'), along with the special rule for compounds (e.g., *sōlstitium* 'solstice' and *mūscipula* 'mousetrap'; see 1.3.3).

[22] In Exon's theory (1906: 128), syncope must have occurred in words with four syllables or morae and accent on a heavy syllable somewhere in the word: **ārid-ŏ́r-is* (or pre-rhotacized **ārid-ŏ́s-is*) > *ardŏ́ris* (gen.sg. of *ardor* 'burning, fire'). According to him, the nom.sg. *ardor* < **ārid-ōs* is a result of analogy. His argument is not compelling for various reasons. For one, the syncope in the nom.sg. is paralleled by *vernus*

does not seem entirely grounded. As far as we learn from the position of podic ictus, the common accent distribution for five-syllable words of ∪∪∪∪⏑# like *pueritia* in iambo-trochaic verse is not something like ***malĕfĭcĭum* but *mălĕfĭcĭum* 'misdeed' (see Allen 1969: 200), which already coincides with the standard norm of the Penultimate Law. Further, the procedure for skipping the light antepenult (as well as the light penult) in *puĕrĭtĭa* would need a supplementary rule not to skip the light pre-antepenult, since the accent movement in Exon's program seems to be so designed as to be halted if it meets a heavy syllable in the word (i.e., the so-called "Weight-to-Stress Principle"); if so, the light pre-antepenult must have been skipped (thus, **púeritia*).

There thus seems to be a limit to any attempt to formulate the *fácilius*-type accent by starting metrical scansion from the end of the word, that is, the right edge of the prosodic word. Actually, a new approach has already been attempted by a couple of scholars to replace the old type of conception. Outcomes of their research are very instructive for understanding how metrical structure was formed in prosodic words in the early period of Literary Latin.

First of all, we should recollect Hayes's brief description (1995: 180), inspired by Allen (1973: 188-190), about early stages of Latin as governed by "[a] stress pattern much the same as that discussed... for Egyptian Radio Arabic (i.e. moraic trochees from

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'vernal' < **u̯ērinos*, whose type of inflection never allows this lexeme to have more than three syllables before syncope took place. In addition, syncope in *ardor* and *vernus* can be perfectly squared with Mester's metrical framework (i.e., *(ā)ri‹dōs› > *ardor*). Further, his insistence that a heavy syllable is required cannot escape some counterexamples like *balineum*. In order to solve this problem, Exon (1906: 140) suggests that accented short vowels in the above forms became long in Imperial Period before the application of syncope: e.g., *bălĭnĕum* > **bălĭnĕum* > *balneum*. This "imperceptible" change of vocalic length (Exon 1906: 140), however, is not entirely justifiable; if his idea was true, *amīcítia* 'friendship', for instance, would have changed into ***amīcítia*, but this is by no means acceptable.

left to right, with optional foot extrametricality)." This might appear eccentric to people who are used to the traditional accentual rule in Classical Latin. But more recently, this idea has been more explicitely represented by Parsons (1999: 122-123), inspired by A. Garrett. Based on left-to-right footing (L → R), the forms like *facilius* would be parsed not as fa(cili)‹us› but (faci)li‹us›, with moraic trochee (∪∪ = LL or — = H) as a metrical podic unit. The only complete foot in the form is hence assigned a trochaic stress as in (fáci)li‹us›, which is the exact accentuation in Pre-Classical Latin. The examination of the *maleficium*-type accent further clarifies the profile of the metrical framework in that period. This word would be footed as (male)(fici)‹um› with two moraic trochees. In order to obtain the correct accent assignment, the rightmost foot is to be counted as the head of the prosodic word.[23] (Actually, the result (male)(fíci)‹um› can also be produced by the standard Penultimate Law, albeit with a difference in footing direction.) The combination of these two factors, left-to-right parsing and rightmost foot priority, would be corroborated by another case of accentuation in Pre-Classical Latin. The adverb *adsŭmĭlĭter* 'similarly' [Pl. *Bac.* 951] (–∪∪∪⏑), cited by Sturtevant (1919a: 241, 243) and Allen (1969: 200), would be footed not as (ad)si(mili)‹ter› (L ← R) but (ad)(simi)li‹ter› (L → R), then provided with stress on the righthand foot, as in (ad)(símí)li‹ter›. This algorithm ends up with an outcome different from the one produced by the standard Penultimate Law, that is, (ad)si(míli)‹ter›.[24] Note, however, that such an accentual

[23] Iambic shortening in forms in Pre-Classical Latin, such as *amĭcitia* 'friendship' (~ *amīcitia*) and *volŭptāte* 'pleasure' (~ *voluptāte*), also strengthens the right foot accent distribution. The underlying forms of these examples are parsed as a(mī)(citi)‹a› and vo(lup)(tā)‹te›, respectively. If leftmost feet were assigned stress as in a(mí)(citi)a and vo(lúp)(tā)te, shortening in accented vowels would hardly be conceivable. See Blumenfeld (2003: 8).

divergence between Pre-Classical and Classical Latin is only emerging in this *adsímiliter*-type and the above *fácilius*-type accent, which share the same syllable configuration ⏑⏑⏑⏑̲# with odd-numbered ⏑'s before the ultima (of course, including words of ⏑⏑⏑⏑⏑⏑̲#, if any). Thus, the pattern ⏑⏑⏑⏑⏑̲# with even-numbered ⏑'s as in *maleficium* should be differentiated from the other types of words.[25]

It is worth mentioning how the accentual system in Pre-Literary Latin, that is, the initial-stress rule, operated from the metrical standpoint. For this purpose, *maleficium* and *adsimiliter* could be useful. Their metrical structure (male)(fici)‹um› and (ad)(simi)li‹ter› are formed by means of left-to-right parsing with moraic trochees, which was more archaic than the right-to-left one and is thus chosen here. In order to lay stress on initial

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[24] An analogous case is offered by Exon (1906b: 32-33), demonstrating ictus assignment in *calamitās* 'disaster', which is not ***calámitātém* but *cálamitātem*. According to him, the latter illustrates a distribution of primary and secondary accents (i.e., *càlamitātem*). Such a state of affairs can be realized by rightward parsing like (càla)mi(tā́)‹tem›, not the opposite one, that is, **cà(lami)(tā́)‹tem›. On the other hand, two options of ictus assignment in another example of his *commoditās* 'opportuneness', that is, *commóditātém* and *cómmoditātem*, can be explained by a fully parsed structure like (com)(modi)(tā)‹tem›.

[25] This outcome tangentially solves the problem of the hapax *puertiae* [Hor. *Carm.* 1.36.8], which is mentioned above. This form is thought of as a syncopated product of the more common *pueritia* by Corssen (1870: 902) and Lindsay (1891: 406-407, 1892a: 373 n. 3), who, like Exon (1906: 140-142), presuppose the stage **puéritia* to have *-i-* in the antepenult syncopated by means of stress in the immediately preceding syllable. In general, this old view is rejected by most scholars. Leumann (1977: 98) maintains that the short form *puertia* is artificially modeled after *puerpera* 'woman who has been *or* is in process of being delivered of a child'. This sort of analogical creation, independent of syncope, for metrical convenience, seems to be right. Sturtevant (1919a: 243) also states that there is no statistical support to the accentuation ***puéritia*, in contrast to *adsímiliter*, though he is still inclined to accept such accentuation. In addition, as *latericulus* [Caes.], a later variant of *laterculus* 'block; cake' [Cato+], disproves a historical development, suggested by Exon (1906: 141-143), like ***latériculus* > *laterculus* due to the reverse chronological ordering of these forms, the chronological relationship itself between *pueritia* and *puertia* may not be as straightforward as it seems. These circumstantial counterarguments are now consolidated from the metrical standpoint. The standard form is parsed like (pue)(riti)‹a› by either the early or standard version of the Penultimate Law, as in (male)(fici)‹um›. In either system, this prosodic word is assigned stress on the rightmost foot, thus being *puerítia*. Therefore, an alleged stage like ***puéritia* would hardly be plausible in this framework. Nor did Pre-Literary Latin under the initial-stress rule allow for such accentuation. If one should want to insist on syncope to obtain *puertia*, such a change might be possible, in the initial-stress stage **púeritia*, in a syllable remote from the stressed one. However, the "non-syncopated" form inherited and reguarly used in both Pre-Classical and Classical Latin would make such a possibility less likely. The alternative of a nonce creation by Horace for his metrics is thus corroborated.

syllables, the leftmost foot must be selected as its host. This factor clearly differentiates Pre-Literary Latin from the later stages, as the parsing direction (i.e., L ← R) separates Classical Latin from the earlier stages (L → R). Such a historical development has already been illustrated by Parsons (1999: 123). Note, however, that there must have been a number of forms which began with #⏑–. Given the current theory, the initial light syllable could not be parsed;[26] the second heavy one (H) would thus host the stress. I should add that stress assignment on the initial syllable in such a pattern can be achieved by the more highly ranked constraint that every prosodic word must begin with a foot (i.e., ALIGN-WD-LEFT; see Kager 1999: 169). This excludes *re(lā́)‹tos›, instead yields *(ré)(lā)‹tos›.[27]

3.3.3. Syncope in Earlier Periods: Removal of Successive Light Syllables

Turning to vowel deletion in *balineum*, we are now in a better position to consider the problem with the knowledge obtained from metrical rules. If the early version of the Penultimate Law is applied to the form, it is parsed as follows: (báli)ne‹um› >

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[26] Thus, Parsons (1999: 134): "Incomplete feet are banned, *except in word-initial position*" (italicized by the present author).

[27] In Pre-Classical iambic-trochaic verses, the following variation of ictus position has been reported: e.g., *fīliolum* vs. *fīlíolum* 'little son' and *ímperium* vs. *impérium* 'supreme administrative power' (see Lindsay 1892b). Both examples are parsed as (H)(LL)‹ ›. The ictus on (H) might be construed as a relic of initial stress. Exon's report (1906b: 33) on *dĭvitiās* 'wealth' also accords with this conception. The statistical result in Sturtevant (1919a: 240-244) actually supports the accentuation **amĭ́citia*, though he lessens this by resorting to the occasional shortening to *amĭcitia*. But such accentuation perhaps presupposes a metrical structure *a(mĭ́)(citi)‹a› (L → R + accent on leftmost foot), which could be construed as an intermediate stage after ALIGN-WD-LEFT was outranked by FT-BIN (i.e., feet are binary under either a moraic or syllabic analysis; see Kager 1999: 156), which came to ban *(á)(mī)(citi)‹a›. I admit that the entire discussion is still open to question, requiring both further theoretical and philological corroboration.

(bál)ne‹um›. The result is still lacking in terms of the metrical interpretation. Again, deletion targets a vowel in the parsed foot and leaves a degenerate syllable. Nor can examples like *dexter* and *pōnō* be rescued by this system. Syncope in the former can only be explained by the initial-stress stage, that is, **déksiteros* (not in the stage of *(dek)(síte)‹ros›). The latter presupposes *(pósi)‹nō› in both the early and standard versions of the Penultimate Law, which still requires the deletion in the complete foot. Thus, metrical factors based on foot parsing cannot be entirely responsible for triggering every case of vowel deletion in Latin.

Given this state of affairs, we are forced to consider a different factor. Recall that we have briefly reviewed in 3.2 a constraint disfavoring two successive light syllables in Arabic. The same or a similar constraint seems to be useful for describing syncope in earlier stages of Latin as well, since these examples contain sequences of two or more light syllables (disregard the weight of final syllables): **deksĭtĕros* > *dexter*, *bălĭnĕum* > *balneum*, and **pŏsĭnō* > **posnō* > *pōnō*. The list is increased by other examples cited in 3.1, which are repeated here with short-vowel markers to help readers visually confirm successive light syllables there: **oi̯nŏ-dĕkem* > *ūndecim*; **k^{w}ink^{w}ĕ-dĕkem* > *quīndecim*; **ău̯ĭ-pŏl-o-* > **au̯-pol-o-* → *ōpiliō*; **sēmĭ-căput* > *sinciput*; **nŏu̯ŏpăros* > *nūper*; **măg-ĭsm̥mo-* > **măgĭsŏmo-* > *maxumus*; **oi̯sŭ-rŭp-ā-* > *ūsurpāre*; **rĕpĕpărī* > *repperī*; **rĕpĕpălī* > *reppulī*; **rĕtĕtălī* > *rettulī*; **rĕkĕkădī* > *reccidī*; **ŏpĭfĭkĭom* > *officium*; **ău̯ĭ-căp-es* > *aucipis*, **pri(s)mŏ-kăp-es* > *prīncipis*, **hostĭ-pŏt-es* > **hospus* → *hospes*, **formŏ-kăp-es* > **formkap-* > *forceps*; **kĕdăte* > *cette*; **pălămā-* > *palma*; *sŭpĕrā* > *suprā*, **sŭpĕrēmos* > *suprēmus*; **fĕrĭkŏlom* > *ferculum*; **ău̯ĭd-ē-* > *audēre*; Gk.

Πολυδεύκης → Lat. **Pŏlŭd-* > **PODLOVQVEI, POLOCES** > *Pollūx*, etc.[28] We could add a prefixed verb *surpiō* 'steal' (< **sub-rapiō*; cf. *surripiō*). Though its pattern of composition is similar to Mester's examples such as **perregō* > *pergō*, *porrigō* > *porgō*, and **subregō* > *surgō* 'rise', it differs in inflectional type. The attested forms in Plautus are all 3.sg.pf. *surpuit* (*Capt.* 8 [in prologus], 760, 1011), whose syncope can be construed as originating from **subrĭpŭit* or the like. Note that many of the potential candidates for deletion are retained by analogical factors (e.g., *mălĕfĭcĭum* ← *male*; 1./2.pl. *sŭbĭgĭmus*, *-ĭtis* ← 1./2./3.sg. *subigō*, *-is*, *-it*, 3.pl. *-unt*; *călĭdus* ← *ārĭdus*; *sŭpĕrus* ← *īnfĕrus*, but cf. *suprā* → *īnfrā*) and/or phonotactic restrictions on resulting consonant clustsers (e.g., ***subgimus*, *-itis*).[29]

Syncope driven by this constraint did not target forms adduced by Mester, such as *lāridī* > *lardī* and **perregō* > *pergō*, since they did not contain a sequence of light syllables. This is the reason that some of the words with syllable patterns like –∪–# could survive in Literary Latin (e.g., *lāridī*, *iūrigō*, *āridō*). The constraint that eliminated –∪–# by means of syncope probably emerged around the end of Pre-Literary Latin or the outset of Literary Latin. These two different constraints may have coexisted with each other for a while in Pre-Classical Latin, maybe depending on styles, dialects, and/or sociolinguistic

[28] There are a couple of possible examples of metrical syncope (with the vowel graphically retained) in a sequence of short vowels: e.g. *cap(i)tibus* 'heads' (Enn. *Ann.* 511Sk) and **FAC(I)LIA** 'easy' (CIL I^2 632.4); see Ernout (1966: 74), Vine (1993: 213 n. 74), and Courtney (1995: 213). But cf. Skutsch's (1985: 664) negative opinion about syncope (and resolution as well).

[29] The combination of these two factors is already found in Ciardi-Dupré (1901: 189 n. 1, 197-199). His view, in short, is that some morphological categories, suffixed by *-co-*, *-tās*, and so forth, were resistant to syncope (thus, *-i-co-*, *-i-tās*, etc.) due to analogical influence from unsyncopated forms (i.e., °C-*i-co-*, etc.; C = [–liquid]) within each category. According to him, though *horitor* [Enn. *Ann.* 346] (> *hortor* 'encourage') morpho-semantically belonged with frequentative verbs, the lack or obsolescence of the base verb **horior* bleached *horitor* of its original marked frequentative meaning (cf. *quaerō* 'search for' vs. *quaeritō*), with syncope of the medial vowel.

factors; thus, for instance, *balineum* > *balneum* by the older constraint and *iūrigō* 'quarrel' > *iurgō* by the younger one. Then it may be that the former constraint was replaced by the latter one:

Pre-Literary Latin		Literary Latin		
		(Pre-Classical Latin)		(Classical Latin)
*Light Light	→	*Light Light, *–⏑–#	→	*–⏑–#

Fig.3.2: Chronology of constraints regarding syncope

This development was perhaps associated with a shift in the nature of accent in Literary Latin (i.e., diminished dominance of stress), but this would require a further typological inquiry.

In light of this tentative chronological differentiation of the dominance of constraints, formulaic descriptions presented by some scholars are understood as mixed ones that lack both synchronic and diachronic sense. For instance, Osthoff (1887: 464) claims that for syncope in the second syllable, the preceding initial syllable can be either heavy (cf. *lār(i)dum*, *ār(i)dus*, *aetās* [< **aiu̯o-*], *ōrnāre* [~ *ōrdināre*]) or light (cf. *valdē* ~ *validus* / *validē*, *audēre* ~ *avidus*). In my theory, *ăvĭdus* was analogically influenced by *āridus* and the relationship between verb *audēre* (< **ăuĭd-ē-*) and the adjective *avidus* became too opaque to incur an analogical reshaping. Much the same problem can be indentified in patterns that Sommer (1900: 4) suggests, such as ⏑́–, ⏑́⏑⏑, ⏑́–⏑, and so forth, believing that a deleted vowel must be followed by more than one mora, and on the other hand, disregarding the weight of the initial syllable. Kent's claim (1932: 96) that ⏑⏑#

was preserved should also be restated.

3.3.4. Syncope in "Closed" Syllables Disfavoring Successive Light Syllables

Vowel deletion generally takes place in open syllables (see Juret 1919: 95 and Rix 1966: 156). However, there is a potential counterexample to this regulation. Syncope in closed syllables with *s*-coda has been well-known as an exceptional case (see, e.g., Skutsch 1913: 196, Juret 1919: 97, Rix 1966: 156),[30] which is actually somewhat paralleled by the further exceptional case in final syllables (e.g., ***agros* > *ager*** 'land', ****mentis* > *mēns*** 'mind', etc.). Examples include the following: [[with -C-]] ****minuscellus* > *mīscellus*** 'hybrid' [Cato+]; ****sēmis-qui-* > *sēsqui-*** 'one and the half' (e.g., ***sēsqui-pēs*** 'distance of 1½ ***pedes***' [Pl., Cato+]); ****sēmis-tertius* > *sēstertius*** (coin or unit of money) [Var., Cic.+]; cf. ***onustus*** 'heavily laden', ***scelestus*** 'calamitous'; [[with -CR-]] ****monestrum* > *mōnstrum*** 'portent' [Pl., Lucil.+];[31] ***fenestra*** 'window' [Pl., Cato+] > ***fēstra-*** [P. F. 80.27L], ***fē(n)stra-*** [read in some manuscripts of Pl.];[32] cf. ***sinistra*** 'left' (not *****sīnstera*) < ****sinisterā̆***.[33] In Meiser's description (1998: 66), -C- is identified with *-t-*, but judging

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[30] Cf. the parallel case in Umb. **etuřstamu** < *****-*tudes-t-ā-*; see Benediktsson (1960: 190).

[31] The nasal segment before *-st-* is presumably restored (at least graphically) based on the preceding nasalized vowel (cf. *Mōstellāria* 'a play about ghosts' [title of one of Plautus' comedies]). See Meiser (1998: 117).

[32] The last example might be explained à la Etruscan (see Rix 1966: 157), but the way that the vowel syncopated is parallel to the other examples. There is thus no need to appeal to external motivation.

[33] Another potential case of such syncope is ambiguous, involving an intervocalic $^{(*)}$-*u̯*- (e.g, **IOVESTOD** 'by right' > *iūsto-* 'rightful' [Pl.+]; but cf. Monteil 1979: 79, appealing to analogy with *iūs* 'law'). Rix (1966: 157), adducing *rūrsus* 'backwards; again' (< ***rou̯o-* < *reu̯orsos*) and *nūndinae* 'market-day' (< ***nou̯endinai̯* < *neu̯e-*), demonstrates two possible developments, both of which evolve

from the above examples, it is not always so (cf. Meiser 1998: 177). Monteil (1979: 79) is skeptical of a regular phonological process in these cases, and so he posits some special circumstances. In regard to *sēstertius*, for instance, he suggests that its frequent use as a term of currency by non-native speakers perhaps helped the violation of the regular phonological principle. Though such a view is theoretically possible, considerations on the basis of phonetic matters, if any, should not be abandoned. Actually, we should ground syncope of this kind to the greatest extent possible by considering the underlying phonetic/phonological matters. My supposition is that *-s-* may be associated with the following consonant to form an onset, thus rendering the preceding syllable open.

The status of *-s-* as the first segment in a consonant cluster is actually not so clear-cut that we may clearly assign it to the syllable coda or the onset of the following syllable. In order to exemplify this unstable situation, a phonetically contrastive pair in English, *discolour* vs. *discover*, adduced by Solé (1995: 24-25), is useful. The composite *discolour* is analyzed as derived from two independent morphemes /dɪs-/ and /kʌlər/, realized as [dɪsˈkʰʌlər], following general phonological processes, such as the aspiration of initial stops of stressed syllables. But *discover*, where speakers are not alert to two separate morphemes and the meanings of each, deviates from general phonological treatments, being resyllabified as [dɪ-ˈskʌvər], with an unaspirated stop. The different procedures can

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with assimilation and subsequent loss of *-u̯-*: *-ou̯i/e-* > *-ou̯o-* > *-oo-* > *-ō-* or *-ou̯i/e/o-* > *-ou̯u-* > *-ou̯-* > *-ū-*. As argued in 2.2.3, *i*-vocalism was rather stable in unstressed medial syllables. It is thus not immediately clear that this vowel can be incorporated into these patterns. On the other hand, the assimilation of *-e-* and *-o-* after *-u̯-* seems likely. Possible transmission of [+round] after [w] has already been exemplified in footnote 28 in Chapter 2 (e.g., *concutiō* and *secundus*). The adjective *faustus* 'fortunate', albeit with *-a-* before *-u̯-* in the proto-form (**fau̯os-* [cf. *onustus*] or **fau̯es-* [cf. *honestus*]; see Leumann 1977: 334), could be explained in a similar way. Even if these forms should exemplify syncope, the addition of them to the discussion does not affect my following argument.

come into being due to the ambiguous feature of *-s-* on the syllabic/morpheme boundary. This is parallel to other clusters that contain sonorants (e.g., *su.bliminal* vs. *sub.lingual*).

Given this opacity, though Cser (1999: 179) and Marotta (1999: 300) claim that the *s* + C cluster in Latin should be heterosyllabic (appealing to, e.g., metrical scansion), the sequence -*s*C(R)- in Latin can be shown to be ambiguous in terms of syllabification. It may be either heterosyllabic as in *in.ces.tus* 'unclean' (instead of ***in.ci.stus*, derived from *castus* 'untouched by'; see Rix 1966: 157 and Marotta 1999: 300-301) or tautosyllabic in some other cases, such as **mi.nu.scel.lus* > *mīscellus* via syncope.[34] In fact, the tautosyllabicity of -*s*C(R)-, usually considered secondary or abnormal, can be identified in reduplicants of perfect forms, as in *stetī* (pres. *stō* 'stand'), *stitī* ~ *stetī* (pres. *sistō* 'set up'), and *spopondī* (pres. *spondeō* 'give a pledge'), which may provide support for -*s*C- as an close-knit onset unit (not ***sestī*, ***sistī* ~ ***sestī*, and ***sospondī*). In regard to this issue, even Marotta (1999: 301) admits that "we might consider the initial clusters of the formant *s* + C as tautosyllabic, at least at some level of representation" (but cf. Rix 1966: 157).

Taken altogether, syncope in forms such as **minuscellus* > *mīscellus* and **sēmisqui-* > *sēsqui-* turns out not to be really exceptional to the rule of syllable structure. The resyllabification that made syllables before -*s*C(R)- open certainly has a

[34] And perhaps **prai.fa.skin*° > *praefiscinē/ī*, via vowel reduction (derived from *fascinus*, but cf. *effascinō, -ās* [Plin., Gell., Symm.], *praefascinō* [Porph., Not. Tir.]). This form is the one that Leumann (1977: 81) considers hard in terms of the change **-a-* > *-i-*. Juret (1919: 95) calls the sequence *-sci-* "le groupe palatal," which he assumes changed the vocalic quality of the preceding syllable; this, however, seems not to be phonetically grounded. Rix (1966: 157 n. 3) explains this case as a result of assimilation (probably triggered by *-i-* the following syllable, though he does not clarify); presumably, this idea must also be grounded in open syllabicity before *-sci-* (cf. *similis* 'similar' < **se.mi.lis*, *cinis* 'ashes' < **ke.nis*).

phonetic/phonological basis.[35] As a result of such a process, we come to have a string of light syllables in the examples: **mĭnŭscellus* > *mīscellus*, **sēmĭs-quĭ-* > *sēsqui-*, **mŏnĕstrum* > *mōnstrum*, *fĕnĕstra* > *fēstra* / *fēnstra*.[36] Syncope under this metrical pattern would indicate its older application, as argued in the preceding section. This agrees with the attestation of these forms in as early as Pre-Classical Latin. The remaining example *sēstertius* [Var., Cic.+] requires some clarification, since it has no sequence of light syllables in its proto-form (**sēmĭstertius*). Notice that the documentation of this form is rather recent; the element *sēs-* may have already been available from *sēsqui-*.

3.3.5. Vowel Deletion and Reduction in Faliscan

As mentioned in 2.4.1, people have traditionally argued that Faliscan demonstrates no vowel reduction. Nor has vowel deletion been posited for the language. Giacomelli (1978: 516-17) denies the presence of syncope in the language. The analogous view is

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[35] Dennison, in his statistical study (1906) on whether consonant clusters are divided or not at the end of lines or by interpunctuation, concludes that they do not follow the onset-maximizing method (thus, e.g., -S | C-, -S | T-) prescribed by ancient grammarians and that this result reflects phonological syllabification understood by inscribers or speakers in daily speech. Though his extensive work is of permanent value, his premise that writing activity reflects a real phonology is not always true (see a similar misgiving in Fortson 1996: 191): e.g., *obtineō* for /-pt-/, cf. *optineō*. The onset-maximizing notation is lurking behind the dominant practice of dividing consonant clusters (19/119 for -| SC-, 51/340 for -| ST-, based on Dennison's study).

[36] Some scholars, such as Ciardi-Dupré (1901: 189) and Juret (1919: 97) etymologize *audiō* 'hear' as **au̯isdiō* < **au̯isdʰi̯ō* (Gk. αἰσθάνομαι 'perceive'). Though it is considered doubtful by Ernout and Meillet (1985: 55), Meier-Brügger (1980: 291) supports it, tentatively equating the first part of the word with Ved. *āvís* 'offenbar' (but cf. LIV 2001: 288 with the verbal root ?**h₂u̯eis-*). The syncope of *-i-* in something like **ău̯ĭsdʰi̯ō* would square with the pattern offered above, unless the verb is based on the *s*-less alternative **au̯i-* (cf. Gk. μέχρι(ς) 'to a given point') suggested by Meier-Brügger.

also found in Wallace (2005: 179 n. 24). However, another remark of Giacomelli's (1978: 513), that "le sincopi sono limitatissime," accidentally reveals that there are perhaps a few cases that exhibit this sound change. There are in fact several forms that have not been seriously considered (other than **cuncaptum**; see 2.4.1) but may prove the presence of vowel deletion, and by extention, reduction in the language.

Several inscriptions carry the same cognomen that corresponds to Latin *Maximus* (< *maxumo-* < **maksomo-* < **magisomo-* < **mag-is-m̥mo-* 'greatest'), though not all of them are unambiguously legible: Ve 286B **maxi̯mo**, 289 **[max]o[m]o**, 300 **[max]o̯mo** (these three are from the old Faliscan capital), 322d **maxo̯mo** (from the new capital), 328 **maxomo** (near Corchiano). Unless one insists that this Faliscan name was a late borrowing from Latin at the specific stage **maksomo-*,[37] syncope in the second syllable of the earlier proto-forms is plausible. Whereas **-o-** after **-x-** in four of them seems to exhibit no vowel reduction, **maxi̯mo** (286B), if the restoration of **-i-** is plausible, may prove its existence. This example is from the old capital and thus regarded as an early form. The other ones, especially those found elsewhere, are probably based on an archaizing orthography, which would be typical for materials of a conservative nature, such as onomastics.[38]

A group of grave inscriptions (Ve 324a-e), found near the new Faliscan capital, is

[37] The possibility of borrowing cannot be entirely excluded. But it is to be noted (see Hirata 1967: 61) that while *Maximus* is not a common cognomen in Latin (mainly used by the *gens Fabia*), the name is used with several gentilicia in Faliscan (Ve 289 **pupelio**, 322d **uecineo**, 328 **nomesina**). It seems less likely that several gentes simultaneously borrowed the same name from Latin.

[38] Giacomelli (1963: 204) tentatively interprets **-i-** as a result of "influenza straniera," which probably refers to Latin, whereas the **-o-** is thought to be indigenous (followed by Hirata 1967: 61). There is no compelling evidence for either proving or disproving her view of **-i-**.

also an interesting case, exemplifying various notations of the same family name; (b) **folcozeo** and (e) **fọlçoṣio** presumably demonstrate the most conservative spelling, whereas **-u-** in (c) **[f]olcuzio** and (d) **folcus(i)o** seems to be a result of vowel reduction in the medial syllable, and the non-realization of the vowel in (a) **fọlczeo** seems to demonstrate syncope. The materials are not unitary in terms of dating. Note, on the other hand, that the most plausible diachronic scenario, the original vowel (b, e) → the reduced one (c, d) → the syncopated one (a), is not applicable to this case, since (a) **sesto fọlczeo** was probably the father of (b) **voltio folcozeo**, as Vetter (1953: 315) assumes; **voltio folcozeo** was a sponsor of the grave for his son, **zextoi fi** 'Sexto filio, for (my) son Sextus', who presumably took his grandfather's name. Such a variety of spellings may rather be construed as stemming from the inscriber's/inscribers' lack of confidence or consensus on the quality of the vowel in the medial syllable, or even the presence/absence of a vowel itself.

This analysis would lead us to the metrical aspect of these words, whose proto-form must have had successive light syllables, that is, **măgĭsŏmo-* and **folcŏsĭos*. The elimination of such a sequence accords with the older type of syncope in Latin, which operated under the initial-stress rule and also the Penultimate Law for a while.[39] This coincidence would thus provide the possibility of syncope in Faliscan with some leverage.

[39] A personal name **uersno** (≈ Lat. *Versĕnus*) in Ve 286 (from the old capital) may relfect the influence of Etruscan (*versni* in CIE 142; Volaterra). For materials, see Schulze (1966: 253).

3.3.6. Remote Syncope in Post-sonorant Position

As we recognized in 3.3.1 within the framework of metrical constraints, the existence of approximants (sonorants and semivowels) that precede vowels targeted by deletion may facilitate syncope (e.g., *lāridī* > *lardī*), because the resulting consonant clusters are so often phonotactically permissible. The same is mostly true of vowel deletion designed for adjusting a sequence of light syllables: **oi̯no-dekem* > *ūndecim*; **ou̯i-* > **au̯i-pol-o-* > **au̯-pol-o-* → *ōpiliō*; **au̯i-cap-es* > *aucipis*; **pri(s)mo-kap-es* > *prīncipis*; **palamā-* > *palma*; *balineum* > *balneum*; **ferikolom* > *ferculum*; Gk. Πολυδεύκης → Lat. **Polud-* > **Pold-* (≈ **PODLOVQVEI**; see 3.1) > **Poll-* (≈ **POLOCES**, with /ll/) > *Pollūx*: but cf. *anculus* 'man-servant' [P. F.] < **ambikolos* < **amb^{h}i-k^{w}olos* (≈ Gk. ἀμφίπολος, Skt. *abhicara-*).

In 2.1, I maintained that unaccented vowels that are not immediately post-tonic are also licensed for sound change. As vowel reduction may take place in syllables that are remote from stress, deletion can likewise operate in such a position in support of approximants, especially sonorants (see Sihler 1995: 70, and also Leumann 1977: 97; cf. Götze 1923: 80):[40] e.g., **fálĭsĭnos* > **falerinos* > *Falernus* 'Falernian' (cf. *Faliscus* 'a kind of sausage; of Falerii'); **pŏ́pŭlĭ-* > *pōpulneus/nus* 'derived from poplar' (if syncope operated under the initial-stress rule; cf. *pōpulus* 'poplar-tree'); **kósĕlĭnos* > **kórĕlĭnos* > **korelnos* > **corulnus* > *colurnus* 'of hazel' (via consonant metathesis) (cf. *corylus* 'hazel-tree'); **u̯étŭsĭnos* > **u̯eterinos* > *veternus* 'morbid state of torpor' (see 2.2.3).[41]

[40] Cf. the preterit active in Etruscan (see Rix 2004a: 957): e.g., *amavunice* > *amavence* 'produced' and *zix(v)anace* > *zixunce* 'had rewritten'.

Syncope in a syllable distant from the stressed one is thus not surprising.[42]

We can add another group with remote syncope driven by foregoing sonorants, that is, diminutive nouns; note that many of them are derived from forms that already contain the diminutive suffix (*-*le/o*-) to end up with the sequence *-*le/o*- + -*le/o*-. These are already dealt with by Leumann (1977: 97), together with forms like *pōpul(i)neus/nus* and *Falernus* (see also Ballester 1990: 42). Some of them start being attested only in Classical Latin, and are perhaps later creations on the model of older ones (though the lack of attestation in Pre-Classical Latin might be by sheer chance). Therefore, for the sake of greater certainty, the following list only includes examples from Pre-Classical Latin onward (following syncope, consonant assimilation is involved in some cases):[43]

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[41] The above intermediate stages that Sihler (1995: 70) reconstructs in each word should be regarded as tentative ones. Relative chronology among syncope, rhotacism, and consonant metathesis requires further inquiry.

Sihler also adduces another example *sepultus* 'interred' [Pl.+] (cf. *sepelītus* [Cato+]), tracing this form back to **sepelitos*. But his reconstruction with *-i-* is not necessarily warranted; cf. Ved. *saparyáti* 'honor' with no vocalic element after the liquid = Lat. *sepeliō* 'bury' (stem **sep-el-* ← root **sep-*; see Meiser 1998: 195-196). Either **sepl̥-to-* with zero-grade root, as laid out by Leumann (1977: 615; but cf. "**sepeli-tos*" in 1977: 98), or **sepel-to-* could directly evolve into *sepultus*.

[42] In *Falernus* and *colurnus*, there is more than one post-liquid position. For the former, analogy with the city name *Falerīī* (or *Faliscus* containing a vowel after *-l-*) is conceivable. The latter is also explainable with recourse to analogy with other adjectives derived from plant names: e.g., *pōpulneus/nus* 'derived from poplar', *ficulneus/nus* 'of the fig(-tree)', *querneus/nus* 'of the oak', and *acerneus/nus* 'of the maple tree' (see Ernout and Meillet 1985: 145).

[43] Rix's reconstruction (1966: 162) of *gemellus* 'paired' (note, however, that it is only [Ov.+]) includes *o*-vocalism as in **gemenolos*. The same timbre is reconstructed in, e.g., *ag^{w}no-lo-* > **ag^{w}inlo-* > *avillus* [P. F.] (*agnus* 'lamb' ≈ Gk. ἀμνός) by Meiser (1998: 125). Whatever their intention, it is not until the following velar *-l-* affected the vowel that we had *o*-vocalism; in view of the Faliscan diminutives **arcentelom** (Lat. *argentum* 'silver'), **hutiçilom**, and **vrnela** with front vowels, it is more likely that *e*-vocalism was underlying, as Leumann (1977: 305) indicates, followed by Schaffner (2006: 37 n. 146). But from the chronological standpoint, it is unclear if such a vocalic (*-*e*- > *-*o*-) change really took place; *-*e*- > [ə] > Ø would be more straightforward). Rix (1966: 162-163 n. 22) very briefly mentions that syncope in *gemellus*, or perhaps in general, is to be explained under the Penultimate Law. However, not all of the diminutives accord with this view (cf. the possible accentuation in **katéne-lā-*(?) > *catēlla* 'light (ornamental) chain', etc.). Note that some of them demonstrate the metrical pattern ∪∪∪⏑#, with stress on the initial syllable under the earlier version of the Penultimate Law (see 3.3.2). Thus, *catellus* 'puppy'

[[*°*el(e)-lo/ā-*]] *catellus* 'puppy' < **kátĕlĕ-lo-* (*catulus*); *vitellus* 'little calf' < **u̯ítĕlĕ-lo-* (*vitulus*); *cistella* 'small box' < **kístĕlĕ-lā-* (*cistula* < **kistelā-* ← *cista*); *ofella* 'cutlet' (with *-f-* degeminated via the *mamilla*-rule) < **offellā-* < **óffĕlĕ-lā-* (*offula* < **offelā-* ← *offa*); *novellus* 'young' < **nóu̯ĕlĕ-lo/ā-* (**nou̯elus* ← *novus*); [[*°*e/or(e)-lo/ā-*]] *misellus* 'poor, mean' < **mísĕrĕ-lo/ā-* (*miser*); *tenellus* 'soft' < **ténĕrĕ-lo/ā-* (*tener*); *ampulla* 'flask' < **ampollā-* < **ámpŏrĕ-lā-* (*amphora*); [[*°*en(e)-lo/ā-*]] *fiscella* 'small basket' < **fiscĕnĕ-lā-* (*fiscina* [Cato+] ← (*)*fiscus* [CIL I^2 583 (123/2 B.C.E.), Cic.+; P. F.]); *patella* 'small dish' < **pátănĕ-lā-* (*patina* ← Gk. πατάνη); *asellus* 'donkey' < **ásĕnĕ-lo-* (*asinus*). The first group would be joined by *tabella* 'tablet' (← *tabula* < maybe **tablā-* < **-dʰlā-*, cf. Umb. **tafle**), for which Leumann (1977: 306) expresses difficulty in laying out a historical derivation. I suggest a proto-form with a double diminutive suffix, and offer the following scenario: **táblĕlĕ-lă* > **tábĕlĕlă* (via haplological dissimilation) > *tabella.* Another example, *fābella* 'story' [Cic.+] (*fābula* 'talk'), might also belong here, unless it is a simple copy of *-ella* in old *tabella*-type diminutives.

It is likely that such an outcome (*-llo/ā-*) or the recurrent process of syncope itself in this morphological category also exerted an analogical effect on other diminutives, which did not have a sequence of light syllables, to incur the same change: [[*°C*n(e)-lo/ā-*]] *columella* 'upright pivot of an oil-mill' < **kólomnĕ-lā-* (*columna*); [[*°V:*ne-lo/ā-*]] *catēlla*

[Pl.+], for instance, can only emerge from **kátele-lo-*. If one claimed that syncope applied at the stage **katéle-lo-*, the accentual basis must have been the post-Plautine system, which contradicts the chronological ordering of accentual rules. Thus, we must have remote syncope as an explanatory option. The deletion process was carried out most likely soon after Process I (see , that is, **kátelelo* > **kát*[ə]*l*[ə]*lo-* > **cat*[ə]*llo-* > *catellus*. Without the distinction between Process I and Process II, one would reach an unattested product: **catelelo* > ***catululo-* > ***catullo-*. The personal name *Catullus* is probably a result of a later application of syncope after the completion of Process II: **kátelelo* > **kát*[ə]*l*[ə]*lo-* (Process I) > **kátululo-* (Process II) > *Catullo-* (syncope). The default of syncope in the earlier stage is most likely due to the generally conservative nature of proper nouns.

'light chain' < **kátēnĕ-lā-* (*catēna*); *corōlla* 'garland' < **kórōnĕ-lā-* (*corōna*); *persōlla* 'little mask' < **pérsōnĕ-lā-* (*persōna*); *catīllus* 'bowl' < **kátīnĕ-lā-* (*catīnus*); *suillus* 'of pigs' < **súīnĕ-lā-* (cf. *bovillus* 'of cattle' [Liv.], perhaps modeled after *suillus*).

The list of analogical forms can be extended by trisyllabic diminutives that also contain no sequence of short syllables: *rāllus* 'sheer, fine' < **rārĕ-lo-* (*rārus*); *vīllum* 'small quantity of wine' < **u̯īnĕ-lo-* (*vīnum*); *ūllus* 'any', *nūllus* 'not any' < **oi̯nĕ-lo-*, **n(e)-oi̯nĕ-lo-*; *paullus* 'little' < **pau̯rĕ-lo-*; *pōcillum* 'little cup' < **pōklĕ-lo-*; *pugillus* 'what can be held in the fist' < **pugnĕ-lo-* (*pugnus*); *tigillum* 'small plank' < **tignĕ-lo-* (*tignum*); *scabillum* 'low stool' < **scabnĕ-lo-* (*scamnum* < **scabno-*); *Sabellus* < **sabnĕ-los* (cf. *Samnium*, *Sabīnus*, Osc. **safinim**). Among such trisyllabic words, *bellus* 'pretty' (*bonus* < **du̯eno-*) may have been a factor promoting syncope like the above quadrisyllabic words with successive light syllables, because its proto-form **du̯ĕnĕ-lo-* included such a sequence, and was in need of remedy.

3.4. Vowel Deletion in Post-rhotic Position

Toward the end of the last section, we saw the analogical operation of quadrisyllabic words with more than two successive light syllables on others that do not meet such a condition. Among the latter, I included examples such as *pōcillum* < **pōklĕ-lo-*; *pugillus* < **pugnĕ-lo-*; *tigillum* < **tignĕ-lo-*; *scabillum* < **scabnĕ-lo-*; *Sabellus* < **sabnĕ-los*. These have sonorants, **-l-* and **-n-*, in post-consonantal position. With this salient phonotactic feature, I still submit that the analogy with quadrisyllabic diminutives

(containing sonorants in post-vocalic position) was one major factor that caused them to undergo syncope. On the other hand, we have the same type of words, which however continue *-C*r*-, such as *agellus* 'small land' < **agrĕ-lo-* (*ager*); *libellus* 'document, pamphlet' < **librĕ-lo-* (*liber*); *sacellum* 'shrine' < **sakrĕ-lo-* (*sacer*); *flagellum* 'whip' < **flagrĕ-lo-* (*flagrum*); *lībella* 'small silver coin' < **lībrĕ-lā-* (*lībra*); *macellus* 'rather thin' < **makrĕ-lo-* (*macer*). The history of these can also be explained by analogy with quadrisyllabic diminutives, like that of the forms above with *-*l*- and *-*n*-. But the historical treatment of *C*r*-clusters + vowels (*C*r*V), though it is technically a sort of syncope, should be considered separately, because of its possible age relative to other sequences. This would lead us to conclude that the development was actually not stress-driven and was earlier than that of *C*l*/*n*-clusters + vowels. If this is the case, forms such as *agellus* < **agre-lo-* could also serve as a model in triggering syncope in **pōkle-lo-* > *pōcillum*, and so forth.

The more detailed phonological development can *roughly* be laid out as follows (we will come back to more details below): **agre-lo-* > **agr̥lo-* > **agerlo-* > *agellus*. This would be parallel to *hībernus* 'of *or* in winter' (< **hībrĭnos* < **ĝ^{h}eimr-ino-*). In view of syncope of *-*e*- in the medial syllable, one might claim that intial stress was responsible for this process, causing unstressed vowels disappear. This conjecture may lead to another one that the syncope in **agre-lo-* was a historical event, almost simultaneous with vowel reduction, which was also triggered under the initial-stress rule. In fact, the presence of **SAKROS** (> *sakr̥s* > *sacer* 'sacred'; see more description below) in CIL I^2 1 ("the Forum inscription" [ca. 600-550 B.C.E.]), which retains the original *C*r*V-sequence

in the final syllable, is possibly compatible with such an idea.[44] However, there is strong evidence suggesting that the phonological change in *C*r*V was older than conventionally assumed. Nussbaum (1999) conducts a detailed analysis of Latin adjectives in *-idus*, and concludes (1999: 394-395) that *acerbus* 'bitter' continues PIE *$h_2e\hat{k}ri$-d^ho- (> *$ak_{\text{r̥}}$ðo- > *akerðo- > *akerβo- > *acerbus*), later restated as *$h_2e\hat{k}ri$-d^hh_1-o- by Hackstein (2002: 17). The phonological change from *-*r*ð- > *-*r*β- is parallel to PIE *$u_{\text{r̥}}erd^ho$- > *$u_{\text{r̥}}er$ðo- > *$u_{\text{r̥}}er$βo- > *verbum* 'word'. This means that *-ð- (< *-d^h-) was still phonemically separate from *-*d*- word-internally (cf. PIE *$b^h\acute{e}i_{\text{i̯}}d^h$-e- > *$fei_{\text{i̯}}$ð-e- > *fīdō* 'trust'; PIE *h_2mord-$é i_{\text{i̯}}$e- > *mordeō* 'bite'), when it came into contact with *-*r*- as a result of syncope of *-*i*-. Given the fact that there is no compelling evidence opposing a prehistoric merger of *-ð- and *-*d*-, the **i*-syncope must also have occurred in a prehistoric period. It is true that it might have taken place under the initial-stress rule already in operation and the syncope might therefore have been stress-driven. But this view has to answer the fact that this syncope appears and occurs at a much earlier period than do the other historical (securely) stress-driven syncopes. For this reason, the deletion of *-*i*- can more likely be considered not to have been motivated by stress distribution. The adjective **SAKROS** mentioned above may thus be an analogically re-made form, as suggested by Nussbaum (1999: 392-396, and esp. 414 n. 98), or simply an archaic form.[45]

In regard to final syllables, syncope just as in **SAKROS** (> *$sak_{\text{r̥}}s$* > *sakers*) is rather

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[44] Cf. some Sabellic parallels: Osc. city or ethnic name **aderl** < (probably) *$ātre$-l^o (≈ Lat. *Ātella*; cf. *āter*, Umb. **atru** 'black < *$ātro$-); Umb. *pacer* 'propituous' < *$pāk$-*ri*-.

[45] For a possible counterexample *sacerdōs* 'priest', see Nussbaum's interpretation (1999: 396-398) with not *$sakro$-d^hoh_1-*t*- but *$sakro$-deh_3-*t*-.

common in Latin in nouns and adjectives in *-C*ri/os*#.[46] This type of syncope always involved loss of final *-*s*, and -*r*- ultimately obtained the preceding -*e*-, as already shown above in **agre-lo-* > **agr̥lo-* > **agerlo-* > *agellus*. Traditionally, syncope has been considered to have predated **s*-loss, which is commonly demonstrated as follows (see Meiser 1998: 74):[47] **agros* (cf. Gk. ἀγρός) > **agr̥s* > **agers* > **agerr* > *ager* 'land'; **ākris* > **ākr̥s* > **ākers* > **ākerr* > *ācer* 'sharp' (cf. m. *ācris* [Enn.] and f. *ācer* [Naev.]).[48] This scenario can be mostly justified by the Germanic cognates of *ager*, such as Goth. *akrs*, OIc. *akr*, and OE *æcer* (see Boutkan 1995: 167-173). Indeed, this pattern is widely found in Latin, though it is not a strict rule:[49] *alacer* < *alacris* 'moving nimbly,

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[46] Unlike Sabellic, phonological contexts that feed final syncope are remarkably limited in Latin. The word-final **-tis* → *-s* (allegedly through **-ts*), particularly common in nom.sg. forms of *ti*-stem nouns, has occasionally been cited as an example of syncope: e.g., *ars* 'skill, art', *cohors* 'farmyard; armed force', *fōns* 'spring', *fors* 'chance, luck', *mēns* 'mind', *mōns* 'mountain', *mors* 'death', *pōns* 'bridge', *sors* 'lot', etc. < **-ti-s*. However, as Leumann (1977: 449) indicates, this process was probably due to an analogy with nom.sg. forms of consonant-stem nouns, such as *dēns* 'tooth' (< **dent-s*) and *nox* 'night' (< **nok-t-s*). For the same reason, ethnicity/geography-related forms, such as OLat. *quoiātis/cuiātis* 'of what country?' [Pl.+] and *nostrātis* 'born in our country' [Hem.] (cf. Classical and Late Lat. *quoiās/cuiās* and *nostrās*), are not considered to prove syncope in **-tis*. As for *Laurentis* 'of Laurentum' [Enn.] (CLat. *Laurēns*), syncope is far less likely since the final *-i-* is convincingly shown to be long by Nussbaum (1973). A legal term *damnās (estō)* 'obliged (to pay, etc.)' < **-tos*(?) is regarded as dialectal syncope by Ernout and Meillet (1985: 164), which is however entirely open to question and further study. Examples cited for syncope after other stops by, e.g., Ciardi-Dupré (1901: 216-220) and Götze (1923: 140), are entirely dubious.

[47] The intermediate stages like **agers* and **agerr* can somewhat be justfied by *ter* 'three times'. As Leumann (1977: 142) and Meiser (1998: 114) assume, this word in Pl. *Bacc.* 1127 is to be scanned as heavy, which can be attributed to its possible development like **tris* > **tr̥s* > **ters* > *terr* > *ter*; that is, the scansion as heavy is based on the stage *terr* (or *ters, underlyingly preserved when the line was composed, and later interpolated with the new form).

[48] Ennius's *ācris* (m.) may perhaps be taken as a relic of the old form. On the other hand, given the fact that a secondary creation *ācrum* [Naev.] in Pre-Classical Latin was modeled after the new form *ācer* (and misanalyzed as a second-declension noun; cf. *ager*, *sacer*), we are led to date the development *ācer* < **ākris* to a very early period, which is far in the past for Old Latin speakers.

[49] Cf. *fūnebris* 'funerary', *muliebris* 'womanly' (< *°*es-ris*; see Weiss 2005: 193); *anclābris* 'sacrificial', *salūbris* 'conductive to a healthy condition', *alebris* 'nutritious' (< PIE **-dʰli*, with **-l-* dissimilated from *-r-* due to the presence of another *-l-* in the word; cf. *stabilis* 'steady'). Based on such examples, Götze (1923: 141) does not believe the phonological change like **-ris* > **-r̥s*, but instead assumes the analogy with *-ros* > *er*.

lively' [Enn., Ter.+]; *aper* 'wild boar' < **apro-*; *āter* 'black' < **ātro-*; *caper* 'he-goat' [Caes., Catul.+] < **kapros*; *faber* 'craftsman' < *fabros*; *imber* 'rain' < **embris* < **ṃb^{h}ris* (cf. Ved. *abhrá-* 'cloudy weather' < **ṃb^{h}ro-*); *integer* 'not touched' < **entagros*; *macer* 'thin' < **makros*; *niger* 'black' < **nigros*; *piger* 'lazy' < **pigros*; *sacer* 'sacred' < **sakros* (= **SAKROS** in CIL I^2 1); *salūber* < *salūbris* 'conductive to a healthy condition'; *uter* 'leather bag' < maybe **utris*.[50]

It is plausible that final syllables became extrametrical, that is, unstressed already in an early stage of Latin. Then, thus syncope in the *C*r*V-sequence in final syllables would have occurred when they were unstressed, just as in medial syllables. However, recall the suggestion above that lack of stress was not a conditioning factor of this type of syncope, as shown by *acerbus*. Moreover, the same development, particularly after **k-* or **t-*, can be found in initial *stressed* syllables: e.g, **krinō* > *cernō* 'sift, distinguish'; **krito-* > *certus* 'fixed, certain'; **tris* > *ter* 'three time'; **tristi-* > **tersti-* > *testis* 'witness'.[51] At this point, we can consider the phonetics of this phenomenon, including the counterpart in medial and final positions.[52] The sequence *C*r*V may have fed an intrusive vowel between *C* and *r* to end up as *C^{ə}*r*V.[53] This type of vowel intrusion involving sonorants

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[50] One can perhaps add **BERBER** ("Carmen Arvale" ad CIL I^2 2104) < **dʰer(-)ĝʰ-ro-s* (cf. Av. *dərəzra-* 'firm, solid'). See Katz (1998: 214-216, 2006: 334-337).

In regard to the sequence *-C*l*__*s*#, as the closest counterpart of *-C*r*__*s*#, we have almost no evidence to generalize from (cf. *pullus* 'young bird' < **putslo-*; see Weiss 2005: 195); cf. apocope in **fakli* > **fakḷ* > *facul* 'easily'.

[51] There are no explicit examples that exhibit how **kro-*/**tro-* developed. Cf. **tresōs* > *terror* in Nussbaum (1999: 394).

[52] Thus, we can dismiss the separate treatment depending on the location, as opposed to Götze (1923: 144), who mostly follows Sommer (1914: 62-63).

[53] Something like this is located in dialectal texts: e.g., CIL I^2 401 **MACISTERATVS** 'authority' ("Lex

typologically behaves as if it does not add to the number of syllables at a synchronic level, as Hall (2004: 3) argues. If Latin underwent a typical intrusion according to Hall's definition, the new vowel would have copied the same quality of the V following *r*. But it may be that Latin never reached such a level, instead amplifying ^{ə} and diminishing V (the latter process would be regarded as syncope in a broad sense).[54] It is a regular process that the schwa-like vowel was realized as *e* before *r*. In its entirety, what is happening is something like a metathetic process between *r* and a vowel (i.e., *C*r*V > *C^{ə}*r*V > *C*ə*r^{V} > C*er*).[55]

As a further ramification, it is worthwhile to examine the development of *V*r*V, as opposed to *C*r*V. In medial syllables, the vowel after *r* was regularly syncopated by the operation of banning successive light syllables, as in *misellus* < **mísĕrĕ-lo/ā-*; *tenellus* < **ténĕrĕ-lo/ā-*; *ampulla* < **ampollā-* < **ámpŏrĕ-lā-*, and also analogically in *rāllus* < **rārĕ-lo-*. In view of this fact, one might similarly appeal to syncope in -V*r*V*s*# as well as -V*l*V*s*# (the second V = *i* or *o*). The following would be potential examples of this patter: [[nouns]] *gener* 'son-in-law' < **generos*; *socer* 'father-in-law' (cf. *socerus* [Pl.], a

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Luceria"; Luceria, early 3rd c. B.C.E.; Oscan substratum interference is assumed by Wallace 1988: 217-219), 59 **MAGISTERE** (Praeneste, maybe 3rd c. B.C.E.), and 1926 **MAGISTERẸ** (Cingulum [Picenum], maybe 3rd c. B.C.E.; for the interpretation and restoration of **Ṃ** and **Ẹ**, see Vine 1993: 157-159). In Standard Latin, when the vowel that follows *-r-* was long, the schwa-like sound did not emerge or even if it did, it did not develop into full-fledged *e*-vocalism.

[54] Hall (2004: 16-17 n. 3) suggests that vowel intrusion is not usual between homorganic consonants (the essential case to us is *t* + *r*), but she admits several exceptions.

[55] As a counterpart of *testis* 'witness' (< **tristi-*), Oscan provides **trstus** (< **tristo-*), which is generally conceived as a mistake for ***tristus**. This is somewhat justified by the presence of **trístaamentud** 'testamentum, will'. If these forms were taken at face value, it could be said that Oscan preserved the sequence *C*ri* in initial *stressed* syllables, and this would allow for the assumption that syncope was motivated by lack of stress. However, Umbrian provides *tertim*, etc. 'third' < **tri-tiio-*, which clearly shows the same development as in Latin (cf. footnote 35 in Chapter 4. The form **trístaamentud** may be a secondary creation and **trstus** possibly stands for ***terstus**, as Buck (1928: 60) deals with this form among other forms with *er*.

'restituierte[m]' form; see Meiser 1998: 73) < **sokeros* < **sụe̯kuros*; *puer* 'boy' (cf. *puerus* [Pl., Caecil.], like *socerus*) < **pueros*; *magister* 'master, commander', *minister* 'servant' < **-teros*; **LEIBER** (CIL I^2 563; Praeneste, ca. 350-250 B.C.E.); *vesper* 'evening' < **u̯esperos* (vs. f. *vespera*); [[pronouns/adjectives]] *alter* 'another' < **aliteros*; *uter* 'which of the two' < **k^wuteros*; [[nominal compounds in *-fer* and *-ger*]] *armi-ger* 'armor-bearer' (cf. *mōri-gerus* 'compliant' [Pl., Naev.+]), *frūgi-fer* 'productive', *lūci-fer* 'light-bringing'; [[adjectives]] *celer* < *celeris* 'moving fast, quick' [Cato] (cf. f. *celer* [Andr.+]); *līber* 'free' < **līberos* (cf. Pael. *loufir* 'free man' < **$h_1$leud^{h}eros*); *miser* 'poor' < **-eros*; *satur* 'well-fed' < **saturos*; *tener* 'soft, tender' < **teneros*; *asper* 'rough, bitter' < **asperos*.

On the other hand, we have a number of counterexamples that retained final syllables: e.g., [[after a short vowel]] *superus*, *īnferus* (cf. *super interque vicinus* [Cato *Agr.* 149]), *exterus*, *posterus*, *properus*, *camurus*, *prosperus* (cf. very later *prosper*), *ferus*, *aemulus*, *pendulus*, *tumulus*, etc.; [[after a long vowel]] *clārus*, *amārus*, *vērus*, *mīrus*, *sevērus*, *dīrus*, *mōrus*, *pūrus*, *mātūrus*, *familiāris*, *mīlitāris*, *populāris*, *aequālis*, *capitālis*, etc. We thus need to reconcile these two patterns (i.e., retention vs. deletion). One might assume that the application of final syncope operated regardless of syllable structure (i.e., **agros* > *ager*, **generos* > *gener*), and then reckon forms, such as *inferus* and *superus*, as analogically restored on the model of the retained -VC*i*/*os* [C ≠ liquid] (e.g., *amātus* 'loved', *brevis* 'short'), as Ciardi-Dupré (1901: 220) and Leumann (1977: 423) maintain.

However, I tentatively submit that what we are dealing with is rather just the opposite; forms that lost final syllables are analogical creations. Note, first of all, that

most of the nouns of such forms denote human beings, either as kinship terms, professions, or social ranks. This semantic character can be referred to other, similar types of nouns, such as *pater* 'father', *augur* 'augur', and *quaestor* [official rank], whose *-r* at the absolute word-final position are historical. Such a semantic association enables us to claim that words in *-V*r*V*s*# were reshaped to resemble those with -V*r*#. This conception could be strengthened by a very plausible explanation that the vocalic change *-u-* > *-e-* in *socer* (< **sụekuros*) is analogical to other kinship terms (see 2.2.3). Some of the compounds in *-fer* and *-ger* can be regarded as denoting professions, such as *armi-ger* 'armor-bearer', like *augur*. Among qualifying adjectives, *līber* 'free (man)' also has this sense by extension. Another basic noun, *vir* 'man, husband' (< **u̯īró-*) similarly agrees with this line of reasoning (see Götze 1923: 124). The phonological condition may have been secondarily extended to *-V*l*V*s*# (→ -V*l*#): e.g., *vigil* 'awake; watchman' < **-ilis*; *pugil* 'boxer' [Ter., Lucil.+] < **-ilis* (cf. *pugilis* [Var.]); *famul* 'servant' [Enn., Lucr.] < **famelos* (cf. *famel* [Fest.], and also Pael. *famel*); *dēbil (homō)* 'powerless' [Enn. *Ann.* 324] ~ *dēbilis*.[56] This process is morphological and enforced by the semantic association; there is no guarantee that it involved a phonological process like syncope. The morphological reshaping may have continued from as early a period as the attestation of the theonym **LEIBER** (CIL I^2 563; maybe also influenced by *Iūpiter*) to a later stage producing *figel* (< *figulus* 'potter'), *mascel* (< *masculus* 'man'), and *barbar* (< *barbarus* 'foreign [n./adj.]'), listed in the Appendix Probi (perhaps, Oscism?; see Leumann 1977:

[56] Götze (1923: 141) suggests that *dēbil* is Oscan, but the appeal to an Oscism is only circumstantially endorsed. Though Leumann 1977: 449) considers it to be modeled after *vigil*, the type of the association (semantic?) is not clearly defined.

94).[57]

This view would render other possible examples of syncope rather isolated. The necessity of positing the sound change only for them is not compelling. To explain the history of such forms, we need to recognize another interesting factor. As already mentioned, forms in *-C*ros* and *-C*ris* ended up as *-C*er* by means of post-rhotic/pre-sigmatic non-stress-driven syncope. In the thematic declension (*-*Cros*), this change particularly affected masculine nom.sg. forms, which thus obtained *-er* as a marker of gender. This is especially conspicuous in adjectives: e.g., *sacer* (< *-*ros*) vs. *sacra*. On the other hand, in regard to the *i*-stem adjective **ākris* (m./f.), the phonologically regular outcome *ācer* must have been shared by both masculine and feminine genders. But except for a few cases of gender interchange in Pre-Classical Latin (see above), the syncopated form *ācer* is selectively used as masculine, while the original shape is reflected in the feminine form as *ācris*. This history indicates that the ending *-er* was identified by Latin speakers as a more or less productive masculine marker. It may well be that this factor gave leverage to the morphological reshaping into *-er* in some forms that originally had *-V*r*V*s*#. There is thus no need to see a (regular) phonological change like syncope in this process.

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[57] This issue concerns and complicates the origin of *lībertās*. Leumann (1977: 98) offers a scenario with successive syncope, such as **leu̯dʰerotās* > **lou̯frotās* > **loi̯fr̥tās* > *lībertās* 'freedom' since he considers it hardly possible to suppose the third syllable was syncopated in **leudʰerotās* (likewise, syncope does not apply to the third syllable according to Sommer 1900: 227). In addition to his misleading notation *-*f*- (cf. Weiss 2005: 92), the unnecessary avoidance of syncope in the third syllable is unacceptable (see 3.3.6); remote syncope of *-o-* after *-r-* could properly yield *lībertās* from the possible Proto-Italic reconstruction **leu̯ðero-tāt-s*. However, one may claim that the secondary product *līber* was directly attached (or re-attached) to *-tāt-* in a later stage, and this option can actually not be eliminated. The same is applicable to the Faliscan counterparts **loifirtato** 'libertatis, freedom' (Ve 253), **[l]o̯ifirta̯** '*liberta*, freedwoman' (Ve 276a), **loferta** 'id.' (Ve 322a); Faliscan may have obtained **loi̯fer* by a process analogous to Latin (*līber* ← * *līberos*), and then suffixed *-tāt-*.

3.5. Vowel Deletion in Final Syllables: Sporadic Apocope

In the preceding section, we saw a type of syncope in *C*r*V that was likely prehistoric, thus not stress-driven, and further not sensitive to the location in the word (e.g, **krinō* > *cernō*; **h₂eĥri-dʰo-* > *acerbus*; **agros* > *ager*). In regard to word-final position, there is no other case of syncope, so that we have no stress-driven syncope in that position.

We briefly consider another type of vowel deletion, that is, apocope. Here particularly discussed is the one from the initial-stress period onward, which is sporadically attested;[58] its relative lateness is highly secure in view of some historical doublets (such as *simul* ~ *simile* and *tribūnal* ~ **TRIBVNALE**; see below). As seen from the discussion above about syncope in medial syllables (3.3), foregoing sonorants (and semivowels) play an important role in facilitating the change. Apocope also seems to have favored and even required such segments. This absolute necessity is opposed to syncope in medial syllables, where liquids, or approximants in general, only serve to promote the change. On the other hand, it is also true that deletion did not always take place in a post-liquid context. Morphological (and perhaps also stylistic and dialectal?) factors make matters rather complicated.

From the observation of very sporadic materials of apocope, it follows that this change seems to have been allowed after both long-vowel and short-vowel penults; this

[58] Benediktsson (1960: 164) insists that underlying causes of apocope are different from syncope, but it seems not necessarily so. The absence of stress may undoubtedly serve as a conditioning factor for both cases.

fact excludes the option of considering the constraint on successive light syllables as in the medial position. Apocope following a short-vowel syllable is typically found in a proto-stage of *simul* 'at the same time': **semh₂-li* > **semali* > **sem*[ə]*l* (≈ **SEMOL** in CIL I^2 1531; near Sora, ca. 150 B.C.E.) > *simul*,[59] whose *i*-vocalism in the first syllable is presumably analogical, stemming from *similis* 'similar' (< **semilis*, via vowel harmonic assimilation; see Götze 1923: 87). On the other hand, its historical doublet *simile* (neut. of *similis*) reflects retention of the final **-i*, which evolved into a more sonorous vowel *-e* for the purpose. Likewise in *mare* 'sea' (< **mari*), the final **-i* was also retained (or restored; see Meiser 1998: 74).[60] Another adverb *facul* 'easily' [Pac., Acc., Lucil.; Fest.] (< **fakli*) might belong here, if **-kl-* is regarded as forming a syllable onset. Otherwise, **fak-* would be considered heavy syllable, and be associated with long-vowel penults. It is the form with a retained **-i*, *facile*, that is common in Classical Latin (see Kent 1932: 98).[61]

Apocope also occurred after long-vowel penults: e.g., *calcar* 'spur' < **kalkāri*; *vectīgal* 'revenue derived from public property' [Cato+] < **u̯ektīgāli*; *animal* 'living being' < **animāli*; *tribūnal* (cf. **TRIBVNALE** in CIL I^2 593) < **tribūnāli*; *exemplar* 'example' (cf. *exemplāre* in Lucr.) < **eksemplāri*; *Bacchānal* 'site where the rites of Bacchus were celebrated' [Pl. +] (cf. **BACANAL** in CIL I^2 581) < **Bakkānāli*; but cf. *mīlle*

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[59] Livingston (2004: 33) assumes that *similis* evolved through **seml̥* from **sem-(i)li-*. She does not explain how *-i-* was deleted. In my theory, the deletion is not required at all.

[60] Maniet (1957: 161) assumes that the final vowel was retained due to its functional value, but its sonority was enhanced to resist deletion.

[61] Non-standard Latin also shows the change in third-conjugation infinitives: e.g., CIL I^2 501 **TANGER** (= *tangere* 'touch'), OLat. *biber* (= *bibere* 'drink'). But given the fact that the penult in this conjugation is syllabically light, the function of a metrical factor cannot be excluded in this case.

'one thousand' < *$m\bar{\imath}li$ < *$smih_2$-$\hat{g}^h$$(e)$$sl$-$ih_2$ (see Meiser 1998: 174, Weiss 2005: 213, 397). These forms are substantivized neuter forms of adjectives in -*āri*- and -*āli*-. Note that neuters that remained adjectives preserved the final *-*i* (> -*e*). The question of the exact relationship between the semantic specialization and the phonological process requires further investigation.

4. Vowel Reduction and Deletion in Sabellic: A Synchronic and Diachronic Interface

We have so far discussed vocalic changes in Latin (and marginally Faliscan). Our attention is now turned to the Sabellic dialects. The same methodology followed in Latin is now applied to them, namely, procedures for mapping out correlations between phonetics and phonology. This enables us to touch on various aspects of vowel reduction and deletion in Sabellic both synchronically and diachronically. Though evidence in Sabellic is fairly fragmentary due to its limited corpus, the pieces of the puzzle will end up allowing us to partially visualize a large picture of the mechanisms, which, I believe, matches general linguistic principles.[1]

In what follows, I will focus on syncope as a representative type of vowel deletion, because there are only a few cases that may exemplify apocope under the initial-stress system (i.e., not the PIE accentual rule; cf. 1.2 and 3.1). In addition, most of the cases concern the loss of final vowels in adverbs capable of being prepositions and prefixes, and it may have taken place when they were followed by nouns or verbs; thus, it was not purely apocope, because vowels are not located at the end of prosodic units. The Umbrian neut. *i*-stem nouns *uerfale* 'place marked off for taking the auspices' and **sakre** 'victim' have traditionally been construed as nom.-acc.sg. (see, e.g, Buck 1928: 133-134, 263, Vetter 1953: 430, 444; but for *uerfale*, cf. Untermann 2000: 843), thus indicating that the final **-i* remained in the language, unlike several Latin forms such as *calcar* 'spur' <

[1] To refer to original inscriptions (represented as, e.g., Pg and Cp) with cited Sabellic forms, I follow the numbering system in Rix's collection (2002).

kalkāri* and *animal* 'living creature' < **animāli* (see 3.5). This retention may perhaps accord with that in Osc. *auti* and Umb. **ute, *ote* 'or' (cf. Lat. *aut*). It is however impossible to form a larger picture than this of (the absence of) apocope in Sabellic.

4.1. Vowel Reduction and Deletion: History and Chronology

4.1.1. Vowel Reduction in Labial Context: From Early to Late Sabellic

Traditionally, whereas vowel deletion as a remarkably common phenomenon in Sabellic has long drawn much attention from scholars, vowel reduction has not been a focus of much discussion, since it has been said to be limited to a small number of forms. However, as we know from Buck's sketch (1928: 55-57) of vowel reduction in Oscan and Umbrian, it is an undeniable fact that the pre-labial position often fed such a vocalic change:[2] e.g., Osc. **últiumam** '*ultimam*, furthest' (< **h₂ol-tm̥mo-*); Osc. *amprufid* '*improbe*, wrongfully' (< **-pro-*); Osc. *pertumum* '*dirimere*, prevent' < **pert-emom* (cf. fut. *pertemest*, fut.pf. *pertemust*); Umb. **prehubia** '*praebeat*, provide' (vs. **prehabia**, cf. simplex forms **habe**, **habetu**, etc.). A simplistic phonetic interpretation of the letter ‹u› at face value as [u] might seem to match its pre-labial position enhancing [+round]. But orthographic arbitrariness shown by the cohort of superlatives like **últiumam** should be taken into account: e.g., Umb. *hondomu* '*infimo*, lowest' (< **ĝʰom-tm̥mo-*); Osc.

[2] This change should not be dissociated from lack of stress. Stressed initial syllables retain *o*-vocalism, whether original or secondary. Contrast Umb. *somo* (< **-u-*) and *onse* with Lat. *summus* 'topmost' and *umerus* 'shoulder'.

nessimas, Umb. *nesimei* 'nearest' (< **ned-tm̥mo-*; see 2.2.2, and footnote 20 there); cf. Lat. personal names **DECIMIO** and **DECEMIA** (CIL I^2 2848; see 2.2.2). See also *atropusatu*, etc. 'perform a *tripudium* (dance)' (< **-tri-* 'three'; cf. *atripursatu*, **ahtrepuřatu**). We are thus more accurately led to assume rounded centralized values of these medial vowels, such as [ɵ]/[ʉ] (represented by ‹o› or ‹u›) and also secondarily [ø]/[y] (represented by ‹i›). As Buck very briefly mentions, post-labial position also allows for such vowel reduction. I pay more attention to this context: e.g., Osc. *praefucus* '*praefectus*, governor' (vs. *facus* '*factus*, be made');[3] Osc. διποτερες 'Iovis, Jupiter' (vs. Osc. **patir** 'father'; see Untermann 2000: 186); Umb. **arputrati** '(according to the) decision' ≈ Lat. *arbitrātū*; Osc. **prupukid** 'by previous agreement(?)' (< **prō-pak̥io-*; see Buck 1928: 56-57);[4] Osc. **aflukad** '?' (cf. **aflakus**), albeit with **-l-** intervening; Umb. *atripusatu*, *atropusatu*, etc. (< **-pod-*; cf. lat. *pēs pedis* 'foot'); Umb. *comoltu* 'grind' (vs. **kumaltu**); and word-finally, Pael. *hospus* 'stranger'.[5] If the association of **aflukad** with **aflakus** is proper, the latter perhaps stands to conceal a real rounded articulation under the etymological **a**-notation.[6]

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[3] Cf. Buck (1928: 56), who also regards the vowel of the following syllable as a triggering factor.

[4] Buck also admits a possibility that **-u-** may continue an ablauting **-ō-* (ultimately **-oh₂-*). But this option is considered less likely by Weiss (1993: 36-39).

[5] Parallels could be supplied from Mycenaean. Vocalization of **n̥* > *o* in the language generally occurred in post-labial position: e.g., *pe-mo*, *e-ne-wo-* vs. *a-ki-ti-to-*, *a-no-we*. On the other hand, there seem to be some pre-labial cases (see Vine 1998: 35, and n. 80): e.g., *o-wi-de-ta-i* (dat.pl.f.) 'invisible (ones)' < PIE **n̥-u̯id-eto-* (cf. Gk. ἄιστος 'unseen' < **n̥-u̯id-to-*); *o-mi-ri-jo-i* (dat.pl.m.) 'immortal (ones)' < PIE **n̥-mr-iio-* or *-ih₂o-* (cf. Gk. ἄμβροτος < **n̥-mr̥-to-*). But for the first example, cf. the objection raised by Palaima (2004: 270 n. 6).

[6] It is well known that the analogy between a simplex and its compound (i.e., "recomposition") is more common in Sabellic than Latin, as Götze (1923: 130) and Haug (2004: 235) point out. Notice that "recomposition" may be either phonetic (depending on style or speech rate) or simply graphic.

We must be more alert to instances of this kind.[7]

Archaic Sabellic dialects, such as Pre-Samnite, Paleo-Umbrian, and South Picene, already demonstrate the same phonological change (though its Proto-Italic status is not necessarily guaranteed), which is orthographically more visible in pre-labial than post-labial context. Obvious cases are Pre-Sam. Ϝολαισυμος (< *°*omōs* < *-°*m̥(m)o-*;[8] personal name?) and Paleo-Umb. **setums** (< **septomos* < **septm̥(m)o-* 'seventh'; personal name). In Pre-Sam. Ϝολαισυμος, the letter ‹u› is used to contrast with ‹o› within the word, which exhibits a certain vocalic change affecting the original *-*o*-.[9] For the Paleo-Umbrian inscription (Um 4) with **setums**, it is to be noted that the letter ‹o›, lacking in the later Umbrian alphabets modeled after the Etruscan orthography (thus, no graphic differentiation between /o/ and /u/ in the older stratum in the Iguvine Tables), is clearly employed in **míom** 'me', another form in the same text. This fact suggests that Old Umbrian speakers in the sixth century B.C.E. were already aware of a reduced feature in the ‹u› of **setums**, which presumably explains why the inscriber did not choose ‹o› for the vowel.

SPi. **estuf** and **estufk** 'here' (as well as Paelig. *ecuf* 'id.' in a later period) can perhaps

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[7] Some Oscan personal pronouns in the dat.sg., such as **tíf[ei]** 'to you' and **sífei** 'to himself' [Cp 36] (cf. Lat. *mihĭ*, *tibĭ*, *sibĭ*), might be considered as cases of vowel reduction in labial context, juxtaposed to Umb. *mehe* 'mihi, to me', **tefe**, *tefe* 'tibi, to you', and Pael. *sef‹e›i* 'sibi, to herself' [Pg 4] with the original vocalism. Though the vowels in question are located in initial syllables, they are practically unstressed when the pronouns are cliticized to neighboring forms. As Buck suggests (1928: 56), the reduced pronunciation underlies the graphic omission of the vowel in Osc. **tfei** 'to you'. But vowel reduction in the pronouns is rather to be considered in a wider perspective as a specific behavior of cliticized words. See the Oscan 'be'-verb **íst**, which does not always follow a labial.

[8] For the reconstruction of the stage with *o*-vocalism before *-m-*, see 2.2.2.

[9] The -υ- in Ϝολαισυμος is most likely not a result of anaptyxis. Buck (1928: 51) asserts that "[i]n fact there is no example of an anaptyctic vowel developed through *m*."

be added here. The segment **f** is reconstructed as an apocopated locational particle **d^he* (cf. Ved. *kúha* 'where' OCS *kŭde* 'id.' Gk.-θε,) or **d^hi* (Gk. -θι); this fact indicates that the vowel **-u-** originally occupied the medial syllable. The stem is undoubtedly **esto-* (Umb. *esto*, etc. 'this'; cf. **eko-* for Pael. *ecuf* as well as *ecuc*, etc. 'id.'). But the South Picene (and Paelignian) forms reveal a certain vocalic modification of **-o-* in ‹u›; if **-o-* was preserved as such, it would be represented as ‹o› by default (e.g., SPi. **okreí** 'citadel' < PIE **$h_2$ôkri-*, **posmúí** 'who' < PIE **k^wo-*).[10] We may have to do with analogy with another unattested locational pronoun cognate with Umb. **pufe**, *pufe*, Osc. **puf** (perhaps, **puh**, as well?) 'where' (< **k^wu-b^hei*, cf. Lat. *ubĭ*) with original *u*-vocalism in the stem; that is, the South Picene (and Paelignian) counterpart **puf(e)* might have transformed **esto-* into **estu-** (as suggested by Vetter 1953: 153, Adiego Lajara 1992: 101, and Baldi 2002: 133; for Paelignian, **eko-* > *ecu-*). Yet, such a relationship between this demonstrative adverb and the interrogative-relative one does not seem to be so close-knit and compelling. Semantically more akin to **esto-* is the far-deictic pronominal adverb, which is in fact attested as Umb. **ife**, *ife* 'there' (< **i-bhei*, cf. Lat. *ibĭ*), and ***estif(k)* (or ***ecif*) would be a more plausible product, with the *i*-vocalism of the far-deictic form as a result of analogy (see Umb. **este**, *este* < **-i(d)* and Osc. **ekík**, Pael. *ecic* [nom./acc.sg.n.]). Selection of *u*-vocalism in favor of interrogative-relative forms seems to be somewhat far-fetched. Given this assumption, the vocalic change in question may well be ascribed to pre-labial (i.e., /__*f* here) vowel reduction.

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[10] SPi. **persukant** might represent /o/ with ‹u›. But this form is hard to etymologize; see various interpretations in Untermann (2000: 712-713). Recently, the reconstruction **soik-ā-* (← **seik-* 'erreichen'; see Meiser 1986: 88) seems to be gaining acceptance; see Maltzloff (2007: 180). If this pre-form is correct, **persukant** should be excluded from our discussion.

If South Picene already underwent pre-labial vowel reduction, a verb **adstaeoms** 'we stand(?)'[11] (< ***-o-mos***) would increase the list of examples, even if it preserves the original letter ‹o›, most likely concealing its phonetic reality, that is, its reduced nature. Further, word-finally, SPi. **esum** 'sum, I am' and perhaps Pre-Sam. **esum** 'id.' (< ***-om***, cf. Lat. **ESOM** on the Garigliano bowl) also exemplify this phonological change (cf. He. *esu* as later Sabellic evidence). We could add South Picene acc.sg. forms of the thematic declension, such as **múfqlúm** 'monument(?)', **meitimúm** 'gift(?)', and perhaps **anaiúm** and **raieimúm**. The original shape of the desinence is ***-om***, which is raised to /-o̯m/ and thus written as not **-om** but **-úm** (see Rix 1993: 338-339, Weiss 1998: 710, 712). I also regard this change as reflecting pre-labial vowel reduction.[12]

[11] Or in the perfect, continuing the thematic aorist; see Rix (1993: 337-338).

[12] The representation **-úm** instead of **-um**, unlike **esum**, is due to graphic analogy with other case forms in the same declension type that are written with **-ú-** by nature: e.g., dat.sg. **mefistrúí**, **titúí**, **brímeqlúí**, **qdufeniúí** (i.e., /-o̯i̯/). In addition, in some South Picene speaking-area, /-o̯i̯/ developed through */oi̯/ to /o̯/, written as **-oh**: e.g., **dúnoh**, **puqloh**, **petroh** (see Weiss 1998: 703-706). The existence of the lower *o*-vocalism in the inflectional paradigm may have phonetically impeded a more drastic vowel raising of */o/ towards the /u/ in the acc.sg.

Much the same can be found in Oscan. The acc.sg. in the thematic declension is represented as **-úm**/*-om* in earlier inscriptions (later, **-um**/*-um*; see Buck 1928: 117), which seems to align with, e.g, dat.sg. **-úí** for /oi̯/ (note the different use of the letter **-ú-** from South Picene), as Buck (1928: 37) suggests. On the other hand, infinitives in **-um**/*-um*, though historically accusative case forms of verbal nouns (i.e., ***-om***), orthographically exhibit vowel reduction from the early stage of the Oscan transmission, since they are morphologically petrified and thus escape paradigm leveling.

Umbrian regularly shows *-om* (not *-um*) in the Latin alphabet for acc.sg. as well as infinitives. The reason why it was not raised by vowel reduction like Oscan can be referred to the fact that the final *-m* in Umbrian is often omitted (e.g., *poplo* ~ *poplom*, as opposed to Oscan; see Buck 1928: 71). This implies that the preceding vowel was nasalized and phonetically lengthened (or lowered) so as to resist vowel reduction, which is somewhat parallel to the treatment of the Latin final *-om* (CIL I^2 9 **OINO**, **OPTVMO**, **VIRO** ≈ *-om*; see 2.1, and also foonote 7 of the same chapter).

4.1.2. Evidence of Vowel Deletion in Early Sabellic

As seen from the observation above, vowel reduction was already operative at a very early stage of Sabellic. Given the fact that it occurred only in non-initial syllables, both word-medially and word-finally, we can reach the conclusion that the initial-stress system was also already in effect.

Turning to vowel deletion in Sabellic, we are led to reconsider its history, particularly its occurrence in the oldest dialects of Sabellic. In regard to final syllables, the evidence clearly proves deletion: e.g., Paleo-Umb. personal name **setums**, SPi. **meitims** 'gift(?)' < *-*os*. We should also pay special attention to medial syllables, because syncope in this position has not been systematically discussed. In light of the positional synchronicity of Latin vowel reduction claimed in 2.1, the question may arise as to whether the Sabellic state of affairs was the same.

Buck (1928: 57) states that "with the limited amount of material before us, it is almost useless to speculate upon the original conditions of the syncope." In spite of this pessimistic remark, Benediktsson (1960) has conducted preliminary but very extensive research on syncope in non-initial syllables in Oscan and Umbrian with a fair deal of success. According to his theory (1960: 197, 230 n. 77, 244-45, 257, 281, 285), syncope in medial open syllables chronologically postdated syncope in final (necessarily closed) syllables. The main evidence he offers are forms like Oscan **túvtíks** 'belonging to the community' < **toutikos*; the vowel (*-*i*-) of the medial syllable was preserved, while the one (*-*o*-) of the final syllable was lost regardless of its closed-syllable structure (1960:

281). This case appears to justify the claim that syncope first removed the vowel in final syllable (*-*o*-), closing the preceding syllable, and thus impeded syncope there (i.e., **-í-** was preserved). Meiser (1986: 61-62) has also reached the same conclusion as Benediktsson.[13] In fact, medial syncope is most likely a common Sabellic (not Proto-Sabellic) phenomenon, judging from Umbrian phonological changes, such as intervocalic palatalization (see Meiser 1986: 131), which certainly preceded medial syncope chronologically: e.g, **struhçla**, etc. [a kind of cake] < ***struu̯i-kelā***. Meiser (1986: 132) concludes that medial syncope occurred from ca. 500 to the fourth century B.C.E. somewhat simultaneously in all the Sabellic dialects. Rix (1998: 254-55) states likewise that while final syncope was prehistoric, the medial case occurred about the fifth century, which is repeated in Rix (1999: 253-54).[14]

Notice, however, that direct evidence for very early medial syncope is available from the oldest Sabellic dialects. Pre-Samnite forms, o(vac.)fρι[-]qτο[δ and (σ)τα[.]ιοσqτοδ,[15]

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[13] One of the examples adduced by Meiser, Pre-Sam. *peracis* (Ps 3), is based on an arbitrary emendation, which is already refuted by Vetter (1953: 87); the exact form is *peraciam*. The emended *peracis* and its interpretation as a part of personal name is less likely; a set of three nom.sg. forms, *vinuχs veneliis peracis*, for a single person, causes an asymmetry with another name in the same text that only consists of two elements, *venilei viniciiu* (dat.sg.). The reason of the retention of *-a-* in the second syllable of *peraciam*, is readily provided by the fact that the form was a nominal composition (*per-ac-*) and was under an analogical control of the simplex (*ac-* in Umb. **peraknem**, etc., probably denoting the age of sacrificial animals). This is based on B. Vine's interpretation suggested in his course on the Sabellic dialects in 2007 at UCLA. The suffixal alternation, such as *-*iio-* vs. *-*no-*, is somewhat parallel to *pius* 'faithful' (< **puH-iio-*) vs. *pūrus* 'pure' (< **puH-ro-*). Thus, the Pre-Samnite form should not be included in our discussion.

[14] See also Rix (1993: 339). Götze (1923: 99, 106) advocates the opposite chronology, presenting a historical development such as **fou̯enis* > **fou̯nis* > Umb. *fons* 'propitious (except after *-s-*, as in Umb. **ferest** 'will bring' < **fereset*; see Götze 1923: 135). His etymologies are mostly obsolete, and thus cannot support his argument. He also oddly appeals to an accentual rule like the Latin Penultimate Law in order to explain syncope in final syllables, justly as criticized by Schmid (1954: 46).

[15] For the equation with Lat. *obescet* [P. F. 207.5L], see Martzloff (2007: 182-183). Though the deletion of the root *-e-* opposes the strong tendency toward recomposition, it is made plausible by Osc.

most likely based on *-*sḱe-tōd* (impv.), seem to show that medial syncope was already completed about 500 B.C.E.

Further, South Picene also provides very early cases of medial syncope.[16] In SPi. **aitúpas** (in TE 5, dated to 550-500 B.C.E.), normally compared with Umb. **eitipes** 'they resolved, decreed, enacted', if its interpretation as a compound is correct, the first element would be reconstructed as PItal. **ag-eto-* (> Proto-Sab. **ag^{y}eto-* > **ag^{y}to-* > SPi. **ait-**).[17] The historical development requires the deletion of **-e-* in the second syllable. TE 5 also provides **múfqlúm** 'monument(?)' as a possible candidate of syncope, if its Proto-Italic reconstruction **mone-stlom* (cf. Lat. *mōnstrum* 'portent') is correct.[18]

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amfret 'surround' (cf. Umb. *amferener* 'lead around').

[16] Though Meiser (1986: 131) interpreted SPi. **matereíh** 'mother' and **patereíh** 'father' as proving retention of medial vowels, their appearance is now generally credited to anaptyxis, parallel to Oscan and Paelignian (see Meiser 1987a: 121-122 n. 89, Vine 1993: 157 n. 69, Mercado 2006: 241-242).

[17] See Vine (1998: 81). Cf. Eichner (1993: 56 and n. 31) with **agit^{o}-* ("etwa mit Annahme von Haplologie **agitātà* > **agità-*"), inspired by G. Meiser, which is misquoted(?) as **aget^{o}-* by Martzloff (2007: 181). Untermann (2000: 73) suggests that the second element continues **habās* (whereby the vowel reduction **-a-* > **-ú-** before the labial is understandable), but given the fact that in TE 5 the letter **b** is in use in a different form (**trebegies** [something about a house? Or a personal name?; see Martzloff 2006: 114]), the use of **p** for this form would be mysterious. Like Umb. **eitipes**, usually reconstructed with **hēpens*, SPi. **aitúpas** might have also reflected **hēpās* with a perfect stem (in the subjunctive mood). But in this case, we need vowel reduction in the long vowel (**-ē-* > **-i̯-* > [ʉ/ɵ] ‹ú›?). This issue is reserved for future study.

[18] Cf. Meiser (1987a: 107, 1998: 66, 117): **mon(i)-/*mone-*, Eichner (1993: 51, and n. 20): **moni-/*mone-*, and Martzloff (2006: 113): **mon(i/e)s(k/t)lom*. Though there is much discussion about the history of Lat. *mōnstrum* (see various views in Eichner 1993: 51 n. 20), the derivational basis **mone-* (or **moni-*), well attested in Latin (e.g., *monumentum*, *monitor*, *monitus*), seems most likely to underlie it. See also footnote 18 in Chapter 2. Vine (1993: 128), adducing **me{n}fistrúí** as a case without syncope before *-st-* (+ liquid), argues that **mon-sklo-* (< **-stlo-*) is preferable to the reconstruction **mone-sklom* (or **moni-sklom*). But if one follows the recent etymology proposed by Martzloff (2006: 115-116 n. 44), that is, **mefi̯o-str-o-* « étendu au milieu » (← PIE **me-dhi̯-o-* 'middle' [Lat. *medius*] + **streh$_3$-* 'spread'), the South Picene reflex should presuppose syncope before *-st-*, as seen before the *-s*C(R)-sequence in later Sabellic (e.g., Osc. *minstreis* 'smaller' < **-is-tero-*; Umb. *eheturstahamu* 'banish' < **ē-tudes-t-ā-*; Umb. **mersus** 'lawful' < **medes-u̯o-s*; Umb. *mersto*, etc. 'propitious(?)' < **medes-to-* or **medos-to-*; and also Osc. μετσεδ < **medestēd*?) and Latin (see 3.3.4). If this is the case, the form can be added as evidence for early syncope to **aitúpas** and **múfqlúm**. Alternatively, if one adheres to the traditional interpretation of the form as a comparative, perhaps substantivized as the title of a magistrate (< **-is-tero-*; see Marinetti 1985: 121, Adiego Lajara 1992: 46; cf. Lat. *magister* 'master, commander', *minister* 'servant'), the retention of **-i-**

The diachronic interpretation of SPi. **mefiín** 'middle' (< Proto-Ital. **meðiio-* < PIE **me-dʰi̯-o-* + postpos. **-en* 'in') is also informative. From the metrical standpoint, this form is likely disyllabic, forming a heptasyllabic sequence with the following **veiat** (2 syll.) and **vepetí** (3 syll.) (see Martzloff 2007: 185-187). If this is the case, the **-i-** in **mefiín** should be consonantal, brought into contact with **-f-** by means of syncope **-Cii̯V-* > **-Ci̯V-* (see Martzloff 2007: 206); that is, something like /mẹfi̯i̯n/.[19]

SPi. **uelaimes** [personal name] can be adduced here, if it can be associated with Pre-sam. Ϝολαισυμος [personal name?] and Osc. *ualaemom* '*optimum*, best' / **valaimas** [personal name]. The historical development of these forms can be outlined tentatively as follows: PIE **u̯l̥H-* (Lat. *val-ēre* 'to be strong, well') > Proto-Ital. **u̯ala-isṃmo-* > **u̯alai̯somo-* > **u̯alai̯sumo-* (≈ Pre-Sam. Ϝολαισυμος) > **u̯alai̯smo-* > Osc. *ualaemom*/**valaimas** (≈ SPi. **uelaimes**). This scenario presupposes syncope of the vowel **-o-* (> **-u-*: [ʉ] or the like). The resulting consonant cluster **-sm-* (> **-zm-*) was simplified in the South Picene form as in the Oscan forms. The regressive nature of Early Sabellic cluster simplification is indicated by Paleo-Umb. **setums**, that is, **-pt-* (Lat. *septumus* 'seventh') > **-t-** (/tt/ may be underlying).

Syncope of such an age in medial syllables would lead us to reconsider the alleged chronological gap with syncope in final syllables. Benediktsson (1960: 281, 285) makes a

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in **me{n}fistrúí** would be explained differently. The base of the form is **me-dʰi̯-o-* (see above). The comparative suffix **-i̯os-*, which underlies **-is-tero-* in the zero-grade, is normally a root formation (cf. Lat. *min-is-ter* vs. *min-us*). Therefore, **me-dʰi̯-*, instead of **me-dʰ-*, may have served as the secondary root for the formation in **-is-tero-*. If this is the case, the Proto-Italic reflex would be **mefii̯-is-tero-* with **CiV* > **Cii̯V*. It contains two identical vowels before and after **-i̯-*; vowel contraction may have yielded *-ī-* (possibly with loss of the intervocalic **-i̯-*). Long vowels are not subject to deletion in Italic.

For SPi. **vepses** as a possible piece of evidence, see 4.2.1.

[19] Cf. Wallace (2005: 4): [meßyẹ̄n].

positional distinction between medial and final syllables — as Bader (1960) does for Latin — in terms of distance from initial stressed syllables. Benediktsson insists that unlike medial syncope, final syncope has no positional relationship to initial stressed syllables, adducing **húrz** 'hortus, protected place' (< **χortos*), **túvtíks** 'belonging to the community' (< **toutikos*), and **luisarifs** [month name] (< **-ifos*), in which syncope occurred in the first, second, and third syllable, respectively, after the initial, stressed one. Prosdocimi (1986: 611), on the other hand, believes in a close relationship between stressed syllables and syncopated final syllables, and assumes a linguistic stage in which stress always fell on penults, prior to the initial stress stage. For him, syncope in final syllables was triggered by immediately preceding, accented penults. Subsequently, when the initial accent came into effect, syncope took place in medial syllables immediately following it.

Though the opinions of these two scholars do not coincide, the adjacency of stressed and syncopated syllables is regarded by both of them as a crucial factor for syncope: by Benediktsson for medial syncope, and by Prosdocimi for both medial and final syncope. But as I said about Latin in 2.1, the assumption that only unstressed vowels directly next to stressed syllables are prone to change is not always true. Rather, we need to put more emphasis on the fact that both medial and final syllables were equally unstressed. This lack of stress seems to have triggered syncope in both positions, just as it caused vowel reduction in Latin. Benediktsson dismisses this factor in the case of final syncope, and thus mistakenly distinguishes this from medial syncope. Prosdocimi, on the other hand, has reconstructed an additional early linguistic stage, particularly in order to explain final

syncope. This is unnecessary because of the very early attestation of medial syncope.[20]

We already know at least one secure case, in which medial syncope operated in a syllable remote from the initial stressed one.[21] Among the rules of syncope Benediktsson proposes, the last one (1960: 234) claims that "... *in quadrisyllabic forms with a long vowel in the second syllable from the beginning and a short vowel in the final syllable, a short vowel in the second syllable from the end was syncopated, if the syllable was open*" (Benediktsson's italics). He neither explicitly explains the phonological background of this process, nor in fact provides any clear example, which might undermine his own statement. However, I proposed a case in which his argument could be to some extent verified (2005: 173-180), though it contains a diphthong in the second syllable (instead of a long vowel as Benediktsson formulated). Recall the following diachronic scenario: Proto-Ital. **u̯ala-isn̥mo-* > **u̯alai̯somo-* > **u̯alai̯sumo-* (≈ Pre-Sam. Ϝολαισυμος) > **u̯alai̯smo-* > Osc. *ualaemom*/**valaimas** (≈ SPi. **uelaimes**). The syncope (and vowel reduction as a preceding stage; see 4.1.1) in this case clearly shows that immediate adjacency between stressed syllables and syncopated vowels is of no importance.

Turning to Benediktsson's example, Osc. **túvtíks** < **tóutikos*, we now see that both unstressed short vowels in the medial and final syllables may be phonologically qualified

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[20] Rix (2004a: 949) seems to follow Prosdocimi's "penultimate" accent in order to explain apocope in Etruscan (including Lemnian and Rhaetic) as a Proto-Tyrsenic phenomenon, adducing **seχi* > *seχ* 'daughter' (cf. gen. *seχi-s*). But the attempt is not entirely well-motivated in view of what we have seen for Sabellic. In passing, disyllabic words like **seχi* are not informative for this matter; penults are in word initial position.

[21] Cf. Osc. cogn./gent. **núvellu̯m** ≈ **nou̯ello-* < **nou̯enlo-* < **nou̯ene-lo-*, based on Götze (1923: 84), with a slight modification. But cf. Benediktsson (1960: 210): **nou̯en-lo-*. Alternatively, it could continue **nou̯e-le-lo-* based on double diminutive suffix (Lat. *novellus* 'young').

The number of instances will be increased when the discussion in 4.3 has been completed. Prefixed nouns or verbs with four syllables or more must be excluded, since their simplex forms could exercise an analogical effect on them: e.g., Umb. **amparitu** 'build(?)' (not ****ampritu**).

for syncope. The selection of the vowel in the final syllable (*-*o*-) would be construed as a pattern of syncope in the case of two successive short vowels in penults and ultimas. The theoretically possible option **-*tks*, with both vowels syncopated, was phonotactically disfavored by the language.[22] It follows that the purported chronological differentiation between medial and final syncope has been made tenuous, and then replaced by a synchronically uniform deletion system. The suggested chronology is that medial syncope and final syncope were both prehistoric.[23]

4.1.3 Relative Chronology of Vowel Reduction and Deletion: Sabellic, Latin, and Etruscan

Such a historical perspective would affect some traditional views regarding a possible

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[22] Recall and compare some Etruscan instances, which result in complex consonant clusters (see Deecke 1878: 176, 186, Rix 2004a: 949): e.g, personal names *clutmsta*, *cluθumusθa* (← Gk. Κλυταιμνήστρα) and *Vestrcna* (< *Vestiricina*).

Also in Sabellic, especially in a later stage, there are several cases that yielded complicated consonant clusters via syncope (with subsequent reduction, if applied): Osc. Ϝερεκο '*publicum*, of the state' < **u̯erko-* < **u̯ergko-* < **u̯ergiko-* (cf. **vereias** < **u̯erĝ-iį̄ā-*); Osc. **prúftú(set)**, **prúftas** 'errected' < **prō-dʰh₁-to-* (cf. nom.sg.m. **prō-fat-s* < **-os*); Osc. **pupdiis** (~ **púpidiis**) / Pael. *popdis* [gentilicium] 'Popidius'. Umb. *todcom-e* (acc.sg.n. + postpos.), *totcor* (nom.pl.n.), *toce* (adv.), etc. 'belonging to the community' (cf. Osc. τουτικες ← nom.sg. **túvtíks**); Umb. **pupřçes** 'public(?)' (< **popliko-*(?); inscriber's mistake? cf. **pupřikes**, **pupřiçes**); Umb. **skalçe-ta** 'vessel, cup' (cf. nom.sg. **skaliks*). Such changes seem to have taken place so haphazardly and marginally that we could not reach a generalization.

[23] Just as Meiser (1986: 131) suggests, Moorhouse (1940: 320) claims that it is not until the Umbrian palatalization was completed that syncope took place (otherwise, we would have ****tiklu** instead of **tiçlu**?). On the other hand, appealing to the fact that Umb. **teitu** 'say' and **feitu** 'sacrifice' (impv.) continue **deiį̄ketōd* and **fakitōd* (cf. **dʰē-iį̄e-* >**fē-iį̄e-*, presented by Untermann 2000: 258-259), respectively, instead of ending up as ****teçtu/façtu** or the like, he argues that syncope in imperatives took place earlier than the palatalization (1940: 320, 326). His entire scenario is as follows: syncope in impv. → **-k-* > **-ç-* → syncope of *(k)elo*-suffix. Though different stages of syncope are theoretically possible, there is no strong evidence for differentiating between these two processes of syncope in conditioning terms. Rather, as Meiser (1986: 124) suggests, palatalization at the stage of **deiį̄ketōd* and **fakitōd*, changing **-k-* into **-k'-* (i.e., [kʸ]), can phonetically explain the later emergence of **-i-** in pre-consonantal position of **teitu** and **feitu** by means of [ʸ]: e.g., **deiį̄ketōd* > **deiį̄kʸetǭd* > **deiχʸtǭd* > **dēχʸtu* > **teitu**.

association between the Italic and Etruscan accentual stress systems. We can maintain that the terminus ante quem for both word-medial and word-final syncope is approximately 550-500 B.C.E.

This chronological evaluation of syncope implies that the Sabellic dialects had the initial-stress system before that period. This may seem to be earlier than has traditionally been thought. The conventional view of a relatively recent origin of the accentual rule is associated with Etruscan. Latin has also been involved with this picture. Among early researchers, Skutsch (1913: 190) is well-known as an advocate of the idea that the initial-stress rule in the Italic languages was a result of Etruscan influence. Sommer (1914: 86), while tentatively suggesting the Proto-Italic origin of the rule, treats Skutsch's historical scenario favorably. More recently, with the premise that the Italic languages originally placed accent uniformly on penults, Prosdocimi (1986: 612, 614) suggests that initial accent was imported from Etruscan, which subsequently caused vowel reduction and deletion. Then, Latin mostly restored its "original" penult accentuation as Etruscan influence gradually faded out, whereas Sabellic did not, due to the more serious impact from Etruscan. The reason why the reference to Etruscan has been appreciated is that vowel reduction in Etruscan is older than Latin, where it was not yet operative around 500 B.C.E. In Etruscan, on the other hand, symptoms of vocalic changes are already observed then. The distinction of its four phonemic vowels, /i, e, a, u/, is clear in initial syllables. In non-initial syllables (especially in medial ones), in contrast, the vowels became continuous targets of reduction and deletion throughout the history of Etruscan. As Cristofani (1991: 40-42) describes, the inscriptions from Crocifisso di Tufo (Orvieto),

which date to the second half of the sixth century B.C.E., show graphic instability in some nouns; *e* ~ *i*, in medial syllables, as in personal names *Larece* ~ *Larice* and *Avele* ~ *Avile*. Cristofani interprets this as neutralization of phonemic contrast, suggesting that the phonological inventory in that position was simplified to *a* ~ *u* ~ *i/e*. Over the course of time, the neutralization of vowels in medial syllables proceeded further. As early as the fifth century B.C.E., orthographic haphazardness became rampant: e.g., *Artam* ~ *Artumes* (← Dor. Gk. Ἄρταμις); *Aχale* ~ *Aχile* ~ *Aχele* ~ *Aχule* (← Gk. Ἀχιλλεύς); *Avile* (orig.) ~ *Avale* ~ *Avele*; *Ramuθa* (orig.) ~ *Ramaθa*. As Cristofani continues, the phonemic differentiation of vowels was entirely lost in medial position, and their phonetic realization was thus most likely a schwa sound. At about the same period (between 500 and 470 B.C.E.; see Rix 2004a: 950 and also 2004b: 437), or shortly afterward (from the fourth century B.C.E. onward; Cristofani 1991: 41-42), reduction took an extreme form as vowels were deleted;[24] e.g, *Aχle*, *Avle*, *Ramθa*. Examples may be supplemented with the following: [[praenomina]] *Aranθ* > *Arnθ*; *Venel* > **Venl* > **Vell* > Cortona *Vêl*; [[family names]] *Larecene* > *Larcna*; *Scanesna*, *Scanasna* > *Scansna*; *Spuriena* > *Spurina*; [[others]] *Leθamsul* > *Leθnsl* [theonym]; *muluvene* > *mulune* 'gives as a gift'; *turuce* > *turce* 'sacrificed'; *vacil*, *vacal* > *vacl* 'then'.

This occurrence of vowel reduction and deletion in non-initial syllables in Etruscan earlier than Latin may indicate that the initial-stress system spread from the former to the latter, where non-initial syllables were subsequently affected by these sound changes due

[24] This argument is commonly accepted; see, e.g., Penney (1988: 723), with some of the above examples. According to Penney (1988: 724), the retention of the vowel as in *Atunes* 'Adonis' suggests that vowel length may have still been distinctive in borrowings in as late as the fifth century B.C.E. (and perhaps in Etruscan itself).

to the new accentual rule. However, the same scenario does not seem to be entirely applicable to Sabellic. As stated above, vowel reduction and deletion in Sabellic may arguably be dated to as early as 550-500 B.C.E. or earlier. It is only around this period of time that vowel reduction came into effect in Etruscan. If one insists that Sabellic also acquired the initial-stress rule from Etruscan, this must have been realized in a far more ancient stage, perhaps the eighth or seventh century, in which case the vocalic changes took place at almost the same time as they did in Etruscan or even earlier. This view is theoretically possible, but chronologically rather less robust than the one for Latin. A more natural scenario would be that Sabellic developed the initial-stress system on its own; though some other substratum or adstratum influence is still possible, this is beyond our knowledge and should be put aside. Vowel reduction and deletion are thus to be attributed to an inner-Sabellic motivation. If this view is correct and the initial-stress rule in Latin is still of foreign origin, not only Etruscan but also Sabellic should be taken into consideration as a candidate, though I assume that it is hardly possible to determine which of these.

4.2. Phonological and Phonetic Background of Vowel Deletion

4.2.1. Phonemic Qualifications for Vowel Deletion

The first step of vowel reduction in Latin (Process I) targets */e, a, o/. Vowel deletion in Sabellic also affects the same phonemes. Examples include: [[loss of */e/]] Osc. **meddíss**

'meddices [title of the chief magistrate]' (< *-*es*); Osc. *actud* (3.sg.fut.impv.) 'plead a case' (< PItal. **ag-e-tōd* < PIE **h₂eĝ-e-*; cf. Lat. *agitō*); Umb. **heries** [with final *t*-loss] < **-s-e-t* 'want' (3.sg.fut.); [[loss of */a/]] Umb. **ampentu**, etc. 'slaughter(?)' < PItal. **-pelnatōd* < PIE **-pel-n-h₂-e-*; Osc. **patensíns** '*aperirent*, open' (3.pl.subj.impf.) < PItal. **pₔt-na-s-ē-nt* < PIE **pt-n-h₂-s-eh₁-nt*; [[loss of */o/]] Osc. **húrz** 'protected place' (< PItal. **χortos*); Osc. **túvtiks** 'belonging to the community', Umb. **fratreks** 'belonging to the brotherhood' (< **-ikos*).

The high corner vowels */i/ and */u/ in Latin were relatively stable until Process II became effective. The behavior of the Sabellic counterparts seems to have been different to a certain extent. The high front */i/ was highly prone to deletion like */e, a, o/: e.g., Osc. **alttram**, etc. 'another' < **ali-tero-*; Osc. *ceus* 'citizen' ← Lat. **ceu̯is* (= CIL I^2 581.7 **CEIVIS**) > *cīvis* (see Untermann 2000: 395); Osc. **puz**, etc. 'ut, that' < Pre-Osc. **puti-s* < **k^{w}ut-* (cf. **k^{w}ut(a)* + *s* in Benediktsson 1960: 222-223); Osc. **serevkid** 'auctoritas, supervision' < **seru̯-ik-iu̯o-*; Umb. **struhçla**, etc. [a kind of cake] < **struu̯i-kelā*; Osc. **vereias** '*res publica*, state' ≈ *u̯ereiu̯ā-* < **u̯eregu̯ā-* < **u̯ergu̯ā-* < **u̯erĝ-iu̯ā-*, Osc. Ϝερεκο '*publicum*, of the state' < **u̯erko-* < **u̯ergko-* < Proto-Sab. **u̯ergiko-* (see Rix 1999: 253-254, 2000b: 219); Pael. *hospus* 'stranger' < Proto-Ital. **χosti-pot-s*. The list can be readily extended.

On the other hand, retention of the high back */u/ has traditionally been posited (by, e.g., Bugge 1854: 421, von Planta 1892: 218, 231, 1897: 158, Brugmann 1897: 216, and Buck 1928: 59; but cf. Bronisch 1892: 192). However, evidence for this claim, especially for */u/ in final syllables, seems to be mostly based on a possible typological parallel in

Germanic, and a more substantial discussion is entirely lacking. In addition, the oft-cited examples, such as Osc. *sipus*, Volsc. *sepu* 'be aware of a fact', and Osc. *facus* '*factus*, be made' should now be explained differently (Proto-Sab. **fak-u̯o-s* and **sēp-u̯o-s*, respectively, as already in Buck 1928: 59; see also Benediktsson 1960: 223, Untermann 2000: 258-259, 677). Whereas it is hardly possible to determine what happened in final position, /u/ in *medial* syllables is of great interest for this matter. Notice that there is no entry for /u/ under "Syncope in Medial Syllables" in Buck (1928: 57-58). In light of our current understanding of historical morphology, considering the texts available at the time of the production of Buck's work, the behavior of /u/ emerges as a phonological peculiarity because of its tendency not to syncopate. The materials are unfortunately all from Umbrian, and thus a generalization to the other Sabellic dialects must be reserved for future discovery of texts or new analyses.

Benediktsson (1960: 223-225), who is skeptical of *u*-retention, explains the reason for the *u*-vocalism in Umb. **tekuries** / *dequrier* (cf. Lat. *decuria* 'group consisting of ten men') by positing a consonant cluster, C + yod (here, the stem-final *-r-* + the suffix-initial *-i̯-* [represented as **-i-**]), which supposedly closed the preceding syllable containing *-u-*. But it seems more likely that this form was derived by means of the disyllabic suffix **-ii̯ā-*, the feminine "Zugehörigkeitsuffix" **-ii̯o-* to create an abstract noun (cf. Lat. *pater* 'father' → *patrius* 'belonging to a father' → *patria* 'one's native land'), with the preceding syllable having remained open (i.e., *$^{\circ}$*ur-ii̯ā-*). If this is the case, we must consider a different factor in order to explain the retention of this vowel. This is particularly so, since the vowel is followed by **-r-**, a sonorant which frequently occasions

vowel deletion (e.g., Osc. **alttram**, etc. 'another' < **ali-tero-*; Umb. **supru**, **subra**, etc. 'above' < **superā-*; Umb. *destram-e*, etc. 'right' < **deĥsi-tero-*). The conservative nature of *-u-* thus comes out to our attention.

An Umbrian theonym *piquier* also seems to prove the persistence of *u*-vocalism. Though it can be ultimately traced back to **piku-* 'woodpecker' + "Zugehörigkeitsuffix" **-iio-* (i.e., **piku-iio-*), it may have been reshaped as **piku-io-* (≈ *piquier*; see Untermann 2000: 556) at some point in the history of Umbrian (this issue will be resumed in 4.3.1; see also footnote 41). We can see that the *-u-* located in open syllable resisted deletion.

The same is perhaps true of an Umbrian adjective of a tribal name **iapuzkum**, etc. (< **iapudisko-*?).[25] It is true that the non-application of expected spirantization of **-d-* (> *-ř-*) is open to question, as Benediktsson (1960: 226) suggests. In addition, the origin of proper nouns is always problematic. However, it is at least plausible that the sequence **-zk-** (voiced + voiceless) is a secondary cluster and was originally separated by a vowel (see Götze 1923: 102). If this is the case, the **-u-* in an open syllable remained intact, while a vowel, perhaps **-i-* in the third (probably closed) syllable was syncopated instead.

Another proper name, Umbrian gentilicium **kastruçiie** (< **kastru-k-iio* ≈ Lat. *Castrucius*; cf. Osc. **kastríkiíeís** ≈ Lat. *Castricius*) also suggests *u*-retention. Benediktsson's explanation for this form is entirely inconsistent (1960: 225, 269, 276); he argues in one place that the vowel is located in an open syllable, while saying the

[25] Muller (1916/17: 206) assumes stress on *-u-* to explain syncope in the following, third syllable. Such a claim on the adjacency between stressed syllables and syncopated syllables has already been refuted several times in this work (see 2.1 and 3.3.6).

opposite elsewhere. The syllable must be open before the suffix *-*iio*-.

All of these examples seem to consistently prove that the phoneme */u/ resisted deletion at least in medial syllables. Because of the scant amount of Sabellic data as well as the intensive attestation in Umbrian, we should not deduce an absolute rule of *u*-retention.[26] In fact, there is a seeming counterexample in South Picene. The form **vepses** in TE 2 has been variously interpreted by scholars.[27] In terms of the syntax and semantics of the text, Meiser (2003: 47-49, 162) properly retains his old view that it is a pf.ppl. (or "so-Partizip") in the gen.sg., continuing **lik^{w}-us-eis* (or **lik^{w}-s-eis*) 'left behind'.[28] More recently, based on a detailed semantic analysis, Martzloff (2007: 182) attractively reconstructs the form with a different root (**uleik^{w}-* 'moisten') as **uleik^{w}-us-/*ulik^{w}-us-* « apprêté pour le tombeau », « paré des atours funéraires ». In any case, this form may justify the claim that the retention of */u/ deduced from the observation above is a mere tendency. This is an entirely acceptable and cautious attitude. But I tentatively raise the question as to whether the deletion of *-*u*- here was really phonological or not. As claimed by Watkins (1995: 131-132), followed by Martzloff (2007: 178, 184-187), TE 2 is a highly poetic text. The use of alliteration is conspicuous: **viam - videtas**, **tetis - tokam**, **alies - esmen** (with vowels), **vepses - vepeten**. And the colometrical aspect is elaborate, with a triple heptasyllabic ("2syll. + 2syll. + 3syll.") sequence. It may thus be possible that the composer of this text needed to compress

[26] The etymologies of Umb. **nuřpener** and **tribřiçu** as **nouo-dupendio-* 'new coin of two pounds' and **tribu-dikion-* 'order referring to three cows', respectively, supported by Heidermanns (2002: 191-193, 197-198), might be possible counterexamples. But these are still speculative reconstructions.

[27] For a concise summary of previous study of this form, see Martzloff (2007: 180-182).

[28] Followed by Rix (1993: 337, 1996: 243).

u̯epuseiṣ* by one syllable to fit in the metrical scheme. In fact, to judge from the persistent presence of *-u-* in morphologically related forms, such as Osc. *sipus*, Volsc. *sepu*, and Osc. *facus* (see above), it is highly expected that the segment was generalized as the marker of pf. ppl. throughout the entire paradigm; i.e., nom.sg. **u̯epu̯os* > **u̯epus* → gen.sg. **u̯epuseiṣ*, instead of **u̯epseiṣ* (cf. nom.sg. **toutikos* > Osc. **túvtíks → gen.sg. **tou̯tikeiṣ* = τουτικες, instead of **tou̯tkeiṣ*; see footnote 36). It seems then that a non-phonological factor, a metrical one, was involved in the deletion in the word. If this argument is somehow correct, **vepses** can be left aside from our discussion, and then the conservative nature of /u/ is solidified.[29]

My tentative suggestion regarding the special behavior of */u/ refers to its inherent phonetic feature, that is, [+round]. This may have served as a factor blocking vowel deletion.[30] One might claim that /o/, prone to syncope, is also a rounded vowel. To avoid this criticism, I appeal to the fact that vowel deletion may be an extreme form of vowel reduction. If deletion of /o/ as well as /e, a/ (and perhaps /i/ as well?) was preceded by reduction process neutralizing these vowels into [ə] or [i̯] and if */u/, on the other hand, remained intact, */o/ would be differentiated from */u/ in unstressed syllables in terms of

[29] The Paleo-Umbrian form in Um 2, usually transcribed as **heduseí**, is tentatively read as **heruseí** by Rix (1996: 242-243). He interprets this as the pf.ppl. of *her-* 'want' (PIE **ǵʰer-*; see footnote 42), which he prefers to the alternative, the pf.ppl. of PIE **gʰed-* 'receive', in case the form is read as **heduseí**. In any case, his interpretation of the form as a pf.ppl. is based on one speculative thought after another, and thus highly dubious. In fact, Rocca (2001: 129) construes **heduseí** as a woman who was given the flask with the inscription. If it is her name, it would be hard to determine even the quantity of the **-u-**.

[30] This assumption may agree with the historical process of Pael. *pros* 'honest' (Lat. *probus*) based on the thematized **-bʰu̯-o-s* (> **-bʰos* > **-fos* > **-fs* > *-s(s)*; for the loss of **-u̯-*, see Buck 1928: 67) not the *u*-stem **-bʰu-s* (cf. Ved. *prabhú-* 'outstanding'). See also Martzloff (2007: 181 n. 11). Note incidentally that in Latin, *u*-stem adjectives were generally remaded as *i*-stems: e.g., Ved. *tanú-* 'long', Gk. τανα(Ϝ)ός 'outstretched' vs. Lat. *tenuis* 'thin'. Thus, it would be less likely that the *u*-stem **-bʰu-s* remained intact in Latin and probably Italic.

the [round] feature. It is noteworthy to cite a question raised and briefly answered by Zuraw (2002: 390): "is a front vowel less sonorous than a back vowel? Under the sense of sonority used here — inherent duration — */o/ and */u/ could indeed be treated as more sonorous than the front vowels, because of their rounding gesture." In the case of Sabellic, only /u/ fits this description, since we hypothesize that */o/ was reduced to [ə] or [ɨ].

4.2.2 Vacillation in Application of Vowel Deletion in South Picene

In 4.1.2, we saw the oldest examples of medial syncope, such as Pre-Sam. o(vac.)fρι[-]qτο[δ, (σ)τα[.]ιοσqτοδ, SPi. **aitúpas**, **múfqlúm**, **mefiín**, and **uelaimes**. Further investigation of the corpus, particularly of South Picene, however, would complicate the state of affairs, because there are a few forms that retain medial vowels; **opesạ[ ]úom** and **erimínú**.[31] Though both of them are not entirely clear in terms of reading or etymology, we should not simply ignore them. A provisional solution will be offered.

The denominative verb **opesạ[ ]úom**[32] 'produce, erect' may be based on the stem

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[31] Only on the condition that the equation of **panivú** with Lat. *quamdiū* '(as) long as' (< *k*^{w}*ām-diu̯ou*?) by Meiser (1987b: 118), tentatively followed by Untermann (2000: 512), is accepted, does the retention of **-i-** matter. Recomposition may be responsible for it. But the initial consonant (*-*d*-) of the second element was completely assimilated to the final one (-*n*- < *-*m*-) of the first element, which had already been assimilated in the reverse direction to become a dental nasal. This can be construed as reflecting close univerbation. Therefore, the retention of **-i-** must be explained by another factor. What I suggest for **opesạ[ ]úom** and **erimínú** may be applied to **panivú**.

[32] Note that the vowel in the final syllable does not show vowel reduction graphically, that is, **-o-** is retained in place of the expected **-u-** (cf. **esum**). This may be due to some phonetic function of the preceding **-ú-** [w]; cf. the notation *servos* (vs. *servus*) that survives in Classical Latin.

opesā-* common in Italic (cf. Lat. *opus* 'work', *opera* 'effort').[33] This is taken over by many of the later Sabellic dialects, but **-e-* is regularly deleted: e.g., Umb. *osatu*, Pael. *upsaseter*, Osc. **úpsannúm. The other form **erimínú** is interpreted as a theonym in the dat.sg. by Meiser (1986: 117), who tentatively and somewhat attractively compares this form with Mars. *erine* (< **-nn-* < **-mn-* < **-m*V*n-*). Though these two forms alone might not represent a strong counter-argument to the presence of medial syncope in South Picene, recall that the evidence for it is also scant, as is expected in studies on Sabellic. In order to explain the retention of the medial vowels in **opesạ**[ ]**úom** and **erimínú**, it may be worth mentioning the phonological context where they are preserved, that is, labial context. As we saw above (4.2.1), *u*-vocalism shows a remarkably low frequency of deletion, probably due to its relatively high prominence caused by the rounding gesture. If the labials in these two forms provided a certain degree of roundedness to the following vowels (cf. the post-labial vowel reduction in 4.1.1), it is not far-fetched to suggest that the deletion process was somehow blocked (or delayed, from a historical standpoint).[34]

One might ask why the vowels are not represented with ‹u› or ‹o›, frequently observed in cases of vowel reduction, as shown in 4.1.1. But we have already seen some haphazard representations of reduced vowels in a labial context. Recall some examples

[33] For the stem structure of SPi. pf. **opsút**, see Rix (1993: 340), appealing to the nomen actionis **h₃ōp-* (← **h₃ep-* 'produce') followed by the aoristic *-s-*. If this interpretation of suppletion between pres. and pf. is correct, the cluster **-ps-** in **opsút** is not a product of syncope.

[34] As a possible parallel, the Palauan data provided by K. Zuraw (p.c.) may be useful; between the high vowels /i/ and /u/, the former reduces to zero more often than to schwa, whereas the latter reduces to schwa more often than to zero. It may be that the [±round] feature is responsible for this distribution. (Note, however, that both mid vowels /e/ and /o/ show the same tendency to reduce to schwa rather than to zero; the vowel height may also play a role.)

with front vowels, as Osc. **nessimas**, Umb. *nesimei* 'nearest', and Lat. **DECIMIO** and **DECEMIA**. Even if these continue a back vowel *-*o*- in medial syllables, they exhibit the reduced reflex by means of ‹i› or ‹e› (cf. Osc. *pertumum* < **pert-emom*). If the medial vowels after labials in **opesa̧[ ]úom** and **erimínú** continue what historically were front vowels, it may be that reduced and rounded outcomes were [ø] or [y], and it is *a fortiori* no wonder that inscribers chose ‹i› or ‹e› for the rounded front vowel.[35]

The discrepancy between complete and incomplete syncope in South Picene may reflect that syncope was an on-going process or completed right before the earliest period of attestation. Thus, in the later Sabellic dialects, vowels in labial context are almost regularly deleted: e.g., Osc. **prúffed** < **pro-dʰe-dʰh₁-e-t*, **prúftú(set)** / **prúftas** < **pro-dh₁-to-* 'errected'; Osc. δεκμας 'tenth' [personal name?] (cf. Lat. *decimus* 'tenth', Sicul. δεκομο, Mess. **dazimas** [personal names]), Osc. **Dekmanniúís** '?' (cf. Lat. *decumānus* 'related to that which is tenth'); Osc. **pustm[as**, *posmom* 'last' < **pos-tm̥mo-*; Osc. **lúvfreís** [epithet for Jupiter] < **h₁leu̯dʰero-* (cf. Pael. *loufir* 'free man'); Osc. *cebnust* 'come there' < **-ben-* ← **gʷem-* (cf. Umb. 3.sg.fut.pf. *benust*); Osc. **embratur** ← Lat. *imperātor* 'commanding officer'; Umb. *onse* 'shoulder' < **omeso-* < **h₃em-es-o-*; Umb. **struhçla**, etc. [a kind of cake] < **struu̯i-kelā*.[36]

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[35] Cf. the development **-ri-* > *-er-* in Latin. As generally stated, it failed to occur in pre-labial context: e.g., *tribus* 'tribe' and *lacrima* 'tear' (see Nussbaum 1999: 394-395). See also Umb. **trifu** 'district' (vs. Umb. *tertim* 'third' < **tri-tiio-*), which is mentioned by Rix (1996: 237-238). Though this change took place regardless of whether it was stressed or not (see 3.4), the retention of the sequence could be attributed to the phonetic function of labial sounds.

[36] Similarly to the forms cited in 4.1.1, we might think that reduced vowels in a labial context tend to be retained, as opposed to the general tendency of syncope in Sabellic, since their original timbres, /e, a, o/, are all qualified for regular loss. However, the retention of medial vowels in other cases is explained by analogy with nom.sg. cases; with **nessimos* > **nessims*, for example, the vowel *-*i*- was then generalized

However, there are several forms with retained medial vowels in the later stage of Sabellic, which have never been considered in connection with vowel reduction and deletion. The phonetic feature [round] may shed light on these relics of old rounded vowels in a labial context, particularly before *-u̯-*, still defying deletion. Umbrian is a resource in such cases. The theonym *fisovio-* and the epithet to several divinities *grabovio-* retain the medial vowel *-o-*. Benediktsson (1960: 268) notes that their oblique cases (except unattested acc.sg. in **-im*) should have had a consonantal cluster *-u̯i̯-* (*°u̯-i̯o-*), which closed the preceding syllable and may have hindered syncope of *-o-*. These forms, however, are better reconstructed with the disyllabic "Zugehörigkeitsuffix" **-ii̯o-*, which makes *-o-* located in open syllable (i.e., *°o.u̯i.i̯o-*). The retention of the vowel can thus be explained by appealing to the labial, which reinforced the foregoing vowel with the [+round] feature. Umbrian ethnonym **ikuvins**, etc. 'Iguvinus' (< **-īnos*) can be added here for the retention of **-u-**.[37] Such a preventive effect of *-u̯-* would

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to the entire paradigm (Osc. **nessimas**, Umb. *nesimei*). This process also explain Osc. **últiumam**, Umb. *hondomu*, Pre-Sam. Ϝολαισυμος, Paleo-Umb. **setums**. (The retention of **-e-** in SPi. **vepet-** 'tomb' may be explained in the same way.) In SPi. **estuf / estufk**, the vowel was also analogically strengthened by the presence of the vowel in cognate pronouns (SPi. **estas**, Umb. **estu**, etc.). For SPi. **adstaeoms**, the thematic vowel was perhaps under analogy with other finite, especially singular forms (e.g., Vest. 3.sg.pres. *didet* 'give'; but cf. Osc. 3.sg.pres.pass. *uincter* 'overcome' < **-e-ter*).

Umb. theonym **puemunes**, if it is reconstructed as **po-emo-*, retains **-e-**. Benediktsson (1960: 262) attributes this retention to an effect caused by hiatus (i.e., **-u-** + **-e-**). However, one might claim that syncope in hiatus is possible, with recourse to Oscan **eestínt** 'exstant, exist' (< **-sta-ē-*). More likely, if **puemunes** was still regarded as a compound by Umbrian speakers, the retention of **-e-** could be due to recomposition.

[37] Umb. *saluuom* (acc.sg.m./n.) and *saluua* (acc.pl.f.) 'unharmed', reconstructed as **sḷh₂-eu̯o-* (> **-ou̯o-*) or **sḷh₂-u̯o-* (see Nussbaum 1997: 186-187) does not lose the vowel (first *-u-*) before the labial (i.e., second *-u-*, which is consonantal *-u̯-*), though **-o-* is realized as *-u-* by either assimilation to or reduction (see Solmsen 1894: 152) before the following *-u̯-*, in contrast to *fisovio-* and *grabovio-* (to which the conservative phonological behavior of proper nouns is probably responsible). Other attestations of these and their different case forms, such as *saluom*, *saluo* (acc.sg.m./n.), *saluam*, *salua* (acc.sg.f.), *salua* (acc.pl.f.), compress the sequence *-uu̯-* (< **-ou̯-*; see Meiser 1986: 116) into a single ‹u›, a common practice of notation in Latin (see Solmsen 1894: 158; e.g., ; e.g., SENATVOS 'assembly' [gen.sg.] in CIL I^2 581, cf.

contrast with the regular vowel deletion before -*i̯*-; e.g., Umb. **plenasier** /plēnasi̯er/ [a specific time of a month?] < **plēnāsii̯o-* (see Meiser 1986: 63), where rhotacism was blocked, assuming that -*s*- was flanked by -*i̯*- via syncope.[38]

4.3. Syncope in the Position Adjacent to *-*i̯*-

4.3.1. Remote Syncope in the Suffix *-ii̯o-/*-ii̯ā-

Throughout the discussion, we have occasionally seen reconstructions with the disyllabic "Zugehörigkeitsuffix" **-ii̯o-* (or f. **-ii̯ā-*). Though Untermann's indispensable etymological dictionary of the Sabellic dialects (2000) offers a lot of forms reconstructed with a monosyllabic **-i̯o-* (or **-i̯ā-*), most of them should be construed as stemming from **-ii̯o-* via medial syncope. Recent studies in Indo-European linguistics have demonstrated that the monosyllabic suffix **-i̯o-* can be reconstructed only marginally in a very limited number of forms, such as Gk. μέσος 'middle' (< **me-dʰi̯-o-*).

Syncope of **-i-* brings **-i̯-* into contact with the preceding consonant, which may

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CLat. *senātūs*), which is also seen in Umb. *castruo* 'heads(?)' (cf. **kastruvu(f)**), *uatuo* 'ribs(?)' (cf. **vatuva/vu**), **ivenka** 'cow' (cf. *iuengar*, *iuenga*), Osc. παϰϝηις [praen.], Mars. *pacuies* [gent.] < **pākou̯ii̯os* (≈ Lat. *Pācuvius* [gent.]; cf. *Pācius* [gent.]; see Schulze 1966: 475-476). Though the labial -*u̯*- may have caused the retention of the vowel, it can also be explained in the same way as nom.sg. **nessimos* > **nessims* → Osc. **nessimas**, Umb. *nesimei*, that is, nom.sg. **saluu̯os* > **saluu̯s* → Umb. *saluuom*, *saluua*.

[38] The retention of the vowel before -*u̯*- is incidentally associated with some other similar cases, where a vowel ‹u› appeared secondarily before -*u̯*- and somewhat perplexed Solmsen (1894: 153) and Brugmann (1897: 233): e.g, Umb. **mersuva** 'lawful' (< **medes-u̯o/ā-*), Osc. *eituam*, **eítiuvam** 'money' (< **ei̯-tu̯ā-*?), Umb. **aruvia** 'viscera(?)' (only in TIg. III 31; cf. more frequent forms **arvia**, **arviu**), and possibly Umb. **kastruvu(f)** and **vatuva/vu** (hard to etymologize in order to know whether **-u-** is secondary or not). Meiser (1986: 196) suggests that such forms are analogical to forms in *-uu̯o/ā-*. This process is more accurately restated in phonotactic terms; that is, -C*u̯*V- came to be disfavored.

trigger a sound change, especially palatalization. This process can be assumed in the following forms: Umb. **vestiçe**, *uestisier* [theonym] < *u̯estik-iio-*; Umb. **vestiçia**, etc. 'libation' < **u̯estik-iiā-*; Osc. *famelo*, Umb. **fameřias** 'household' < **famel-iiā-*; **uhtretie** 'auctorship' < **-et-iio/ā-*; Osc. **pumperias**, etc., Umb. **pumpeřias** [a certain day of the month] < **-er-iiā-*; **tehteřim** '?' < **-ed-iio-*(?) (cf. Benediktsson 1960: 197-198 n. 44); Osc. **vítel(l)iú** [toponym], **fistelú** [ethnonym] < **-el-iiā-*; Osc. **teremen(n)iú**, 'boundary-stone' < **-men-iiā-*.[39] We can also add Umb. **plenasier** /plēnasi̯er/ [a specific time of a month?] < **plēnāsiio-* as an example that shows rhotacism blocked by the contact of **-s-* and **-i̯-* (see also 4.2.2). It is often stated that syncope took place in second syllables immediately after initial stressed syllables, but these examples prove the existence of distant syncope as in Latin (see 3.3.6).[40] For some of them, the retention of vowels in second syllables might be ascribed to analogy with base forms (e.g., **fameřias** ← cf. Pael. *famel*; **pumperias** ← **pompe* 'five'). But this did not block syncope in syllables further on in the word. The choice of the **-i-* for syncope may have partly been due to the avoidance of resulting complex consonant clusters (cf. ***u̯estkiiā-*). In view of the way **-iio-*/**-iiā-* developed, it is plausible that after syncope, the suffix became productive in the form of **-i̯o-*/**-i̯ā-*.[41]

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[39] Lat. *Marsī* [ethnonym] reflects a dialectal, i.e., Sabellic treatment of *-s-*, as Ernout and Meillet (1985: 388) suggest: **martiei̯* '(people) belonging to Mars' > **martī̯* > *Marsī*. The contact between **-t-* and **-i̯-*, presupposed for *-s-*, was realized by means of syncope of *-i-*.

[40] Cf. Lat. *prāndium* 'meal eaten about midday' (< **prām-ed-iio-* ← PIE **pr̥h₃-mo-* 'first' + **h₁ed-* 'eat') with syncope in the second syllable. See Rix (1996: 236, and n. 13).

[41] This sequence beginning with yod seems also to be retained after a vowel. Osc. **kerríiúí** 'belonging to Ceres', though reconstructed as **keres-i̯o-* by Untermann (2000: 390), would rather continue **kersē-i̯o-* (reshaped from **keresē-iio-*), based on the presence of **-í-** (< **-ē-*). Benediktsson (1960: 170 n. 10) suggests that the segment **-ē-* was analogically taken over from Osc. **kerrí** 'Ceres' (< dat.sg. **-ēi̯*); this noun had

We may now associate this process with the remote syncope that we already saw in Proto-Ital. *u̯ala-isn̥mmo- > *u̯alai̯somo- > *u̯alai̯sumo- (≈ Pre-Sam. Ϝολαισυμος) > *u̯alai̯smo- > Osc. *ualaemom*/**valaimas** (≈ SPi. **uelaimes**). It reinforces our take that immediate adjacency to stressed syllables is not pivotal to vowel deletion and reduction.

4.3.2. Double Syncope in the Sequence *-ii̯o- *in Word-medial Position*

The suffix *-*(t)*ii̯on- in Proto-Italic (see Weiss 1993: 3), producing abstract nouns (cf. nom.sg. in *-f#* < *-n-s#*: e.g, Osc. **tríbarakkiuf** 'aedificium, building', **fruktatiuf** '*fructus*, profit'), clearly altered its shape in oblique cases to *-*(t)i̯n*- (e.g., Osc. *tanginud* '*sententiā*, opinion', Osc. *medicatinom* '*iudicium*, legal proceedings', Osc. **leginum**, **leginei** '?', Umb. **natine** '*gente*, patrician family', Umb. *duti* 'a second time', Umb. *tertim*, etc. 'third').

It is *-*(t)īn*- that has traditionally been reconstructed (by, e.g, Buck 1928: 182, Livingston 2004: 6, Nussbaum 2005, 2006), which could eventually end up as *-*(t)i̯n*-. Yet, given that syncope of both unstressed vowels in *-*(t)ii̯on*- is theoretically possible, we do not have to consider *-*(t)īn*- as an obligatory proto-stage, since double syncope would yield *-*(t)i̯n*- directly. Note that the remaining *-*i̯*-, occupying the syllable nucleus and being vocalized (known as *samprasāraṇa*), made no complex cluster with adjacent

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secondarily changed into a *ē*-stem noun from a cons.-stem noun (cf. dat.sg. *-*ei̯*). Recall also the Umbrian theonym *piquier*; its proto-form **piku-ii̯o-* may also have been remodeled as **piku-i̯o-* (≈ *piquier*), retaining *-*i̯o*- in post-vocalic position. Further, Umb. **kvestretie** 'quaestorship' was probably a later creation and based on the syncopated *-*i̯o/ā*- to begin with, given the fact that the base noun **kvestur** was borrowed from Lat. *quaestor*. An Oscan praen. **marahis** may also exemplify post-vocalic *-*i̯o*-; see further details in my forthcoming paper on this personal name.

consonants (cf. **tóutikos* > ***-tks*).

Certainly, this line of reasoning may seem speculative, but there is an interesting case that may prove it. If Untermann's interpretation (2000: 541) of Umb. **persnimu** 'pray' as denominative, something like **perk-sk-iḭon-/-iṇ-* (with my slight modification; ← PIE present stem **pṛ̂k-sḱe/o-*), is correct, it is likely that the derivation was based on the weak stem **perk-sk-iṇ-*, to judge by **persnimu**. Based on the fact that **-i̯-* was ultimately syncopated, the vowel must continue a short vowel. This would thus make **-(t)īn-* (> **-(t)iṇ-*) less plausible, and syncope of two vowels in **-(t)iḭon-* leaving the vocalized **-i̯-* (> *-i̯-*) would become rather realistic.[42]

4.4. Pre-rhotic Vowel Reduction

In 4.1.1, we examined how vowel reduction in Sabellic operated in a labial context. Here we consider another case of phonological contexts, that is, the position adjacent to *r*. It seems that South Picene was also involved in the change, as already mentioned by Rix (1993: 338), which suggests that it was a rather old phonological process. But we cannot tell if it was already Proto-Sabellic, due to the small corpus size and ambiguous orthography that impede generalization of patterns for the proto-stage.

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[42] The Umbrian inf.pass. **herifi** (cf. 3.sg.pres.pass. **herter** 'want' < **herer*; see Nussbaum 1976: 253) probably continues Proto-Ital. **her-iḭo-* (< PIE **ĝʰer-i̯o-*; see footnote 29). To explain the retention of the medial **-i-**, Benediktsson (1960: 166 n. 5) reconstructs a long vowel **-ī-*, which is also suggested by Nussbaum (1976: 243). But the vowel **-i-** may be analogical to some other forms in the same verbal paradigm; in **heris**... **heris**... 'either... or...' (← 2.sg.pres.; = *heri̯s* < **heriḭes*), **-i-** was morphologically retained as the vocalic part of the stem (cf. Osc. **faamat** 'order' < *-ā-t* [1st conj.]), and it spread to other forms, perhaps including p.p. **eretu** / *heritu* / *hereitu* 'intended'.

The segment *r* is commonly found in passive (deponent) endings,[43] corresponding to 3.sg. *-tur* and 3.pl. *-ntur* in Latin (< **-to* and **-nto* + **-r*). The Latin set is used for both primary and secondary endings, that is, it is common to all tenses and moods. Traditionally, as in Buck (1928: 177-179), Meiser (1987a: 107, 1998: 219), Jasanoff (2003: 52-54), Weiss (2005: 416-417), and Wallace (2006: 4), a formal contrast between primary and secondary endings has been claimed for Proto-Sabellic (further, Proto-Italic), which is thought to have been preserved in Umbrian: e.g., primary *-ter* (< **-tro*)[44] in Umb. **teřte** 'is given' [3.sg.pres.pass.] vs. secondary *-tur* (< **-tor*) in Umb. **emantur** 'are taken' [3.pl.pres.pass.subj.]. On the other hand, such a distinction is thought to have been neutralized in Oscan in favor of *-ter*, and in South Picene in favor of *-tor*, which is found in **qolofítúr** 'is erected'[45] [3.sg.pres.pass.] (< **-tor*) and appears with a reduced vowel (i.e., /-tǫr/; cf. **esum, múfqlúm** < **-om* in 4.3.1).[46]

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[43] For peculiar *t*-less 3.sg. forms (pass.subj), such as Umb. *dia* 'give; be possible(?)', *ier* 'go', *f‹e›rar* 'bring', Osc. *loufir* 'or' (← 'es sei geliebt'; see Untermann 2000: 437), see Villanueva Svensson (1999). These forms continue a modal suffix, long vowel **-ē-* or **-ā-*, which is thus not subject to vowel reduction before *-r#*.

[44] Götze (1923: 135, 130) already presents **-tro* as a precursor of *-ter*. Meiser presents a different reconstruction; see footnote 48.

[45] For the semantics of this verb, see Vine (2006c). In passing, it seems that Vine succeeds in equating the South Picene form and Gk. κολοσσός [something like a statue], by reconstructing the derivational process of the latter as **kl̥h₃-dʰh₁-ó-* « mettant debout » → **kĺ̥h₃-dʰh₁-o-* « action de mettre (qqch.) debout » > **kólotʰo-* + **-i̯o-* = **kólotʰi̯o-* > κολοσσός, or the like (2006: 511). Given this **i*-less derivation, his reconstruction **kol[H]-dʰo-* for SPi. **qolofítúr** may be preferable to his other option **kolH-i-dʰo-* with *-i-* syncopated later (2006: 504). Further, if this adjectival form **kolH-i-dʰo-*, corresponding to Latin *idus*-adjectives (e.g., *calidus* 'hot'), had existed, then the expected reflex of the nom.sg. would have been ***qolifs**, and the retained **-i-** may have spread throughout the paradigm, as discussed in 4.3.3 (cf. SPi. **vepetí**, etc. 'monument(?)', with **-e-** unsyncopated (loc.sg.) ← ***vepes** (nom.sg.) < **vepets* < maybe **leped-s*); thus, ***qolif-** is expected to have also been the basis of derivation.

[46] Umbrian offers some other materials for this matter: [[pass.ind.: "primary"]] *ostensendi* 'be prepared(?)' (3.pl.fut.); [[pass.subj.: "secondary"]] Umb. **terkantur** 'be examined *or* confirmed' (3.pl.pres.); Umb. *tursiandu* 'be chased away' [3.pl.pres.]. Interpreted at face value, the distribution of these forms agrees with the traditional view of the contrast between primary and secondary endings. But recall

In contrast to Umbrian, the Oscan orthography is informative for determining vocalism, particularly height difference. For the reflex of *-*ter*, the evidence from the language does not show homogeneous vocalism: **lamatir**, *lamatir* 'should be exiled *or* killed(?)' [3.sg.subj.pres.pass.]; **karanter** 'feed themselves' [3.pl.pres.pass.]; *comparascuster* 'be advised' [3.sg.fut.pf.pass.]; **sakarater** 'sacratur, be consecrated' [3.pl.pres.ind.pass.] / **sakraítír**, **sakahiter** 'sacretur, should be consecrated' [3.sg.pres.subj.pass.]; *uincter* 'convicted' [3.sg.pres.pass.].[47] These examples are distinct in their vowel quality, that is, between /ę/ vs. /į/.[48] This observation implies a phonetically unstable status of the vowels in question, which presumably perplexed Oscan inscribers, and it is likely that vowel reduction was responsible for this fluctuation.[49]

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that we are discussing unstressed syllables, and note again the orthographic fluctuation in such a position between Umb. *hondomu* '*infimo*, lowest' (< *$\hat{g}^h$om-tṃmo-) and Umb. *nesimei* 'nearest' (< **ned-tṃmo-*); the influence of neighboring vowels is conceivable to some extent, which actually holds true for passive endings (stem - ending: ‹e› - ‹e/i/ei› vs. ‹a› - ‹u›). Such graphic variation in pre-rhotic context may also be pointed out in Umb. **ukar**, *ocar* 'citadel', which continues PIE **h*$_2$*ok̂-ri-s* (Ved. *aśrí-*, Gk. ὄκρις). But cf. Pael *lifar* < **lifri-* '*scriptum*' (Lat. *liber*)?; see Villanueva Svensson (1999: 258 n. 25). The more familiar outcome of the sequence *-C*r*V*s* via syncope (and subsequent *s*-loss) would be -C*er*, as seen in Umb. *ager* 'land' (< **ag̑-ro-s*) and *pacer* 'propitious' (< **pāk-ri-s*). See also the Umbrian pronoun *irer* = *erer* 'eius' (unstressed when cliticized), which shows another possible type of graphic variation. Further, cf. Lat. **MVLIAR**[in the Corcolle Altar (= CLat. *mulier* 'woman'?) and **MARMAR** / **MARMOR** [theonym] (CLat. *Mārs*) in Carmen Arvale. Without evidence for the opposite distribution of endings (i.e., *-*tor* for primary and *-*ter* for secondary classes), we may follow the traditional view. But it should be borne in mind that this issue would face some potential problems with a new discovery.

[47] Cf. Marr. *ferenter* 'be carried, obtained' [3.pl.pres.pass.]. For the dubious verbal status of Osc. **krustatar** 'cruentetur(?)', see Untermann (2000: 404).

[48] The attempt to assign /ę/ for primary and /į/ for secondary endings (Meiser 1992, followed by Villanueva Svensson 1999: 261) is undermined by the vacillation of **sakraítír** vs. **sakahiter**. The influence from the pres. **sakarater** on **sakahiter** (based in the Agnone Tablet), proposed by Meiser (1992: 295), is theoretically possible, but the motivation to distinguish two types of endings is questionable. Given the possible tendency in Capua of vowel raising before *-r*, pointed out by Meiser (1992: 294 n. 22), the validity of **lamatir** (Cp 37) for his argument is also questionable.

[49] For this argument, Umb. *ostensendi* 'be prepared(?)' (3.pl.fut.) < *-*ter* would also be of import.

Nouns also provide us with evidence for this idea. The kinship term **patir** 'father' (cf. **patír** < *-*tēr* vs. Lat. *pater*; see Meiser 1992: 294 n. 22) includes **-i-**, which perhaps represents an /i̯/, even more closed than /i̯/, since in the inscription (Cp 2) with this form, the letter **-í-**, usually used for /i̯/, was already in use (cf. **miínieịs** [praen.]). We may juxtapose *loufir* 'free man' (= Lat. *līber*) in Paelignian, genetically akin to Oscan, though the Paelignian orthography, based on Latin, is not as elaborated as Oscan with regard to vowel representation.[50]

This mode of vowel reduction, raising vowels before *-r#*, may be parallel to a word-internal vocalic change in pre-rhotic position in Oscan and Paelignian. Oscan forms, such as μαμερτει / **mam(e)rt(eí)** 'Mars', μαμερτινο / μαμερτινουμ 'of the Oscan-speaking mercenary group', and **mamerttiais** / μαμερττοι 'Martius, of Mars', have not drawn sufficient attention. Peruzzi (1963: 92) regards the *e*-vocalism in these forms as original, then argues that *-a-* in Etr. *mamarce* was due to vowel harmony in the language. On the other hand, Ernout and Meillet (1985: 388) note "osque *Mamers*, issue par dissimilation de **Marmart-s*." The earliest attestations of the (relevant) form in Latin are found in **MAMARCOM** [personal name] (Gnade and Colonna 2003; ca. 575-525 B.C.E.) and **MAMARTEI** 'Mars' (CIL I^2 2832a; ca. 500 B.C.E.), whose *a*-vocalism, probably original, favors the latter's view. But I propose that the phonetic process that changed the vowel was not dissimilation but a pattern of vowel reduction in pre-rhotic position; in such a phonological context, the lowest vowel may have been raised more than in other contexts (cf. Osc. **esskazsiúm** 'pier(?)'< **eks-skand-ti-* ≈ Lat. *escēnsiō* ← *escendere*

[50] Cf. Falsc. **loifirtato**, **loifirta**, but **loferta**. See footnote 57 in Chapter 3.

'disembark'). The same vocalic change is also conceivable in Pael. *pperci* (*p-* = *pe-*).[51] Rix (2000a: 222), following A. L. Prosdocimi's interpretation, sees the PIE verbal root **preḱ-* in this form. Umb. **pepurkurent** [fut.pf.] 'sie werden verlangt haben' more clearly reflects this root in the reduplicated form, that is, **pe-pr̥ḱ-* > *pepork-*. As for the *e*-vocalism in Pael. *pperci*, Rix, followed by Meiser (2003: 187), explains it as resulting from the introduction of a full grade in the wrong place like Umb. **persnimu** 'pray' and **persklum** 'prayer' (i.e., **perḱ-* in PIE terms). On the other hand, we have Osc. *comparascuster* 'be advised' [fut.pf.pass.] (< **kom-pr̥ḱ-sḱ-us-ter*)[52] and **kú]mparakineís** 'council' (< **kom-pr̥ḱ-iio̯n-*), though the development of **-r̥-* into *-ar(a)-* is an intriguing problem (see Untermann 2000: 530). Given the close linguistic affinity between Oscan and Paelignian, it seems more likely that the same *a*-vocalism underlies Pael. *pperci*. If this is the case, we would have another example of vowel reduction, patterned roughly as **a* > *e* /__*r*. The retention of *a*-vocalism in *comparascuster* is most likely due to recomposition.

4.5. Reduction of Medial Unsyncopated Vowels in Umbrian

Aside from the composite forms with medial vowels retained but reduced in labial contexts, discussed in 4.1.1 (e.g., Osc. διποτερες 'Iovis, Jupiter' vs. Osc. **patir** 'father'), there are several forms that contain unsyncopated but somewhat reduced vowels in

[51] This is the so-called "syllabic notation". See Ernout (1905-1906: 310-311), Vine (1993: 333), and also footnote 2 in Chapter 2.

[52] For the morphological peculiarity of this word, see Villanueva Svensson (1999: 254 n. 11).

non-labial contexts in Umbrian. Haug's recent work focuses on this type, with the hypothesis that "*a* was weakened to *e* in *open, medial* syllables and in *closed, final* syllables" (2004: 236), which is "a natural and possible pre-stage of syncope" (2004: 248). Despite his elaborate scenario, however, some of the evidence he offers is based on disputed material. Perfect passive participles of the second conjugation, such as Umbrian **taçez** 'tacitus, be silent', are taken to include the zero grade of the stative suffix *-eh_1- (*th_2k-h_1-*tó-s* > **takatos* > *tacetos* > **taçez**; **-e-** escaping from syncope in a "medial" syllable, since the change *-CV̆*s*# > *-C*s*# pushed this vowel into the final, closed syllable [thus, **-ez**]). But it is highly controversial whether such an ablaut (*-h_1- ~ *-eh_1-) was really operative in this suffix (see Jasanoff 2002: 132 et passim).

On the other hand, the historical scenario Haug (2004: 236, 243-244) postulates for Umb. p.p. **pruseçetu** 'cut off' (acc.pl.n.) by comparing it with **prusekatu** (3.sg.fut.impv.) is worth mentioning, which is something like nom.sg. *-*sekh*$_2$-*tó*- > *-*sekato*- > *-*seketo*-. That is, he denies the replacement of *-a-* by *-e-* (the thematic vowel?) in the 1st conj. verb *prusekatu* (< **sekā*-), a pattern presented in Meiser (2003: 30 n. 10). I welcome Haug's attempt based on phonological factors, and it agrees with the most plausible phonetic interpretation, which I present in what follows.

For Umb. **prusekatu** vs. **pruseçetu** (cf. Lat. *sectum* < **sekH*-; I follow LIV 2001: 524 for the underspecification of the kind of laryngeal), we can posit the following historical development: *s_e*kH*-i̯*e*- > **sekai̯e*- > **sekā*- (via loss of intervocalic *-i̯-* and maybe subsequent vowel contraction; cf. Osc. **faamat** 'order') and **skH*-*to*- → **sekato*-, respectively. The **sekā*- is seen in **prusekatu** as it is. The p.p. **sekato*-, on the other

hand, underwent vowel reduction in both medial and final syllables to reach **sek*[ə]*t*[ə]*s* (nom.sg.). Following the pattern of vowel deletion I argued for in 4.1.2, it removed the second [ə] to result in **sek*[ə]*ts*. Other case forms, reduced as **sek*[ə]t^{o}, could thus retain the medial vowel, following **sek*[ə]*ts*. Here we confront the problem of the phonemic status of the remaining [ə]. Recall the significant raising of *-a-*, as assumed for Latin (see 2.4.3). The lowest vowel, raised far from the phonetic value [a], did not retain its original phonemic status /a/. The orthographic selection of ‹e› for [ə] (**pruseçetu**) was presumably due to similar tongue height[53] and graphic analogy with other *eto*-forms (e.g., *tasetur*) that inherently continue /e/. Such a state of affairs would also be plausible in Umb. *oseto* 'facta, errected' (but cf. Osc. **upsatuh**) vs. *osatu* 'facito, dedicate' (3.sg.fut.impv.).[54]

There is another case of similar reduction of *a*-vocalism in Umbrian. Several forms have long been known for the change, but need to be sorted out chronologically. While **prehabia** vs. **prehubia** '*praebeat*, provide' and **kumaltu** vs. **kumultu** / *comoltu* 'grind' already show graphic variation of vowels within the old native alphabets, the contrast in **prestate** vs. *prestote* [theonym] and **tesenakes** vs. *tesenocir*, *tesonocir* [ethnonym] is distributed in the different — old and new — notations. Therefore, pace Meiser (1986: 268-271), it seems that the sound change in the latter was relatively late and should be separated from that in **prehabia** vs. **prehubia** and **kumaltu** vs. **kumultu** in my theory;

[53] Cf. footnote 51 in Chapter 2, where we saw Latin speakers using the letter ‹e› for [ə], a mid unrounded vowel, which is close to /e/ in terms of phonetic features.

[54] The same explanation may perhaps be applied to Osc. **genetaí** [theonym], also adduced by Haug (2004: 243), if this form continues **genato-* (< **ĝenh₁-tó-*). Haug refutes the replacement of *-a-* by *-e-* (see Meiser 2003: 30 n. 10, 53 n. 61) for this form as well, which seems to me legitimate. On the other hand, according to Vine (1998: 56-59), Osc. **genetaí** is equated with Lat. *Genita* (*Māna*) [theonym: '(lit.) good birth'], Gk. γενετή 'birth' < **ĝenh₁-etó-* '(good) to be born', 'worth bearing' (→ 'born well' 'born legitimately'); cf. Welsh *geneth* 'girl' < **genettā-* < **ĝenh₁-eteh₂-*. If this is the case, the Oscan form does not inherit *a*-vocalism and is thus irrevalant to our discussion here.

the vocalic development in these forms can be attributed to early reduction in a labial context commonly found in various Sabellic dialects. Meiser's observation (1986: 270), though, that the vowel in **prestate** vs. *prestote* and **tesenakes** vs. *tesenocir*, *tesonocir* was unsyncopated is worth mentioning.

For the compound **prestate** ~ *prestote*, recomposition must have served to preserve the original vocalism. But in contrast to some other compositions that were still related to their constituents (e.g., Umb. *procanurent*, semantically associable with 'sing'), **prestate** ~ *prestote* may have lost such a connection, used as a divine name (i.e., with the literal meaning 'stand' of *sta-t-*, or p.p. *sta-to-*, bleached out). The failure of recomposition may have provided leverage for representing a vowel that is reduced, but retained, in the unstressed medial syllable as such. The expected phonetic entity in the position was [ə], and its height, remarkably reduced from [a], did not allow the sound to be represented by ‹a›, but ‹o›, which is similar to [ə] in height. The persistent use not of ‹e› (cf. *-eto*-adjectives) but ‹o› for *prestot-* (26 times in the Iguvine Tables) is most likely due to the clear qualitative difference, as perceived by the inscriber, between stressed /e/ (ideal for ‹e›) in the initial syllable and the following [ə]. Thus, Meiser's description (1986: 33, 268-271) of "Verdumpfung" of *a* ‹a› to *å* ‹o› in medial syllables has been restated both phonetically and orthographically.

For the toponymic adjectives **tesenakes** ~ *tesenocir* ~ *tesonocir*, a certain degree of conservatism is always conceivable as they are proper nouns. For this reason (and others, if any), these forms presumably retain unsyncopated [ə]'s in medial syllables. The mechanism of vowel reduction in the third syllable, where [ə] is no longer represented by

‹a›, is much the same as we have seen about **prestate** ~ *prestote*. Note that the second syllable also shows a graphic fluctuation, ‹e› ~ ‹o›. The choice of ‹o› for the reduced vowel in this position as well as in the third syllalble, though Meiser (1986: 33) posits "Verdumpfung" into *å* ‹o› even for non-low front vowels like *-e-*,[55] can be attributed to exactly the same reason as argued for **prestate** ~ *prestote*, that is, the qualitative difference between stressed /e/ in the initial syllable and the following [ə]'s.

[55] The other Meiser's example **atropusatu** 'perform a *tripudium* (dance)' (< **-tri-*) should be referred to the labial consonant **p**, which is responsible for the rounding of the preceding **i* (see 4.3.1).

5. Conclusion

5.1. Retrospective

We have discussed various problems of vowel reduction and deletion in the Italic languages. Solutions have been offered, some of which are provisional. Here we look back at what we have considered and summarize what we have concluded.

In Chapter 1, we started by discussing the qualitative aspect of Italic accent (1.1). This problem — whether it was stress or pitch — has long been insufficiently treated, with recourse to internal evidence alone, such as some sound changes, which have not been considered substantially from a typological standpoint, and ancient grammarians' remarks, which have been variously interpreted by scholars. But based on the knowledge archived by phoneticians and phonologists, it was rather clear that a language with vowel reduction and deletion (esp. the former) typically has strong stress. Though the issue of stress in Classical Latin is still controversial, there is no doubt that Pre-historic Latin and Sabellic were characterized by a salient stress accent. In terms of the placement of stress, I simply endorsed the initial-stress theory, which is now commonly agreed upon by many scholars (1.2). The nearly total absence of vowel reduction and deletion in initial syllables in Italic is best understood by assuming that stress was assigned to those syllables. Objections to this theory have been (and will be) presented (1.3), but none of them has succeeded in offering a proper phonological model or an overall mechanism that accounts for morphologically disparate forms (i.e., not only composite ones). With

this said, the accentual system with *stress* on *initial* syllables has been the basis of my analysis. The last section of this chapter describes how our knowledge of phonetics and phonology is applicable to various issues of vowel reduction and deletion in Italic (1.4).

Chapter 2 is devoted to vowel reduction in Latin (marginally Faliscan). The first topic we discussed was the chronological distinction between the change in word-internal and word-final position (2.1). Some inscriptions seem to show that vowel reduction first occurred in medial syllables and only later in final syllables. We also reviewed Bader's hypothesis (1960: 238) that the change happened only in one place in the word. Recall that the traditional interpretation of examples has more or less relied on the supposition that it is unaccented vowels immediately after stress alone that licensed for these sound changes. The attempt to chronologically differentiate vowel reduction according to positions in the word also seems to have been influenced by this notion. But as Crosswhite (2001, 2004), Flemming (2005), and Barnes (2006) indicate, more than one unstressed syllable may undergo the change. A simplistic division, like that between medial and final syllables, does not properly capture the one mechanism underlying the changes in both of these positions. I also presented some inscriptional evidence for a very early application of vowel reduction in final syllables. This led us to assume that the process targeted both medial and final syllables synchronically in an early stage of Latin. Though these two positions showed different types of reduction processes, the co-existence of more than one pattern was typologically unsurprising.

More detailed mechanisms of vowel reduction were pursued in Chapter 2. We first considered the process in medial syllables (2.2). Though the frequent final result of vowel

reduction in this position was phonemic merger with *-i-* (e.g., *conficiō* 'complete' < **-fak-*), Rix's suggestion (1966: 161-162) that unstressed vowels were first reduced to schwa-like sounds (Process I) was important. To posit [ə] or the like as an intermediate outcome facilitates our understanding of how and why reduced vowels in Latin were influenced to such an extent by adjacent phonological segments, such as labials, laterals, and vowels in preceding syllables. Labials provided preceding, reduced, and centralized vowels with a more or less rounding, which was clearly reflected in the orthography (e.g., *maxumus* 'greatest' < **-om-* vs. *hospitem* 'host' < **-pot-*), though haphazardly to some extent (influenced by neighboring vowels). Reduced vowels were also affected by a following *-l-*, depending on its quality, that is, palatal or velar. The fact that the original /i/ did not participate in the coloring process (e.g., *mutilus* 'mutilated') was taken to reflect that it may have been phonemically preserved at the completion of Process I. The same reasoning was provisionally applied to /u/. I then concluded that it was the mid and low vowels that basically reduced to [ə], a pattern theoretically predicted by Crosswhite (2001). The intermediate stage [ə] was also useful for explaining the presence of doublets, such as *anat-* ~ *anit-* 'duck'; the former retained the association between the surface [ə] and underlying phonemic /a/ due to the preceding /a/, while in the latter, the association was lost and the vowel was further reduced in terms of [front][back] to merge with /i/ (i.e., Process II).

Next, the pattern of vowel reduction in final syllables was described (2.3). In contrast to that in medial syllables, the reduction process did not neutralize the [front]/[back] contrast, as seen from the developments *-o-* > *-u-* and *-e-* > *-i-*. Another pattern of change

-a- > *-e-* was tentatively interpreted along the same line of reasoning. The phoneme /a/ in final, unstressed syllables resulted not in [e] directly, but [ə], consistently preserving [−front]/[−back]. Unlike in medial position, the /i/-merging Process II did not take place in final position, so [ə] was no longer raised. On the other hand, it was phonemically dissociated from /a/, merged with /e/, which is close to [ə] in terms of tongue height and [−round]. As already mentioned, the co-existence of two different patterns of vowel reduction depending on the location in the word (medial vs. final) is typologically normal. That the reduction was more restrained in final syllables than in medial ones is also cross-linguistically natural.

Turning back to medial syllables, we considered Process II, that is, the total neutralization in the contrast of [front]/[back] (and [round]) features (2.4). In Latin, this process caused most of the vowels that underwent Process I to merge with the high front /i/. Though Faliscan seldom reflects Process II, this should not be taken as evidence that the language lacked vowel reduction entirely. The idea of a two-step process allows for Faliscan at least to have experienced Process I, which helps explain the sporadic occurrence of vocalic changes, including deletion, in the language, which were resumed in Chapter 3. The trigger of Process II in Latin is not easy to identify, but the presence of /i/ in adjacent syllables may have contributed somewhat to a preference for the same vocalism in the outcome of reduction, by a sort of assimilatory effect (e.g., *-cipiō* < **-kapiō*). Eventually, Process II became a fairly wholesale process, affecting vowels in unstressed syllables in general. However, not all of the vowels potentially targeted by the process merged with /i/. The difference between application (e.g., as *subigō* 'drive under'

< **sub-agō*) and non-application (*concrepō* 'crack', *ēdoceō* 'instruct thoroughly') of Process II in prefixed verbs was especially conspicuous. To solve this discrepancy, we adopted Garrett's standpoint (2005), which took the formation not only of present but also of perfect stems into account. In addition, we paid special attention to Rix's prediction (1966: 160-162) that in the beginning of the literary period, reduced vowels had not yet merged phonemically with /i/. This was entirely compatible with positing two steps (Process I and Process II) through the entire development of vowel reduction. That is, though Process I was already completed, Process II reduction was still on-going at that period. The absence of phonemic merger with /i/ in forms such as *concrepō* and *ēdoceō* was thus explained by analogy with perfect forms: around the onset of the literary period, the Penultimate Law came into effect, whereby we obtained *concrépuī* and *ēdócuī* with stressed root vowels; this accent accordingly prevented *cóncrepō* and *édoceō* from evolving into ***cóncripō* and ***édiceō*. On the other hand, prefixed verbs with, e.g., *rapiō* 'snatch', *habeō* 'have', *taceō* 'be silent', underwent the regular Process II reduction, though they formed perfect forms in the same manner as *concrepō* and *ēdoceō*. To solve this problem, we identified the phonetic feature of *a*-vocalism as being common to these examples, and suggested that it must have already been so significantly raised by Process I that the association with /a/ could not be retained at Process II. In medial closed/diphthongal syllables, though realized as schwa-like sounds under the initial-stress rule, vowels could preserve their original phonemic status due to the stress newly assigned to them under the Penultimate Law. For the special treatment of **-a-* (> *-e-*) in such syllables, I highlighted the same phonetic feature of *a*-vocalism.

The division into Process I and Process II and the chronological extension of the whole process into the beginning of the literary period had a ramification for the consideration of vowel contraction (2.5). In the traditional framework of simple formulaic patterns of vowel reduction (*a/e/o* > *i/u* in medial open syllables, etc.), if vowel reduction preceded contraction, the proto-form **ko-agō* of *cōgō* 'drive together' would have developed into ***coigō* and later ***cūgō* or ***coegō*. One might thus chronologically order vowel contraction before reduction. But we considered some cases that suggested a relatively late application of vowel contraction to sequences of unlike vowels (e.g., *mālō* 'would rather' < OLat. *māvolō* < **magu̯olō* < **mag(i)s-u̯elō*); the reverse order, reduction → contraction, then emerged as a possible chronology. By means of an intermediate stage like [ə] before Process II, vowel contraction in *cōgō* was analyzed as follows: **kó-agō* > **kó*[ə]*gō* > *cōgō*.

Finally in Chapter 2, we discussed the non-application of vowel reduction in nominal compounds (2.6). In many examples of this morphological category, the second member tended to resist vowel reduction in its first syllable (e.g., *sacri-legus* 'temple-robber'). I proposed that compounded elements preserved their own stress (with a possible difference of primary and secondary stress) for awhile after composition: that is, **sákro-légos*. Though there were some examples that appeared to contradict this hypothesis, we were able to solve them by positing some phonological and morphological conditions. For third-declension compounds in *-ceps -cipis* (< **-kap-*), and so forth, the nom.sg. could have phonemic reduction (*-e-* < **-a-*) because the vowel was located word-finally and thus prosodically extrametrical. I concluded that oblique case

forms underwent vowel reduction secondarily (*-i-* < **-a-*), since "apophonie", such as the alternation of *-e-* in the nominative singular and ***-a-* (i.e., lower than *-e-*) in other cases, was synchronically banned. In derivatives from compounds, the application of Process II was optional; thus some underwent reduction due to a change in the stress domain concomitant with derivation (e.g., *du-pondius* 'the sum of two *asses*' vs. *du-pund-* [Lucil.+]).

In Chapter 3, we turned to vowel deletion in Latin (again, marginally Faliscan, together with vowel reduction). This sound change seems to have occurred a number of times throughout the history of Latin. Thus, it is reasonable to assume that different factors were involved in vowel deletion at different periods. To prepare for this complex state of affairs and further inquiry in the chapter, we first sorted out examples of syncope according to when it took place; roughly, in Pre-Literary Latin, Pre-Classical Latin, and Classical Latin (3.1). We also considered apocope and concluded that the number of cases of "real" apocope, driven by initial stress, was rather small.

As another preliminary to the rest of Chapter 3, I introduced two different factors, both independently relatable to how syncope operated: sonority (prominence) of vowels and metrical constraints (3.2). Considering the former, /i/ (and maybe /u/) and [ə] were the most likely to delete. The latter were designed to disfavor and remove certain syllabic configurations, which were discussed in more detail in the following section (3.3).

We then started by reviewing some previous studies, which clarified what we already know about a pattern of syncope caused by metrical constraints and the history of accent distribution rules, which must have had ramifications on metrics. With the premise that

Latin is metrically based on bimoraic trochees, Mester (1994: 37-43) explained post-tonic syncope in Pre-Classical and Classical as "trapping" deletion (e.g., (lā́)ri‹dī› > (lar)‹dī› 'bacon'). On the other hand, there were a number of cases of syncope that could not be explained in the same way in Pre-Classical Latin and earlier (e.g., *(dek)(síte)‹ros› > (dex)‹ter› 'right', ba(líne)‹um› > (bal)ne‹um› 'bath'). Then, we considered what metrical system operated in that period, following the view of Hayes (1995: 180) and Parsons (1999: 122-123) that the parsing direction changed from left-to-right to right-to-left between Pre-Classical and Classical Latin and that the leftmost foot in the word hosted the stress in Pre-Literary Latin as opposed to the rightmost one in later periods. With this knowledge, however, we failed to define syncope in early stages as trapping deletion. Then, we invoked another metrical constraint disfavoring two successive light syllables as proposed for Arabic, which could account for the motivation of early syncope. This discovery led us to posit a historical shift in the targets disfavored by metrical constraints. The older constraint against a sequence of light syllables was in competition with the new one described by Mester in Pre-Classical Latin, and eventually replaced by it. The constraint banning successive light syllables is reflected in three groups of forms; those with *s*C(R)-clusters, which were usually taken to close preceding syllables, but we saw that the clusters were actually syllabified as onsets; the major group of forms which reflect this constraint consists of diminutives, many of which showed syncope in syllables distant from initial stressed ones, facilitated by the presence of foregoing sonorants; it was also tentatively suggested that some Faliscan forms reflected the constraint against sequences of light syllables.

We further discussed "syncope" in phonological contexts involving *-r-* (3.4). The task was twofold: (1) to clarify whether syncope in the sequence *C*r*V was truly driven by the initial-stress rule and (2) to determine whether alleged cases of post-vocalic syncope in final syllables were really syncope or not by definition. For the former question (1), based on Nussbaum's analysis (1999: 394-395) of *acerbus* 'bitter', we concluded that the phonological change in *C*r*V (> C*er*), though it was "one pattern" of syncope, was certainly older than other cases of syncope. Moreover, the same development could be found in initial *stressed* syllables (e.g, **krinō* > *cernō* 'sift, distinguish'; **tristi-* > **tersti-* > *testis* 'witness'). It could thus be defined not as stress-driven syncope but rather as something like a metathetic process between *r* and a vowel (i.e., *C*r*V > *C^{ə}*r*V > *Cə*r*^{V} > C*er*). For the latter question (2), we saw that it is quite possible that the development *-V*r*V*s*# > *-V*r*# was semantically motivated; nouns that denote human beings, either in kinship terms, professions, or social ranks (e.g., *pater* 'father', *augur* 'augur') and masculine adjectives in *-er* (e.g., *sacer* 'sacred' < **sakros*, *ācer* 'sharp' < **ākris*) would have served as models for the reshaping.

As a result of the consideration in the preceding section, vowel deletion in final syllables that could be defined as phonological was limited to sporadic apocope (3.5). No metrical constraint could be identified, while semantics may have partly contributed to the change.

Chapter 4 focused on both vowel reduction and deletion (particularly syncope) in Sabellic; chronological problems were treated first (4.1). Despite the traditional view that vowel reduction was a rather restricted phenomenon (which is partly true), we concluded

that this vocalic change was already operative in early Sabellic, such as Pre-Samnite (e.g., Ϝολαισυμος [personal name?]), Paleo-Umbrian (e.g., **setums** [personal name]), and South Picene (e.g., **estuf** and **estufk** 'here'), and that it was particularly noticeable in labial context. The application of vowel reduction to non-initial syllables, both word-medial and word-final, was construed as reflecting that the operation driven by initial stress was already active in these early dialects. In regard to syncope, scholars have traditionally assumed that final syncope predated medial syncope, which was dated as late as the fifth century B.C.E. However, we observed several instances of medial syncope that could go back to the sixth century: e.g., Pre-Samnite imperatives ο(vac.)fρι[-]qτο[δ, (σ)τα[.]ιοσqτοδ, SPi. **aitúpas** (Umb. **eitipes** 'they resolved'), **múfqlúm** 'monument(?)', **mefiín** 'middle', **uelaimes** [personal name]. The antiquity of medial syncope in these forms led us to reconsider the alleged chronological gap between syncope in medial and final syllables. Osc. **túvtíks** 'belonging to the community' < **toutikos*, for instance, preserved the vowel (*-*i*-) in the medial syllable, while it lost one (*-*o*-) in the final syllable. This case appeared to justify the claim that syncope first removed the vowel in final syllable (*-*o*-), closing the preceding syllable, and thus impeded syncope there (i.e., **-í-** was preserved). But I suggested that only one of the types of syncope applied to **túvtíks**, since the consonant cluster **-*tks* would have resulted if both vowels had been syncopated, and I tentatively concluded that syncope in Sabellic targeted both medial and final syllables synchronically. Further, this antiquity of vowel reduction and deletion in Sabellic (prior to 550-550 B.C.E.) was taken to reflect that Sabellic probably developed the initial-stress system on its own, independently from Etruscan, where vowel reduction

came into effect only later than ca. 500.

Next, we identified which vowels were subject to syncope (4.2). The phonemes */i, e, a, o/ were regularly deleted in unstressed syllables, unless paradigm leveling intervened (cf. Osc. nom.sg. **túvtíks** → gen.sg. τουτικες). On the other hand, the particular retention of the high back */u/ drew more attention than before, after the examination of some examples in Umbrian, such as **tekuries** / *dequrier* (cf. Lat. *decuria* 'group consisting of ten men'), *piquier* [theonym], **iapuzkum** [adj. of a tribal name], **kastruçiie** [gent.]. My tentative suggestion as to this special behavior of */u/ was that its retention was due to the inherent phonetic feature [+round], which enhanced its prominence sufficiently to resist deletion. (Recall that */o/ was reduced as [ə] or [ɨ], losing roundedness.) For retained vowels in SPi. **opesa̯**[ ]**úom** 'produce, erect' and **erimínú** [theonym?], I also tentatively suggested that they obtained a certain degree of roundedness from adjacent labials to resist (or delay) deletion. And as possible parallels, some Umbrian forms, such as *fisovio-* [theonym.], *grabovio-* [epithet to several divinities], and **ikuvins** [ethnonym], were offered, where vowels were preserved before the labial *-u̯-*.

In the following three sections, we discussed more specific issues of vowel reduction and deletion in Sabellic. First, syncope in the position adjacent to *-i̯-* called for some consideration (4.3). The *-i-* of the disyllabic "Zugehörigkeitsuffix" *-i̯io-* (or f. *-i̯i̯ā-*) was deleted so regularly that it occurred even in syllables distant from initial stressed syllables. Then, with reference to the development of Umb. **persnimu** 'pray', that is, the syncope in the second syllable of Proto-Sab. **perk-sk-i̯n-*, I tentatively proposed that the targeted *-i̯- was consistently short and that the underlying suffix *-(t)i̯i̯on-* did not evolve into *-(t)īn-*

but directly into *-*(t)i̯n*- by means of syncope of both *-*i*- and *-*o*- followed by the vocalization of *-*i̯*-.

The phonological context before *r* was also a locus of discussion (4.4). We focused particularly on vowel reduction in Oscan and Paelignian, starting with the position before word-final *-r#*, where the raising effect was conspicuous (e.g., Osc. **sakraitír** 'be consecrated' < *-*ter*, **patir** 'father' < **pater*, Pael. *loufir* 'free man' < *-er*). Further, in word-internal position, we recognized a similar effect, formulated as *-*ar*- > *-er*- (e.g., Osc. μαμερτει 'Mars', Pael. *pperci* 'prayed').

Lastly in Chapter 4, we discussed some Umbrian examples that retain unsyncopated but somewhat reduced vowels (4.5). For *a*-vocalism, the phonetic and phonological mechanism was parallel to vowel reduction of *-a-* in Latin, where the lowest vowel tended to undergo the significant reduction-induced raising toward [ə], so that the reduced vowel could not retain its original phonemic status /a/. We also saw a marginal case where even the mid vowel *-e-* underwent vowel reduction. The phonetic contrast between vowels in initial stressed syllables and others was preciously reflected in orthography.

5.2. Miscellaneous Problems on the Italic Accentuation

This work is chiefly devoted to vowel reduction and deletion caused by stress accent, not accent itself. Hence, we have not discussed the essence of accent or its rules at length, except as to the nature of accent (1.1), the initial-stress theory (1.2, 1.3), and the early

version of the Penultimate Law (3.3.2), which were all required to understand several aspects of vowel reduction and deletion. In what follows, I will briefly survey two separate problems in Italic accentuation, which are of great interest, but not so pivotal in considering vocalic changes.

5.2.1. Secondary Accent and Shift in Accent Location in Latin

Scholars have long noticed the systematic difference in the initial-stress rule of Pre-Literary Latin and the Penultimate Law of Literary Latin. The question as to what diachronic process motivated the accent shift from initial syllables to penults or antepenults has thus been repeatedly raised, and is regarded by several scholars as the crux of the initial-stress theory.[1]

Among scholars who advocate the initial-stress rule, the concept of so-called "secondary accent" has commonly been held to illustrate the shift in accent from Pre-Literary Latin to Literary Latin, since as early as Dietrich (1852: 554), Thurneysen (1883-1885: 313), and Brugmann (1897: 973).[2] Secondary accent is thought of as a phonetic counterbalance to primary accent, for the purpose of facilitating pronunciation of the entire word. Its occurrence is generally conditioned by the number of syllables in a

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[1] See Juret (1919: 102-104), Nyman (1983: 31), and Ballester (1990: 33). Von Planta (1973: 589) also acknowledges a huge discrepancy between these two types of accent, suggesting "zwei so grosse Accentumwälzungen [= the initial stress and the Penultimate Law] aber kaum bald nach einander stattgefunden haben werden."

[2] Though Pedersen (1905: 338) does not accept the initial-stress theory, he nevertheless posits a secondary accent that emerged under the PIE accentuation (1905: 339). According to him, it always fell on antepenults, and the Penultimate Law was established in a later period by the processes ⏑—x > ⏑—́x and —́ — x > — —́ x. His view is, however, hard to justify.

word. Historically speaking, primary and secondary accent may have switched status. Dietrich suggests that Greek influence triggered the exchange.[3] Lindsay (1891: 406, 1894a: 158-159, 1894b: 409), on the other hand, seeks an internal motivation to Latin, positing historical scenarios such as **sápièntia* > *sàpiéntia* 'wisdom', **témpestàtibus* > *tèmpestátibus* 'a portion of time; weather', **pótestàtibus* > *pòtestátibus* 'power, control', **lánificium* > *lànificium* 'spinning, weaving', and **béneficium* > *bènefícium* 'service, kindness'. Kent (1932: 66) also assumes that the change was internally motivated. Ballester also (1990: 38-39), based on his own accentual rules for Pre-Literary Latin (see 1.3.5), illustrates the process as **dépertre* > *dèperíre* 'to perish'. Leumann (1977: 248) elaborates the mechanism, by assuming that secondary accent first occurred in heavy antepenults, and then replaced initial primary accent in terms of status: **sápièntia* > *sàpiéntia.*[4]

What is crucial to this theory is determining the location of secondary accent. It is worth citing Allen's discussion (1969: 200-202) based on foot structure. We have already reviewed basic facts about the shape of the foot and its parsing process in Latin (see 3.3.1 and 3.3.2), elaborated by Mester (1994), as well as Hayes (1995) and Parsons (1999). Following the premise that the "cadence" of stress on a light syllable is carried over into a following light syllable, Allen (1969: 197) posits the metrical foot ◡◡ as "an indissoluble unit" (also in Allen 1973: 164-178; already in Kuryłowicz 1958: 382), or in recent

[3] See also Poccetti (1999: 77-78). But the question arises as to why Latin did not come to have oxytones like Greek. We can hardly prove, or disprove, some sort of sociolinguistic factor.

[4] Deroy (1981: 234) argues that initial stress did not become obsolete, but survived as a dialect feature which caused consonant gemination in some Italian forms, e.g., *pellegrino, tollerare, seppelire, rettorica.* But the possibility that initial secondary accent newly emerged under the Penultimate Law cannot be excluded.

terminology, a moraic trochee. He also posits equivalent status for a heavy syllable, which may carry both the peak and cadence of stress (in short, LL = H).

Allen's framework correctly predicts the difference in the location of stress between *adsímiliter* 'similarly' (later, *adsimíliter*) and *maleficium* 'misdeed' at the Pre-Classical stage. We can roughly restate his explanation as follows: *(ád)(sìmi)li‹ter›, *(mále)(fìci)‹um› → (àd)(símì)li‹ter›, (màle)(fíci)‹um›; namely, secondary accent which emerges in a foot different from the one that hosts primary accent later becomes primary. Whereas *maleficium* already has the Classical accentuation and thus requires nothing further, we need an additional step to get from *adsímiliter* to its Classical form *adsimíliter*. There is, however, no straightforward explanation by means of secondary accent. See the following process by Allen: *ădsímìliter* > *âdsimĭ́lìter* (˝ = peak of primary accent]; ˊ = peak of secondary accent; ˋ = cadence; ˆ = peak + cadence). A similar pattern is seen in *făcìlius* > *facĭ́lìus* 'easily'. In these scenarios, it is the cadence in left-hand stages that came to host primary accent in right-hand ones, which is theoretically difficult to accept based on his hypothesis. Allen (1969: 202) is therefore constrained to regard such examples as "special cases" (see also Allen 1973: 189-90), a definition which amounts to an abandonment of systematic account of them. As regards the examples adduced above by Lindsay, Ballester, and Leumann, which are all favorable to the theory of secondary accent,[5] their selection seems intended to avoid the problem of *adsimiliter* and *facilius*.

There is yet another potential difficulty in the theory. Oniga (1990: 201) maintains

[5] In his study on the Saturnian, Parsons (1999: 132) provisionally supports "a re-analysis whereby speakers swapped the primary and secondary accents," citing forms across two dipodes, ... **TEMPES|TÁTEBVS**, ... *ministrā|tŏ́rēs*, ... *expedīti|ŏ́nem*, where the head of a dipode (metrically strong position) coincides with the syllable that would bear primary stress under the Penultimate Law. On the other hand, Parsons notes that "[t]hese lines are the only concrete evidence I know of for this hypothesis."

that two accented syllables cannot be adjacent to one another. Though we require further research regarding Latin, this is now known as a constraint on stress clash, widely found in many languages.[6] The adverb **ádsìmiliter* (→ *àdsímiliter*) violates this constraint. The same is true of **rélàtus* (p.p. of *referō* 'bring back'), as reconstructed by Allen (1969: 200-201) and Liénard (1977: 611). The fact that they are composite forms may partly account for marked accent distribution (cf. Eng. *backwoodsman* [˗ ˊ ˊ –, ˋ ˊ ˗ –]), but we must confront the enormous amount of non-composite forms (*amātus* 'loved', *amāre* 'to love', etc.).[7]

What we can assume at most is indirectly suggested by Kuryłowicz (1958: 382-384), in an attempt to illustrate the historical process of stress movement based on the foot structure of Latin. Though the development ⏑–× > ⏑˗ˊ× (e.g., *relātus*) which he posits, and his abrupt appeal to ⏑⏑ = ˗ˊ (i.e., moraic trochee) to explain the shift are only vaguely defined,[8] we can see his intention to base movement solely on the metrical shape of the word. As seen in 3.3.2, an initial light syllable cannot be parsed if moraic trochees

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[6] See brief descriptions on this matter in Myers (1991: 318) and Kager (1999: 165).

[7] Cf. Deroy's list of examples (1981: 233-234) to prove the theory of secondary stress; alongside *maledīxit*, *disciplīna*, *perīculōsus*, *officīna*, *cīvitātēs*, *gubernāculum*, and *mancipāre*, it contains *perīculum* (< **pérìklom*?), an analogous pattern to **rélàtus*.

Liénard (1977: 615-616) focuses on the cadence such as *cónde sepúlcrō* (cf. *cóndere géntem*). According to him, *sepúlcrō* resulted from **sépùlcrō*, by means of the change of accent status. It seems that he intends as if hexameter induced such a change, but the point of departure, that is, the existence of *sépùlcrō* is not entirely guaranteed.

[8] Cf. the misgivings in Oniga (1990: 202-203 n. 27) and Ballester (1990: 38-39). Nor is Mignot (1980: 305) convinced by Kuryłowicz's theory, but his attempt to refute it by using *voluptās* is not successful. Based on Kuryłowicz's framework, the original "syllabification" (something close to metrical parsing) and accentuation **vó-lup-tās* should have changed through *vólup-tās* to *volúp-tās*. Mignot argues that *vólup-* is syllabically as well-formed as *sénex* and must have been stable as an "un bloc accentué," then without necessity of accent reassignment. But he entirely ignores extrametricality in *sénex*, which then cannot be compared to *vólup-* in word-internal position.

are strictly assigned. But the constraint that every prosodic word must begin with a foot (ALIGN-WD-LEFT), if ranked more highly, can produce a stress distribution like ◡́—× (thus, *(ré)(lā)‹tos›). The shift to ◡—́× may have been caused by outranking of the metrical constraint on foot-binarity that banned a degenerate foot like a single ◡ from being parsed.[9]

The metrical framework seems to better account for the diachronic shift of accent location. Note, on the other hand, that I do not deny the possible synchronic existence of secondary accent. Rather, I have suggested the notion in the discussion on the separate accentuation of compounds (2.6). Further, in tangentially mentioning pre-tonic syncope in Literary Latin, we have seen the possible involvement of secondary accent, which may have been assigned to the syllable prior to the one targeted by syncope (3.3.1, 3.3.2). Such synchronic phenomena, however, are not readily recast for purposes of diachronic considerations. We should bear in mind that secondary accent is often post-phonological and is itself not easy to analyze.

5.2.2. Accentual Rule in Late Sabellic

The problem as to whether or not Sabellic acquired an accentual system like Literary

[9] Lepschy (1962: 221-226) tries to strengthen Kuryłowicz's framework by adducing experimental phonetic studies, in which tape-recorded words, for example, *sŏ̆rră̆gĭs* and *ă̆ttīgă̆s* with marked intensive accent in the initial position, when reproduced in reverse, sound respectively like *sĭgă̆rrŏs* and *să̆gĭttă̆*. Following this data he arrives at a conclusion that syllable weight contributes to the perception of accent. His idea, however, does not help in understanding the development ◡́—× > ◡—́×, with accent shifted to the heavy syllable, since Lepschy provides no account of how words with a configuration like ◡́—×, with accent on the light syllable, could exist as the starting point in the development of words such as *sĭ̆gărrŏs* and *să̆gittă̆*.

Latin has often been discussed by scholars. In the earlier literature, there is less agreement on the matter. For example, von Planta (1892: 592), Hirt (1895: 41), Muller (1917), and Kuryłowicz (1958: 381), assume a parallel development between Latin and Sabellic, while Thurneysen (1908: 241; at least in Oscan) argues that Sabellic retained initial stress throughout its history. Brugmann (1897: 976) cautiously notes that whether Sabellic came to have a different accentual rule, similar to the Latin Penultimate Law, is hard to decide; in his opinion, though, evidence presented by von Planta is not convincing (von Planta's view is concisely summarized and also refuted by Benediktsson 1960: 283-284; also already by Solmsen 1894: 151-158); see also Buck's neutral and pessimistic statement (1928: 101).

Disagreement among scholars continued well into the middle of the last century. Schmid (1954: 43, 46) argues for the same innovation in Sabellic as in Latin, though he presents an implausible accentual rule, which ignores the weight of penults and invariably lays accent on antepenults. On the other hand, Benediktsson's work (1960) on Sabellic syncope seems to have played a determinative role in enforcing the thesis that Sabellic was conservative about accent location (1960: 230 n. 77). Still, Bottiglioni (1954: 23) takes a neutral stance, noting that whether Sabellic developed like Latin in terms of accent location and quality remains unclear due to the scant amount of data.

The most informative and relatively reliable piece of evidence on this issue remains the orthographic device for representing long vowels. While conjectures of accent location based on sound changes such as syncope and anaptyxis must be more or less indirect, the evidence of notation possesses a more direct power, given general agreement

that, because they are accented, vowels written as long are phonetically long (phonologically in initial syllables?). The idea is entertained by Thurneysen (1908: 241-242), particularly in regard to Oscan, where the long-vowel notation, that is, the gemination of vowels, appears mostly in initial syllables: e.g, **fiisnú** '*fanum*, temple' < Proto-Sab. **fēs-nā-* < PIE **dʰeh₁s-* (cf. Lat. *fānum* < Pre-Lat. **fasno-* < PIE **dʰh₁s-*); **biitam** '*vitam*, life' < PIE **gʷih₃-teh₂-* (= Lat. *vīta*); **ḍuunated** '*donavit*, gave' < Pre-Osc. **dōn-ā-* (cf. **dúnúm**, Lat. *dōnum*, *dōnāre*); **tríibarak[avúm]**, **tríbarakavúm** '*aedificare*, build' < Pre-Osc. *trēb-ark-ā-* (cf. **tríibúm** 'house'). The last two examples are particularly important for detecting the location of accent, since they historically contain two long vowels, both in the initial syllable and elsewhere; the former position shows the orthographic length. From this observation, it can be said that accent consistently fell on initial syllables, causing vowels there to become phonetically long. Potential counterexamples, such as αfααματτεδ 'order' (< **ā-fāmā-*) and **trístaamentud** '*testamento*' (< **tri-stā-mento-*; cf. **trstus** 'testes' < **tristi-*), are not so compelling as to weaken the argument, since in form they are either derivative or inflected from a composite word; hence, the gemination of the vowel in the second member can be attributed to their free-standing forms (cf. **faamat**, **faamated**, Umb. *stahmei*, *stahmito*). Modern Italian reflexes of Latin toponyms in Oscan-speaking areas seem to agree with this line of reasoning, such as *Ótranto*, *Bríndisi*, *Pésaro* (cf. Lat. *Hydrūntum* ← Gk. Ὑδροῦντ-, *Pisaurum*, *Brundisium*), as cited by Leumann (1977: 239). He explains them as caused by 'epichorische Anfangsbetonung', which I assume is referring to that of Oscan.

An orthographic device for the treatment of long vowels is also well-established in Umbrian as either ‹Vh›, ‹VhV›, or ‹VV›. This sequence is frequently found in initial syllables of either single words or the second member of composite forms: e.g., **sehmeniar (furu** ~) '*(forum) seminarium*' < Proto-Ital. **sēmen-iį̄ā-*; **sahta** '*sanctam*', *sahatam*, *Sahata* < **sānχto-* < **sank-to-* (cf. Lat. *sānctus*; see Meiser 1986: 55, 92, 106, 139); *seipodruhpei* '*seorsum utroque*' < **sē-poterōd-pid* < **-k^{w}o-terōd-*; *meersta* '*iustam*' < **mede/os-to-*. This seems parallel to the Oscan material and thus strengthens the argument that Sabellic generally preserved the old initial accent. Note that Umbrian possesses several examples in which ‹Vh› or ‹VhV› is not employed in initial syllables of simple words or the second members of composite forms, but medial syllables that are part of suffixes: e.g., *eheturstahamu* 'exterminato' < **-tudes-ā-*; **persnihmu** 'precator' < Proto-Sab. **perk-sk-įn-į-* (see 4.3.2); *anouihimu* 'induitor' < **-oų-į-* < PItal. **-oų-ē-* < PIE **-oų-eį̄e/o-*. But the appearance of ‹Vh›/‹VhV› is limited before the imperative marker *-mōd*, which might have served as an enclitic and dragged accent onto the penult (cf. *seipodruhpei* 'seorsum utroque' < **sē-poterōd-pid*); as a result, these examples cannot be regarded as sufficient evidence for accent shift.[10]

The fact that vowel reduction and deletion in Italic are overwhelmingly common in non-initial syllables enables us to securely posit the initial-stress *system*. Even if it is not stress itself, but lack thereof, triggers a decrease of duration in the articulation of vowels, and is thus responsible for their reduction and deletion, the relationship between the

[10] Meiser (1986: 33), based on the observation that "Verdumpfung" in **tesenakes** ~ *tesenocir* ~ *tesonocir* (see 4.5) is seen in non-initial syllables, argues that initial stress was retained in a much later stage of Umbrian. But this quadrisyllabic word is reminiscent of the *fácilius*-type accentuation in Pre-Classical Latin, if the vowel in the initial syllable is short. Other types of words might behave differently with regard to accent location.

location of stress and these phonological changes is straightforward. We might be able to entertain some sporadic sound change in Sabellic that suggests a shift in stress position, but we do not have enough evidence to visualize a new *system*.

References

Abbreviations

- Collections of ancient texts and inscriptions (mentioned in the text):
CIL = *Corpus Inscriptionum Latinarum*; **K** = Keil 1855; **L** = Lindsay 1933; **LIV** = Rix, ed. 2001; **Pocc** = Poccetti 1979; **Sk** = Skutsch 1985; **trag.** = Ribbeck 1897; **Ve** = Vetter 1953.

- Books:
Dialects of Italy = 1997. *The Dialects of Italy*, ed. M. Maiden and M. Parry. London/New York: Routledge.
Cambridge Encyclopedia = 2004. *The Cambridge Encyclopedia of the World's Ancient Languages*, ed. R. D. Woodard. Cambridge/New York: Cambridge University Press.
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Fs Murpugo Davies = 2004. *Indo-European Perspectives: Studies in Honour of Anna Murpurgo Davies*, ed. J. H. W. Penney. Oxford: Oxford University Press.

- Periodicals:
AGI = *Archivio Glottologico Italiano*; ***ALL*** = *Archiv für lateinische Lexikographie und Grammatik mit Einschluss des älteren Mittellateins*, ed. E. Wölfflin; ***BSL*** = *Bulletin de la Société de Linguistique de Paris*; ***CPh*** = *Classical Philology*; ***CR*** = *The Classical Review*; ***HS*** = *Historische Sprachforschung*; ***IF*** = *Indogermanische Forschungen*; ***JASA*** = *The Journal of the Acoustical Society of America*; ***JIES*** = *The Journal of Indo-European Studies*; ***KZ*** = *Zeitschrift für vergleichende Sprachforschung*; ***MSL*** = *Mémoires de la Sociéte de Linguistique de Paris*; ***RPh*** = *Revue de philologie, de littérature et d'histoire anciennes*; ***SE*** = *Studi Etruschi*; ***TPhS*** = *Transactions of the Philological Society*.

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